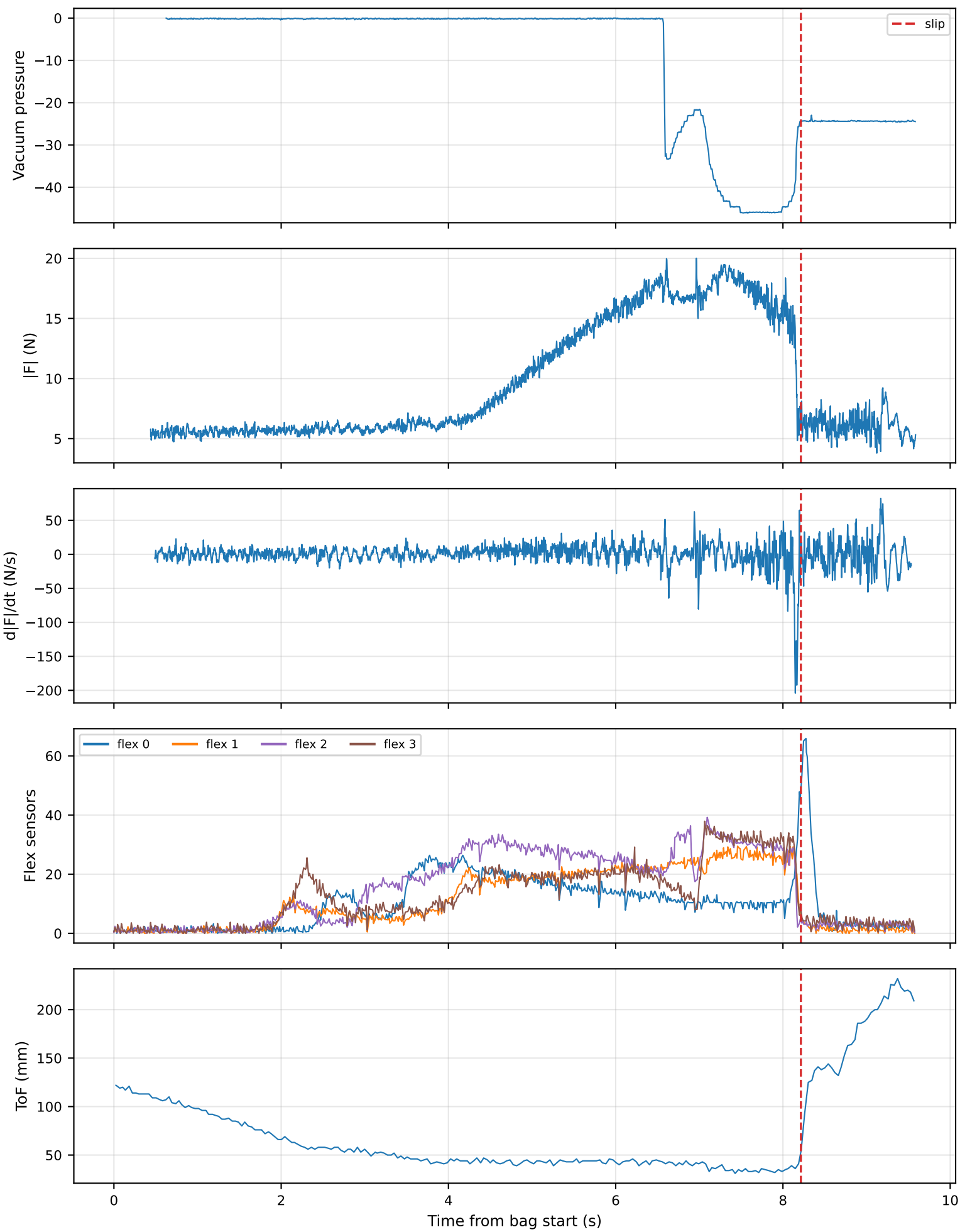
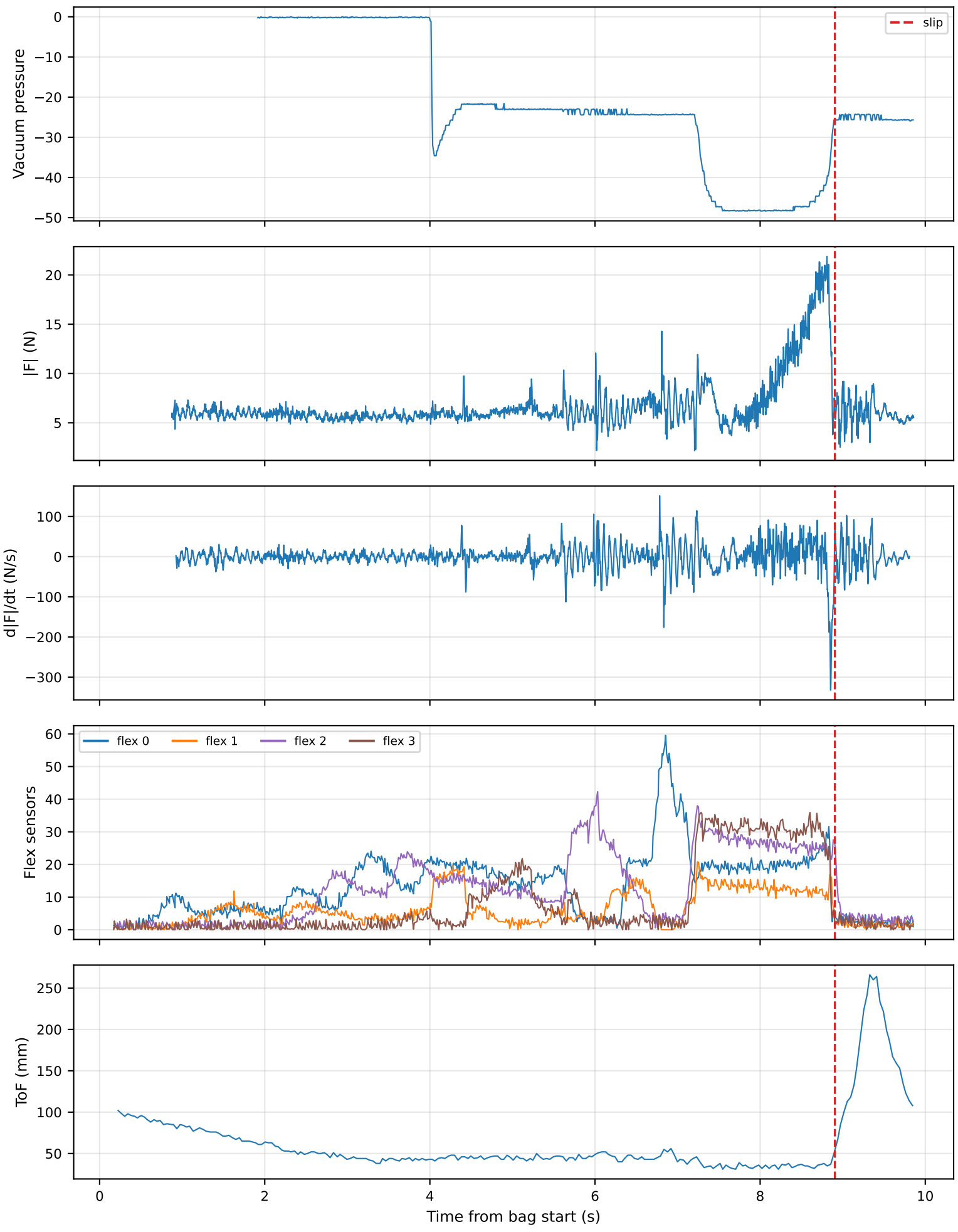
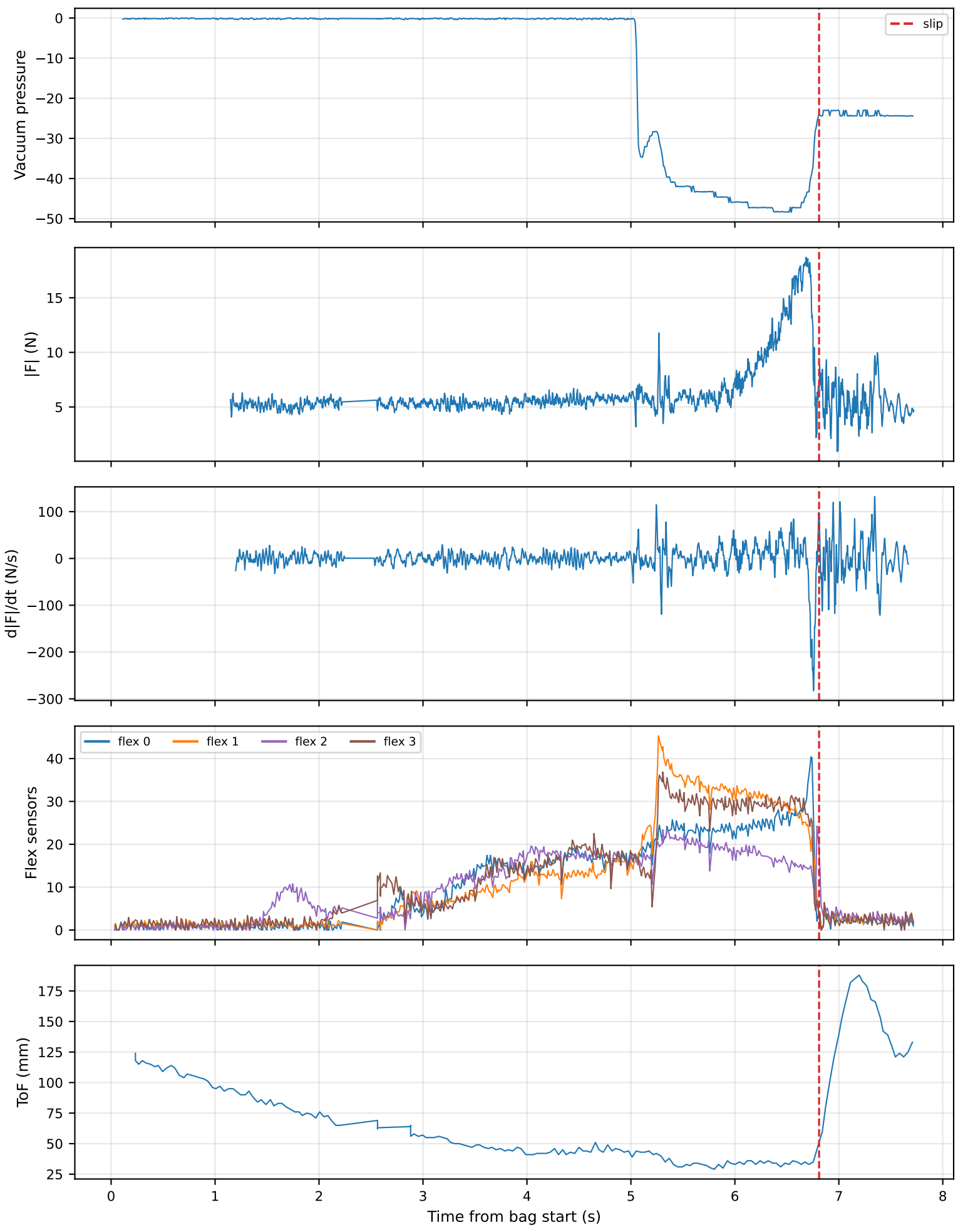
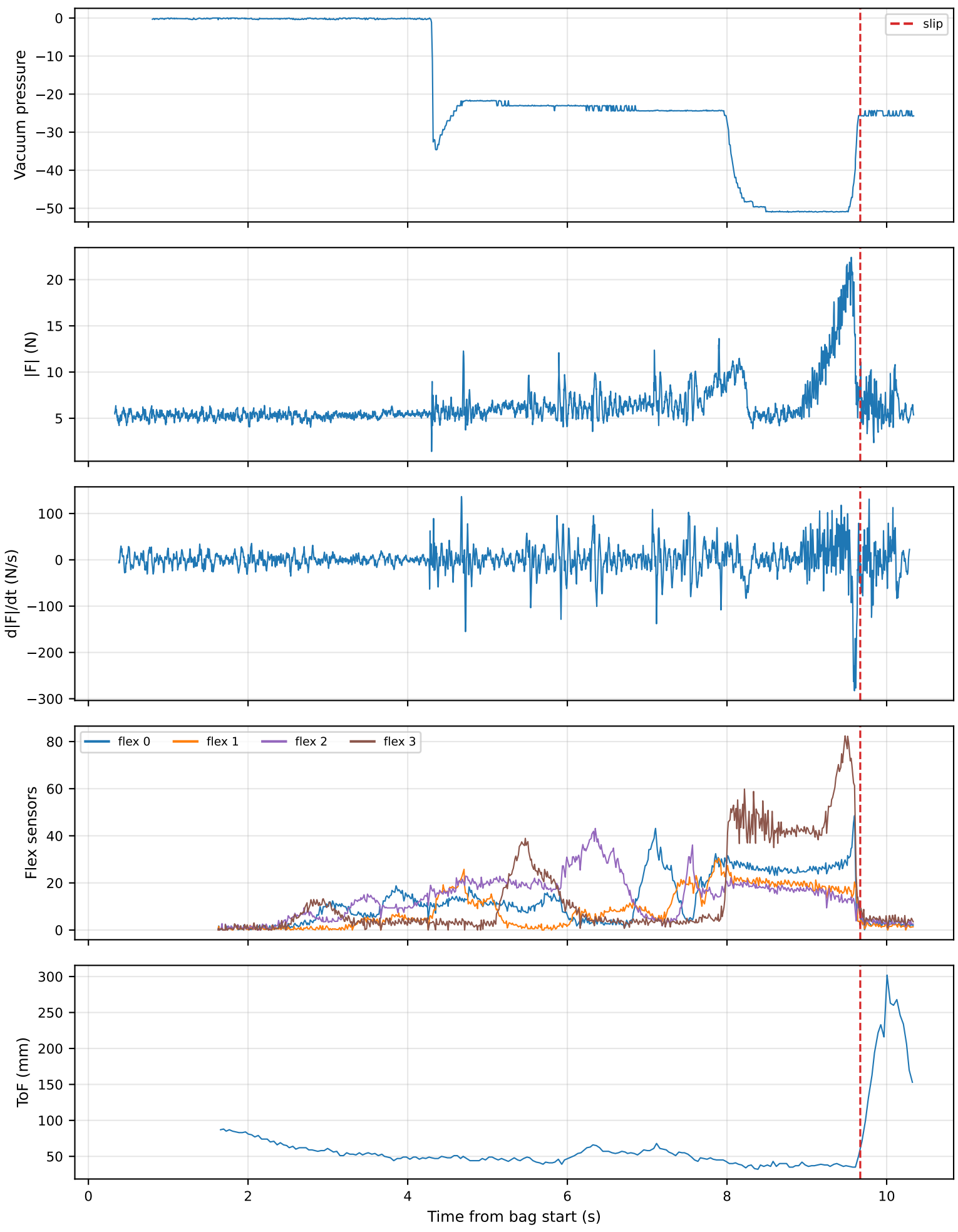
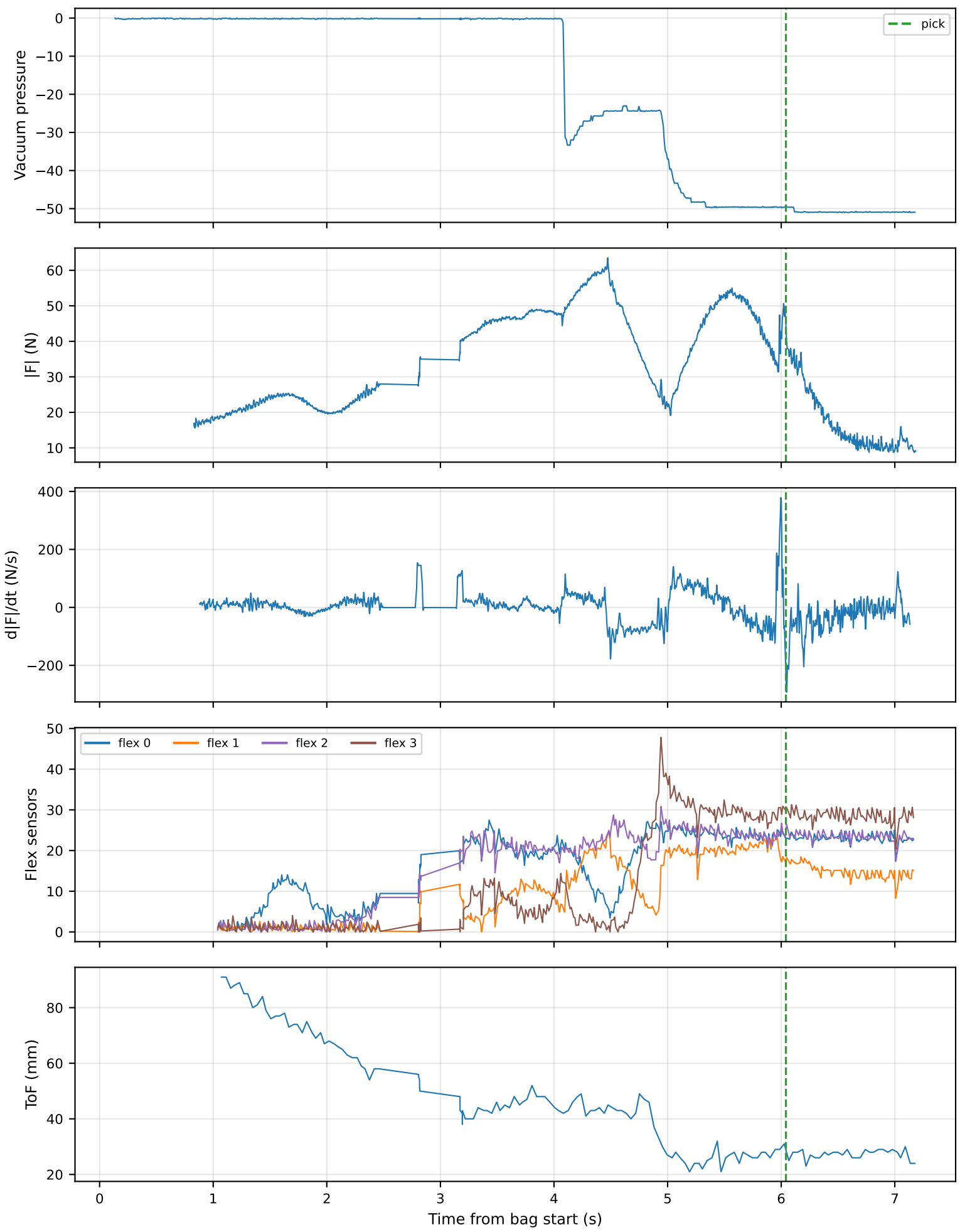
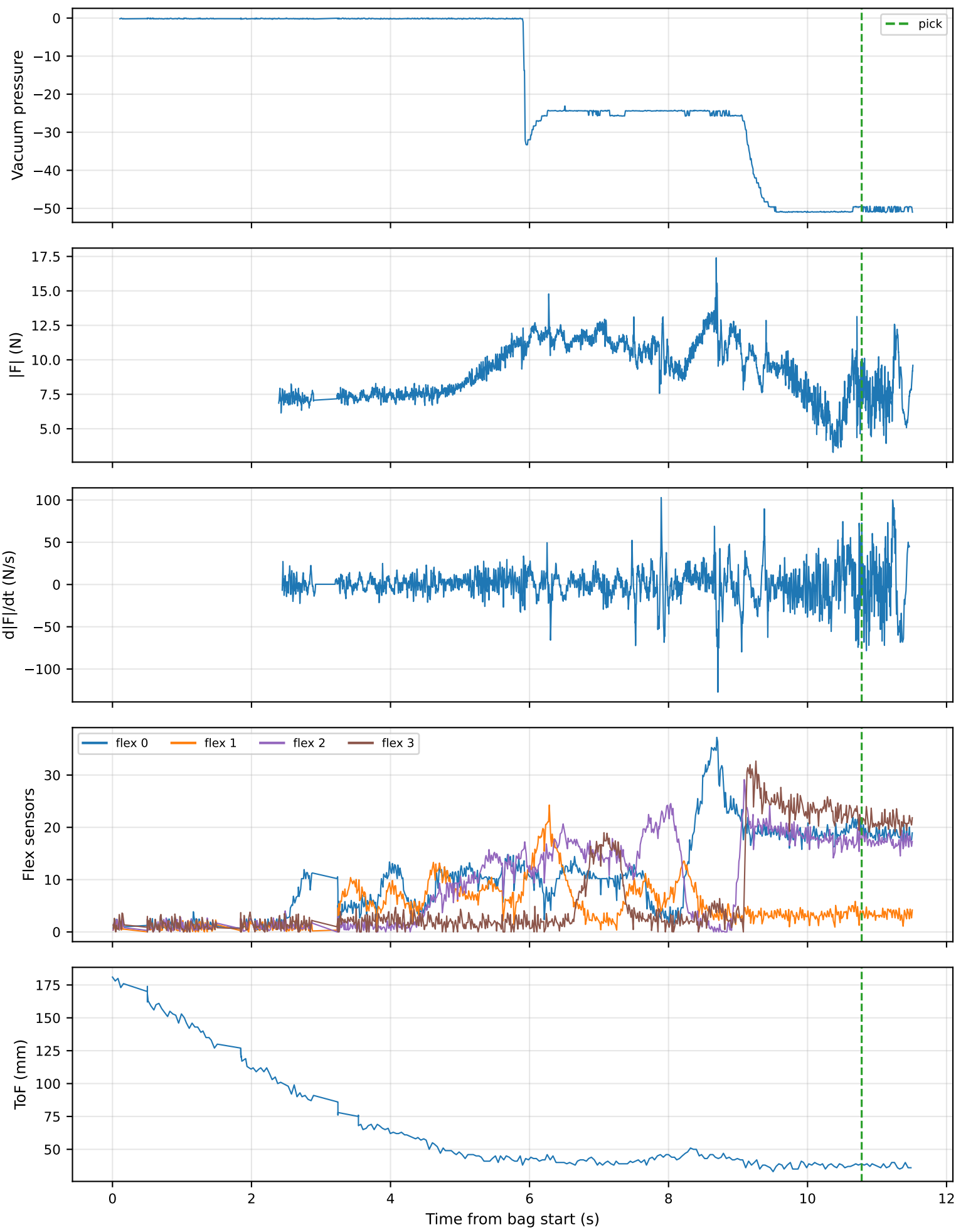

final_approach_and_pick_20261006_094454.db3
SLIP at 8.21 s | 2026-10-06 09:44:55 | batch 32, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

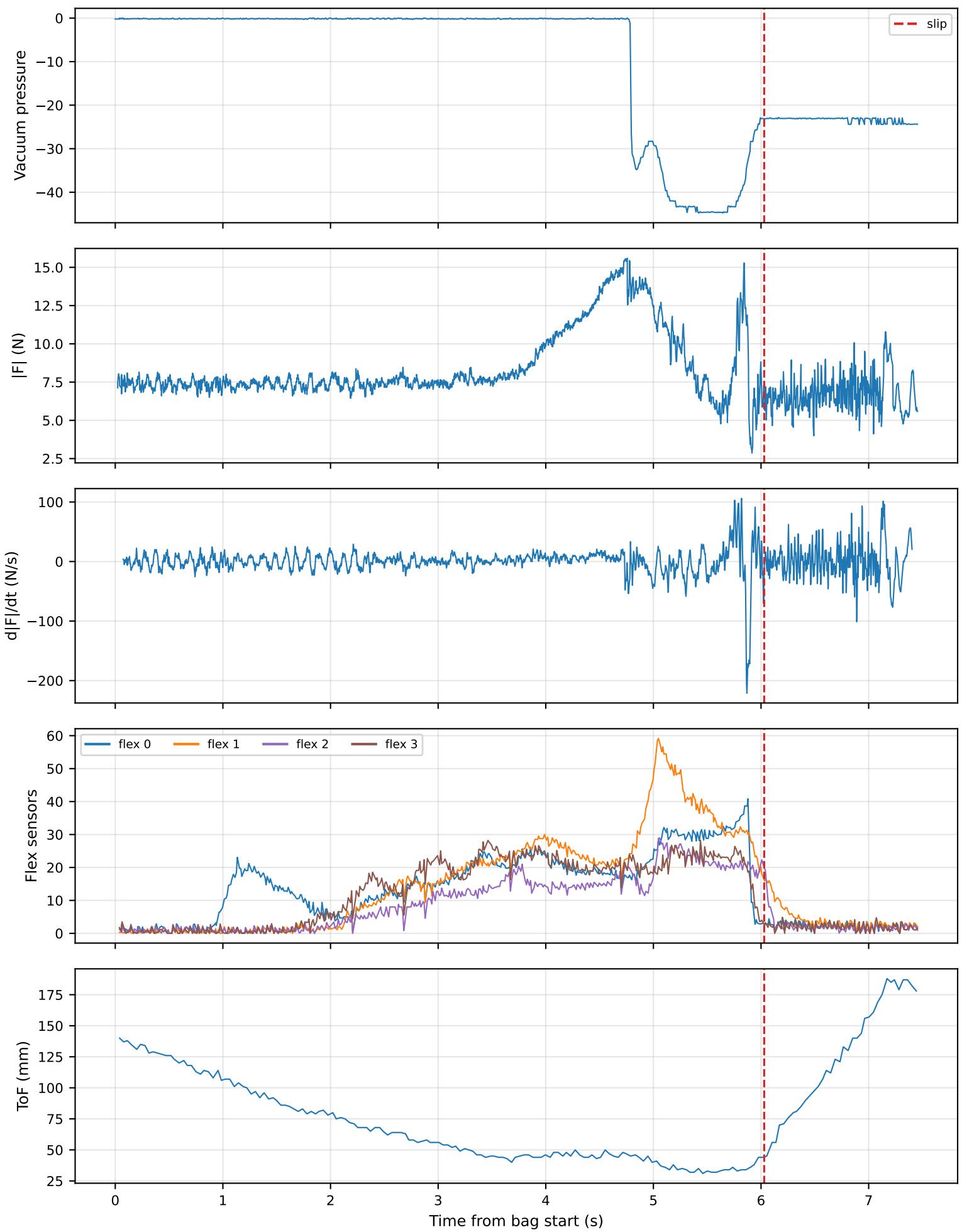

final_approach_and_pick_20261006_094615.db3
SLIP at 8.91 s | 2026-10-06 09:46:16 | batch 33, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

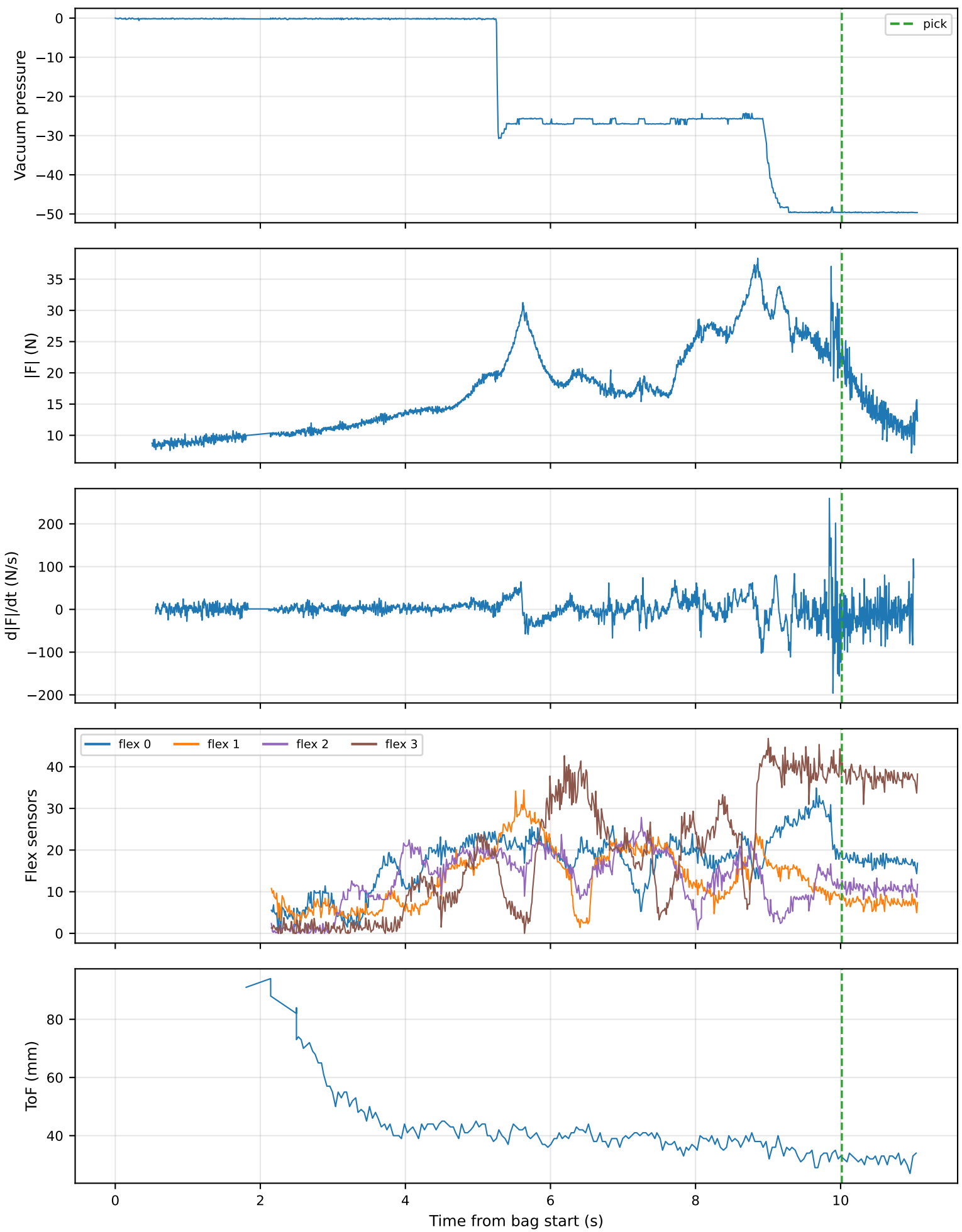

final_approach_and_pick_20261006_094815.db3
SLIP at 6.81 s | 2026-10-06 09:48:17 | batch 34, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

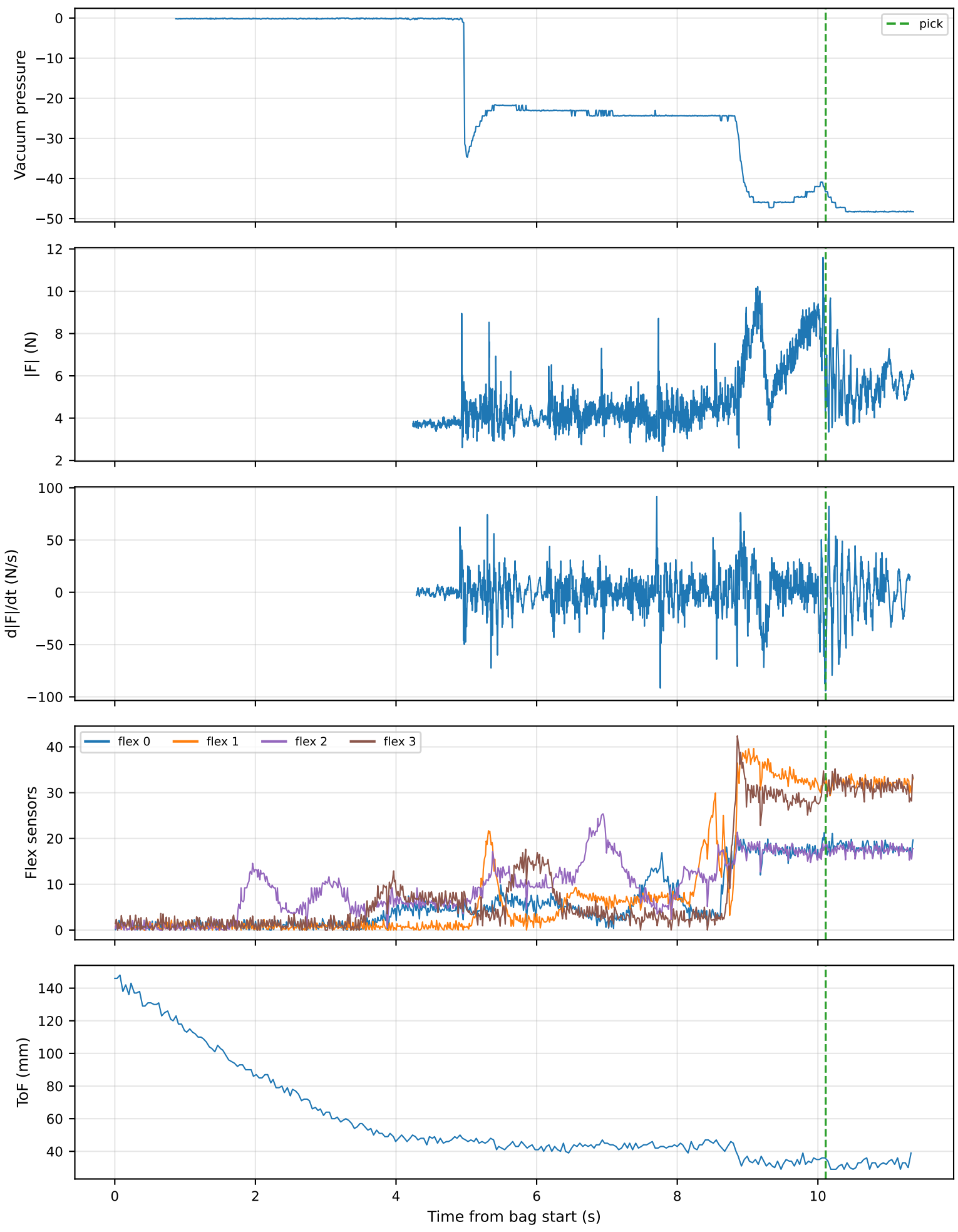

final_approach_and_pick_20261006_094935.db3
SLIP at 9.67 s | 2026-10-06 09:49:36 | batch 35, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

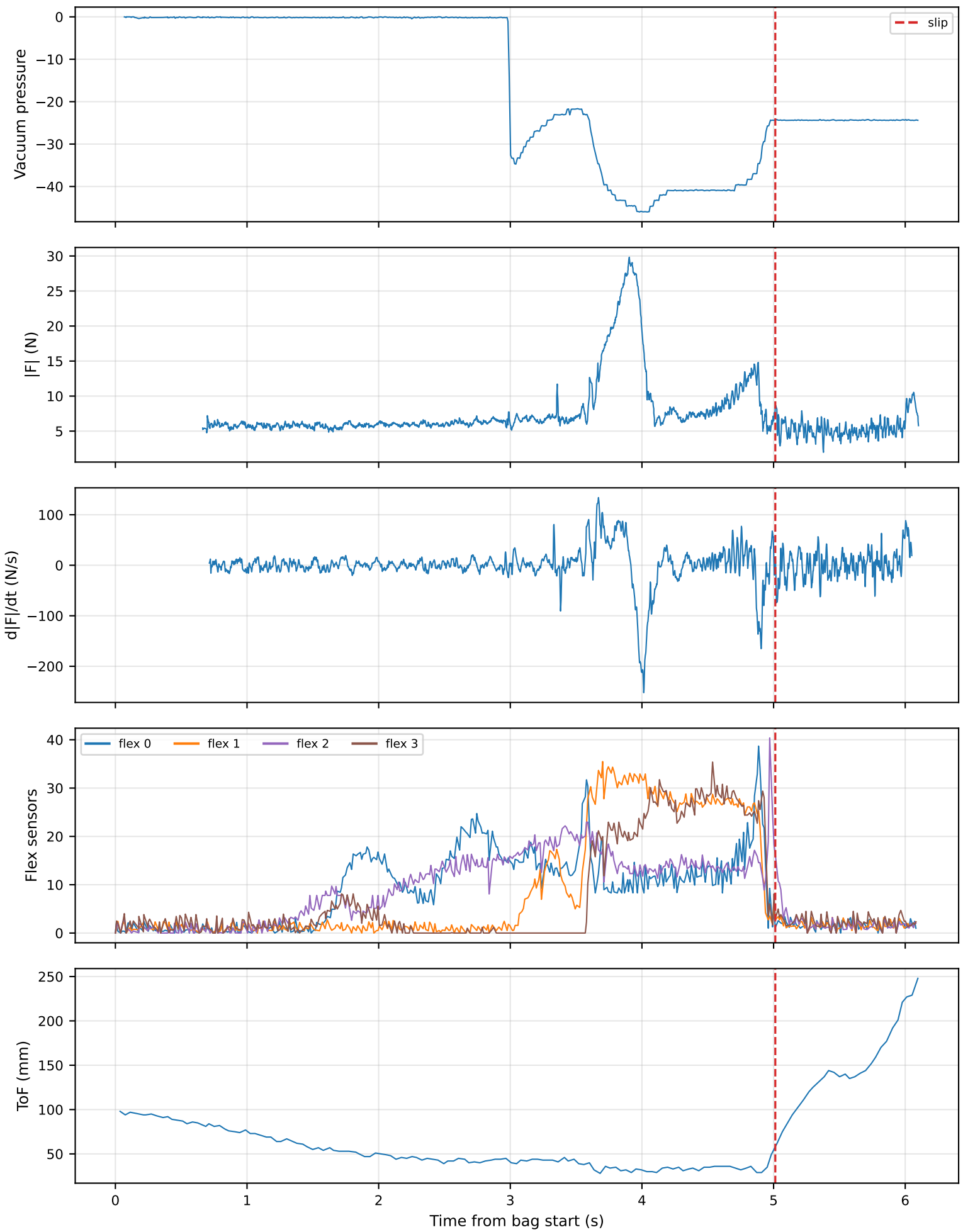

final_approach_and_pick_20261006_095220.db3
PICK at 6.04 s | 2026-10-06 09:52:21 | batch 37, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

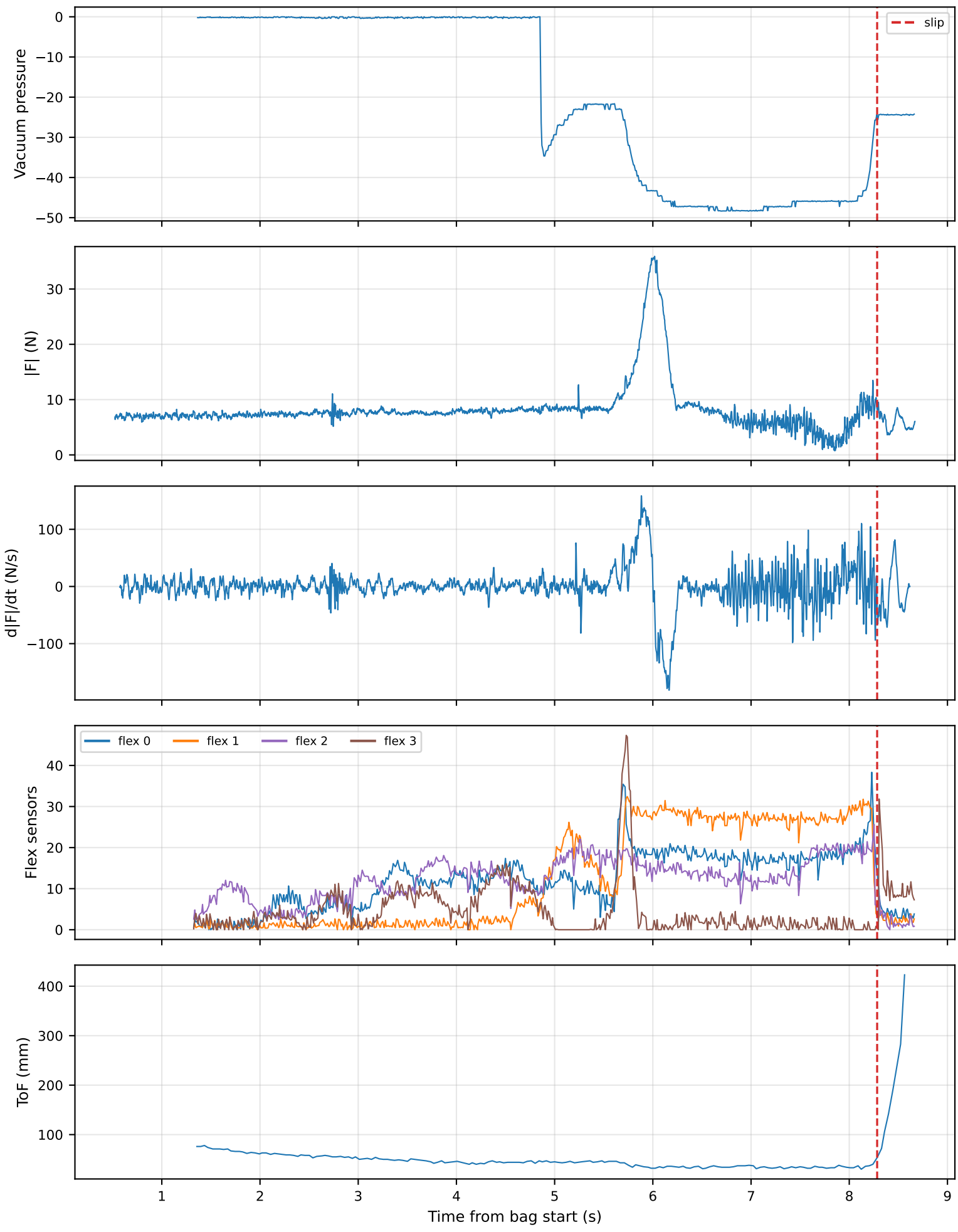

final_approach_and_pick_20261006_095653.db3
PICK at 10.78 s | 2026-10-06 09:56:54 | batch 39, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

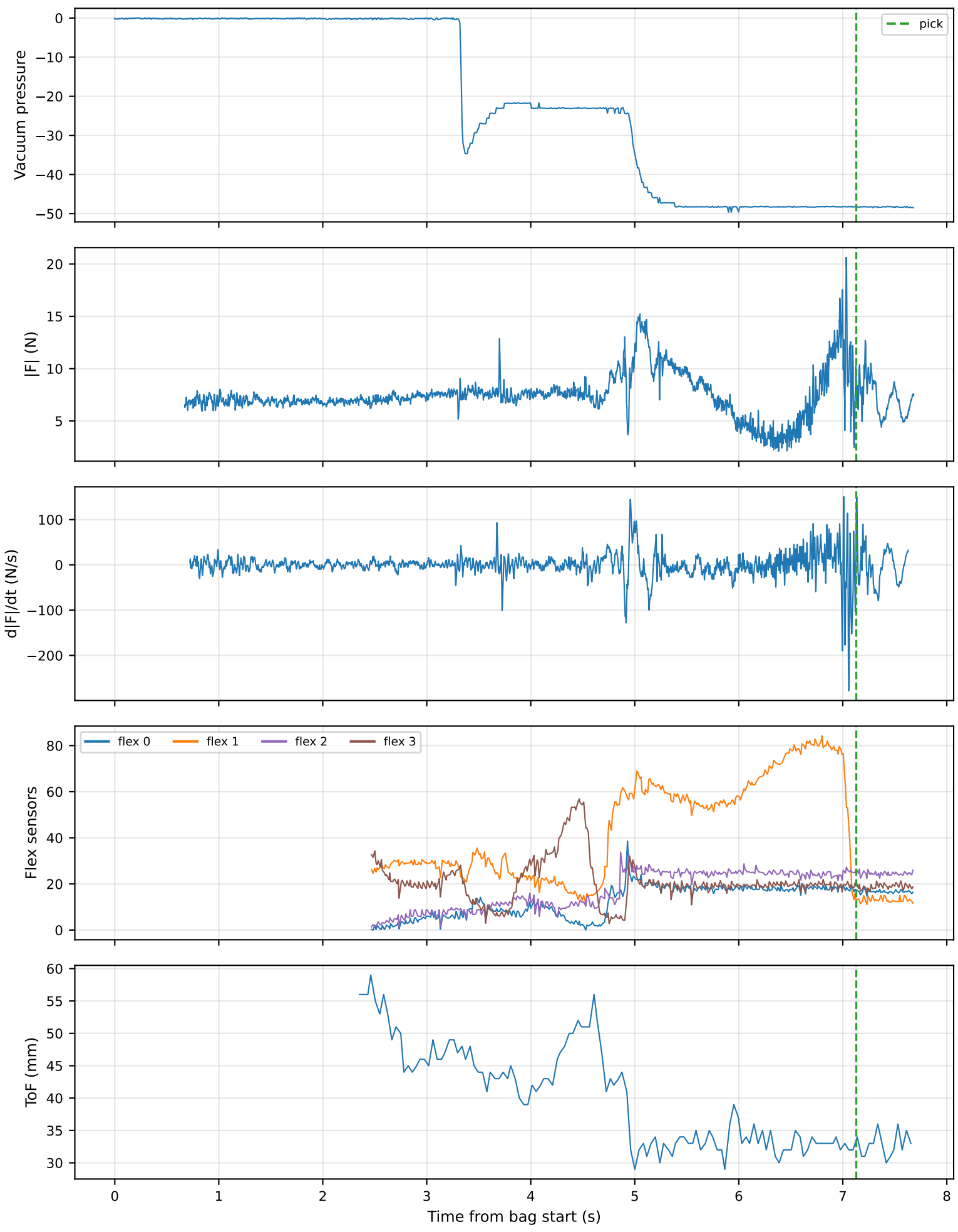

final_approach_and_pick_20261006_095937.db3
SLIP at 6.03 s | 2026-10-06 09:59:39 | batch 41, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

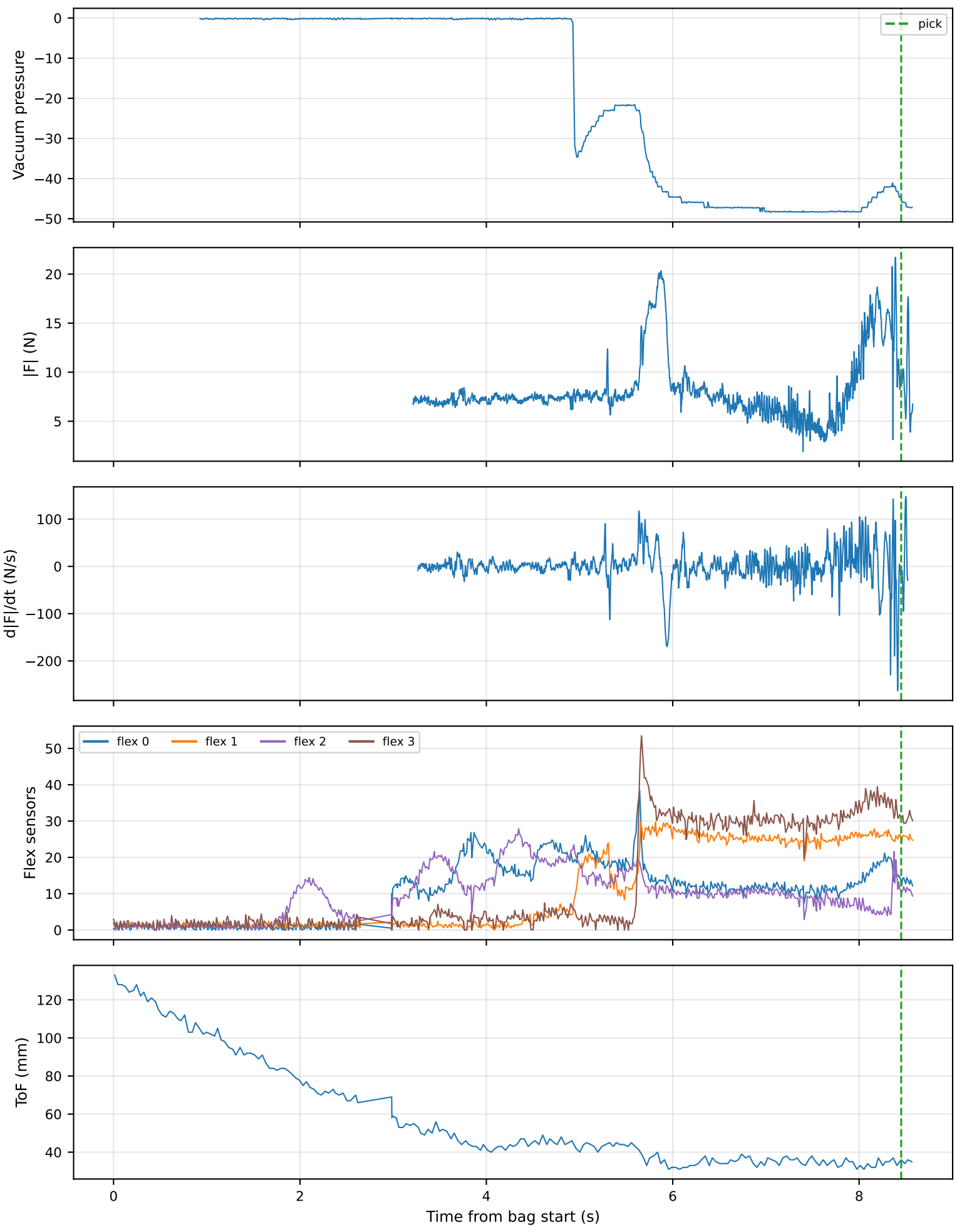

final_approach_and_pick_20261006_100237.db3
PICK at 10.02 s | 2026-10-06 10:02:38 | batch 43, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

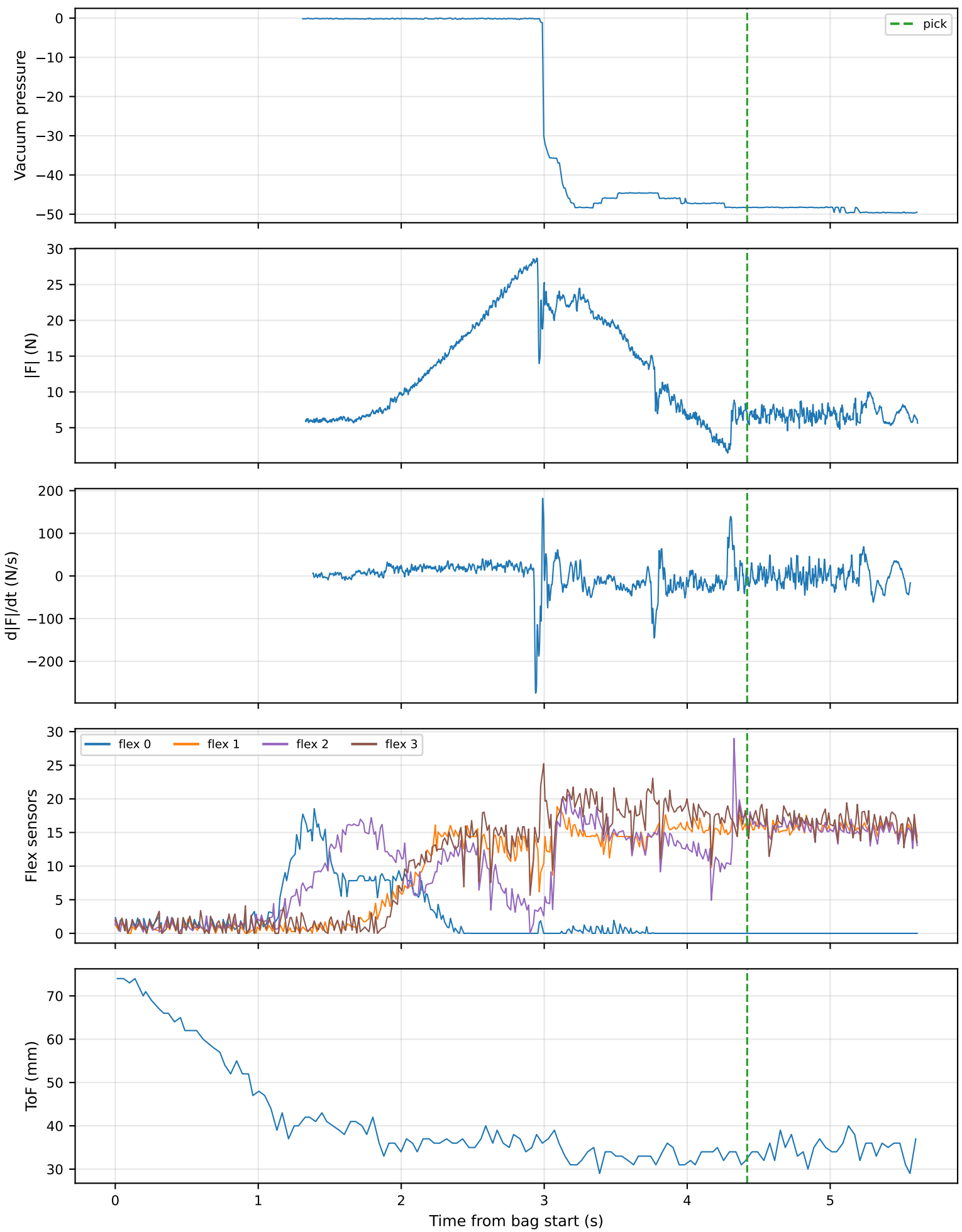

final_approach_and_pick_20261006_100419.db3
PICK at 10.11 s | 2026-10-06 10:04:20 | batch 44, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_100814.db3
SLIP at 5.01 s | 2026-10-06 10:08:15 | batch 47, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_100927.db3
SLIP at 8.28 s | 2026-10-06 10:09:29 | batch 48, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

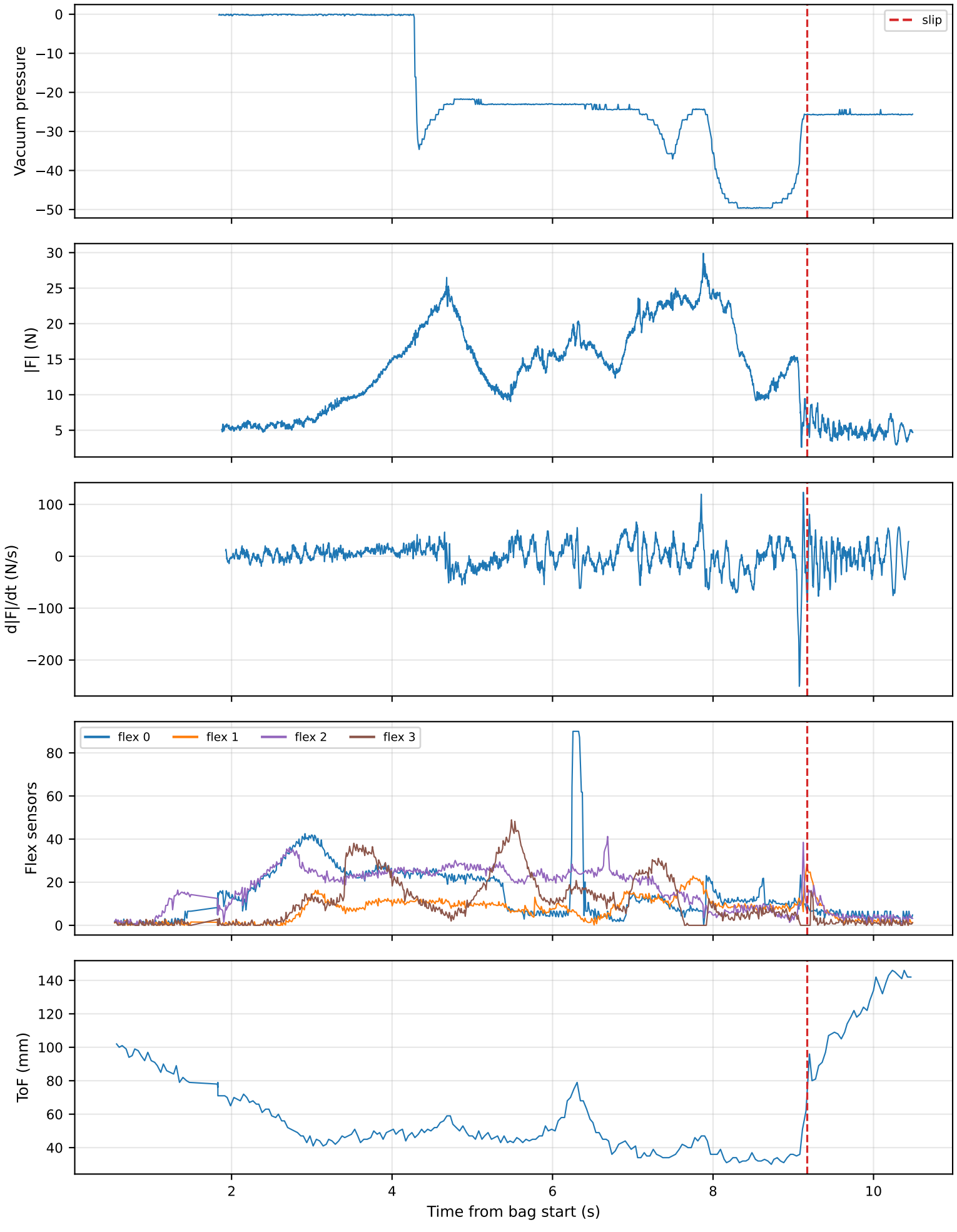

final_approach_and_pick_20261006_101053.db3
PICK at 7.13 s | 2026-10-06 10:10:54 | batch 49, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

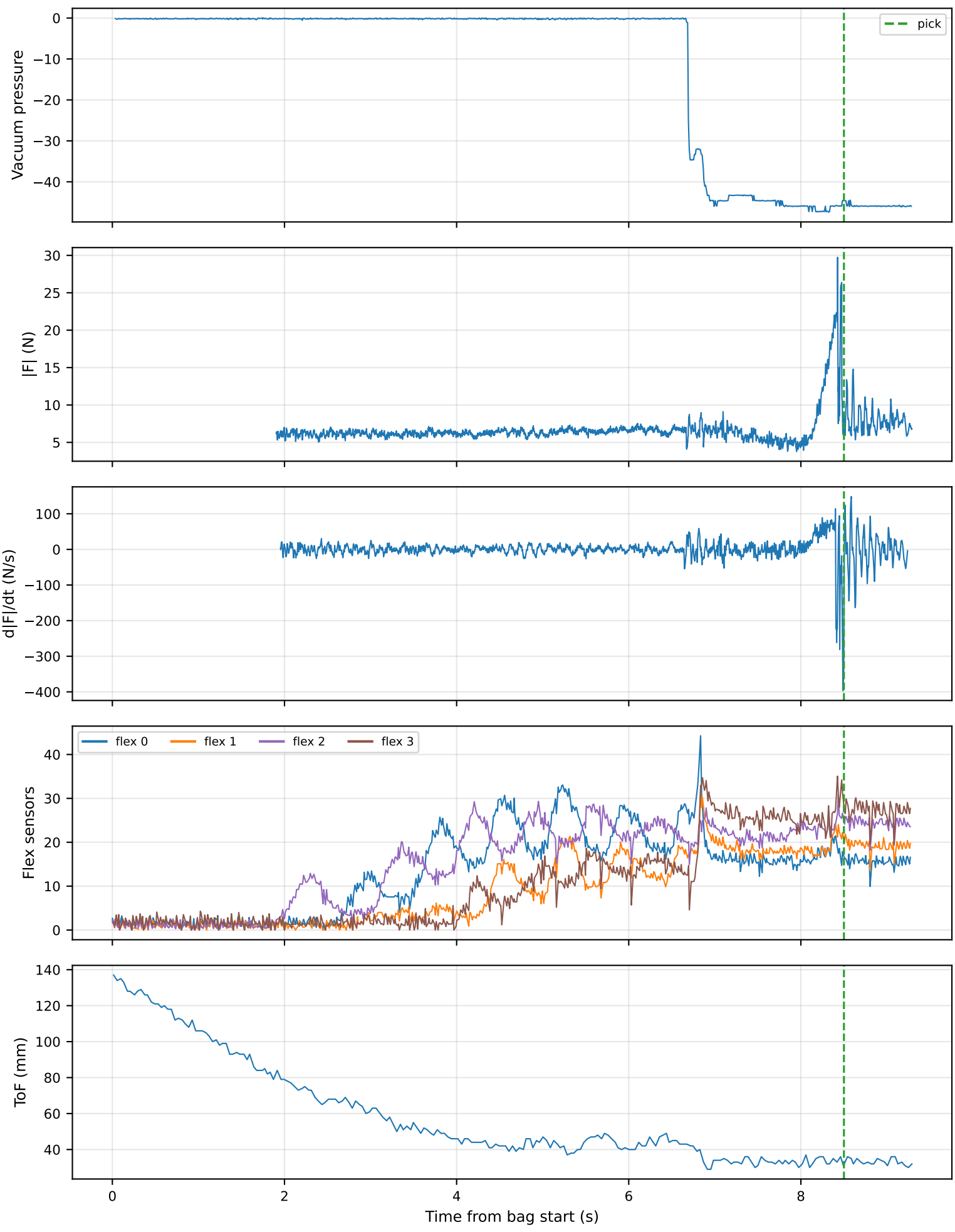

final_approach_and_pick_20261006_101153.db3
PICK at 8.45 s | 2026-10-06 10:11:54 | batch 50, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

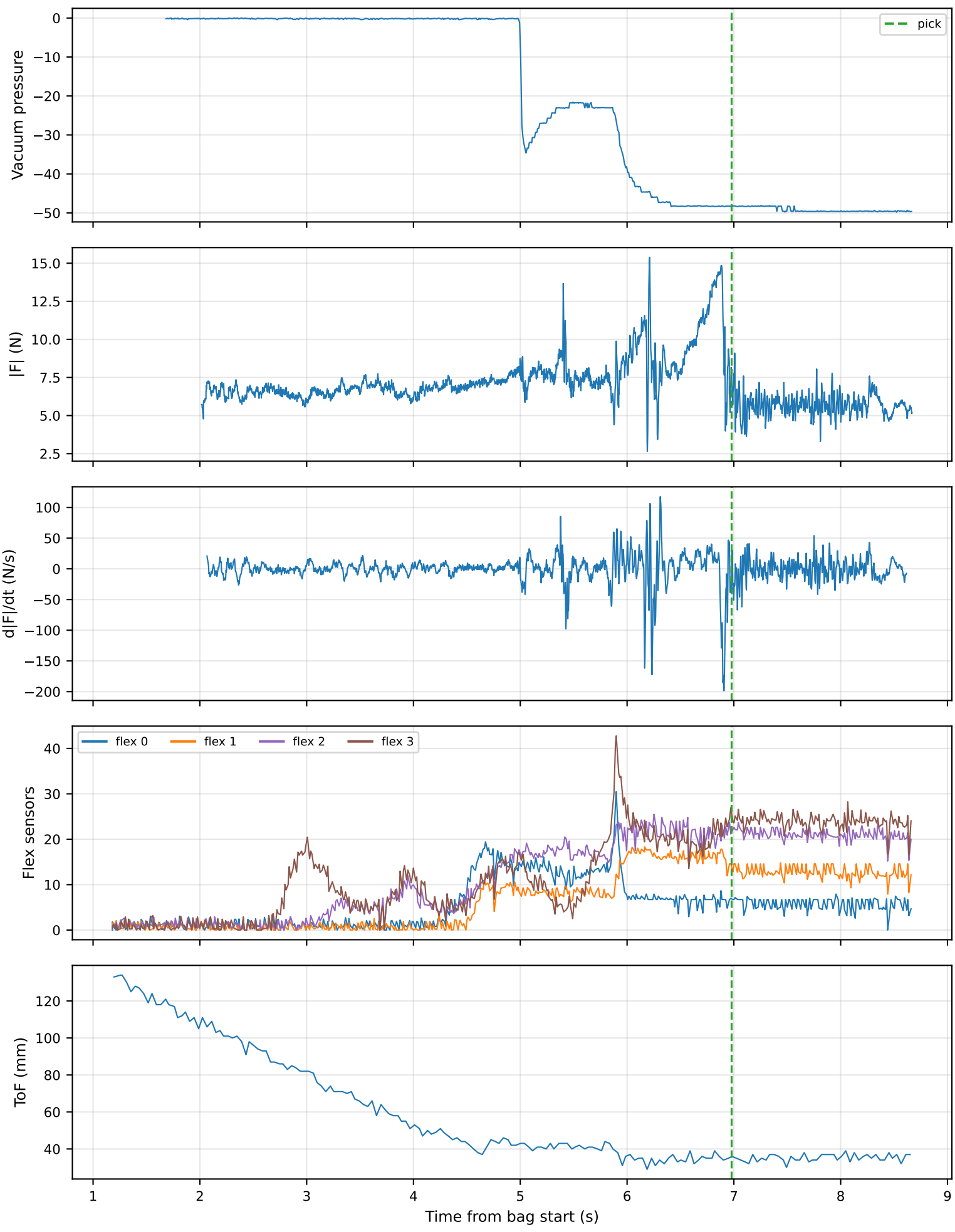

final_approach_and_pick_20261006_101308.db3
PICK at 4.42 s | 2026-10-06 10:13:09 | batch 51, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

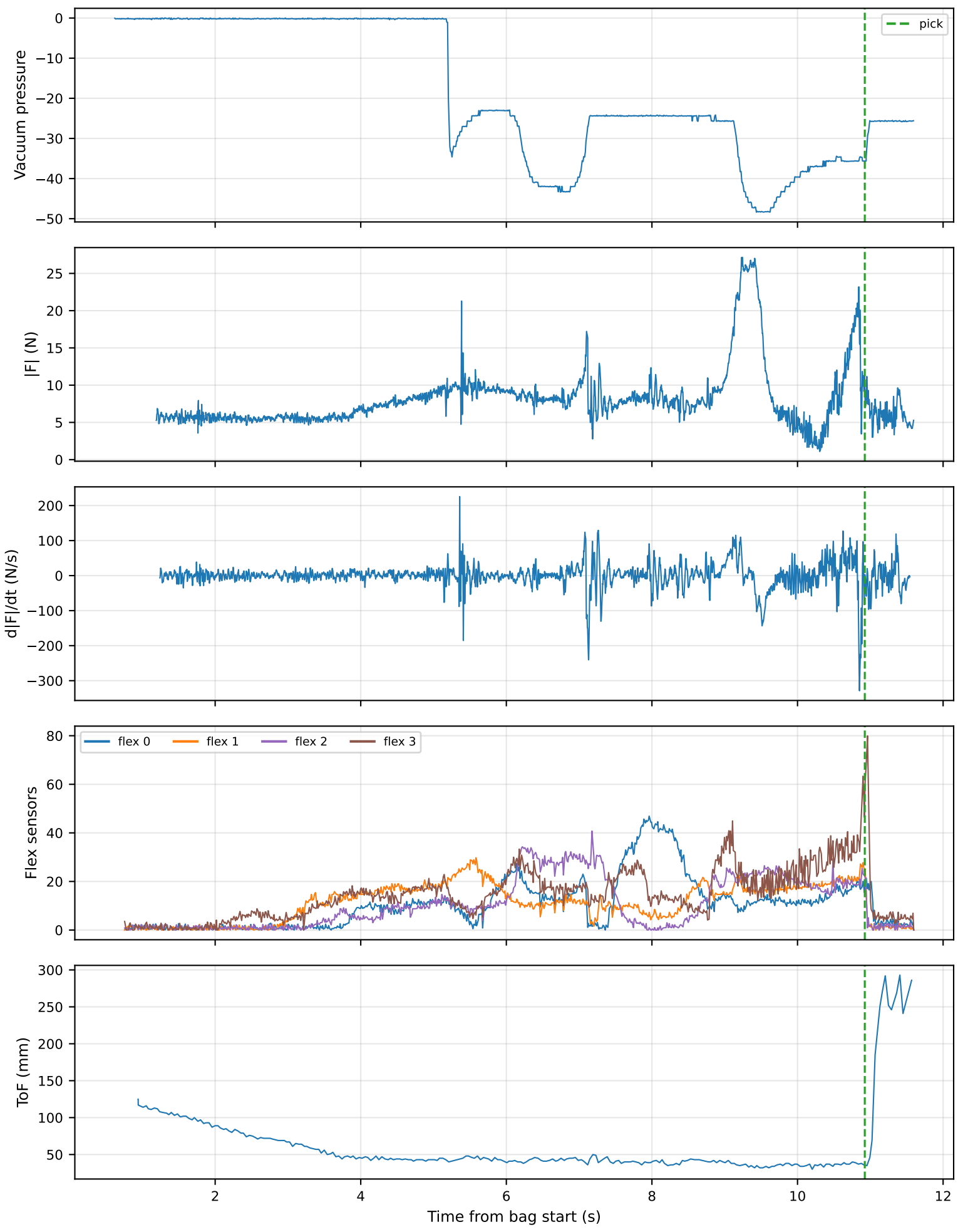
final_approach_and_pick_20261006_101451.db3

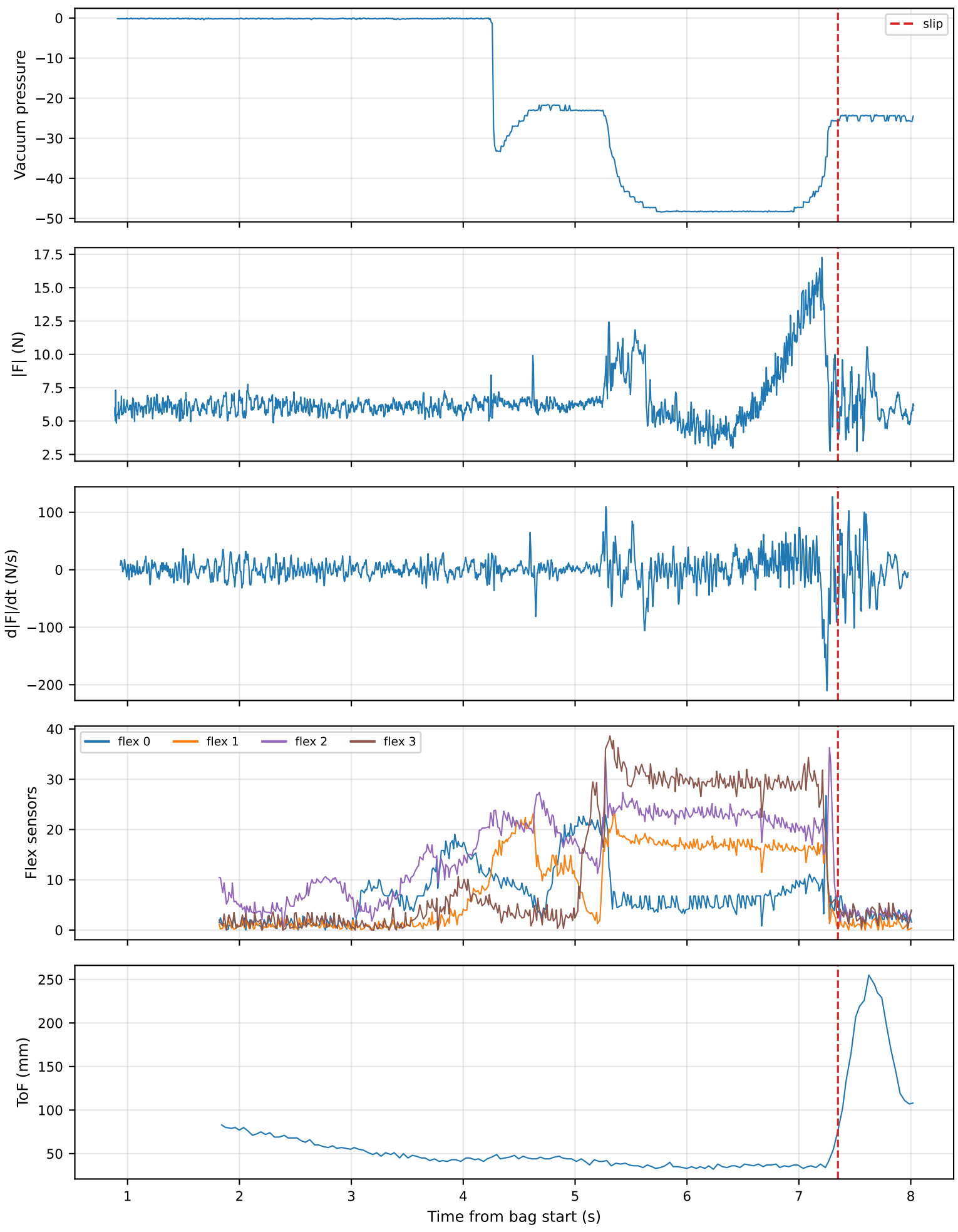
SLIP at 9.17 s | 2026-10-06 10:14:52 | batch 52, apple 0 | pull_twist_controller

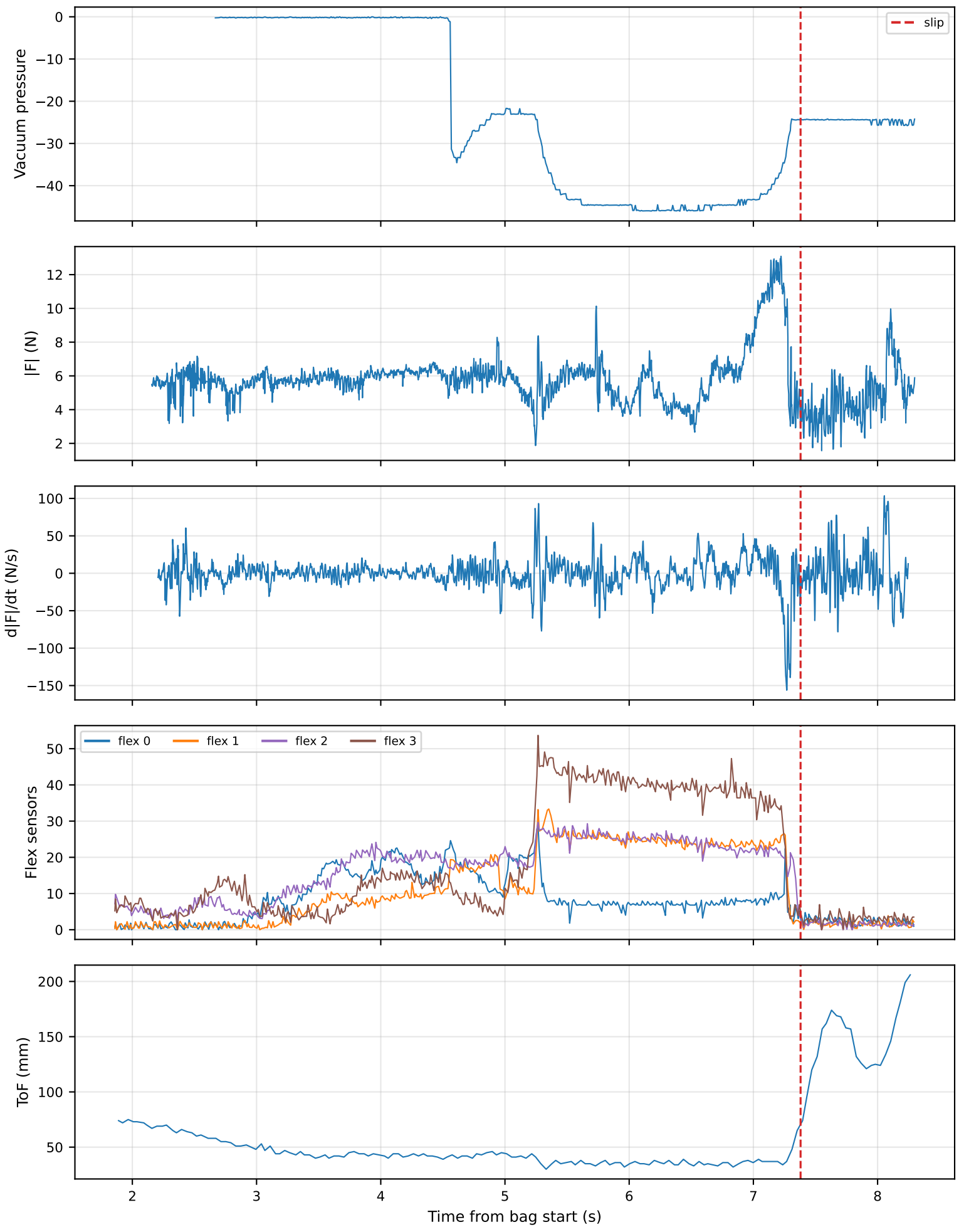


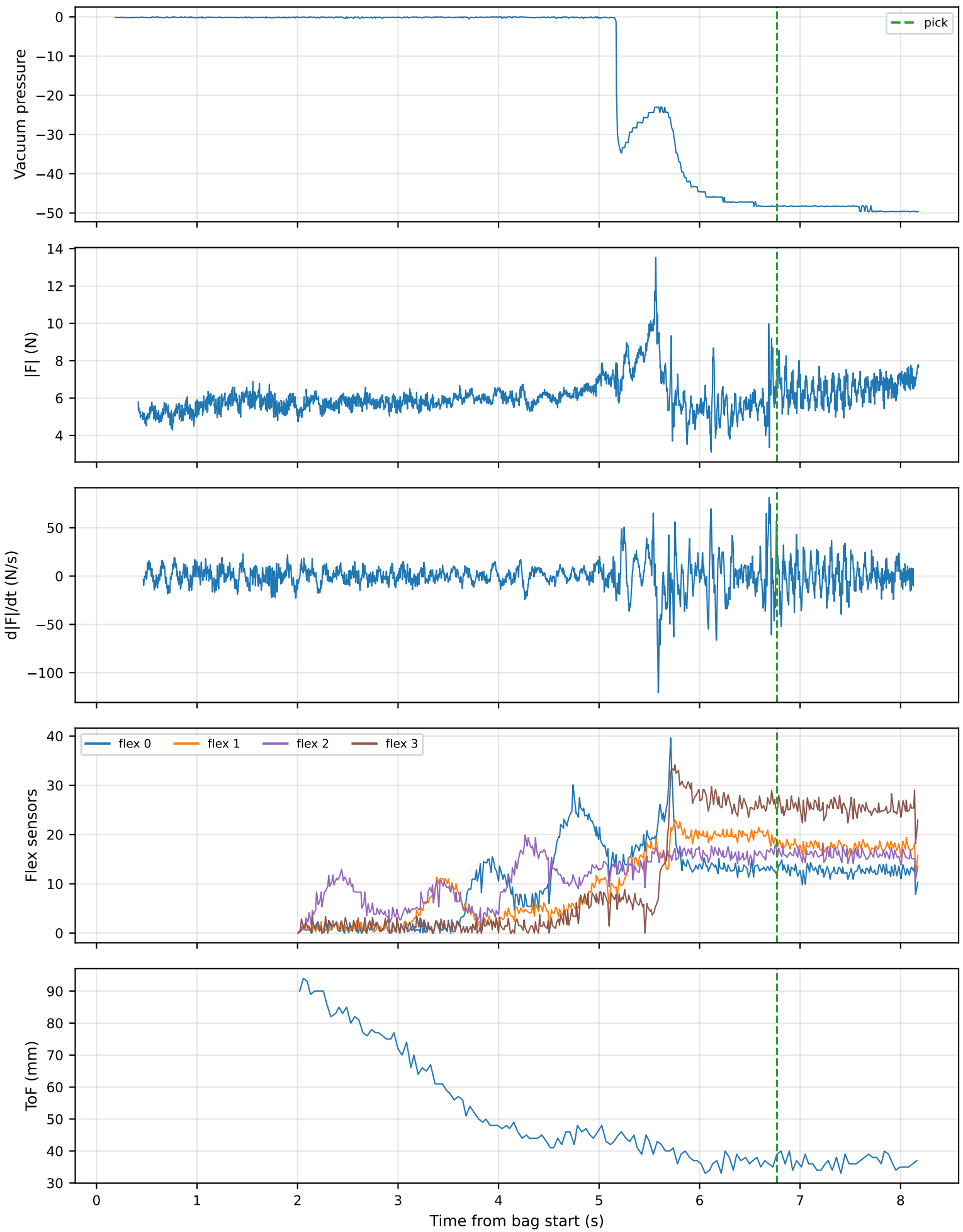

final_approach_and_pick_20261006_102227.db3
PICK at 8.50 s | 2026-10-06 10:22:28 | batch 55, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

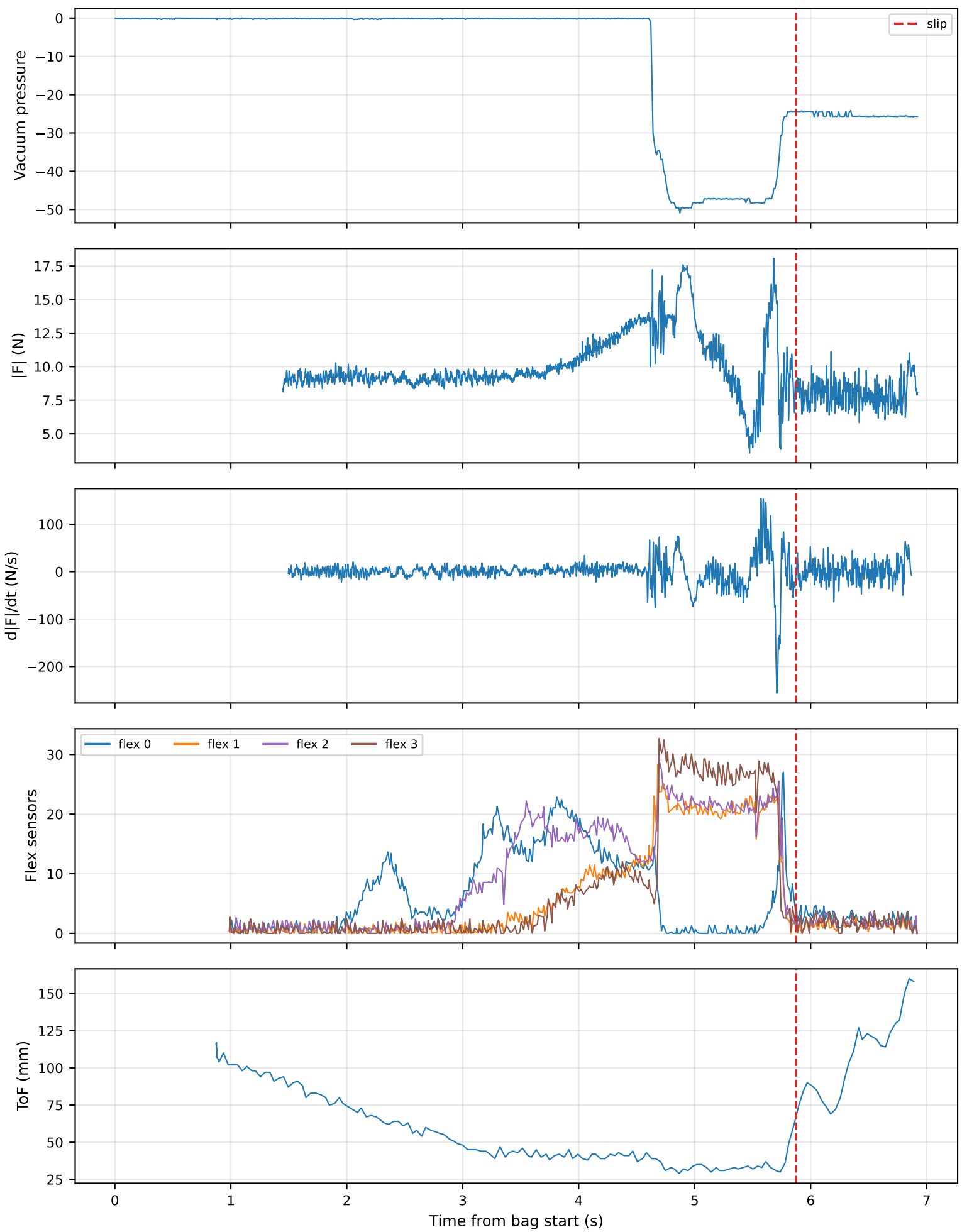

final_approach_and_pick_20261006_102359.db3
PICK at 6.98 s | 2026-10-06 10:24:00 | batch 56, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

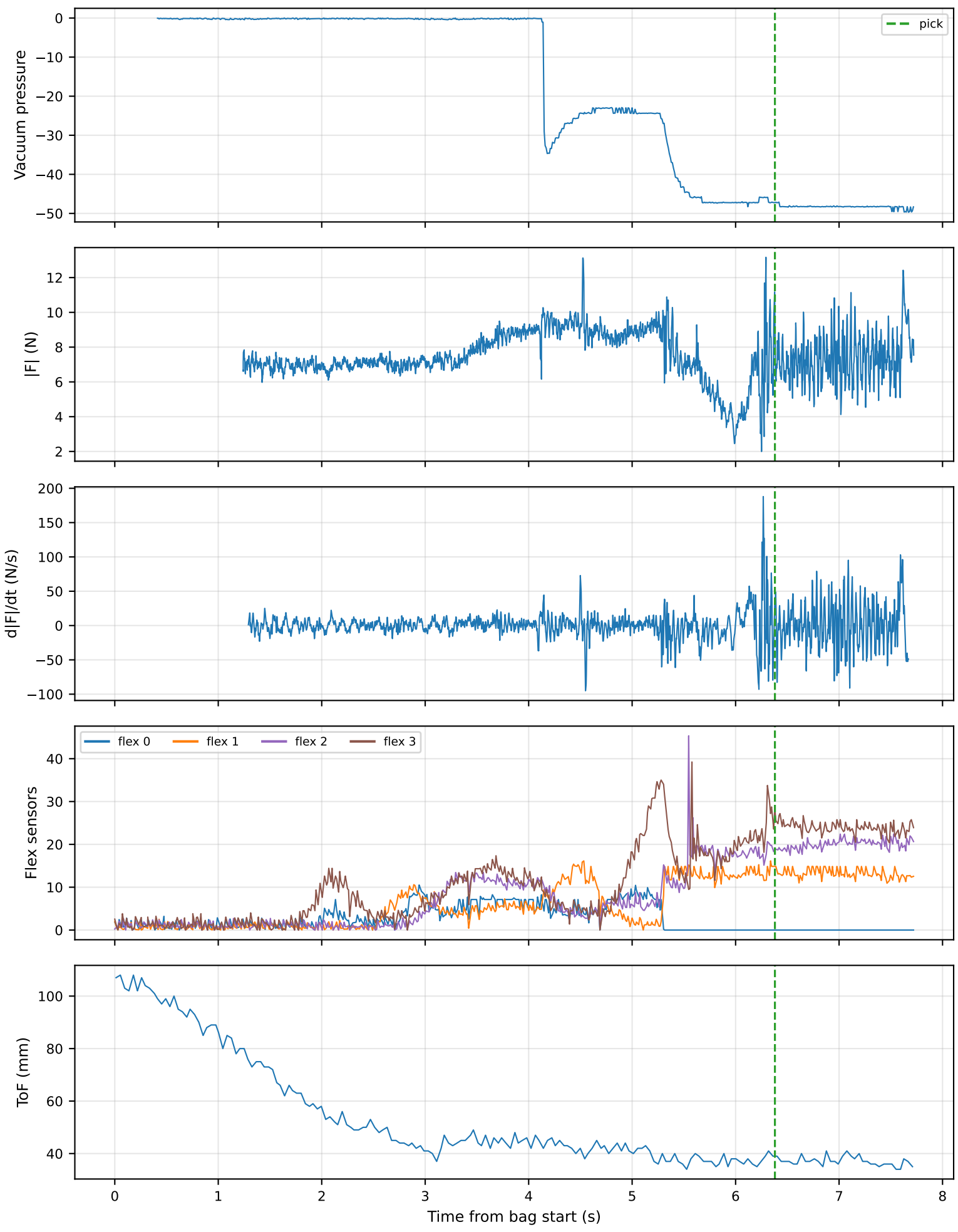

final_approach_and_pick_20261006_102615.db3
PICK at 10.92 s | 2026-10-06 10:26:17 | batch 58, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

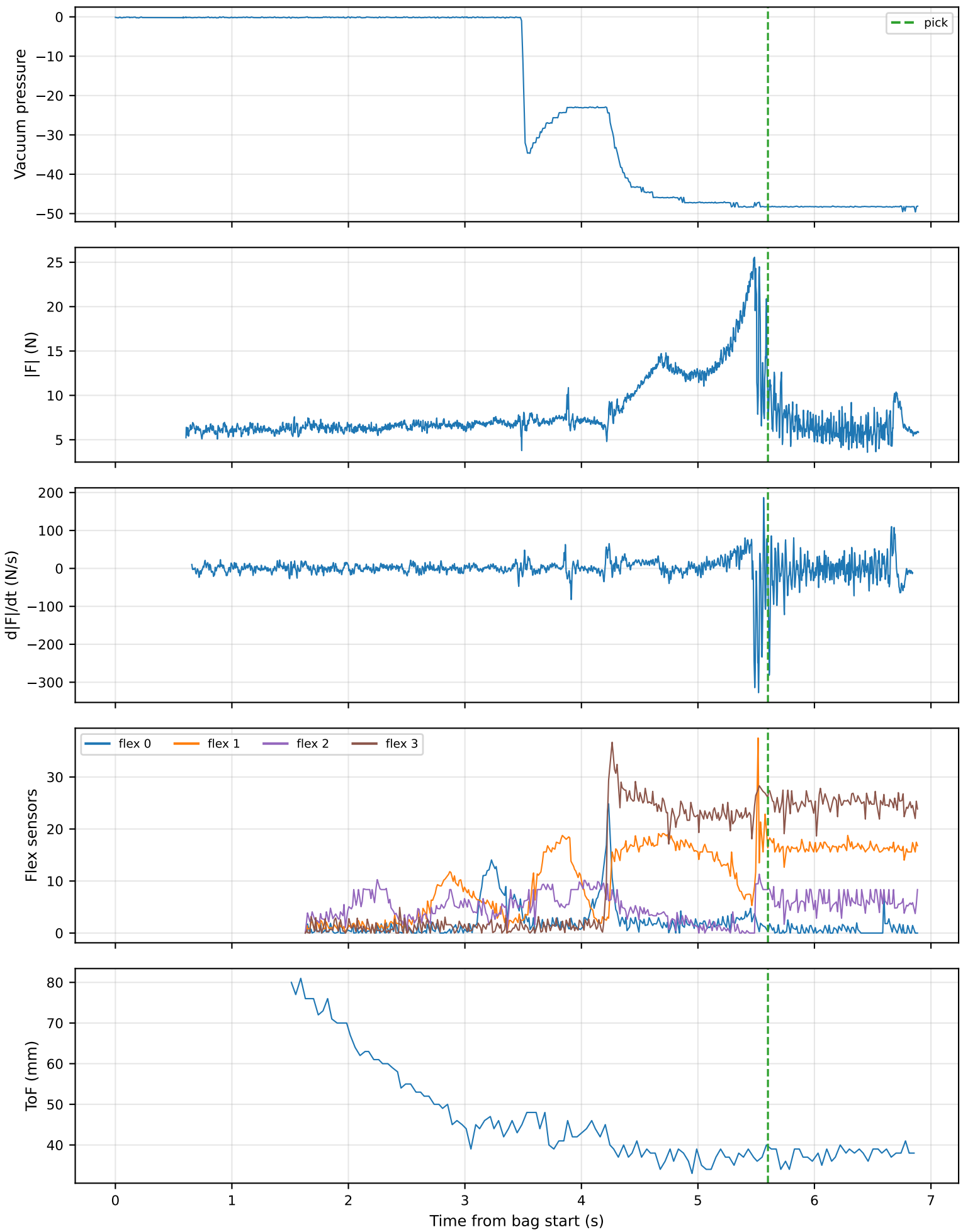

final_approach_and_pick_20261006_102745.db3
SLIP at 7.35 s | 2026-10-06 10:27:46 | batch 59, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

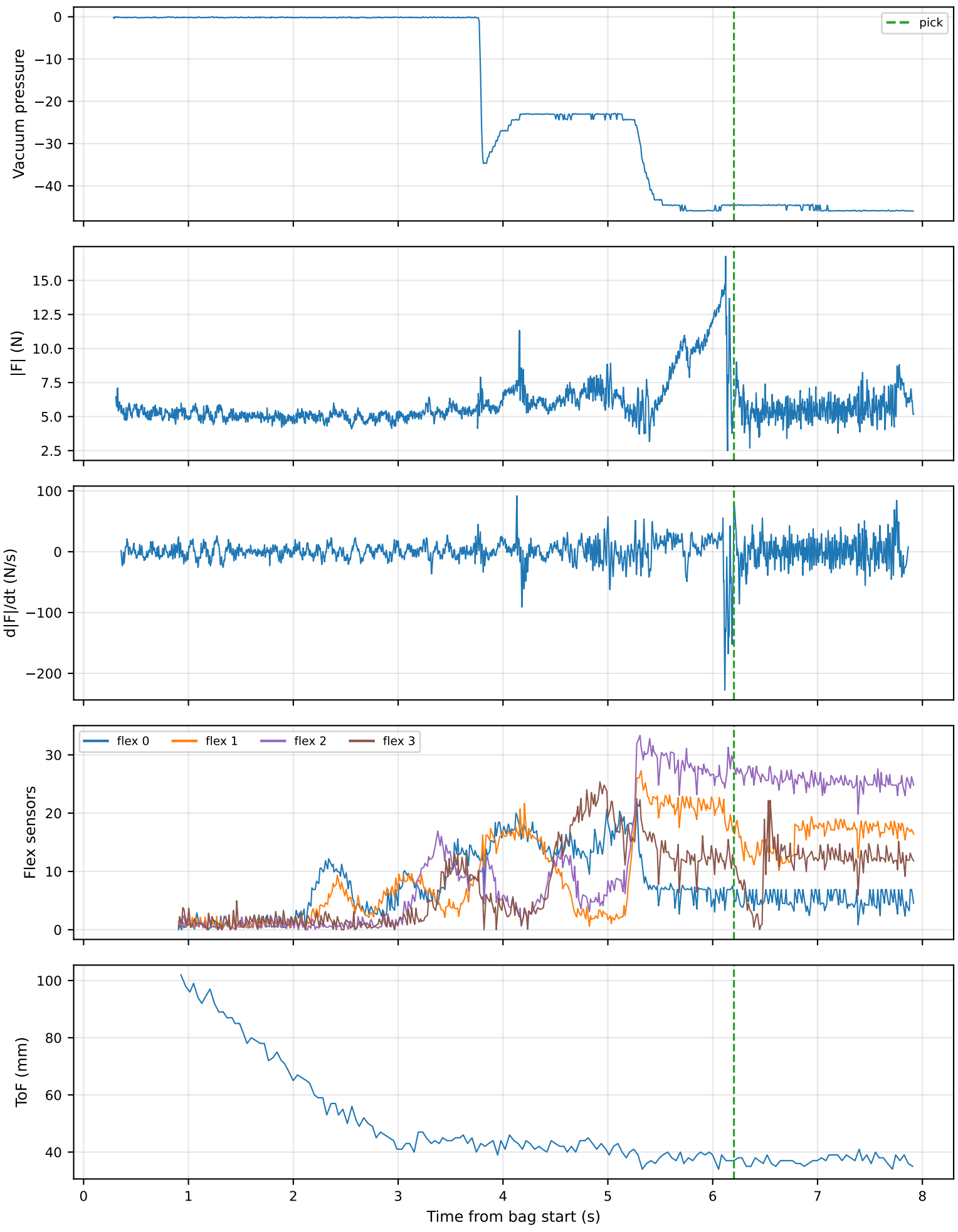

final_approach_and_pick_20261006_102912.db3
SLIP at 7.38 s | 2026-10-06 10:29:13 | batch 60, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

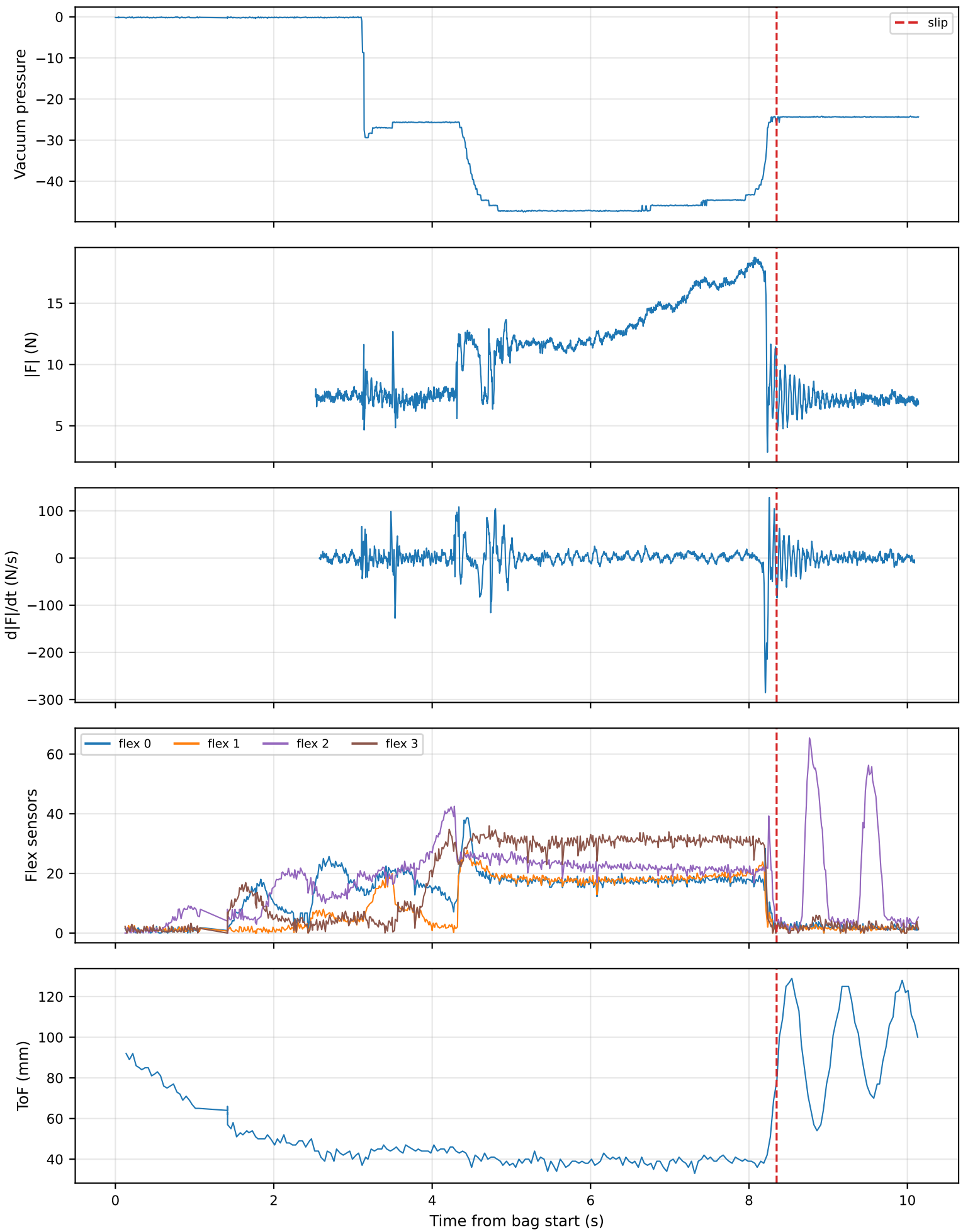

final_approach_and_pick_20261006_103403.db3
PICK at 6.77 s | 2026-10-06 10:34:04 | batch 63, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

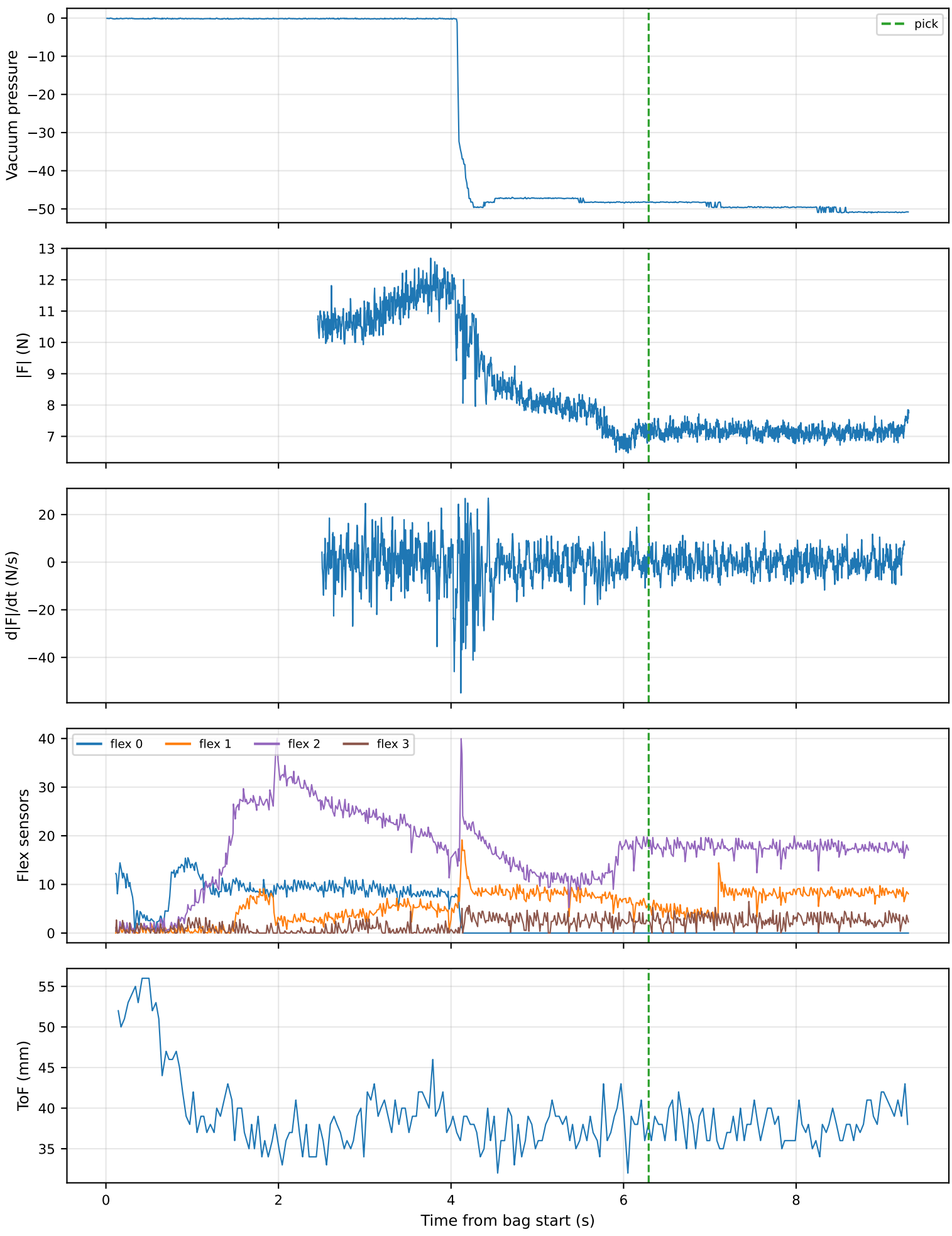

final_approach_and_pick_20261006_104020.db3
SLIP at 5.87 s | 2026-10-06 10:40:22 | batch 68, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

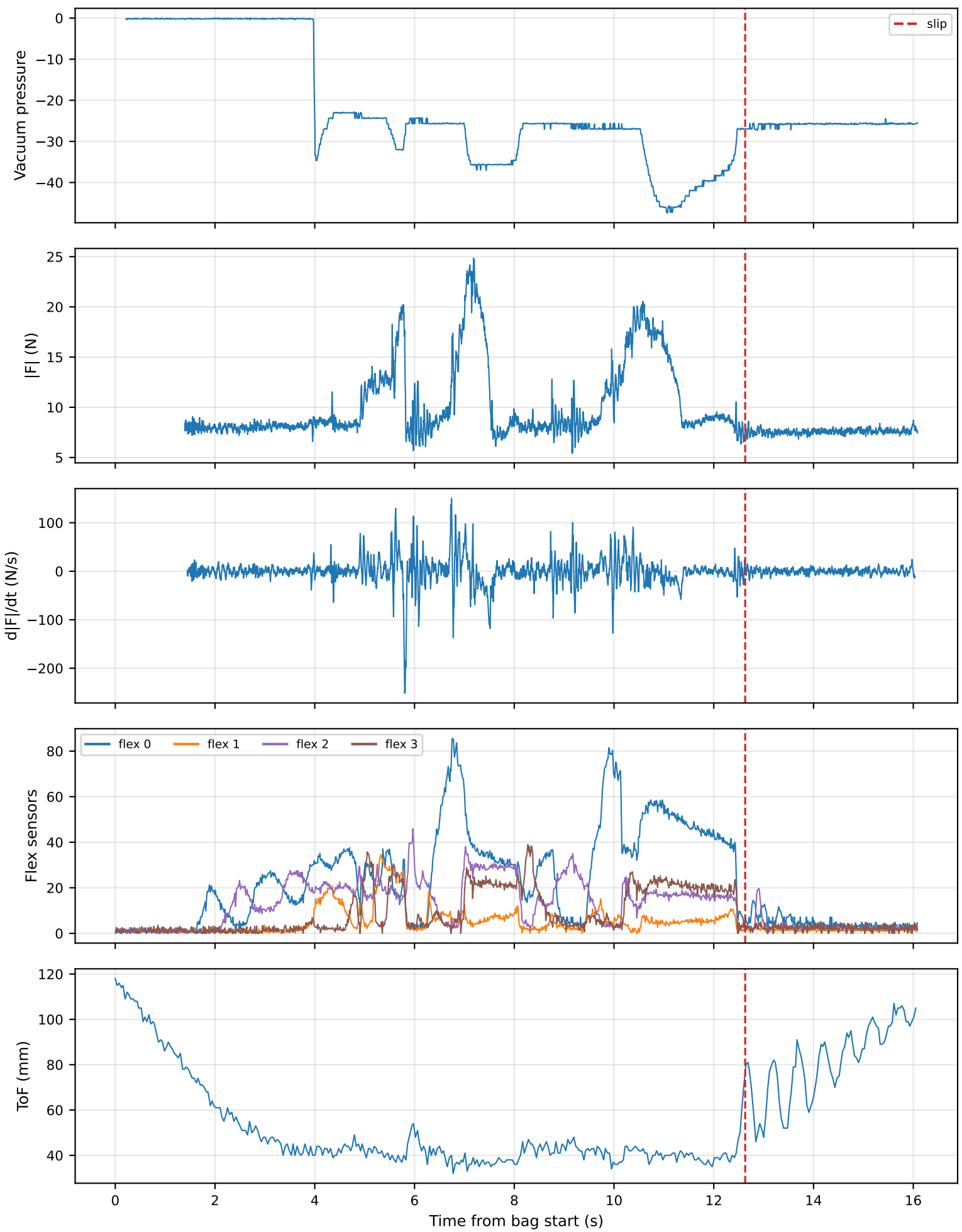

final_approach_and_pick_20261006_104331.db3
PICK at 6.38 s | 2026-10-06 10:43:33 | batch 70, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

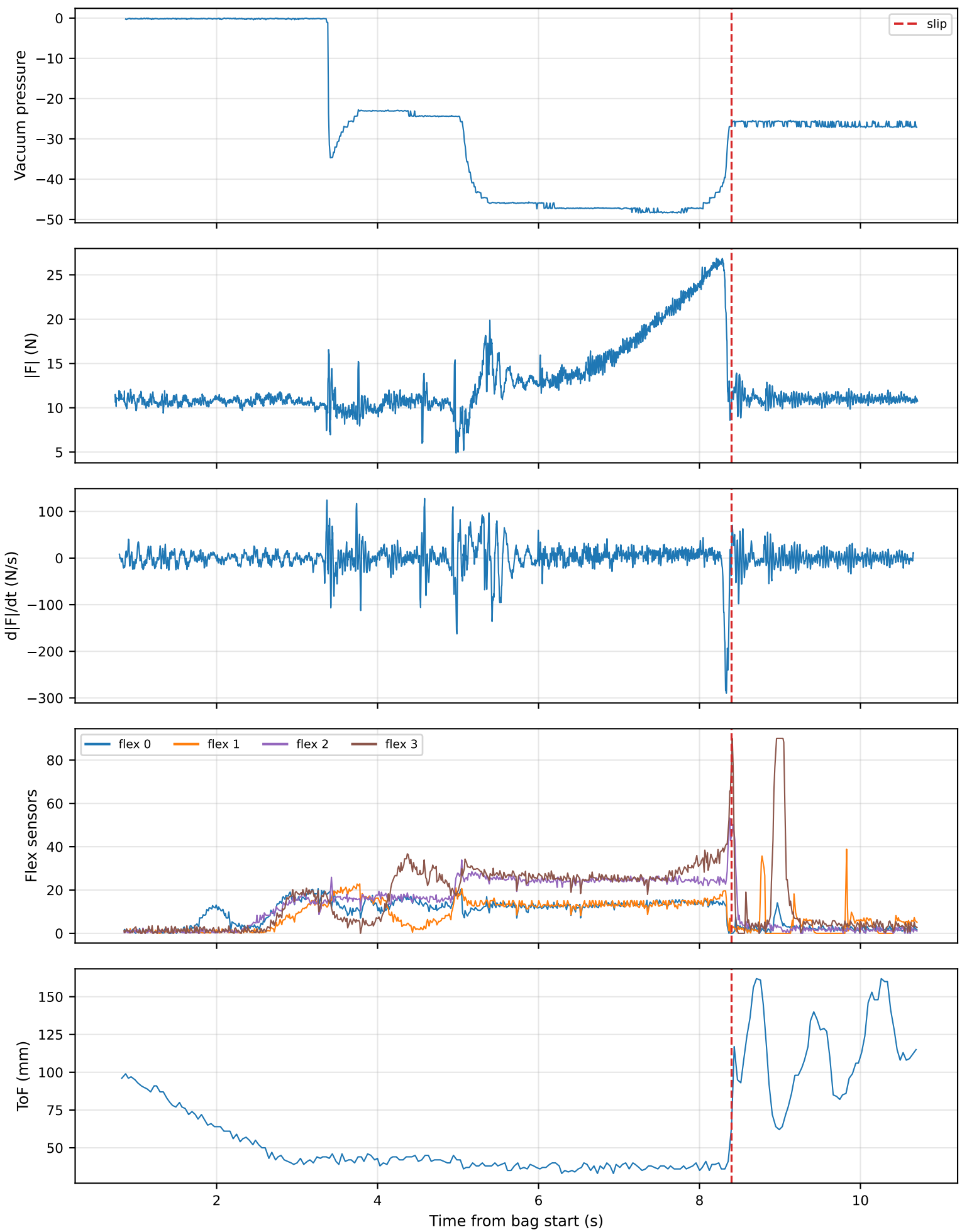

final_approach_and_pick_20261006_104435.db3
PICK at 5.60 s | 2026-10-06 10:44:36 | batch 71, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

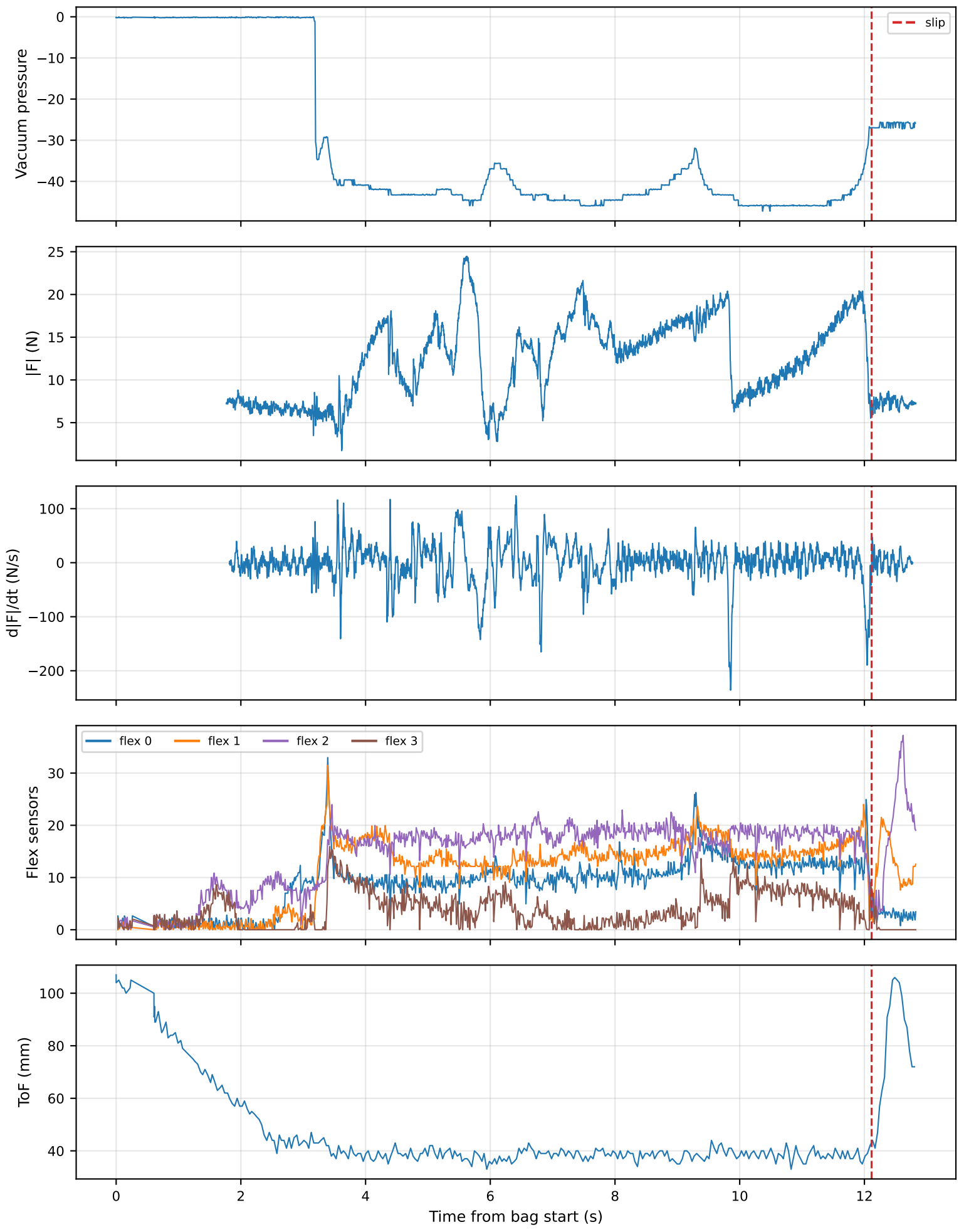

final_approach_and_pick_20261006_104535.db3
PICK at 6.20 s | 2026-10-06 10:45:36 | batch 72, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

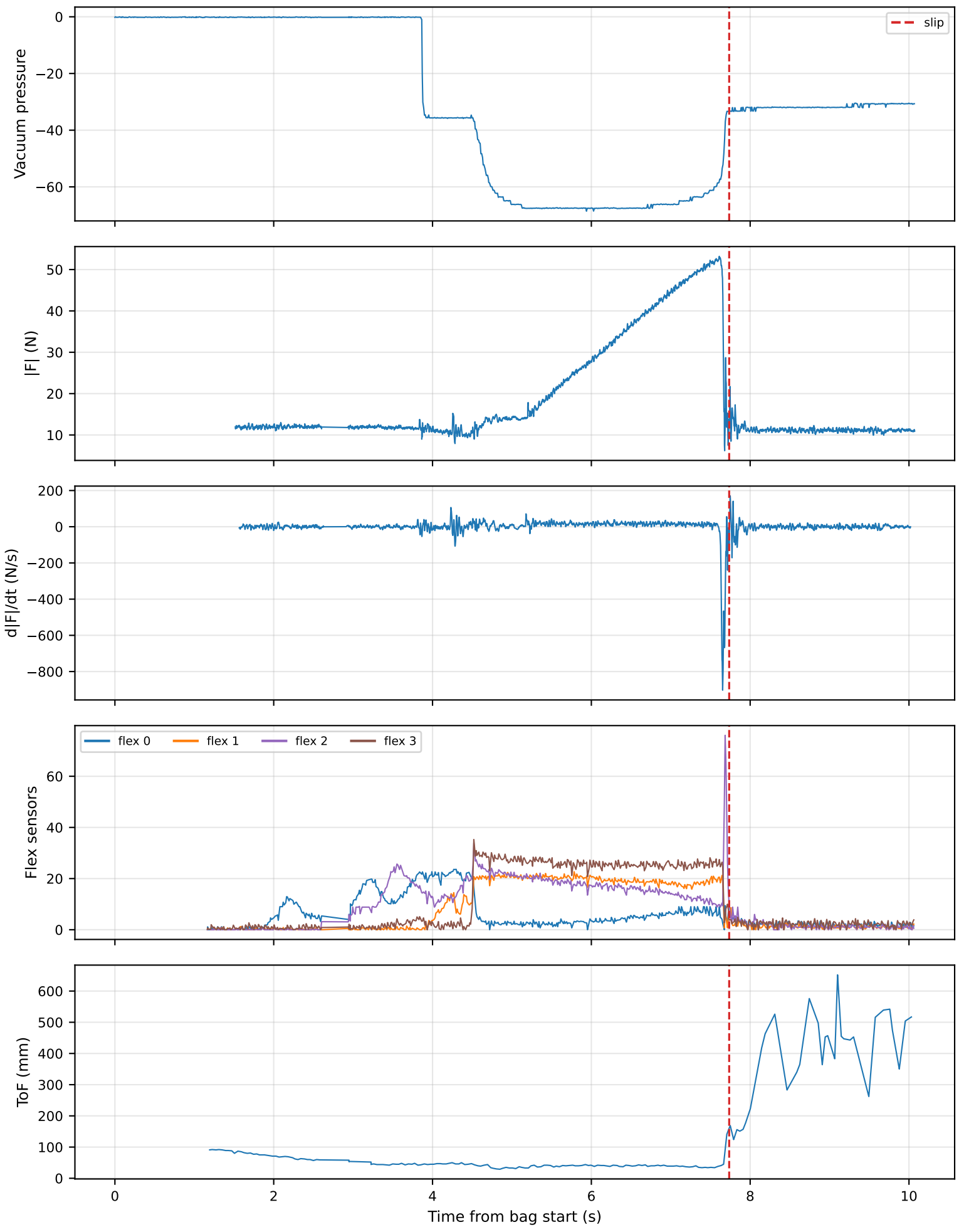

final_approach_and_pick_20261006_105130.db3
SLIP at 8.35 s | 2026-10-06 10:51:31 | batch 74, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_105243.db3
PICK at 6.29 s | 2026-10-06 10:52:46 | batch 75, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_105403.db3
SLIP at 12.63 s | 2026-10-06 10:54:04 | batch 76, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

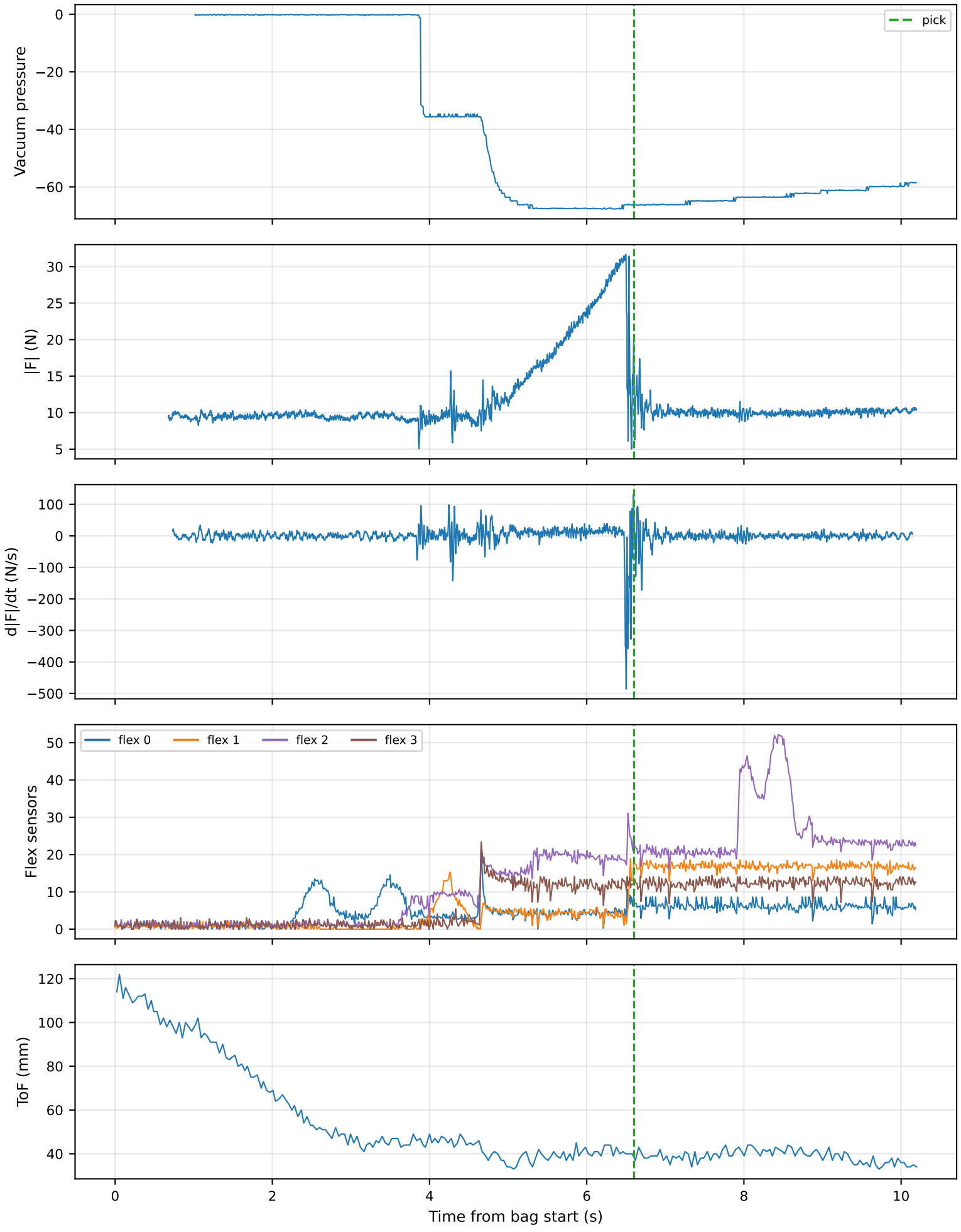

final_approach_and_pick_20261006_105949.db3
SLIP at 8.40 s | 2026-10-06 10:59:50 | batch 79, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

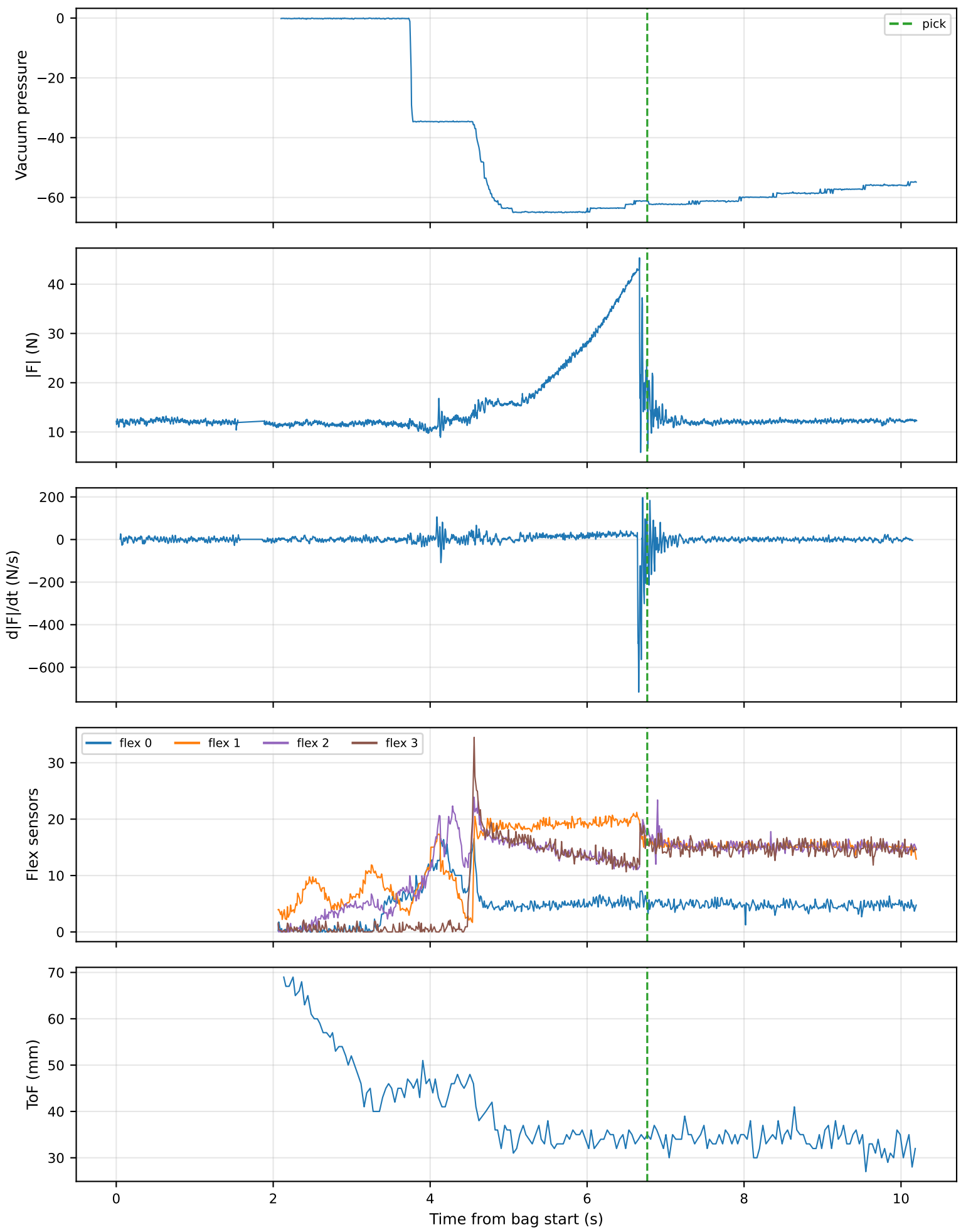

final_approach_and_pick_20261006_110128.db3
SLIP at 12.12 s | 2026-10-06 11:01:29 | batch 80, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

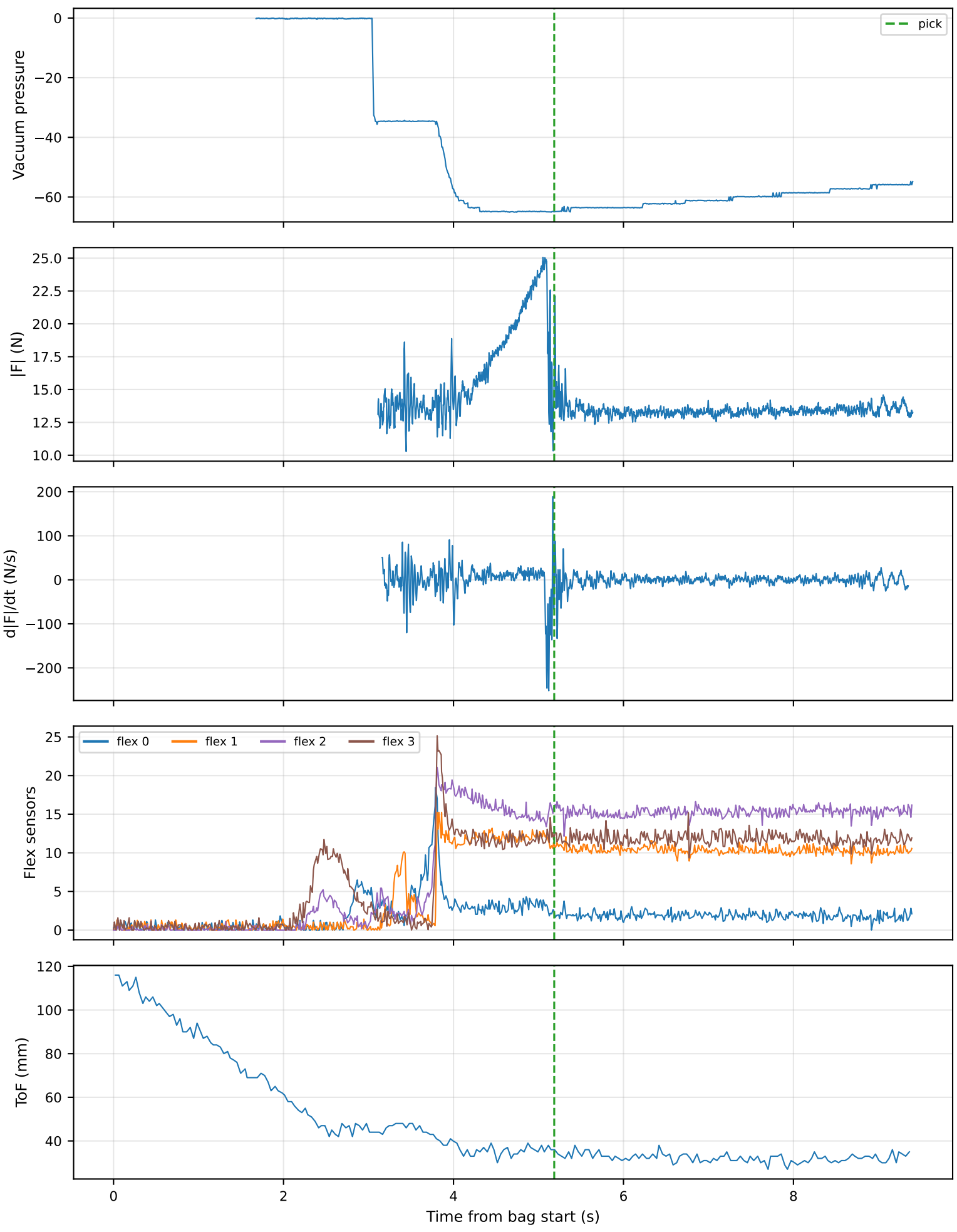

final_approach_and_pick_20261006_113139.db3
SLIP at 7.74 s | 2026-10-06 11:31:40 | batch 81, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

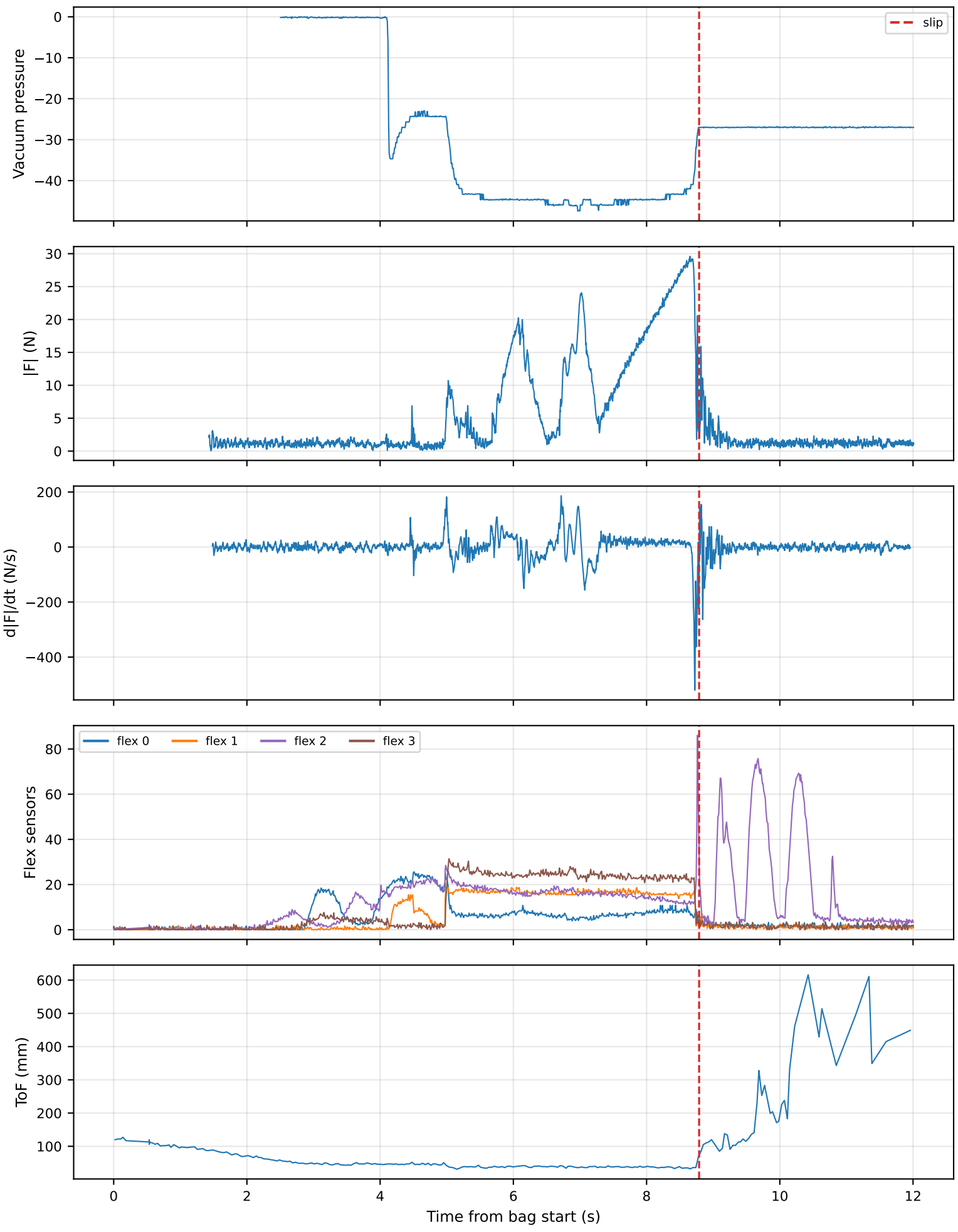
final_approach_and_pick_20261006_114224.db3

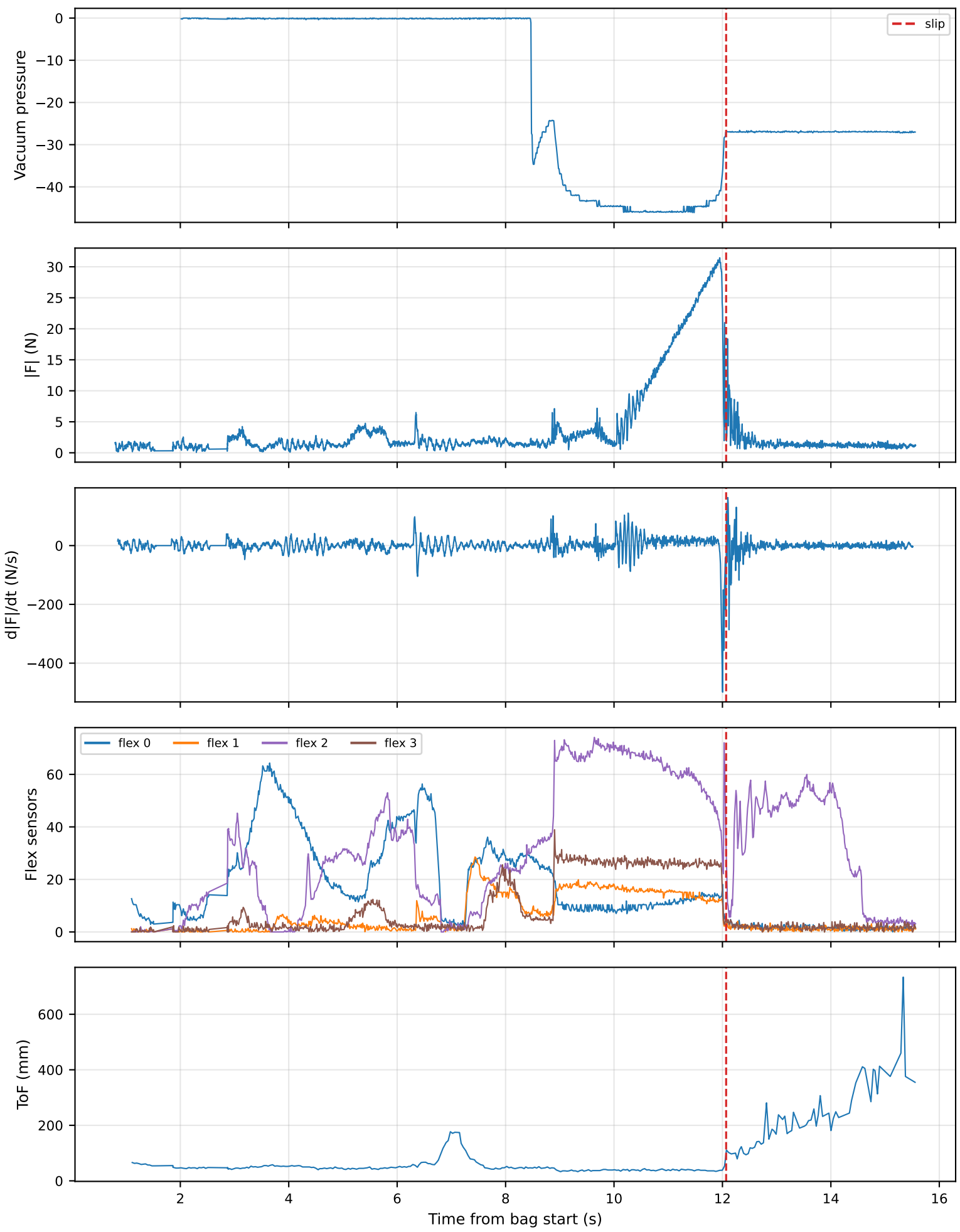
PICK at 6.60 s | 2026-10-06 11:42:25 | batch 83, apple 0 | stiffness_controller

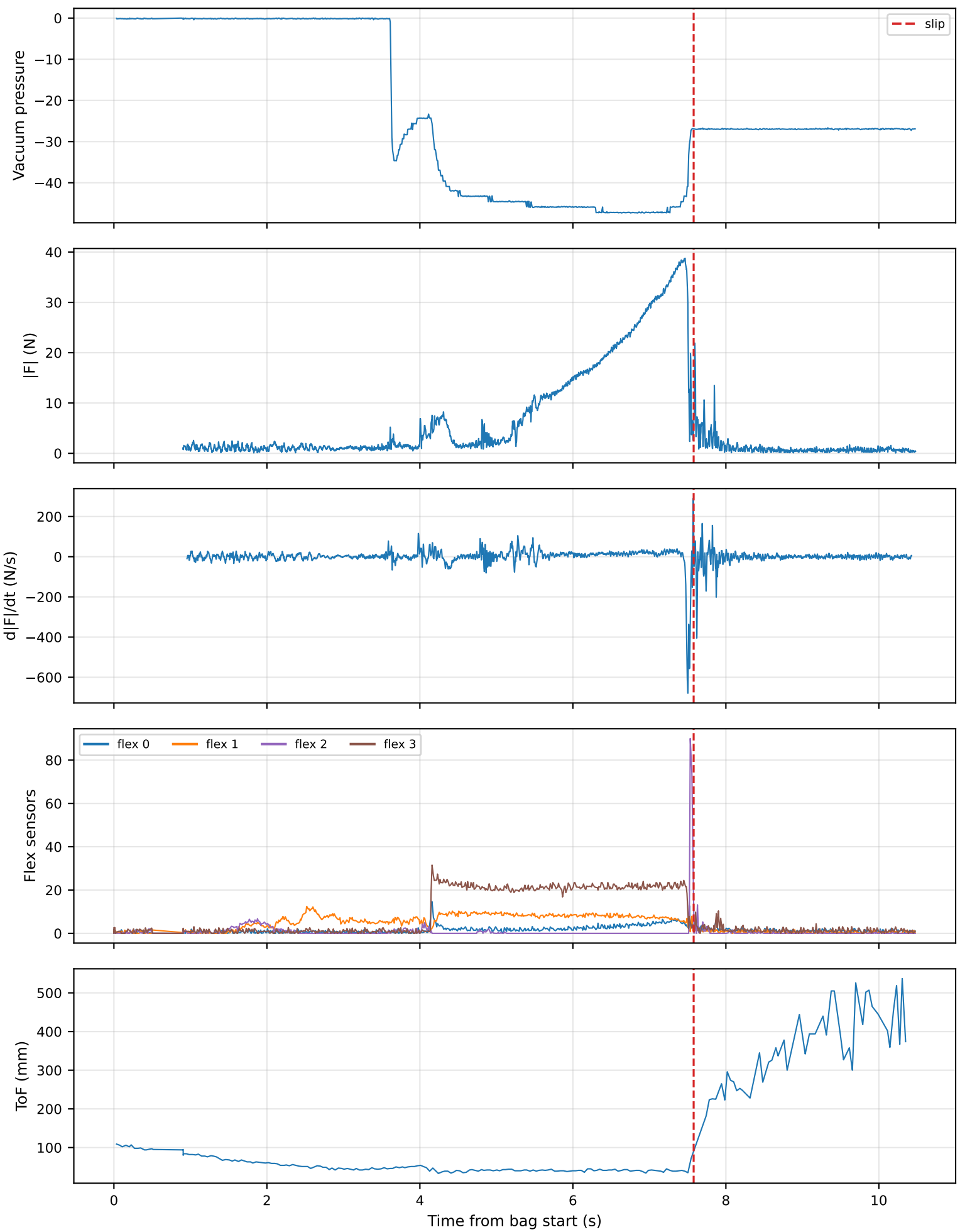



final_approach_and_pick_20261006_120649.db3
PICK at 6.77 s | 2026-10-06 12:06:51 | batch 87, apple 0 | stiffness_controller
pick
Vacuum pressure
0
−20
−40
−60
|F| (N)
40
30
20
10
d|F|/dt (N/s)
200
0
−200
−400
−600
flex 0
flex 1
flex 2
flex 3
Flex sensors
30
20
10
0
ToF (mm)
70
60
50
40
30
0
2
4
6
8
10
Time from bag start (s)


final_approach_and_pick_20261006_120835.db3
PICK at 5.19 s | 2026-10-06 12:08:36 | batch 88, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

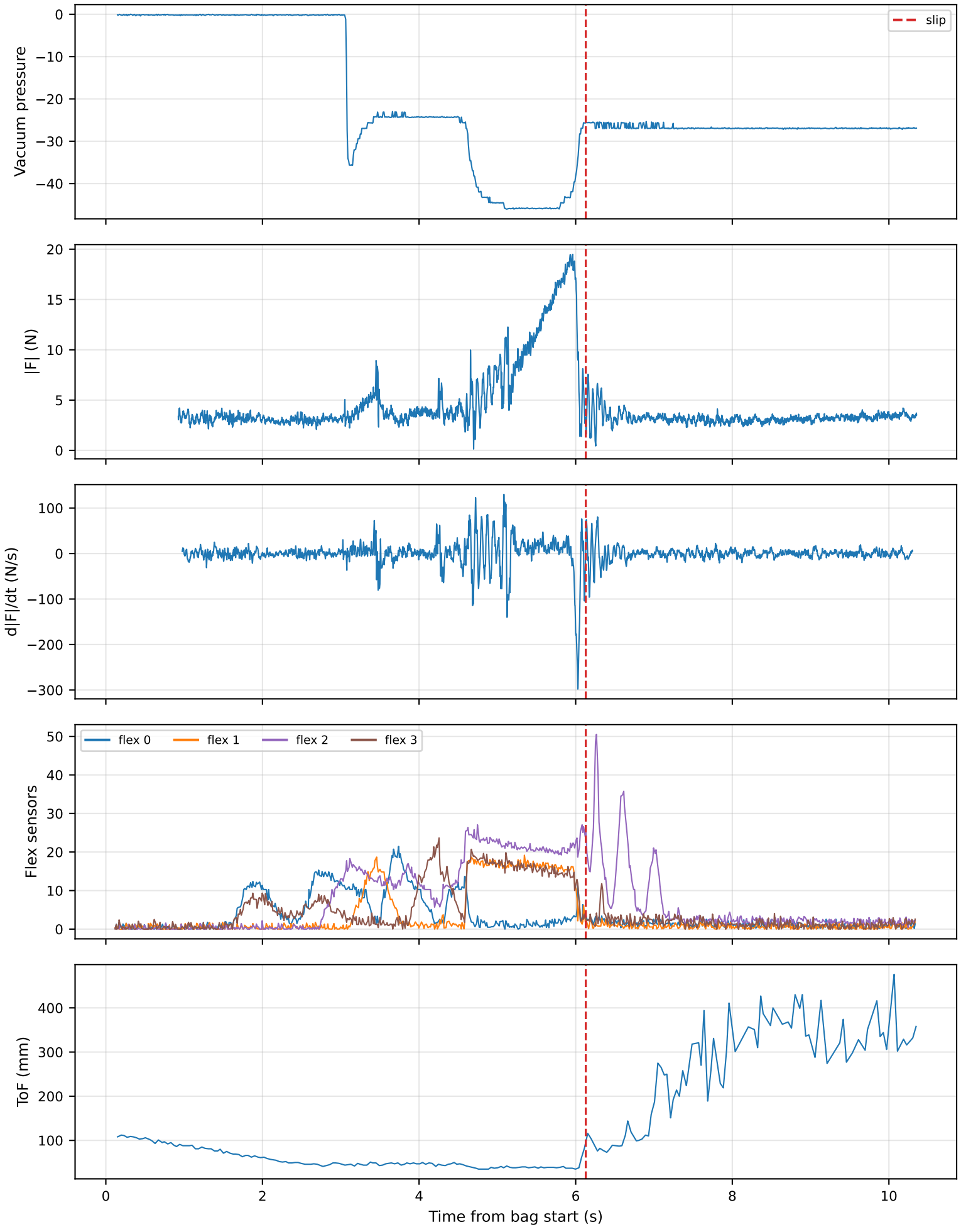

final_approach_and_pick_20261006_132241.db3
SLIP at 8.79 s | 2026-10-06 13:22:42 | batch 91, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

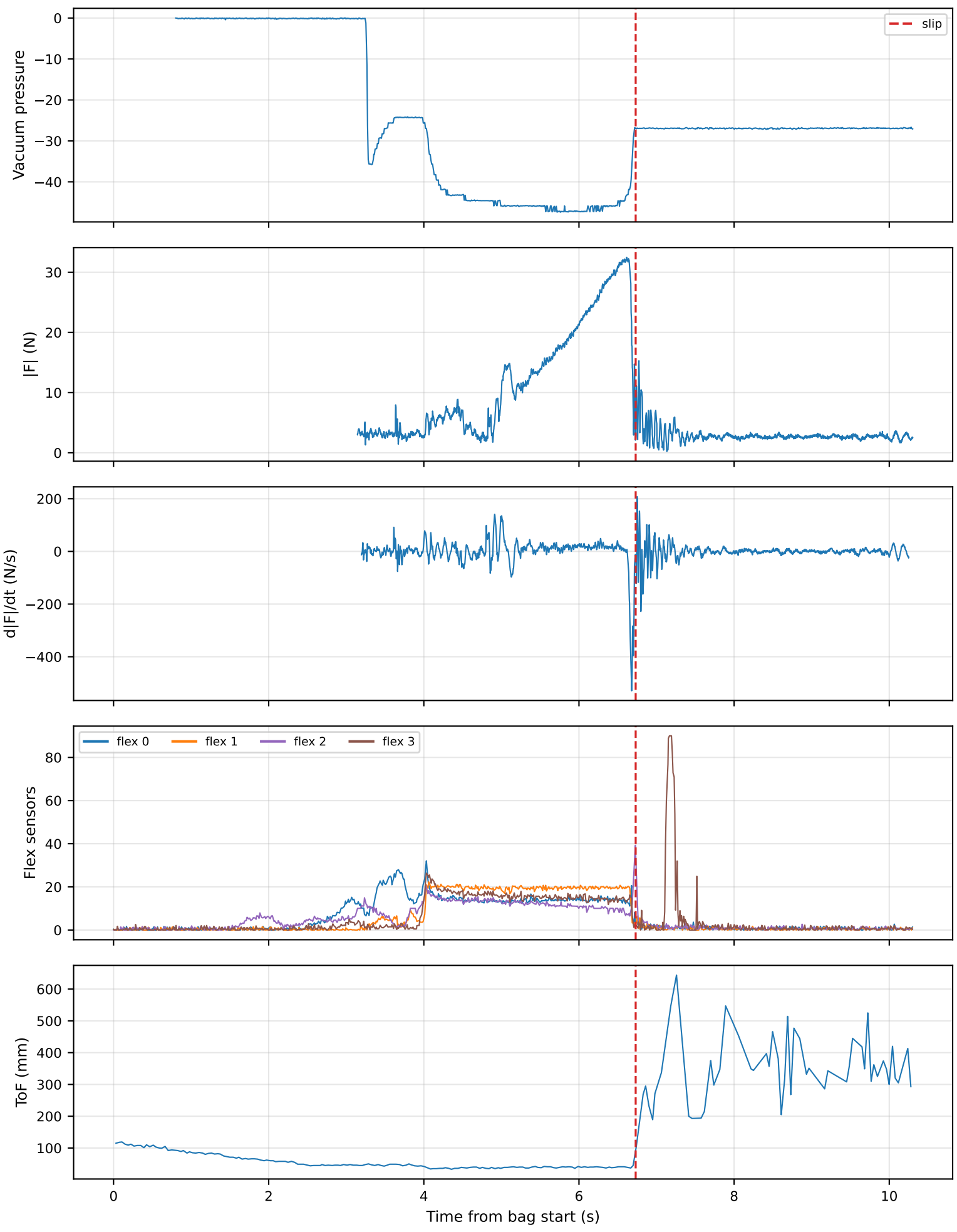

final_approach_and_pick_20261006_132356.db3
SLIP at 12.07 s | 2026-10-06 13:23:58 | batch 92, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_132507.db3
SLIP at 7.58 s | 2026-10-06 13:25:08 | batch 93, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261006_132632.db3

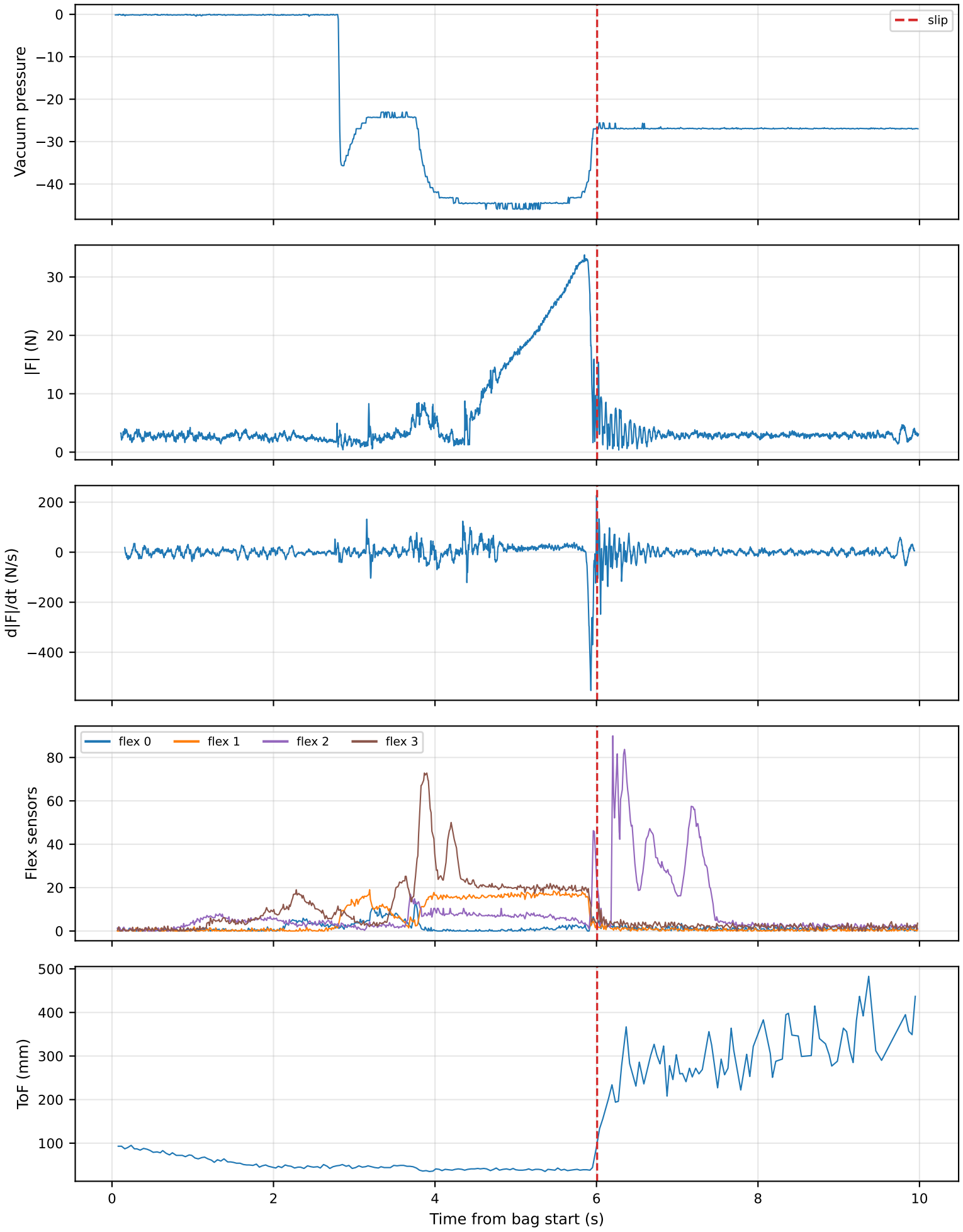
SLIP at 6.13 s | 2026-10-06 13:26:32 | batch 94, apple 0 | stiffness_controller

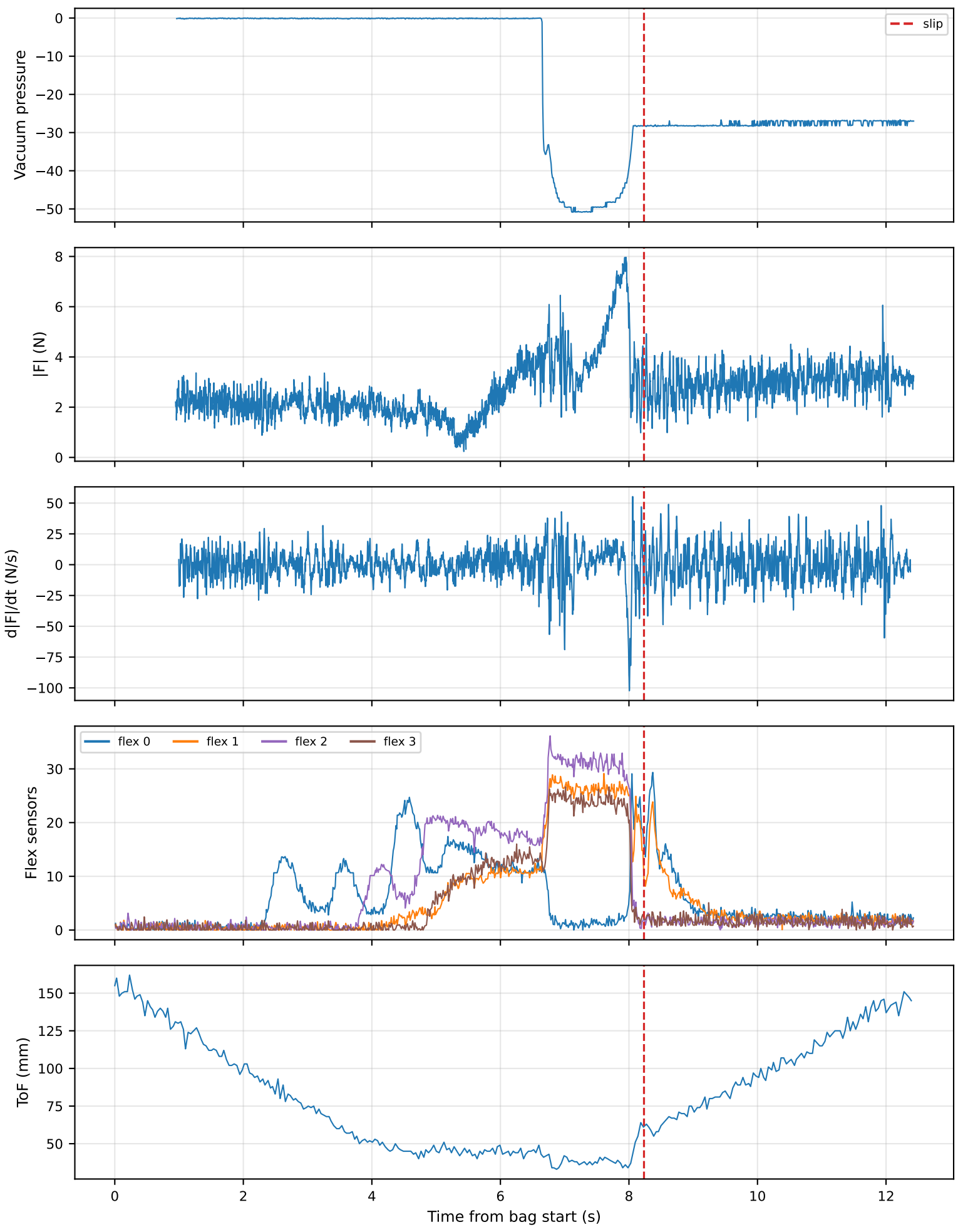


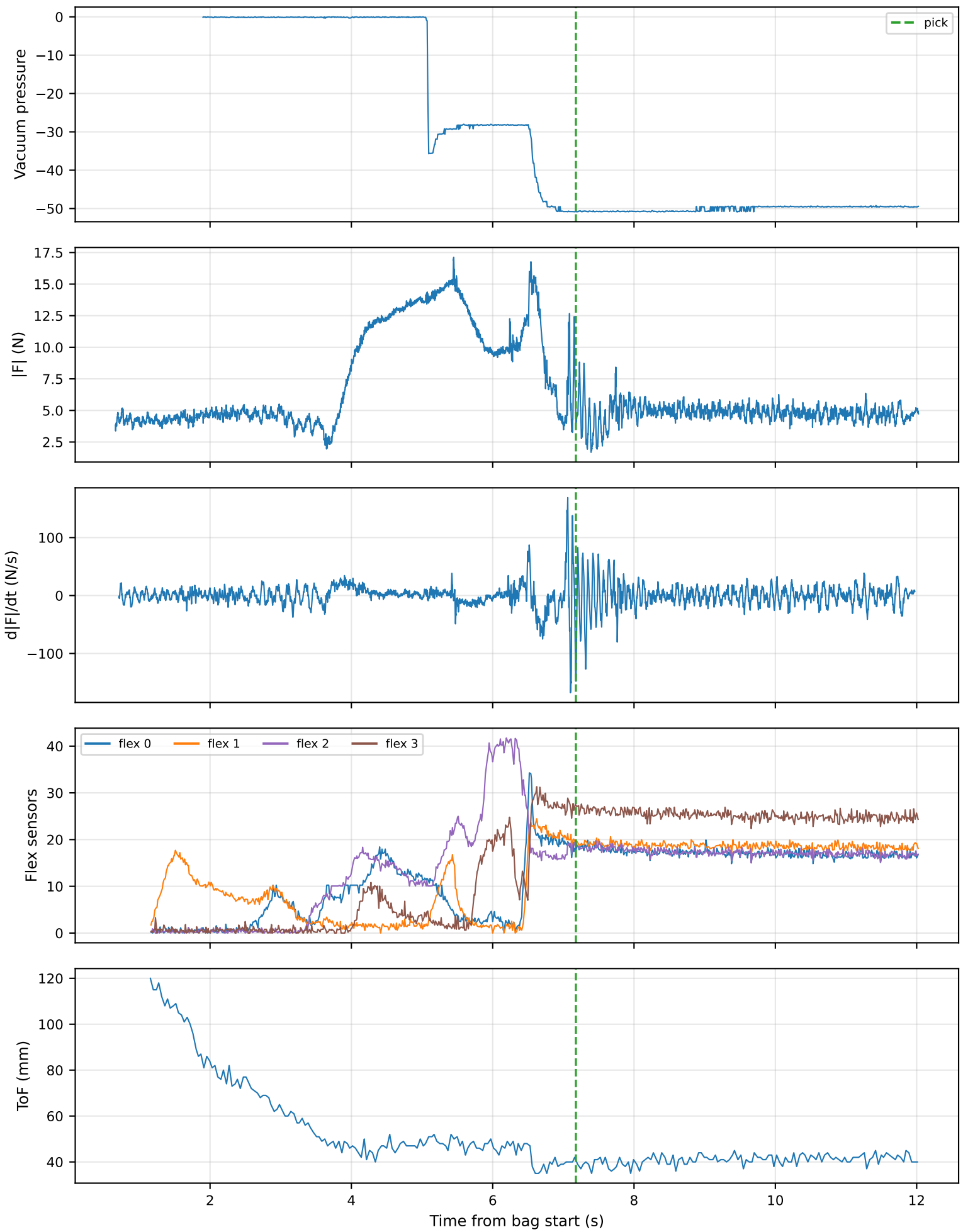

final_approach_and_pick_20261006_132858.db3
SLIP at 6.73 s | 2026-10-06 13:28:59 | batch 96, apple 0 | stiffness_controller
slip
Vacuum pressure
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

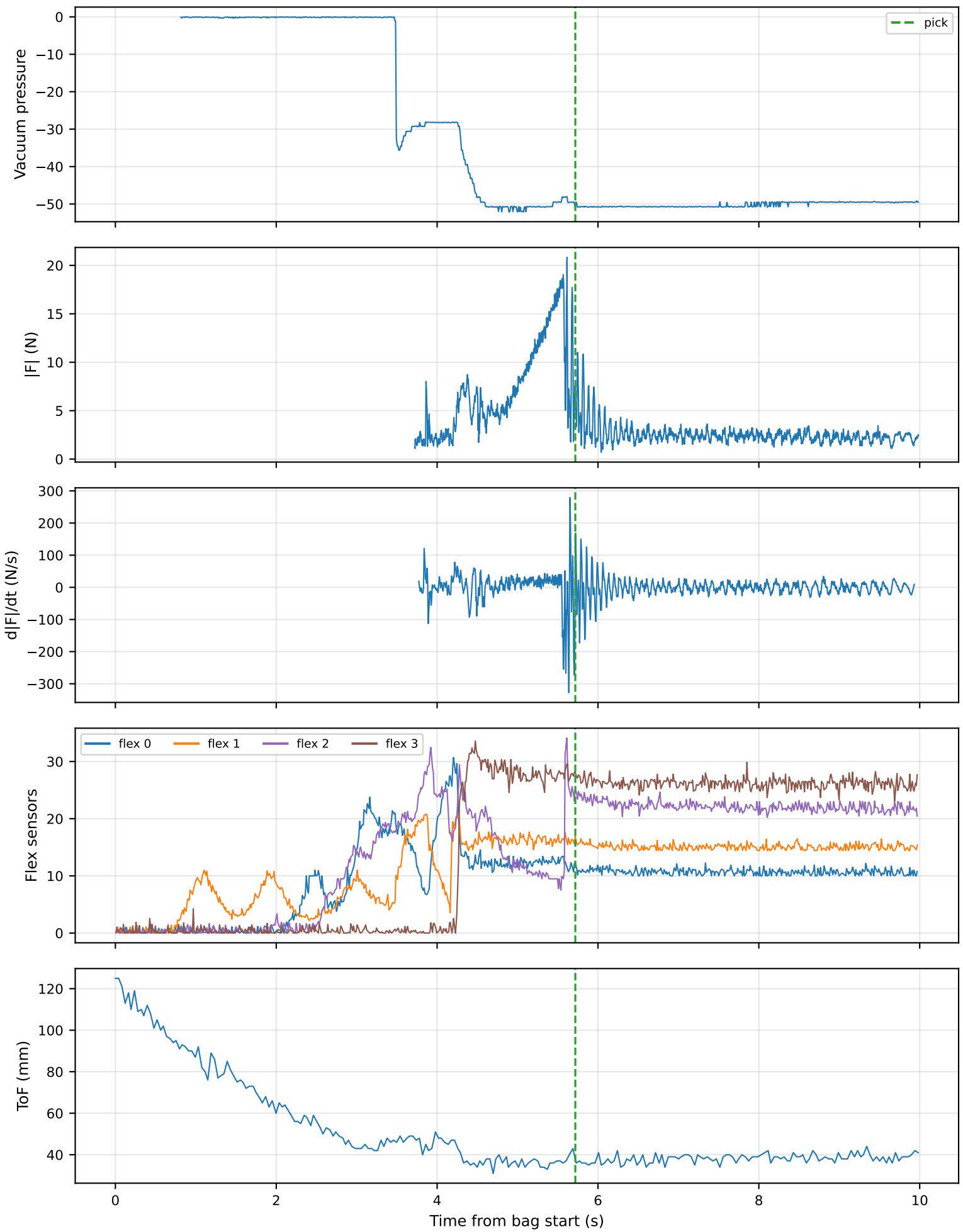
final_approach_and_pick_20261006_133024.db3

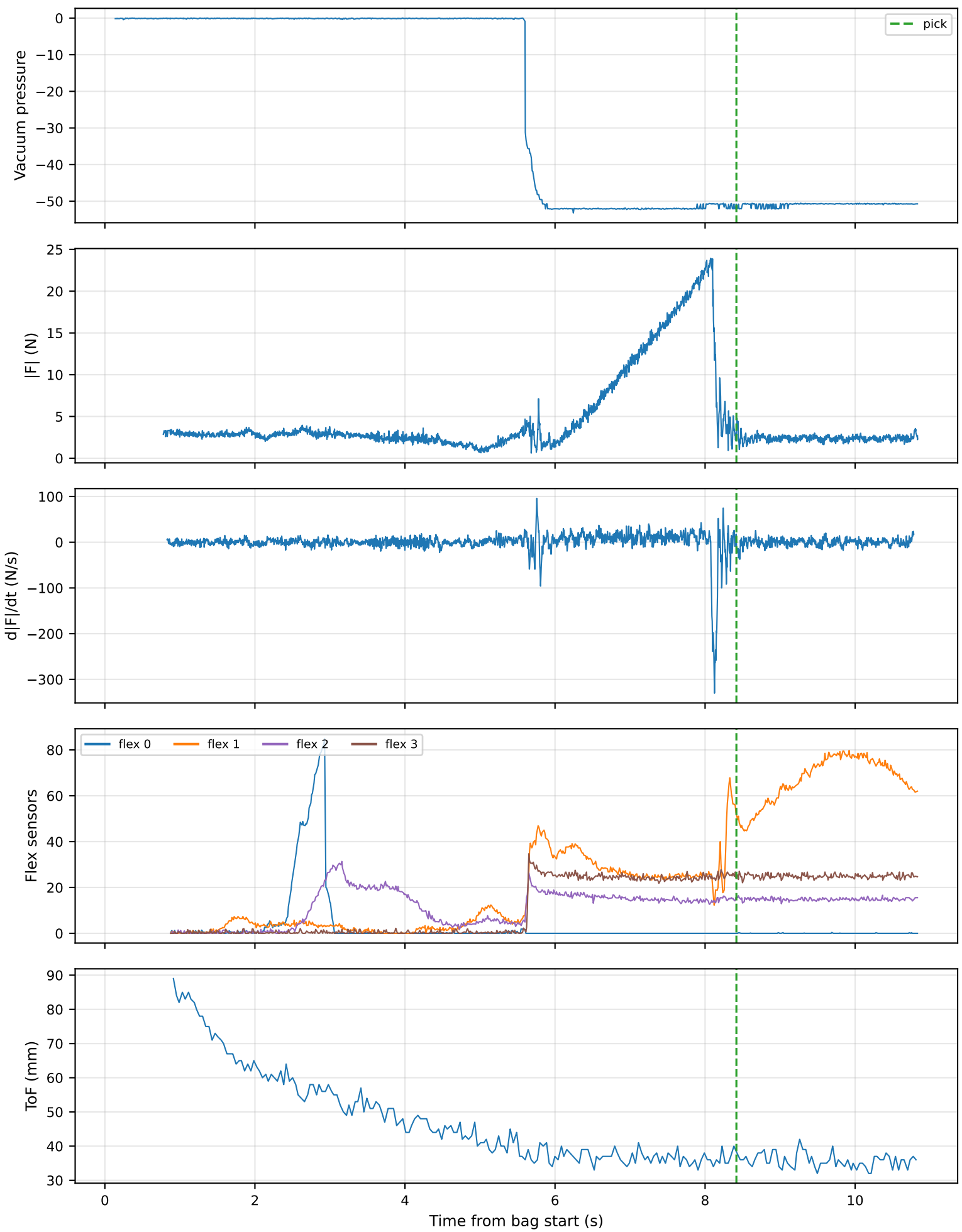
SLIP at 6.01 s | 2026-10-06 13:30:26 | batch 97, apple 0 | stiffness_controller

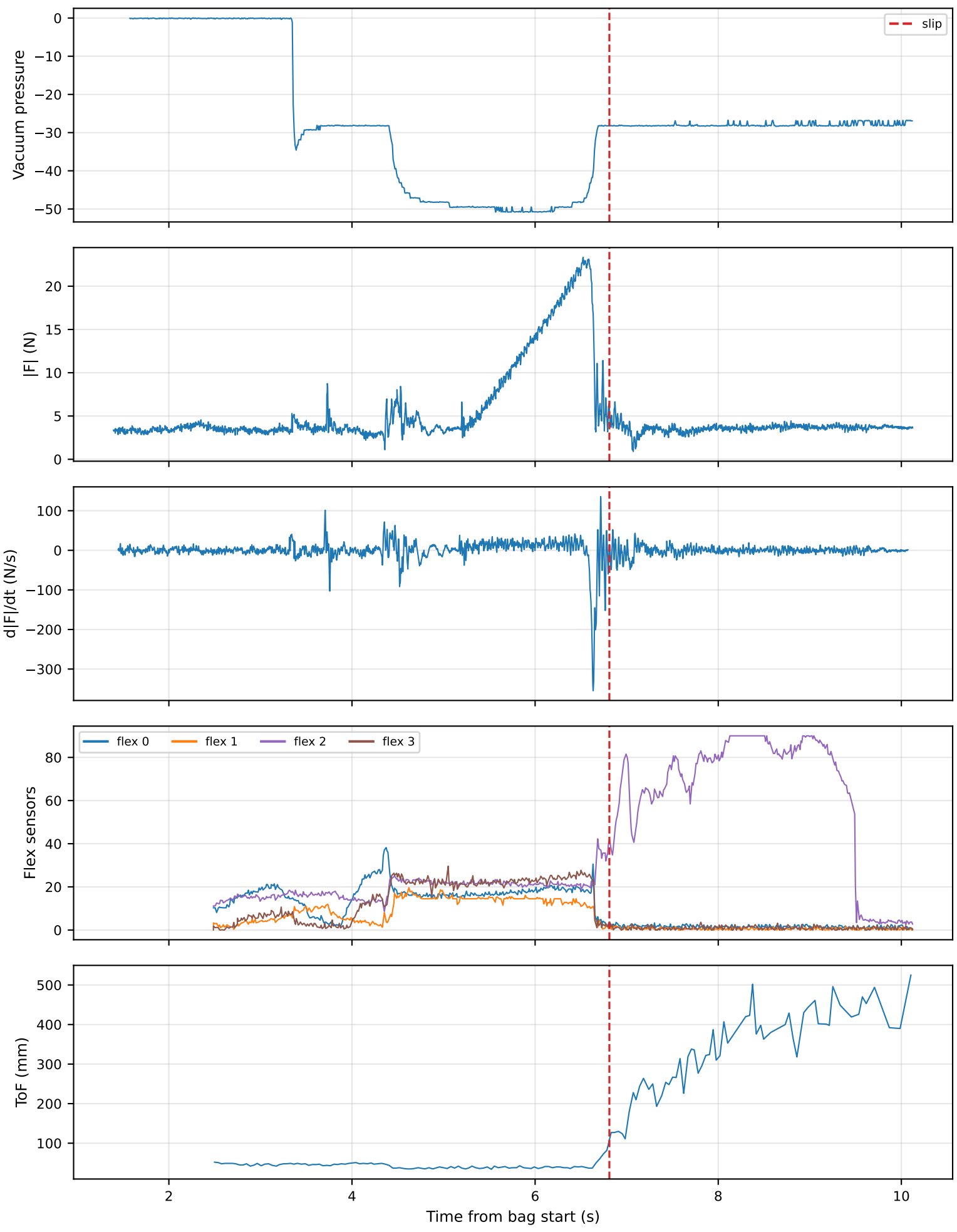


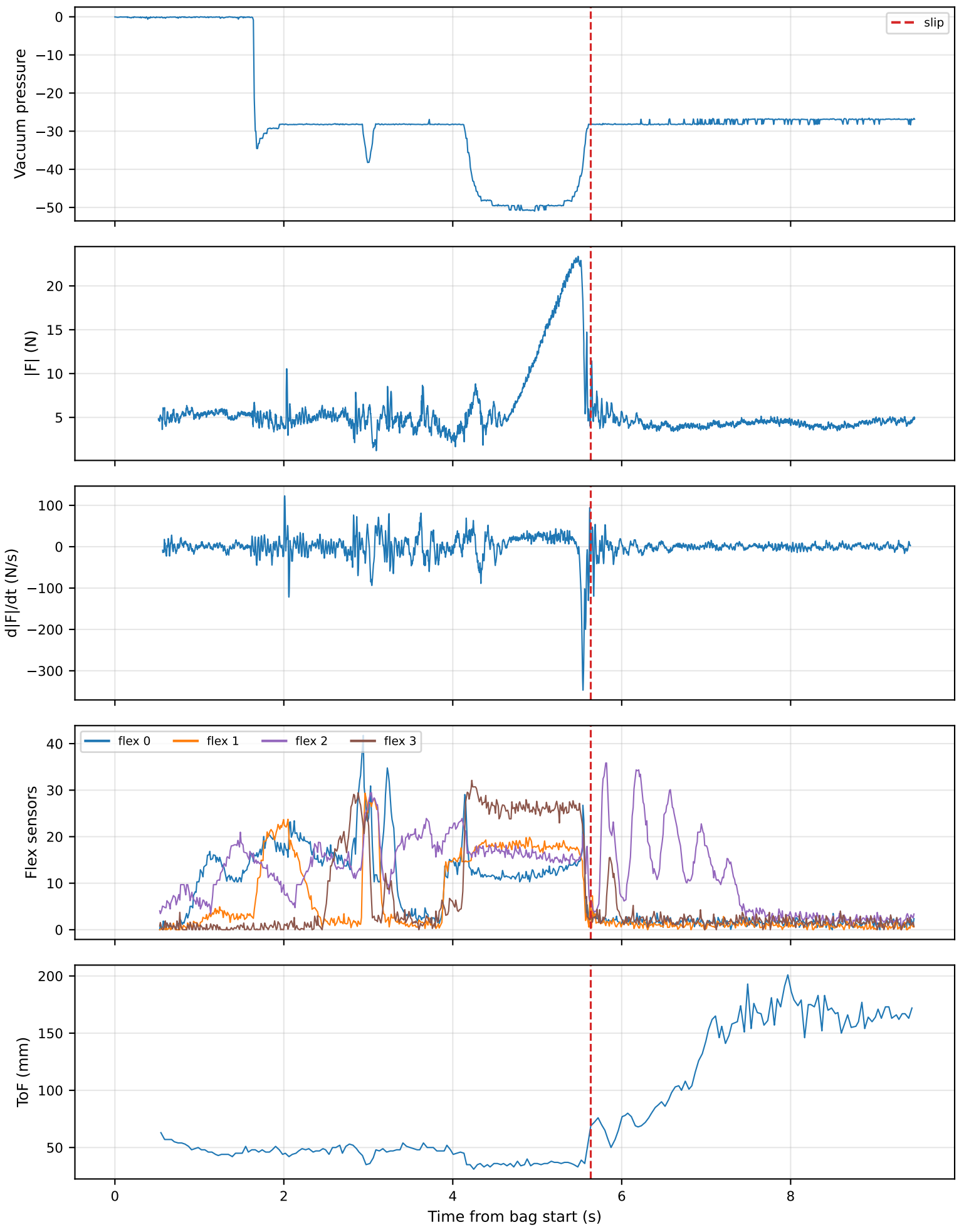

final_approach_and_pick_20261006_133504.db3
SLIP at 8.23 s | 2026-10-06 13:35:04 | batch 98, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

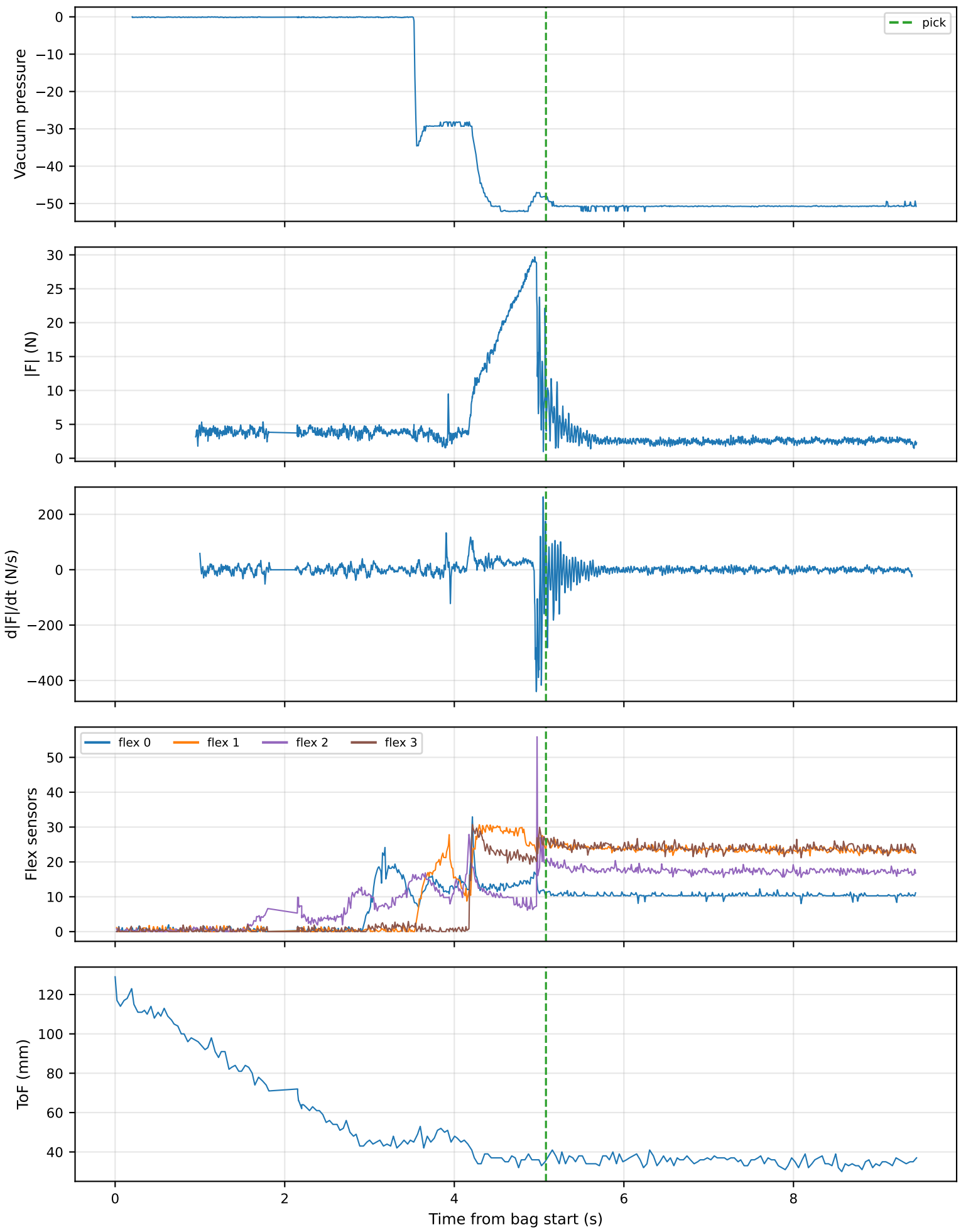

final_approach_and_pick_20261006_133626.db3
PICK at 7.18 s | 2026-10-06 13:36:28 | batch 99, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_133721.db3
PICK at 5.72 s | 2026-10-06 13:37:23 | batch 100, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_133822.db3
PICK at 8.42 s | 2026-10-06 13:38:23 | batch 101, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

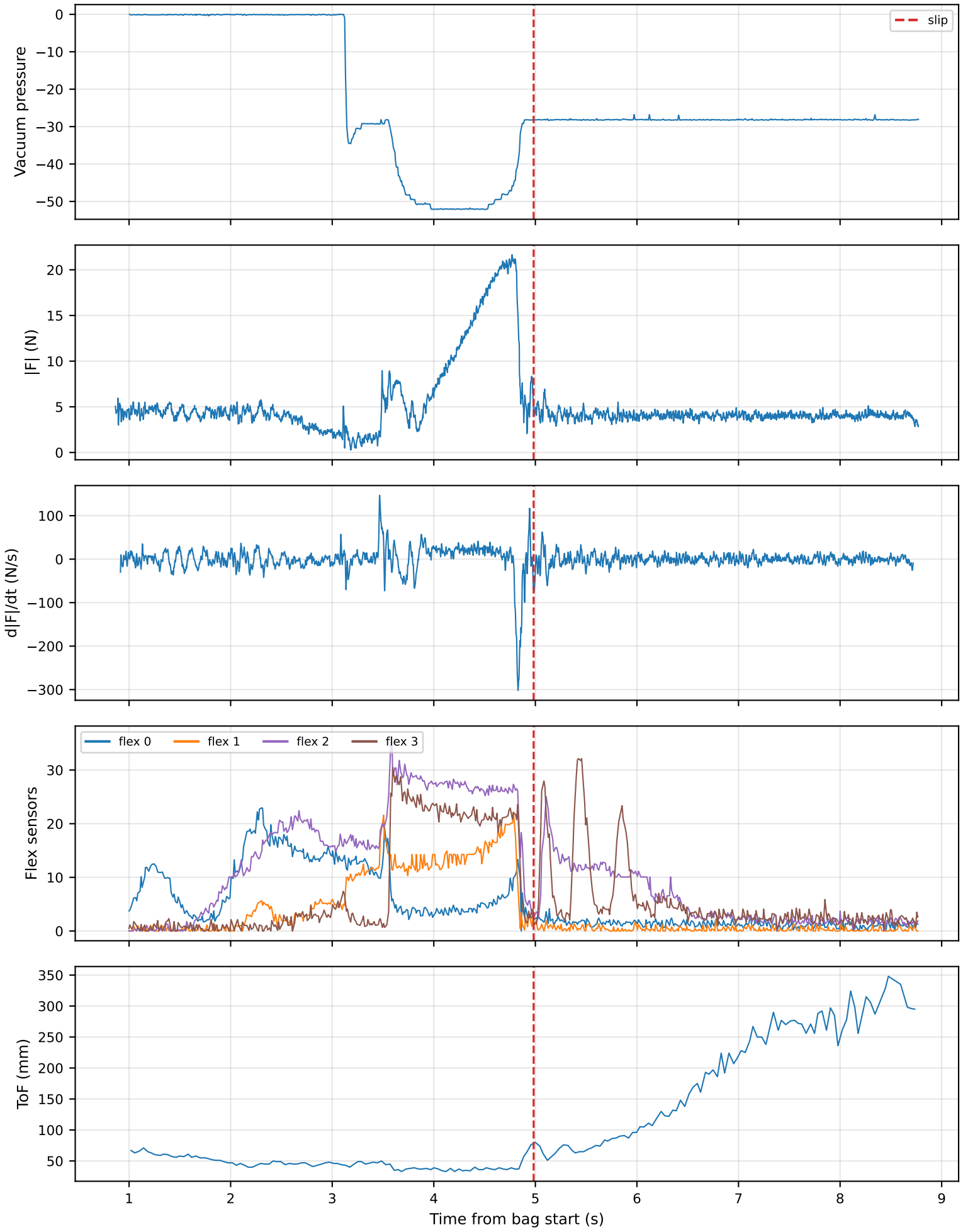

final_approach_and_pick_20261006_134010.db3
SLIP at 6.81 s | 2026-10-06 13:40:12 | batch 102, apple 0 | stiffness_controller
slip
Vacuum pressure
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

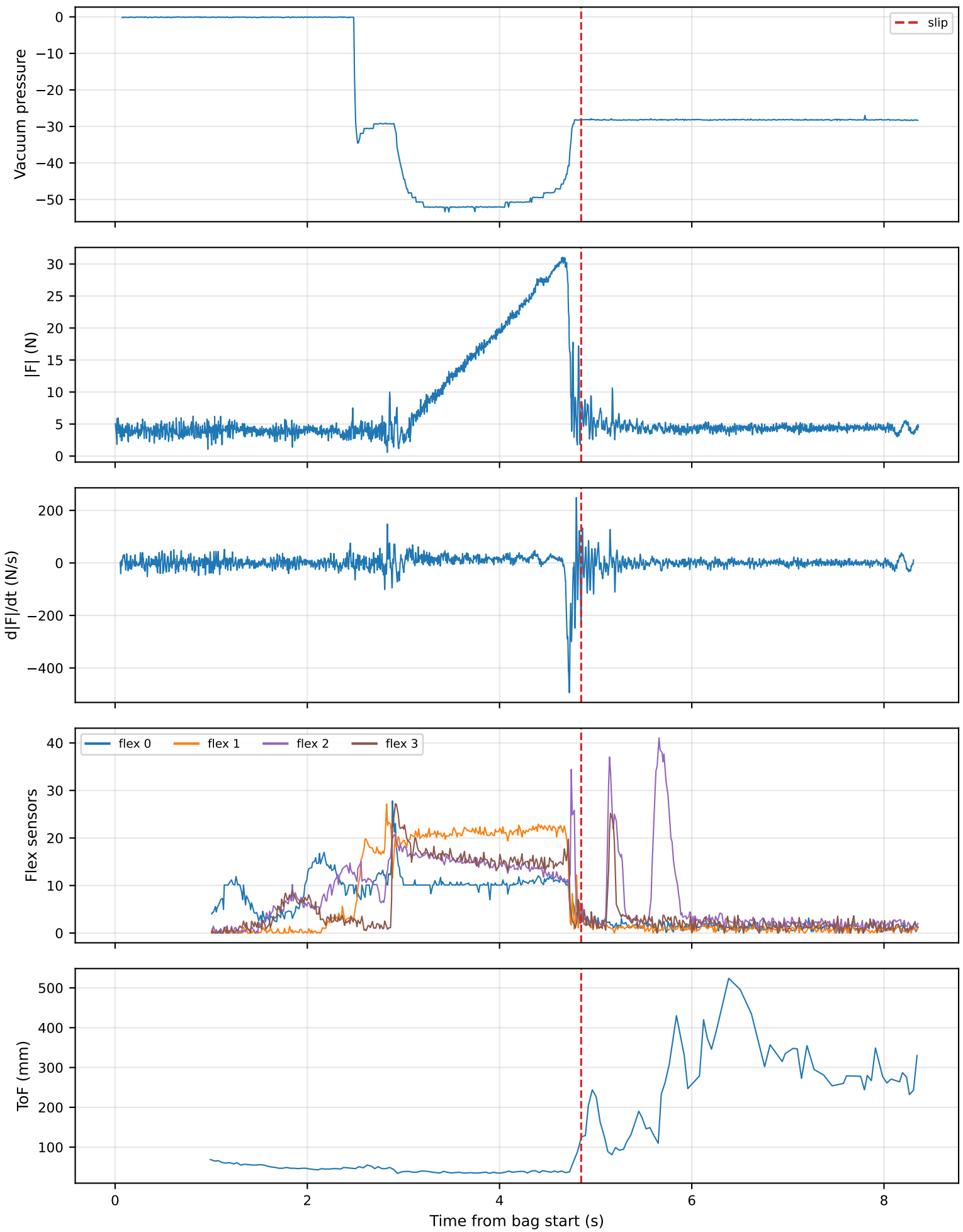

final_approach_and_pick_20261006_134114.db3
SLIP at 5.63 s | 2026-10-06 13:41:16 | batch 103, apple 0 | stiffness_controller
slip
Vacuum pressure
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

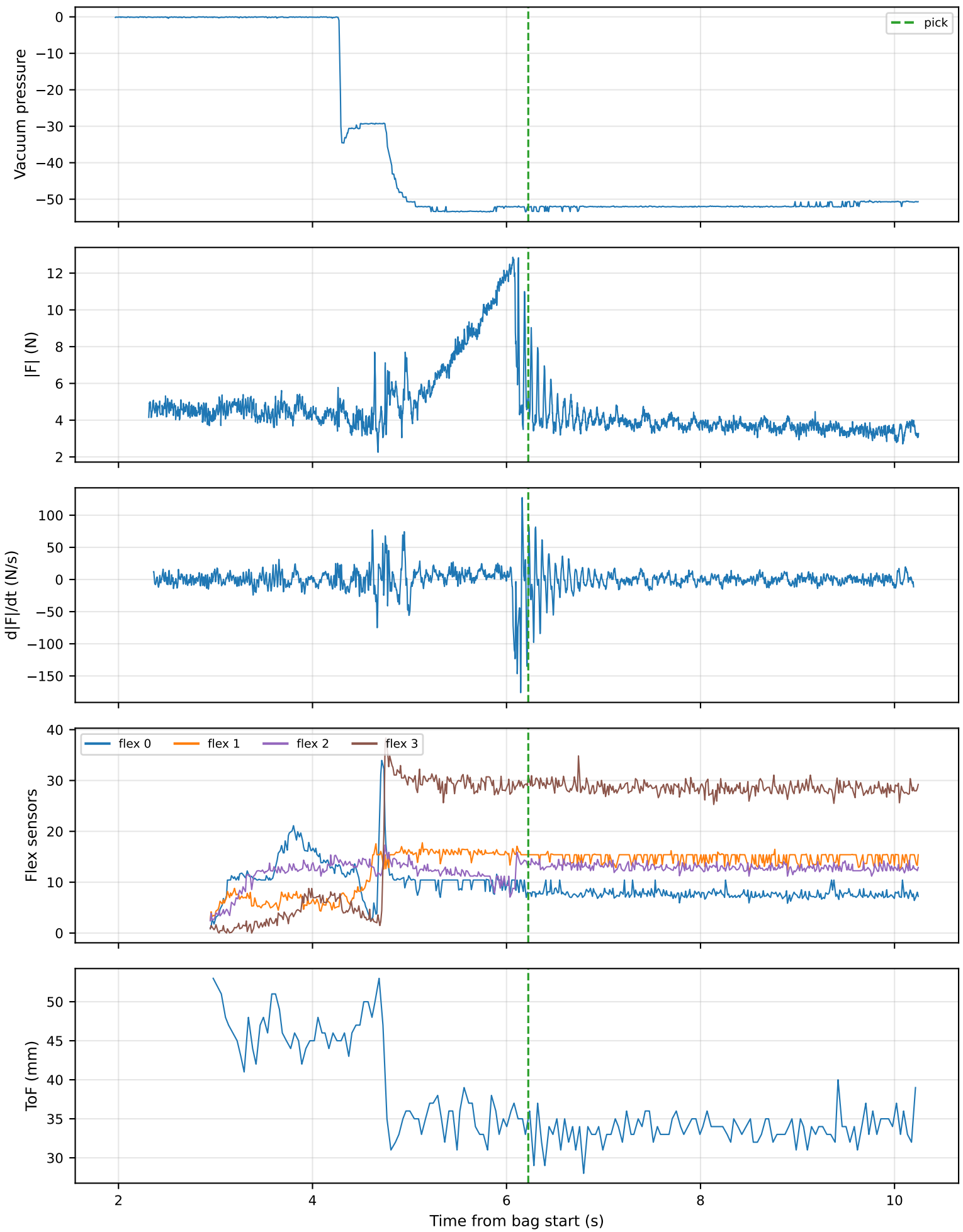

final_approach_and_pick_20261006_134247.db3
PICK at 5.08 s | 2026-10-06 13:42:48 | batch 104, apple 0 | stiffness_controller
pick
Vacuum pressure
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261006_134349.db3

SLIP at 4.98 s | 2026-10-06 13:43:50 | batch 105, apple 0 | stiffness_controller

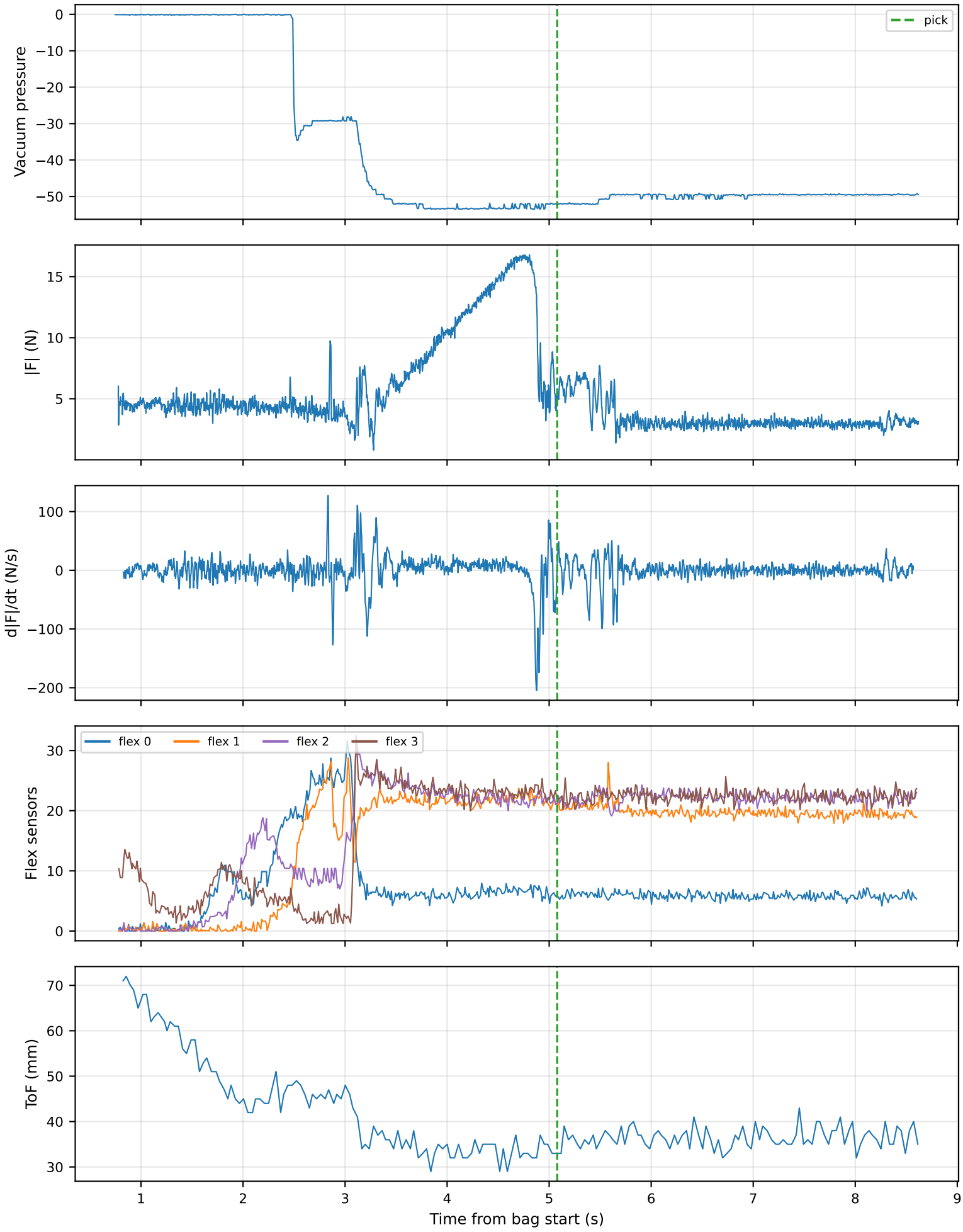



final_approach_and_pick_20261006_134516.db3
SLIP at 4.85 s | 2026-10-06 13:45:18 | batch 106, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_134634.db3
PICK at 6.22 s | 2026-10-06 13:46:35 | batch 107, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

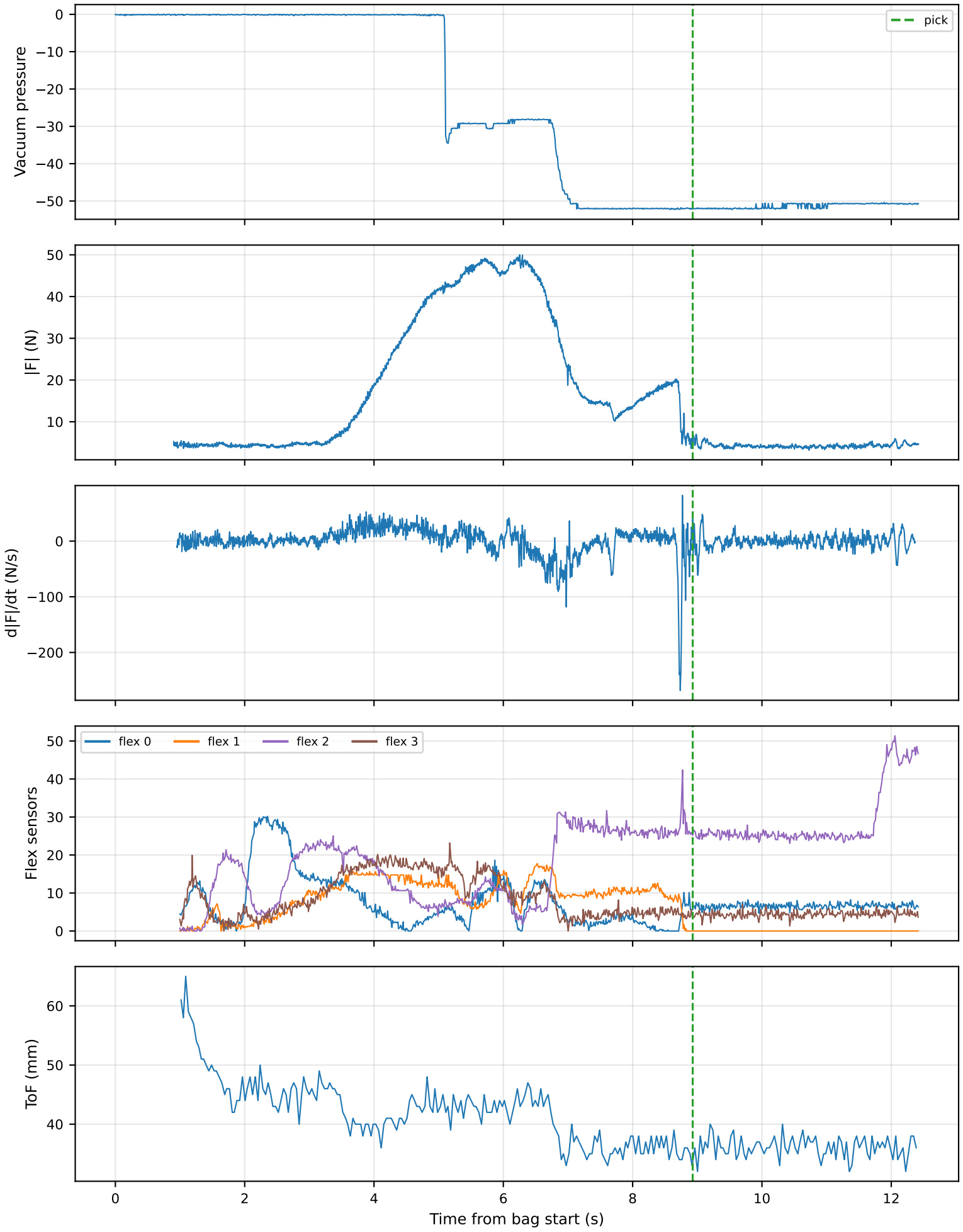
final_approach_and_pick_20261006_134744.db3

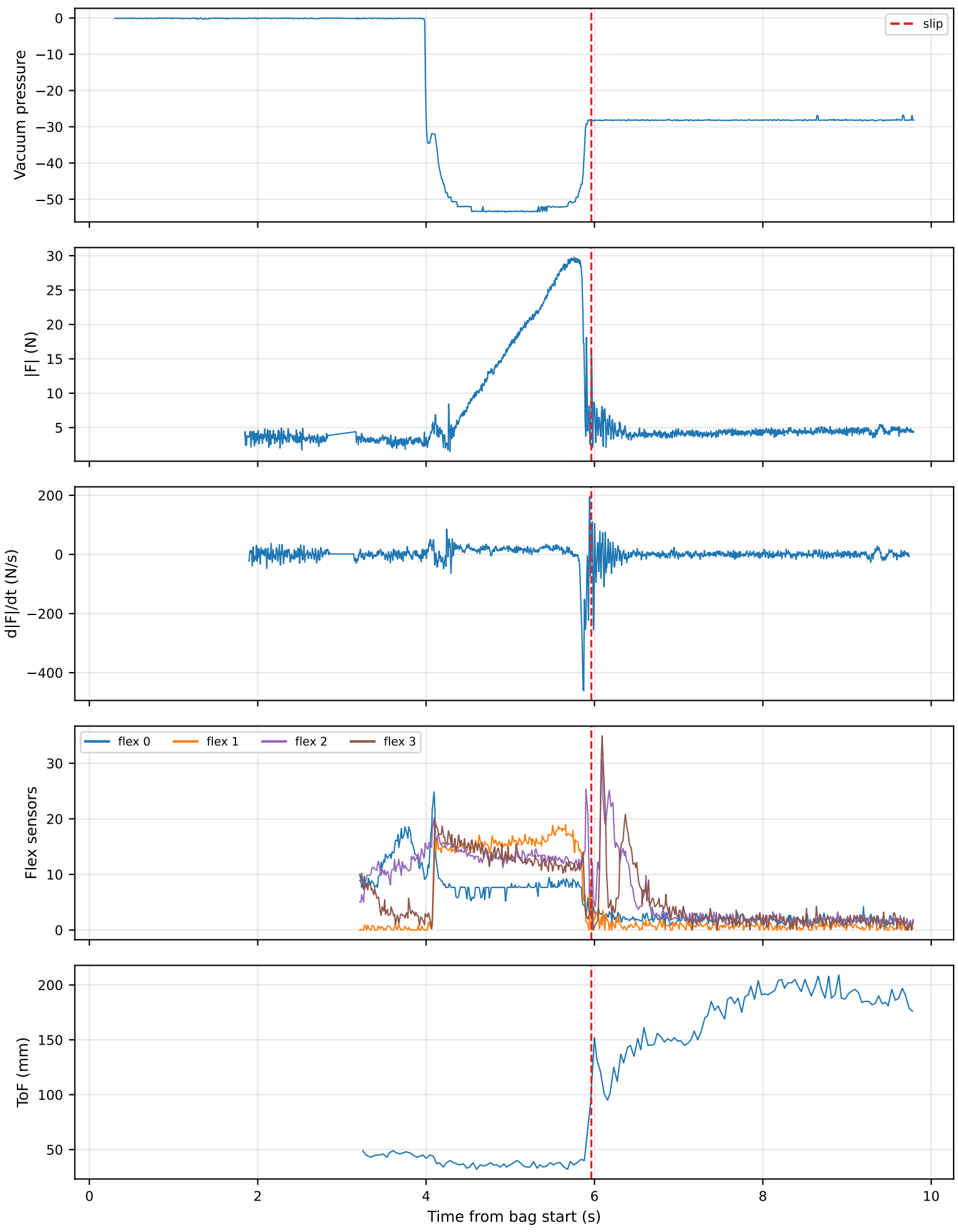
PICK at 5.08 s | 2026-10-06 13:47:45 | batch 108, apple 0 | stiffness_controller



final_approach_and_pick_20261006_134905.db3

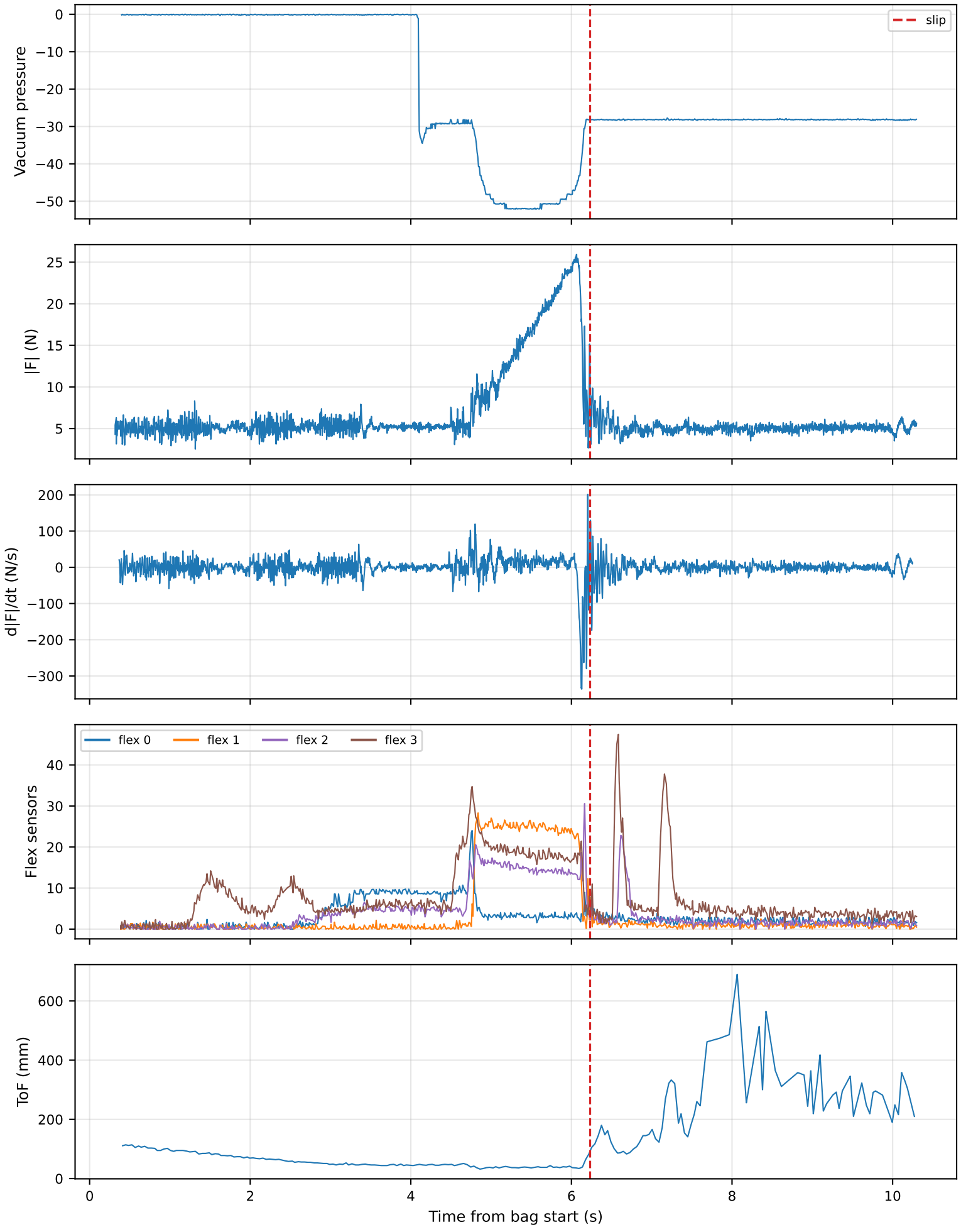
PICK at 8.93 s | 2026-10-06 13:49:07 | batch 109, apple 0 | stiffness_controller

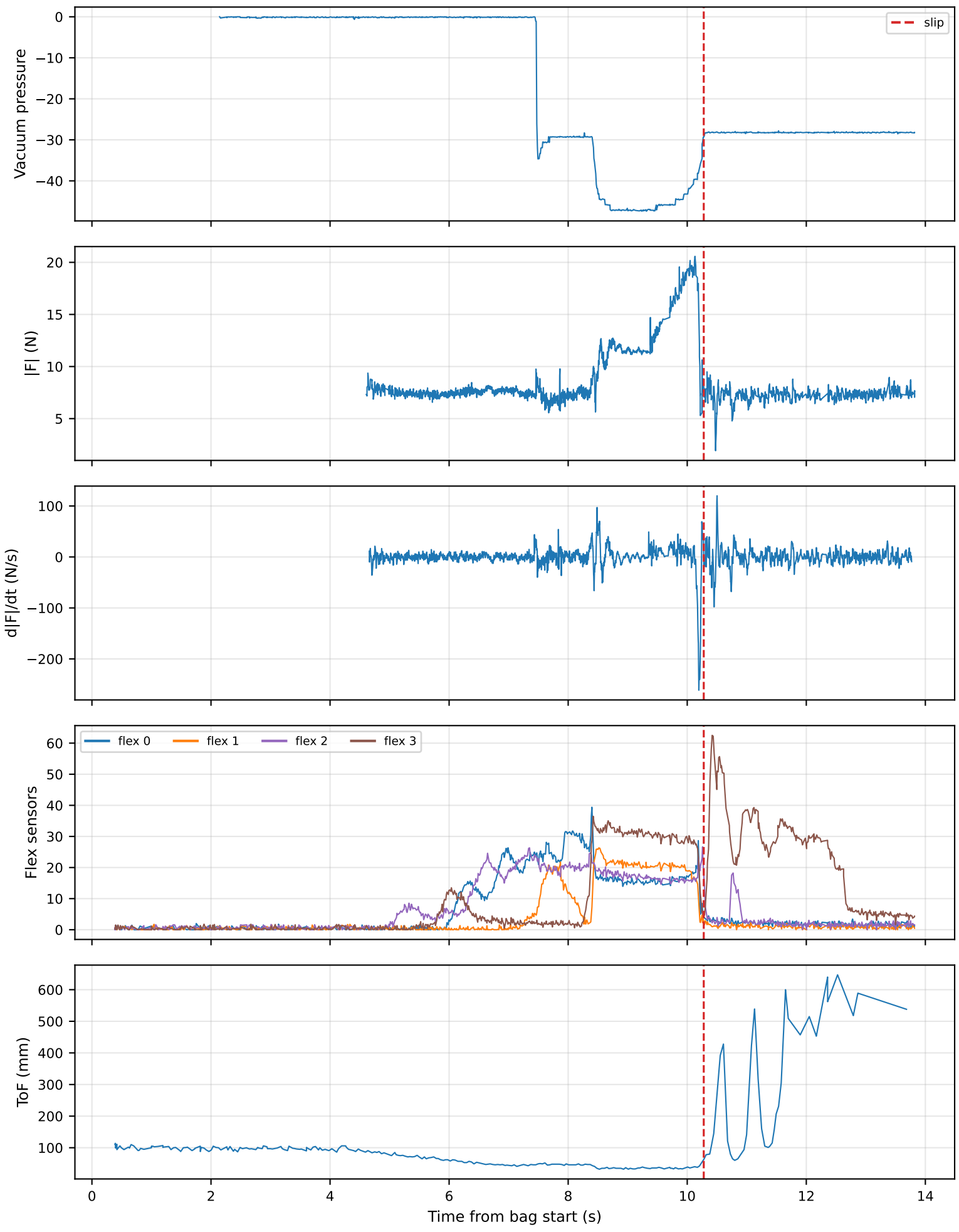


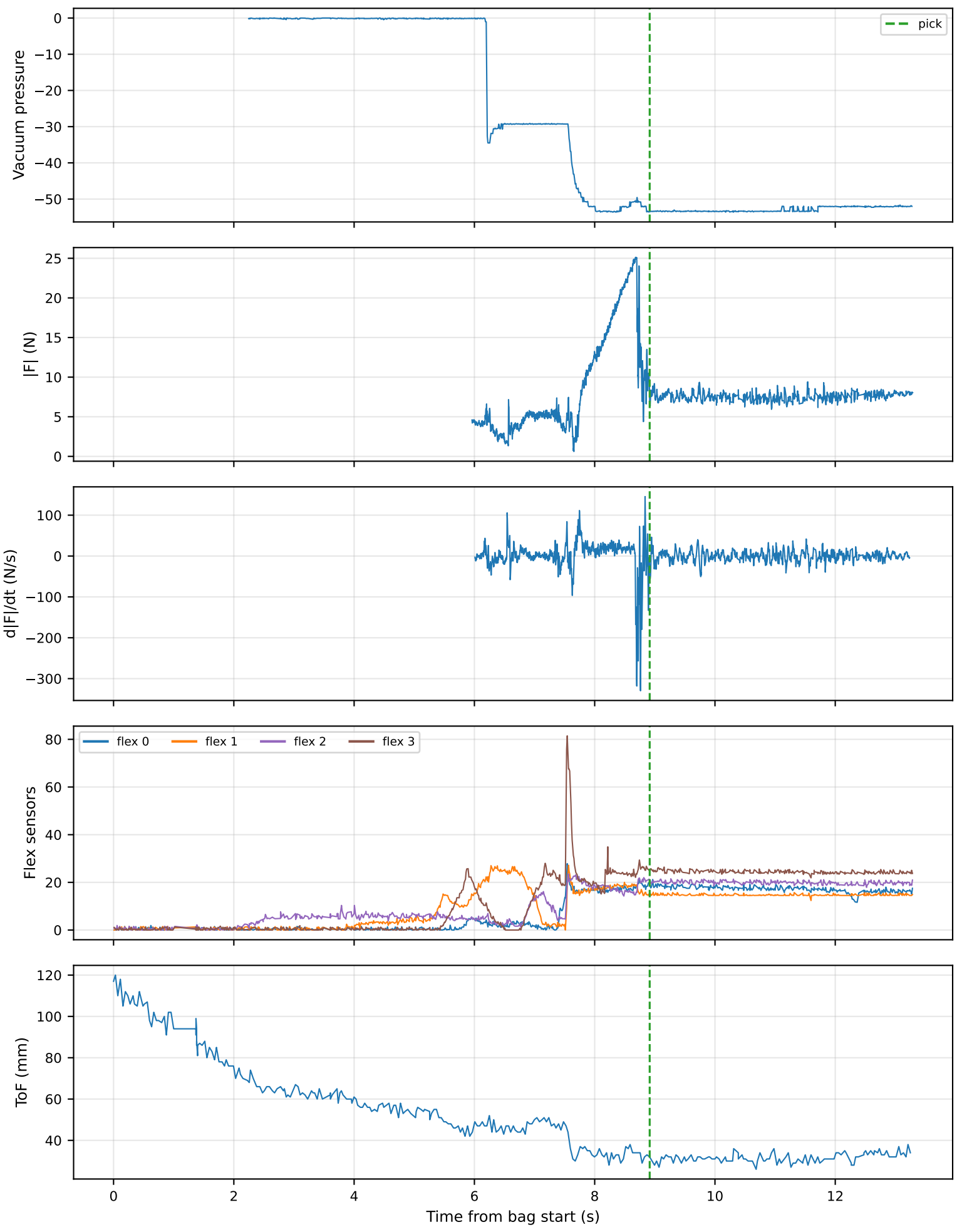

final_approach_and_pick_20261006_135019.db3
SLIP at 5.96 s | 2026-10-06 13:50:20 | batch 110, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

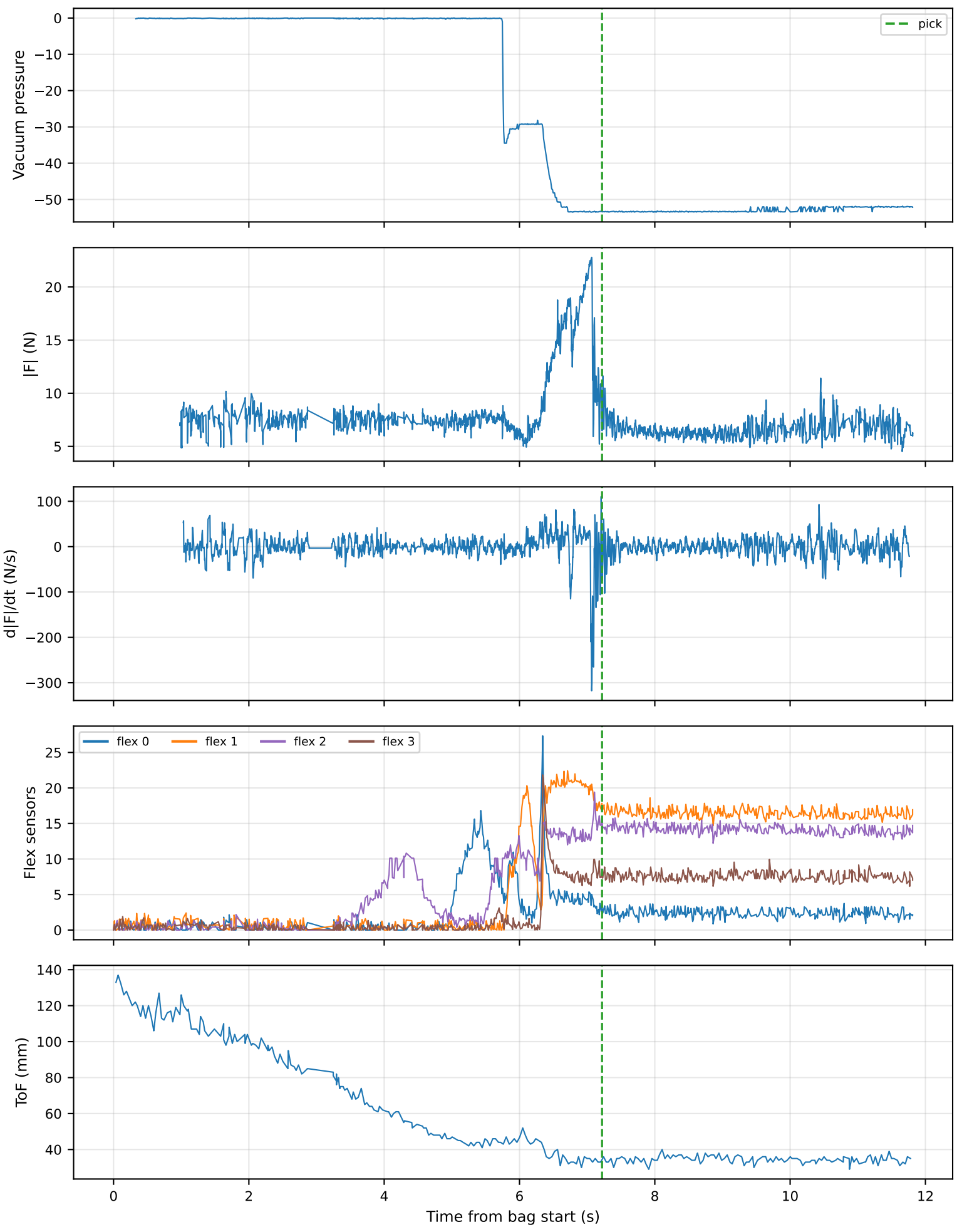
final_approach_and_pick_20261006_135224.db3

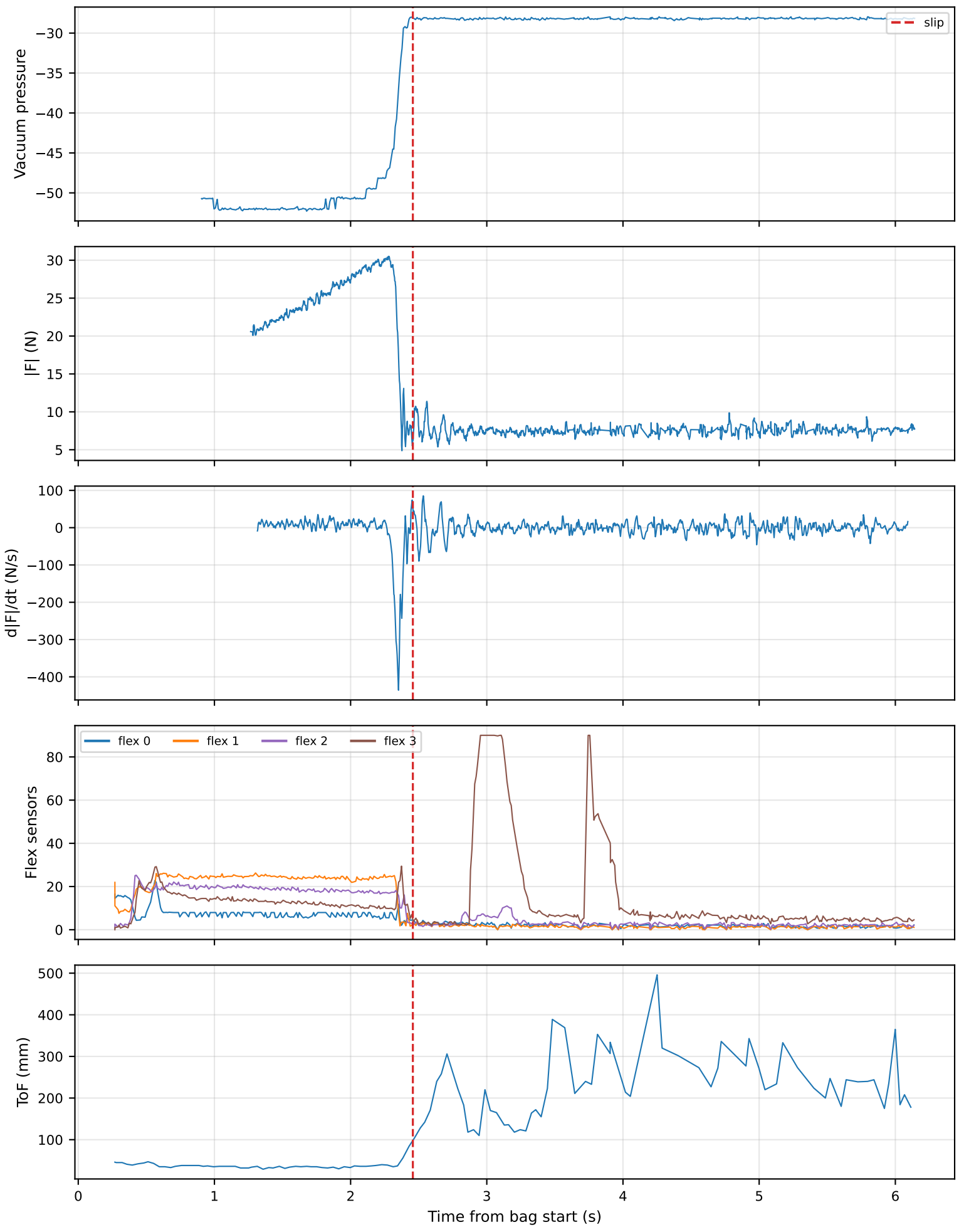
SLIP at 6.23 s | 2026-10-06 13:52:26 | batch 112, apple 0 | stiffness_controller

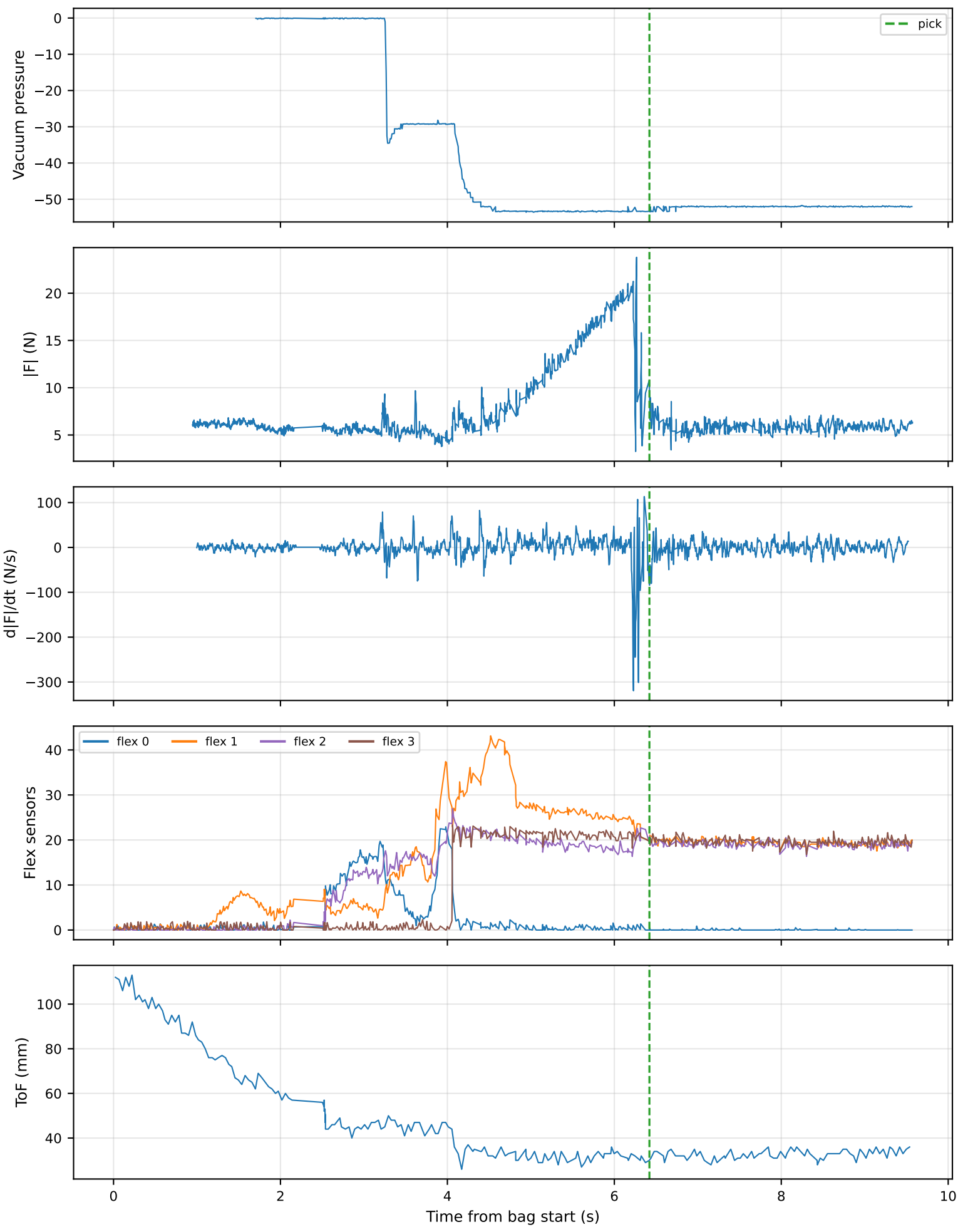


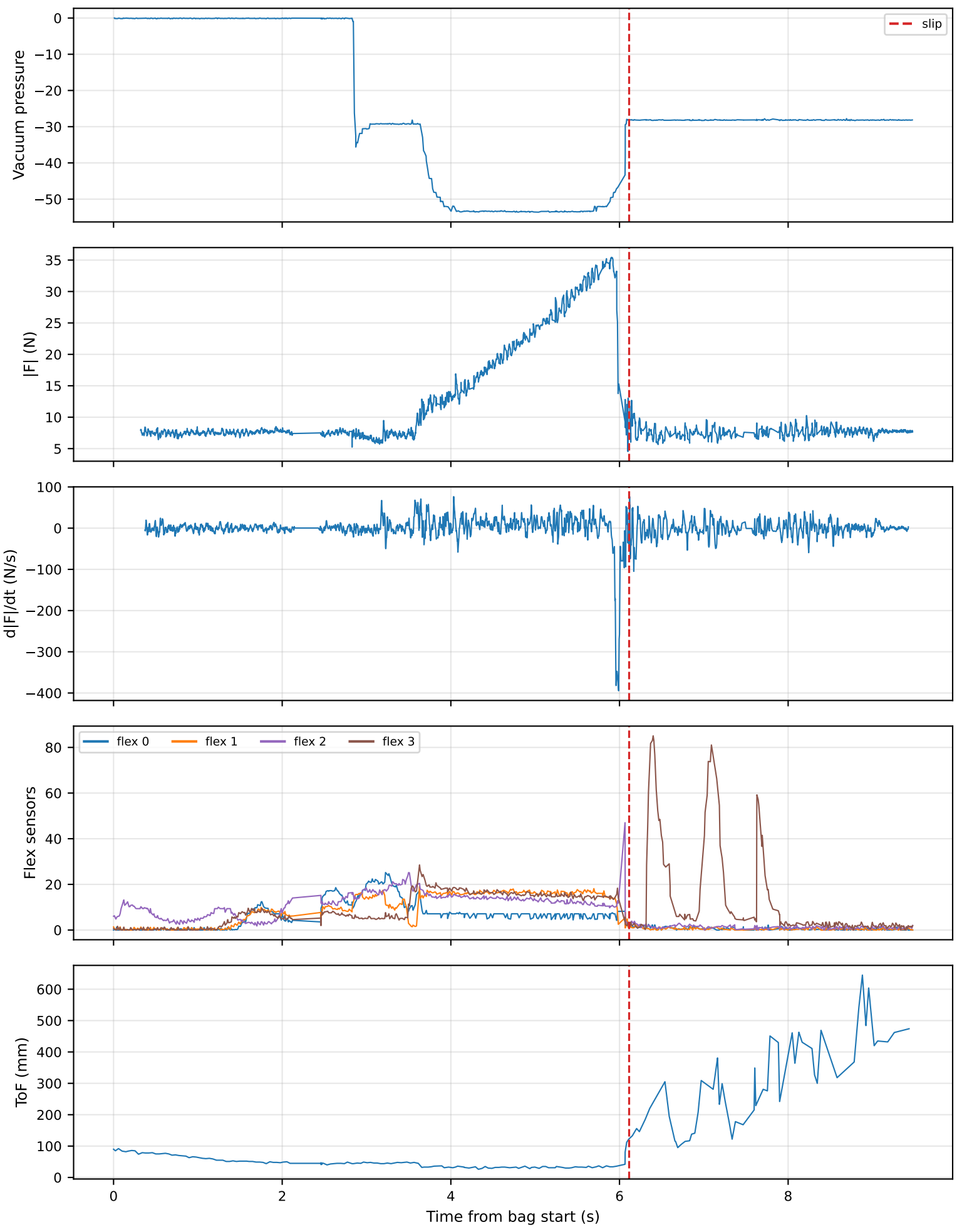

final_approach_and_pick_20261006_140032.db3
SLIP at 10.28 s | 2026-10-06 14:00:34 | batch 114, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

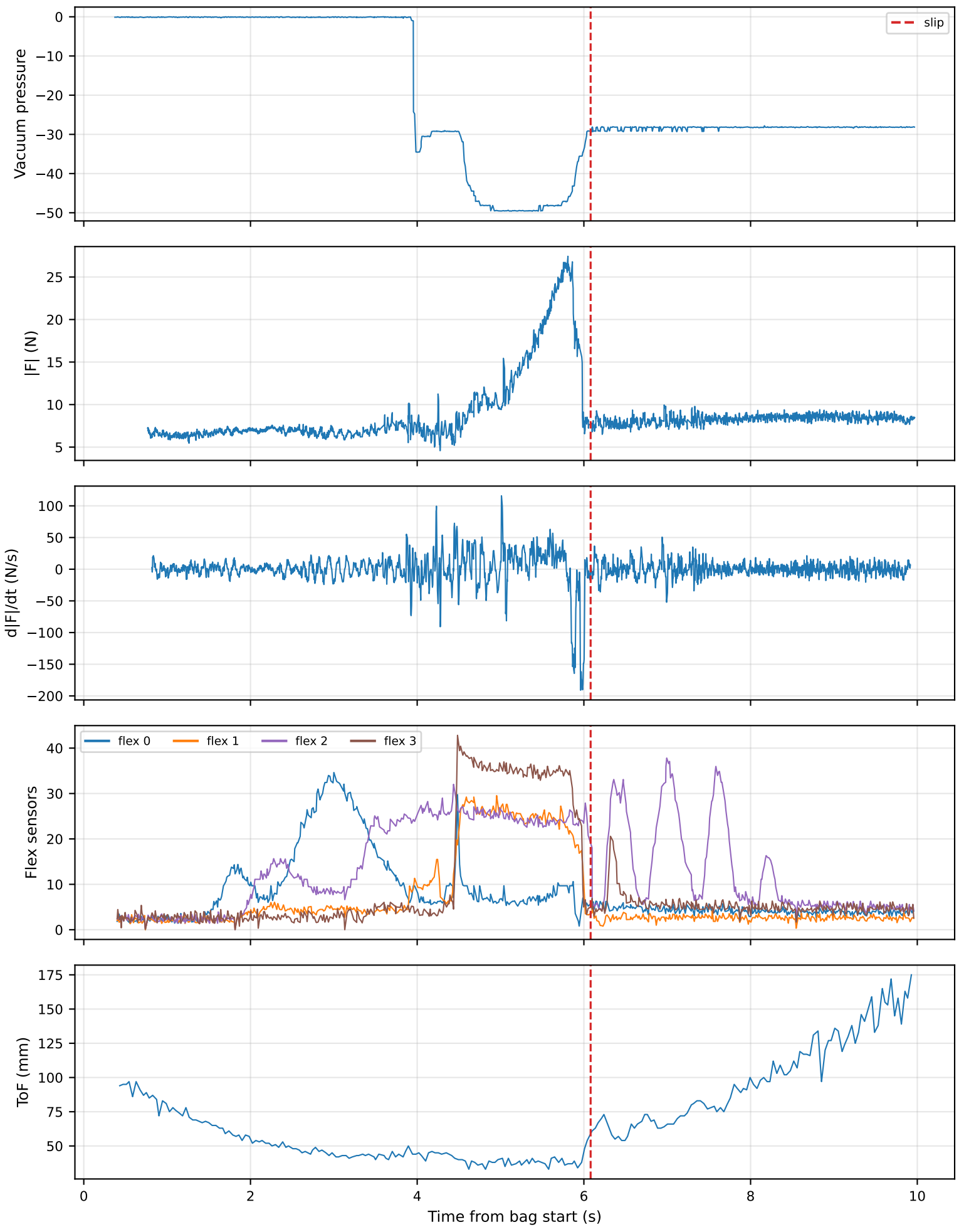

final_approach_and_pick_20261006_140146.db3
PICK at 8.91 s | 2026-10-06 14:01:46 | batch 115, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

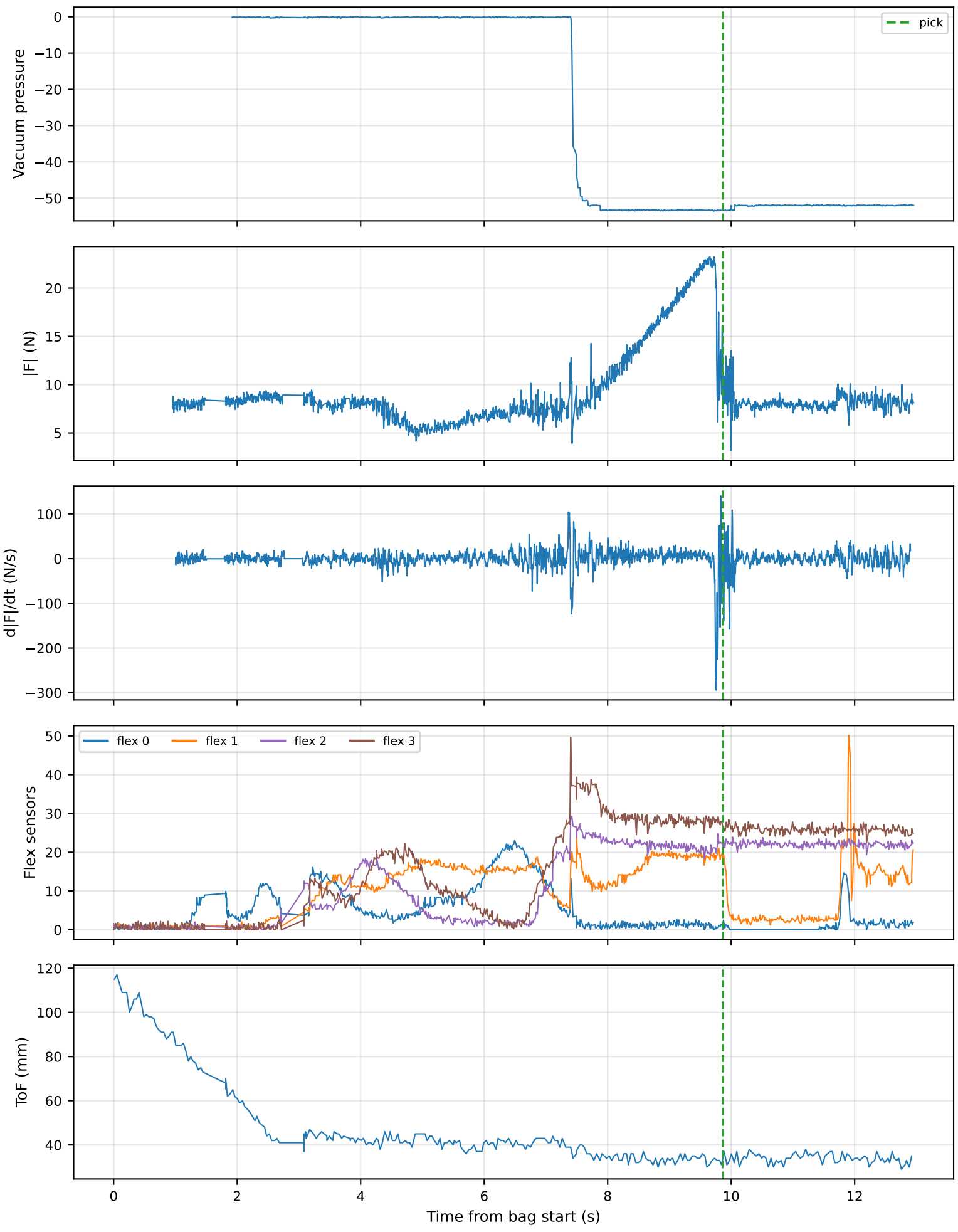

final_approach_and_pick_20261006_140258.db3
PICK at 7.22 s | 2026-10-06 14:02:59 | batch 116, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_140423.db3
SLIP at 2.46 s | 2026-10-06 14:04:28 | batch 117, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_140528.db3
PICK at 6.42 s | 2026-10-06 14:05:31 | batch 118, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

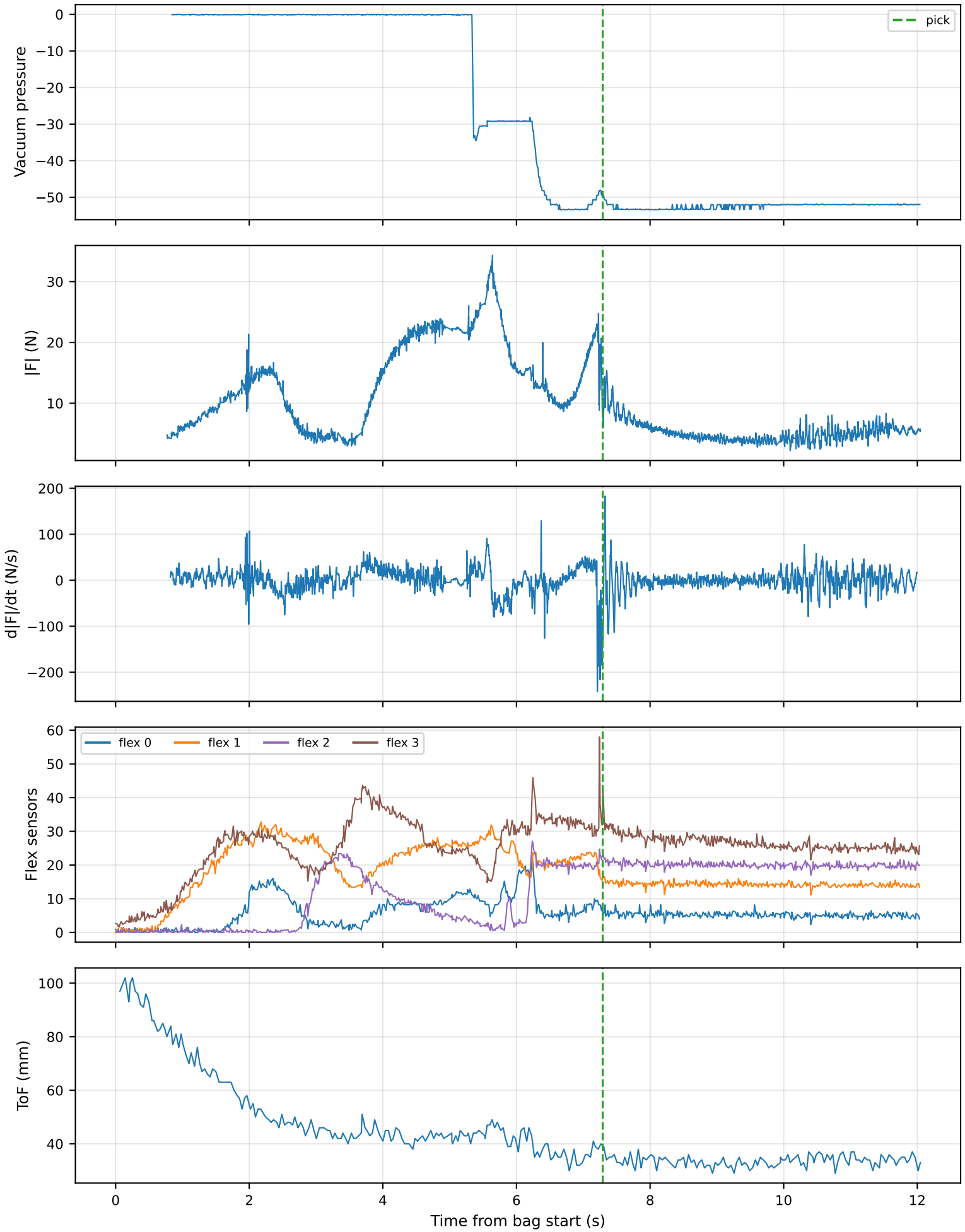

final_approach_and_pick_20261006_140622.db3
SLIP at 6.11 s | 2026-10-06 14:06:25 | batch 119, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_141022.db3
SLIP at 6.08 s | 2026-10-06 14:10:23 | batch 120, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_141154.db3
PICK at 9.87 s | 2026-10-06 14:11:55 | batch 121, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

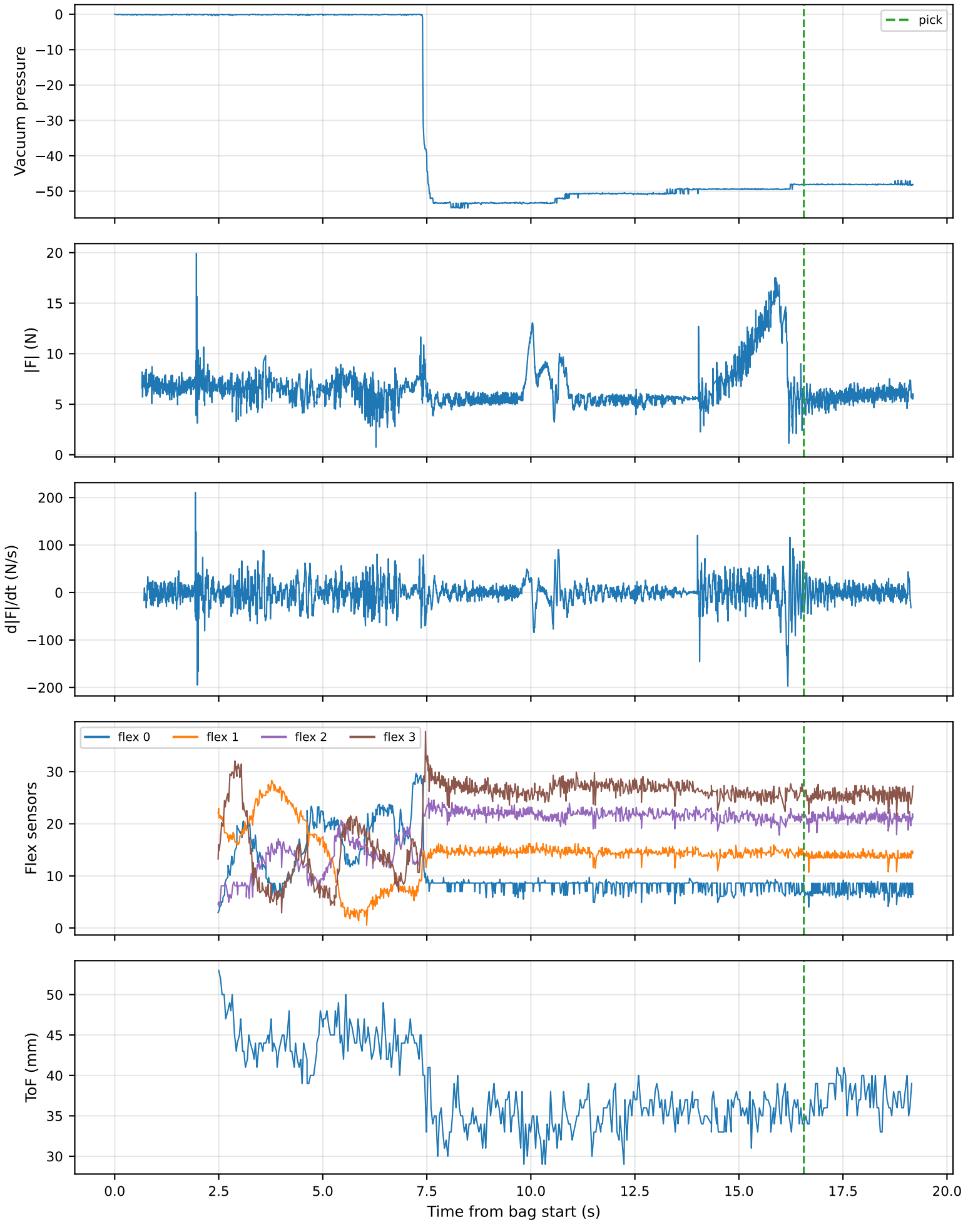
final_approach_and_pick_20261006_141300.db3

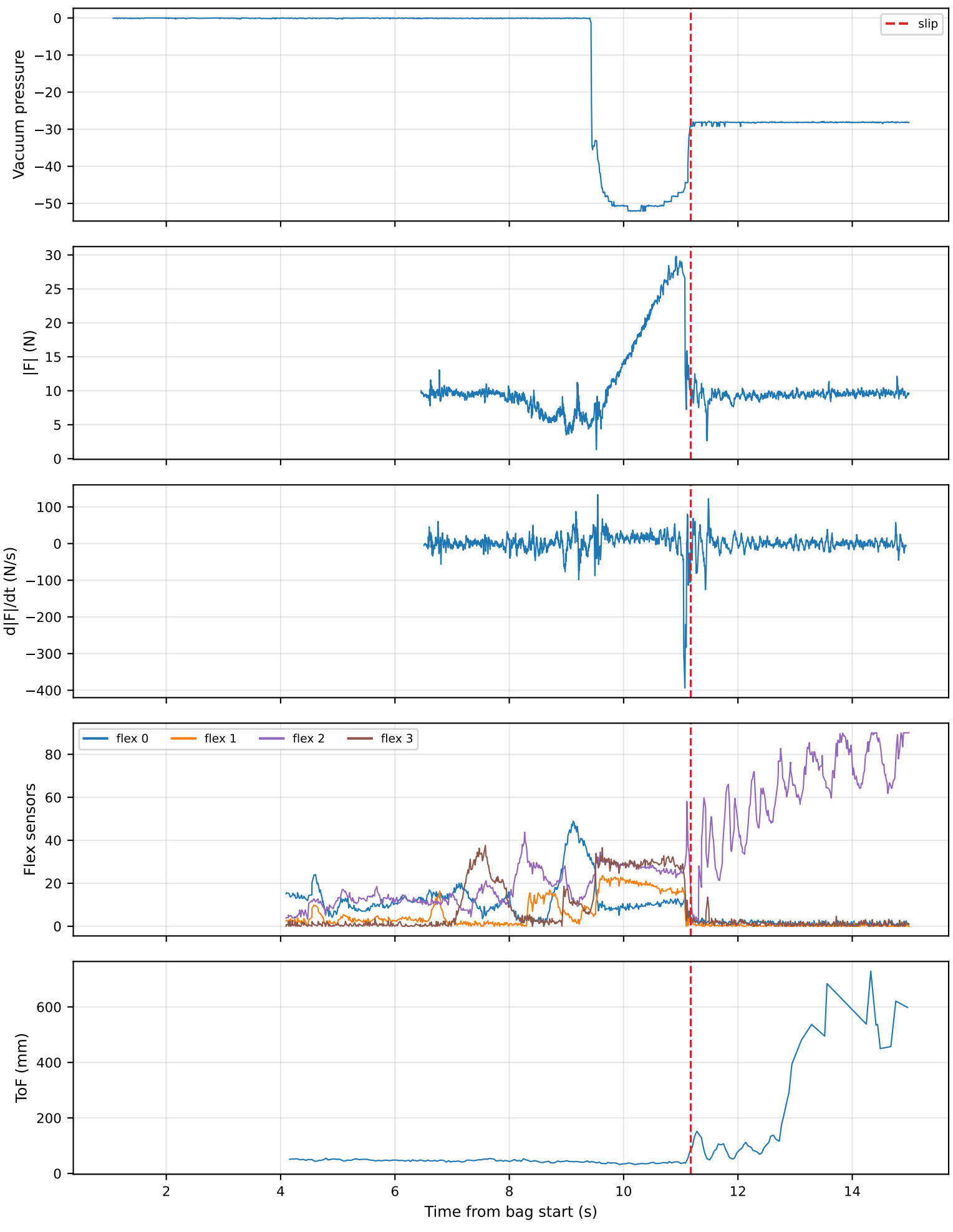
PICK at 7.29 s | 2026-10-06 14:13:01 | batch 122, apple 0 | stiffness_controller



final_approach_and_pick_20261006_141528.db3

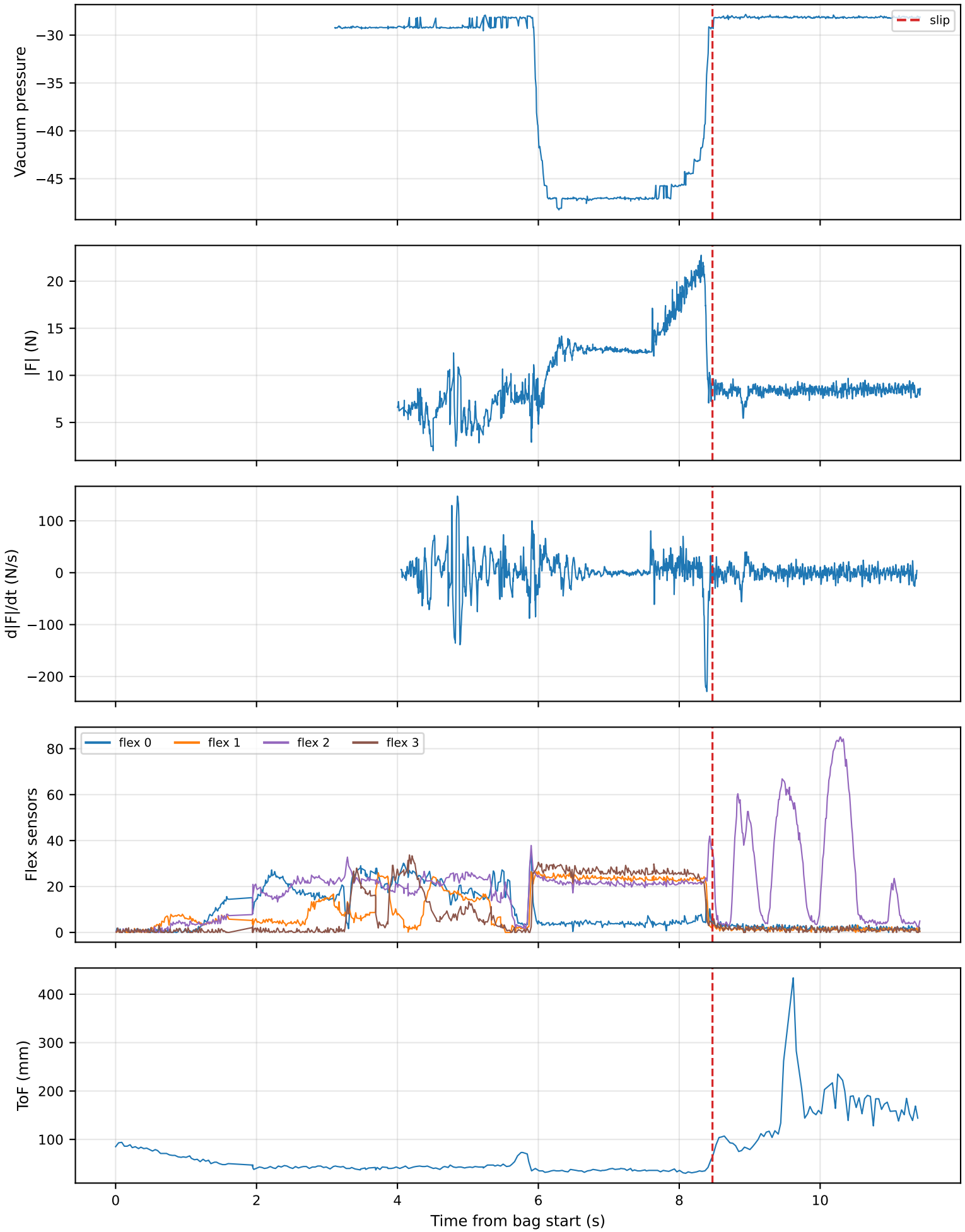
PICK at 16.56 s | 2026-10-06 14:15:29 | batch 123, apple 0 | stiffness_controller

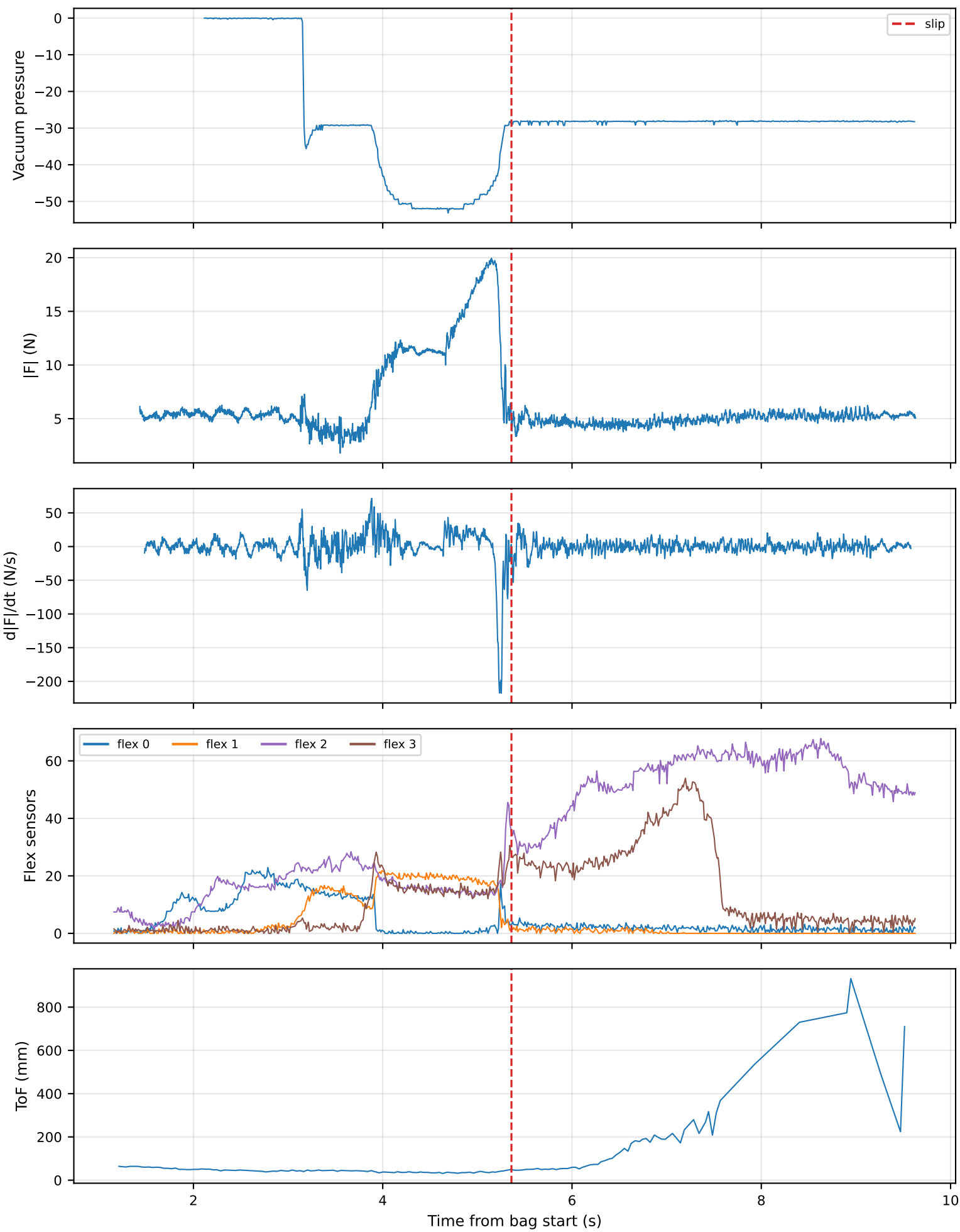


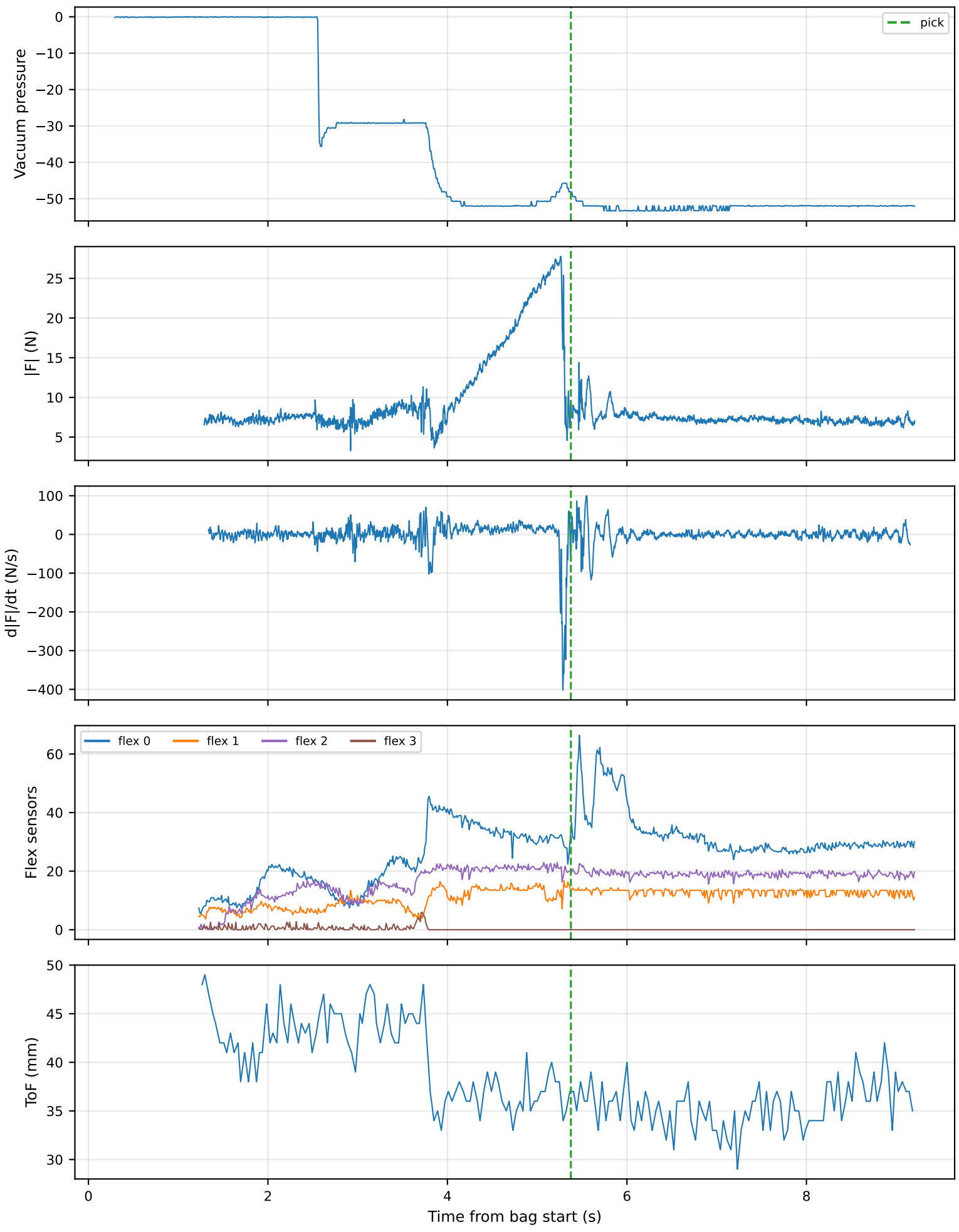

final_approach_and_pick_20261006_141642.db3
SLIP at 11.18 s | 2026-10-06 14:16:43 | batch 124, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

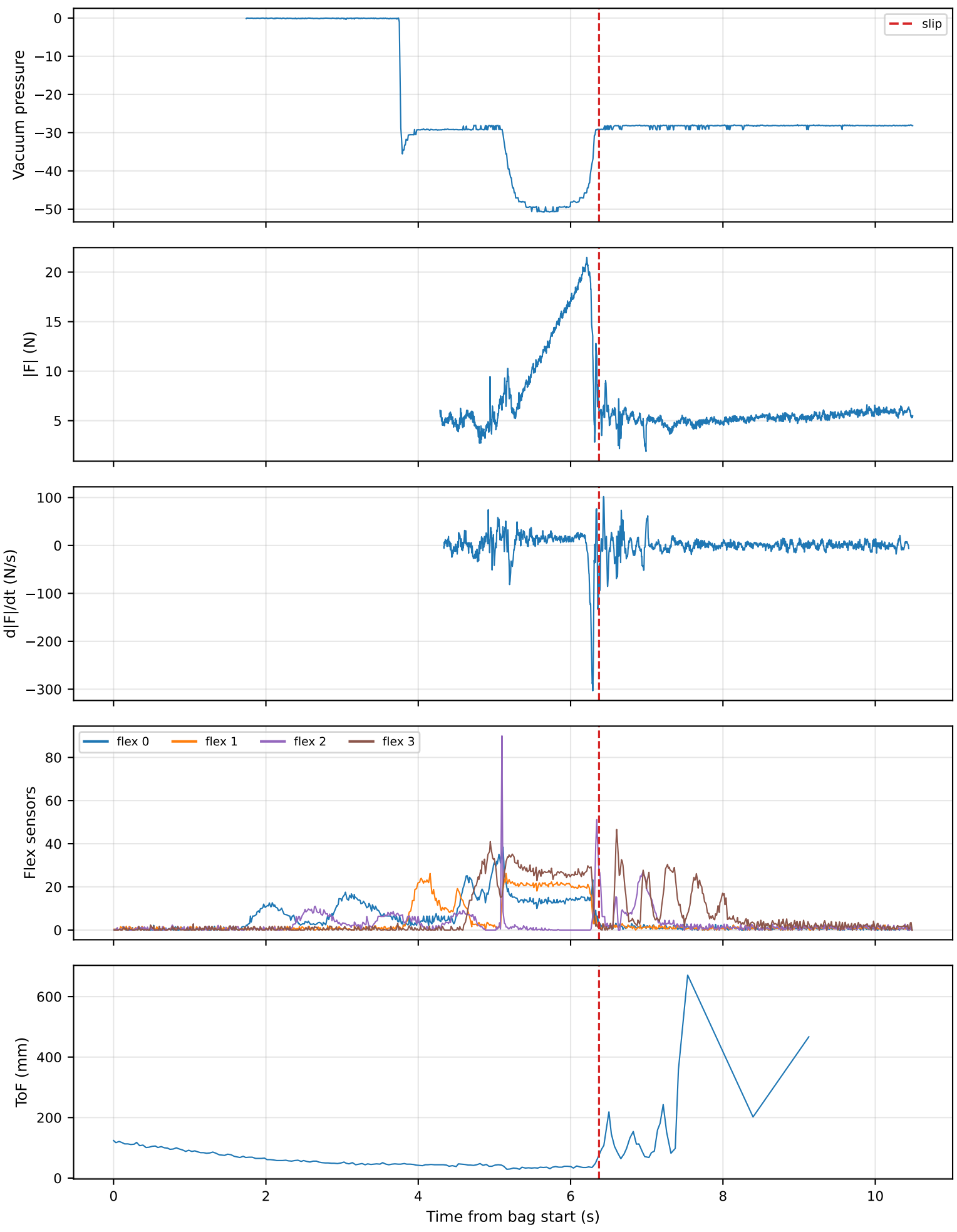
final_approach_and_pick_20261006_141814.db3

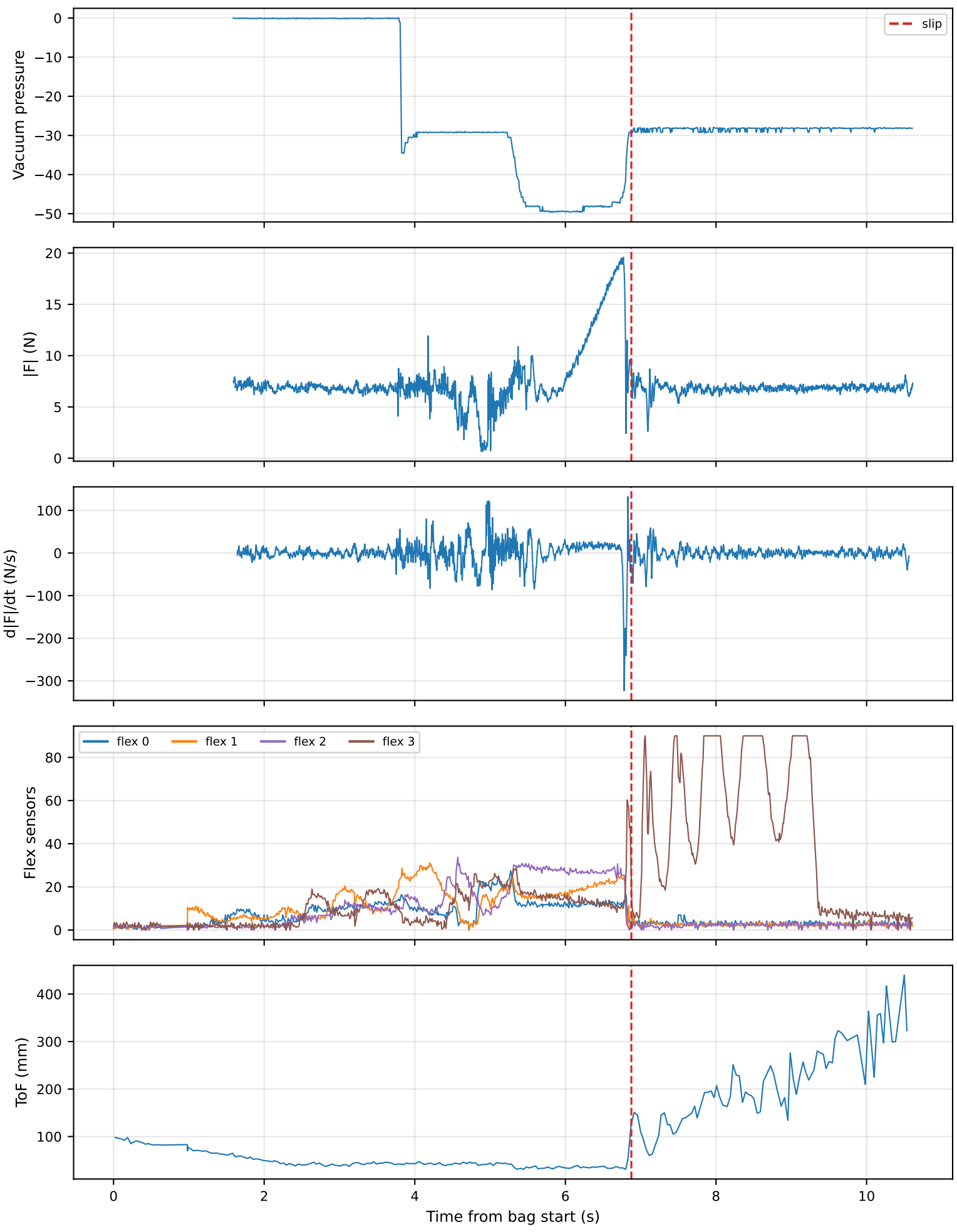
SLIP at 8.47 s | 2026-10-06 14:18:15 | batch 125, apple 0 | stiffness_controller

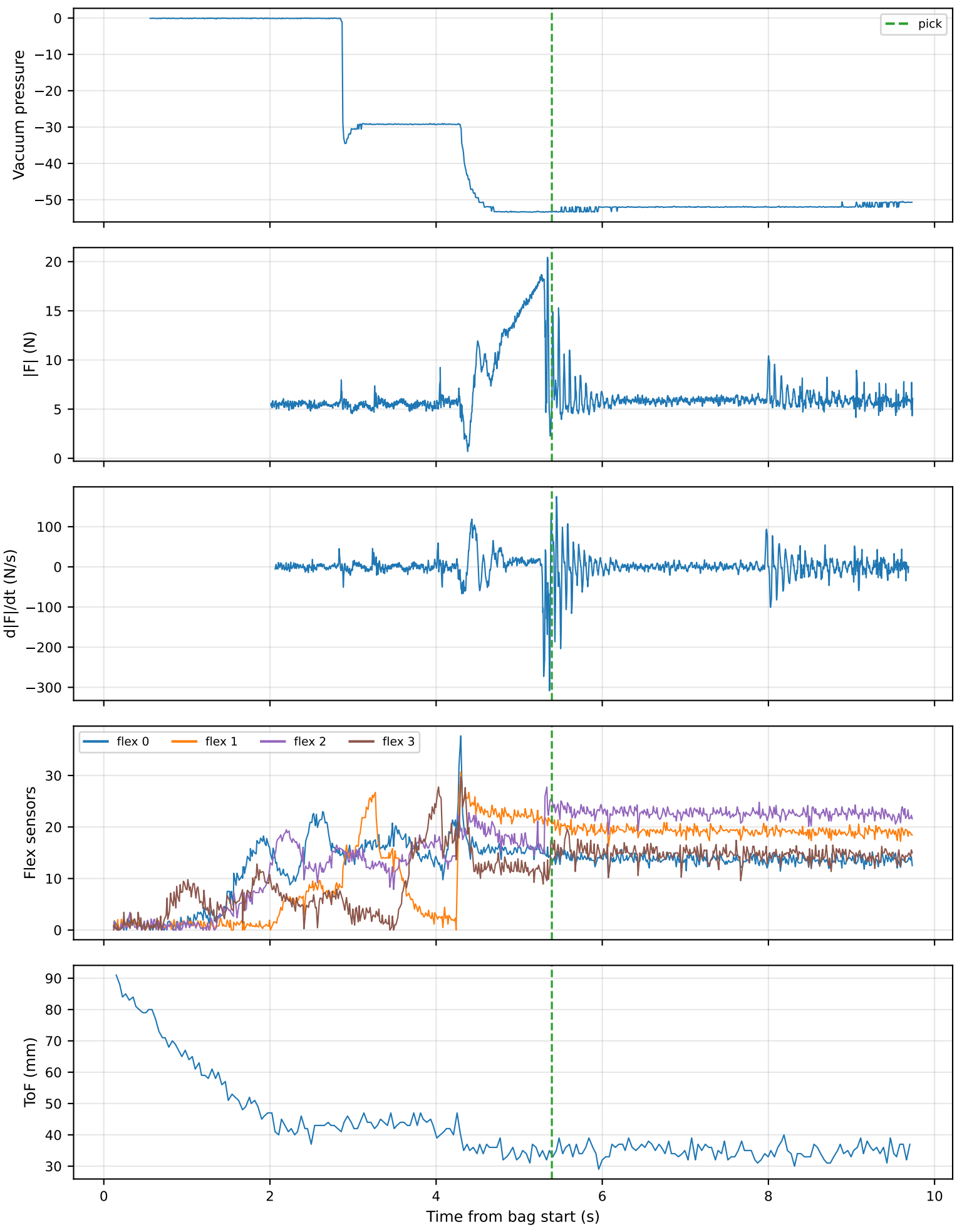



final_approach_and_pick_20261006_142205.db3
SLIP at 5.36 s | 2026-10-06 14:22:06 | batch 126, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_142311.db3
PICK at 5.38 s | 2026-10-06 14:23:12 | batch 127, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

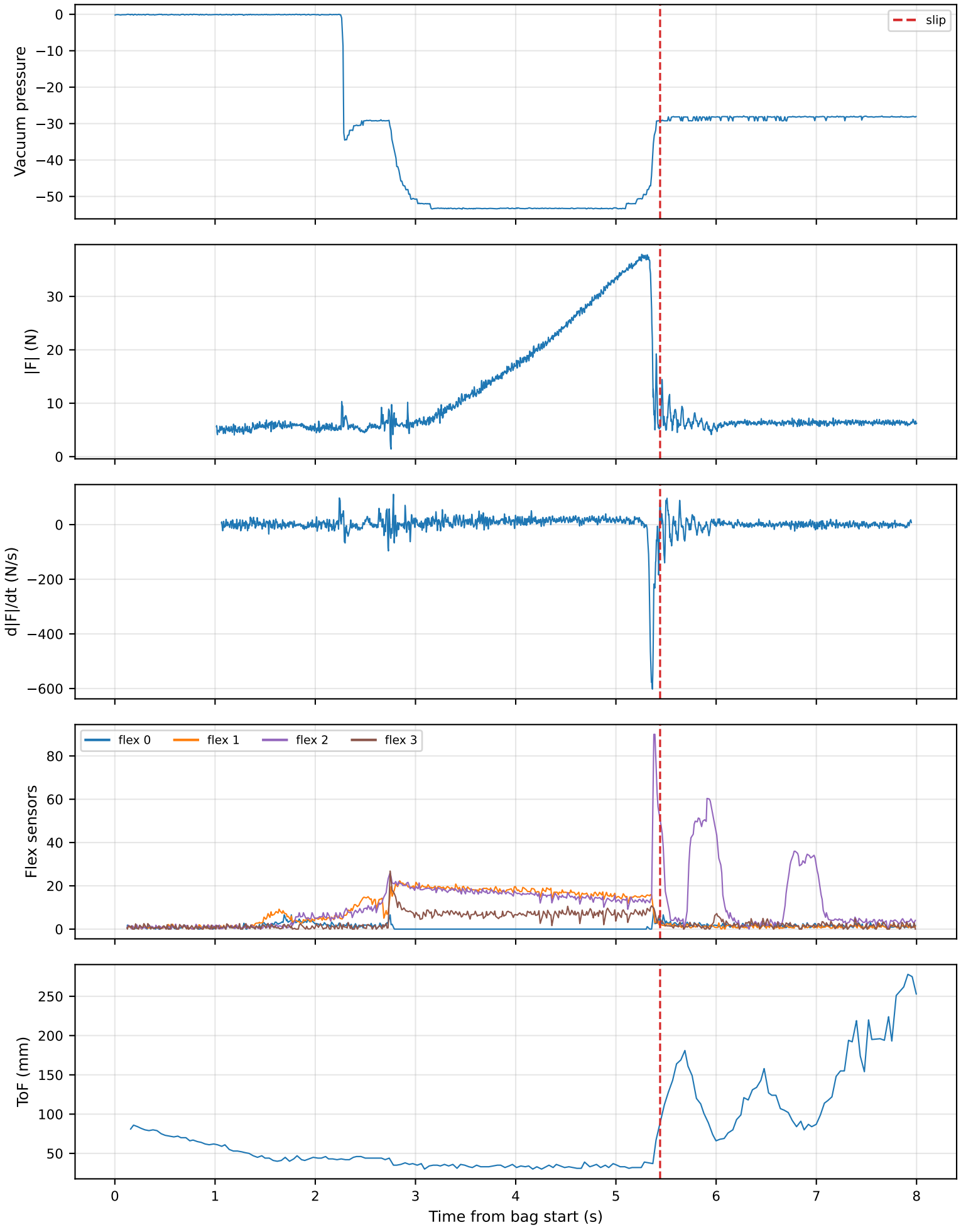

final_approach_and_pick_20261006_142424.db3
SLIP at 6.37 s | 2026-10-06 14:24:25 | batch 128, apple 0 | stiffness_controller
slip
Vacuum pressure
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

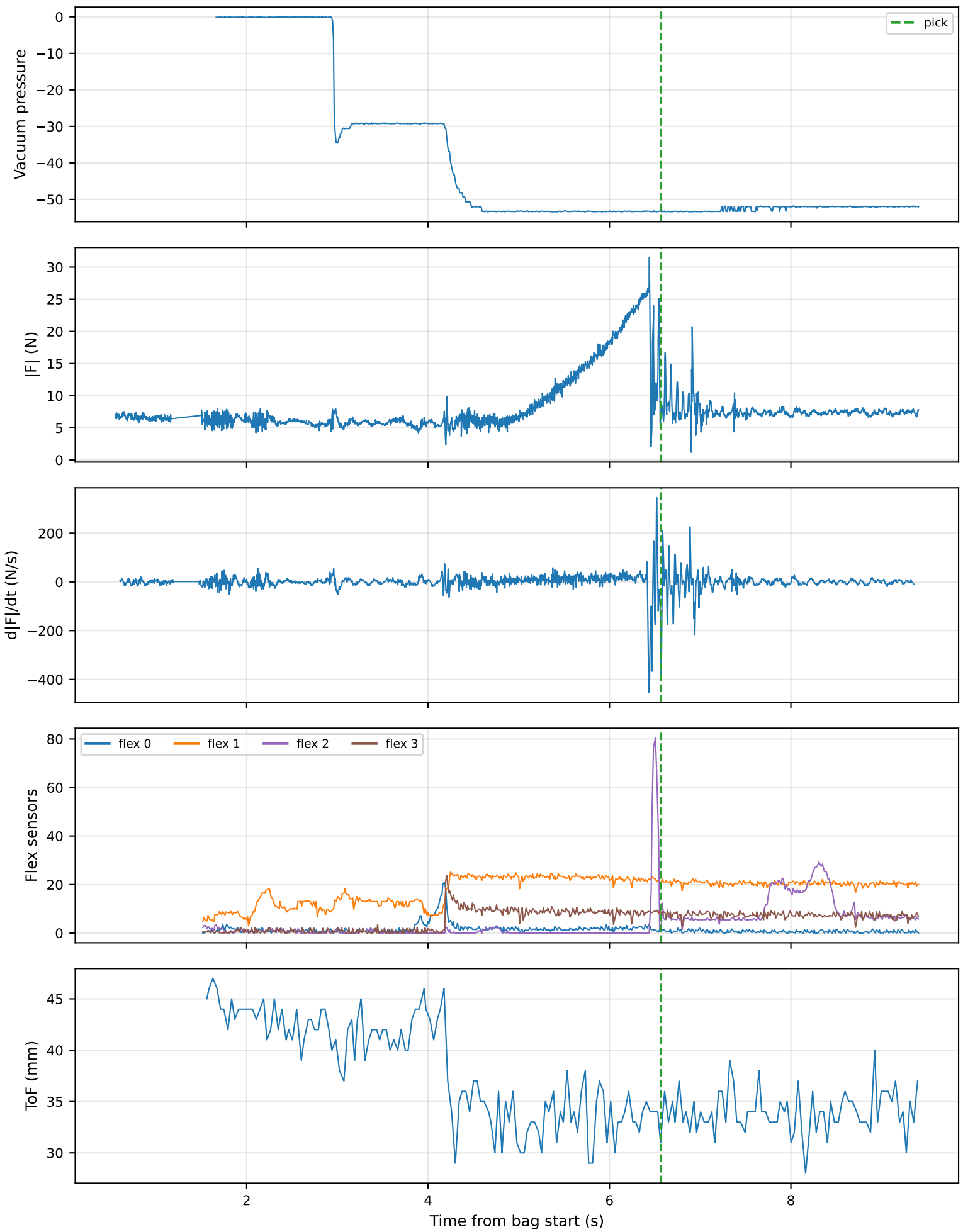

final_approach_and_pick_20261006_142609.db3
SLIP at 6.88 s | 2026-10-06 14:26:11 | batch 129, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

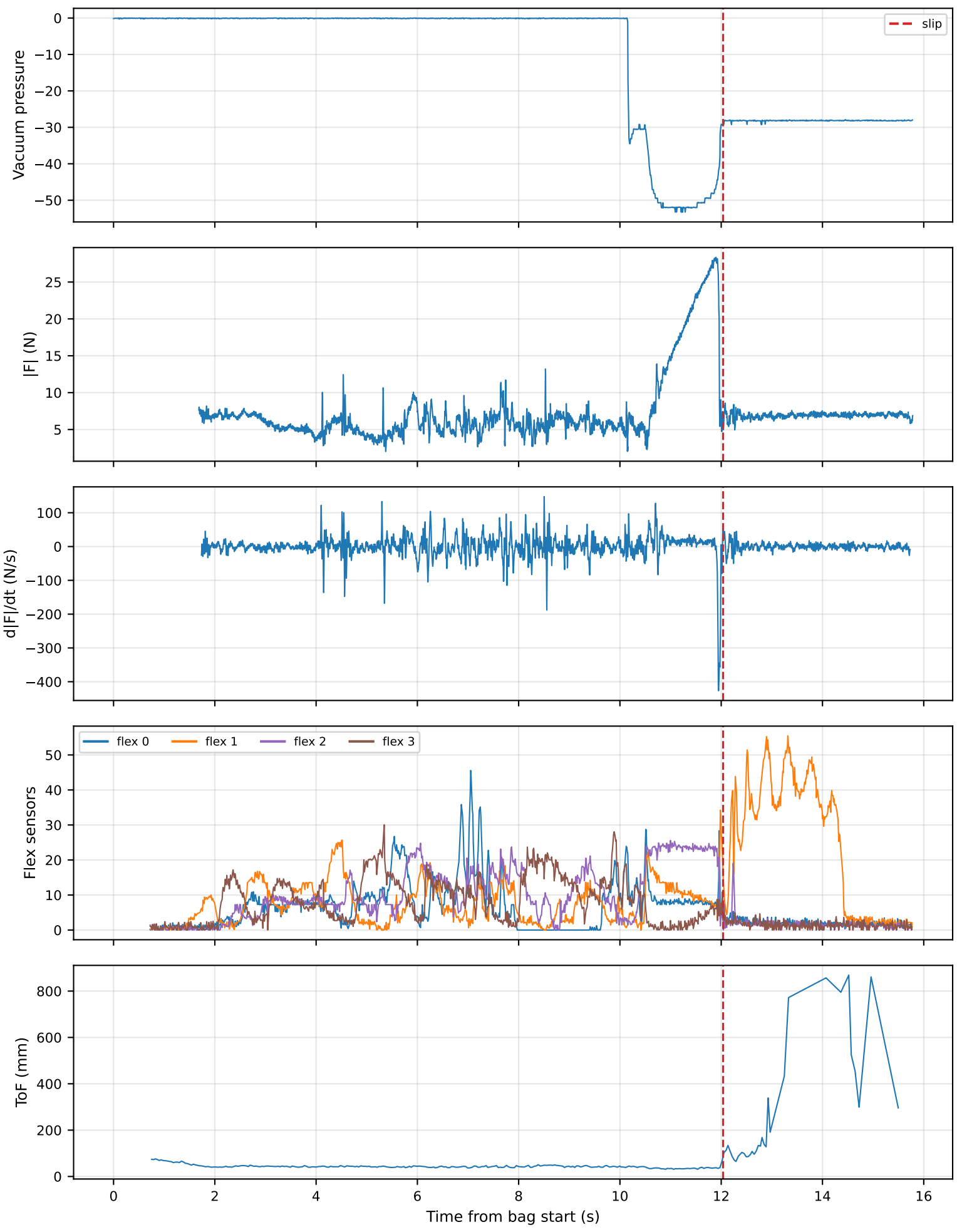

final_approach_and_pick_20261006_142730.db3
PICK at 5.39 s | 2026-10-06 14:27:32 | batch 130, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

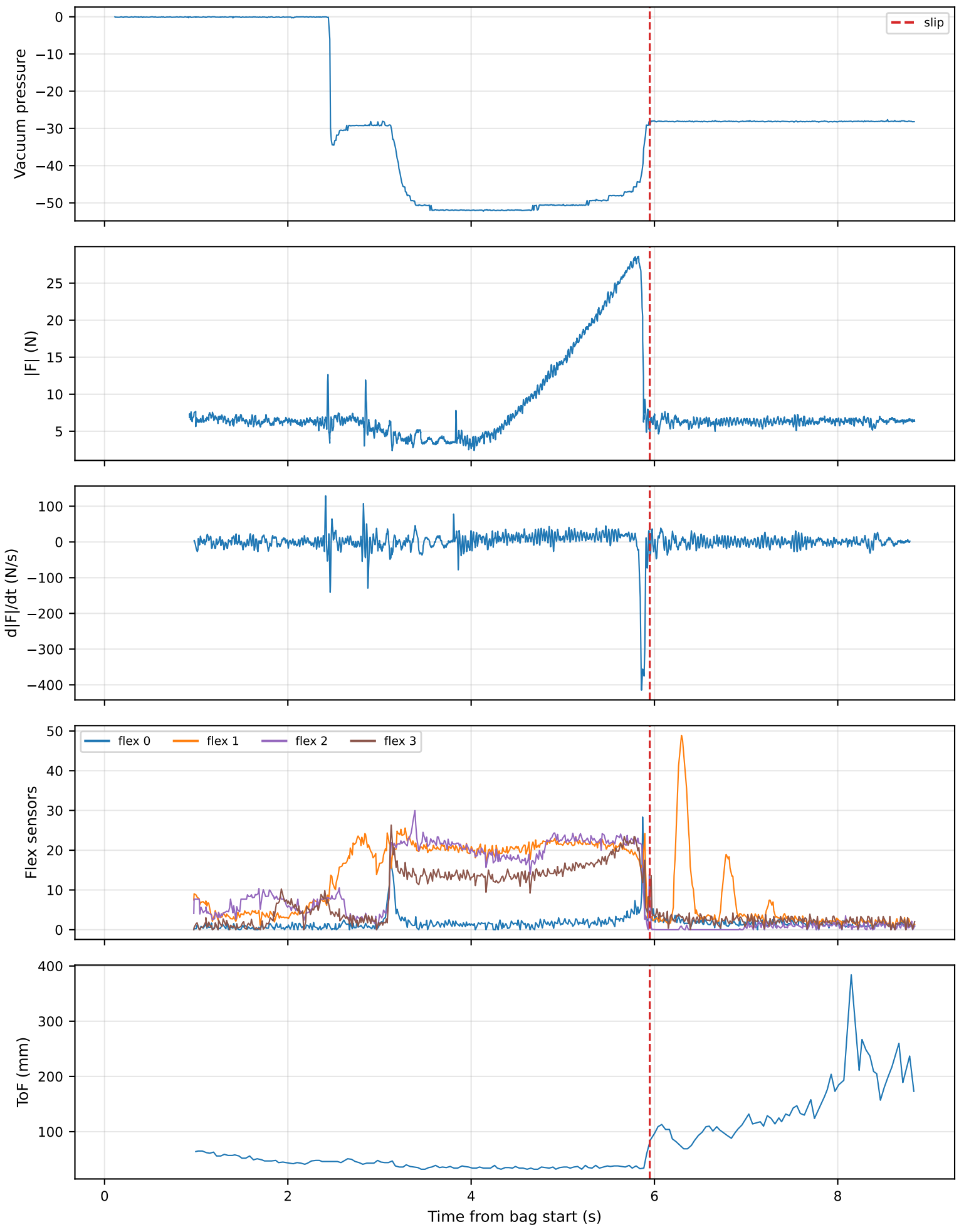
final_approach_and_pick_20261006_142840.db3

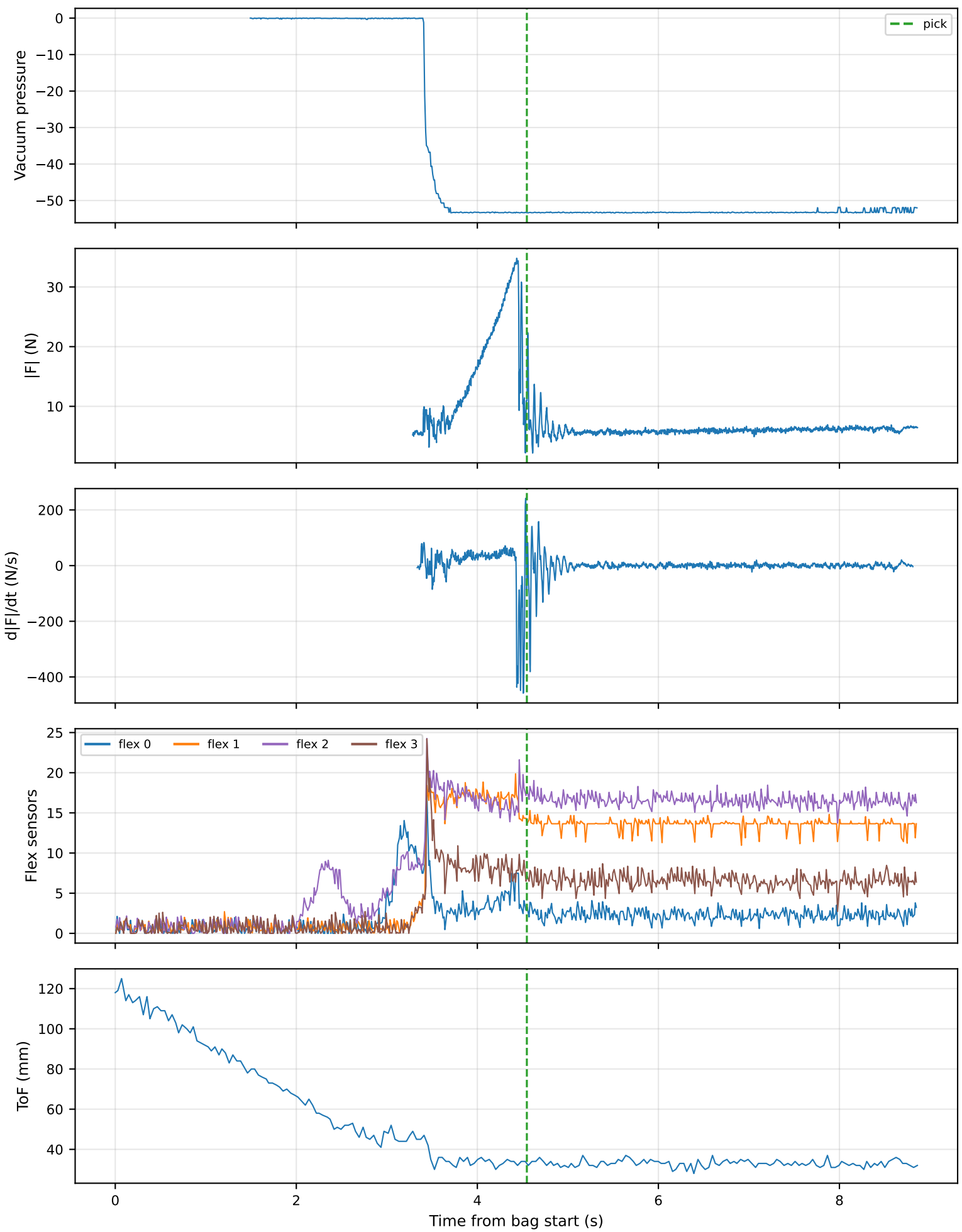
SLIP at 5.44 s | 2026-10-06 14:28:41 | batch 131, apple 0 | stiffness_controller

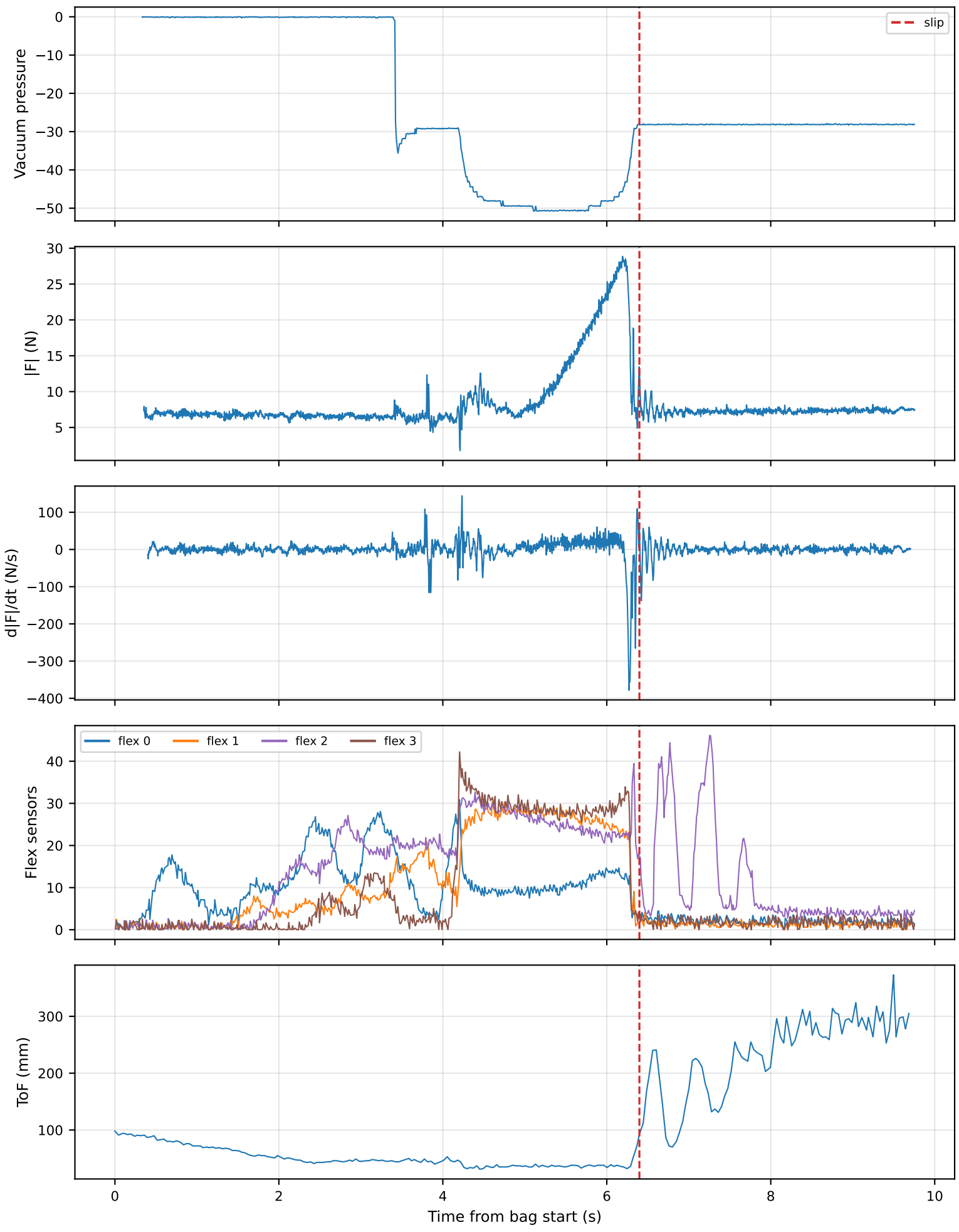



final_approach_and_pick_20261006_142930.db3
PICK at 6.57 s | 2026-10-06 14:29:31 | batch 132, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_143129.db3
SLIP at 12.04 s | 2026-10-06 14:31:30 | batch 33, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

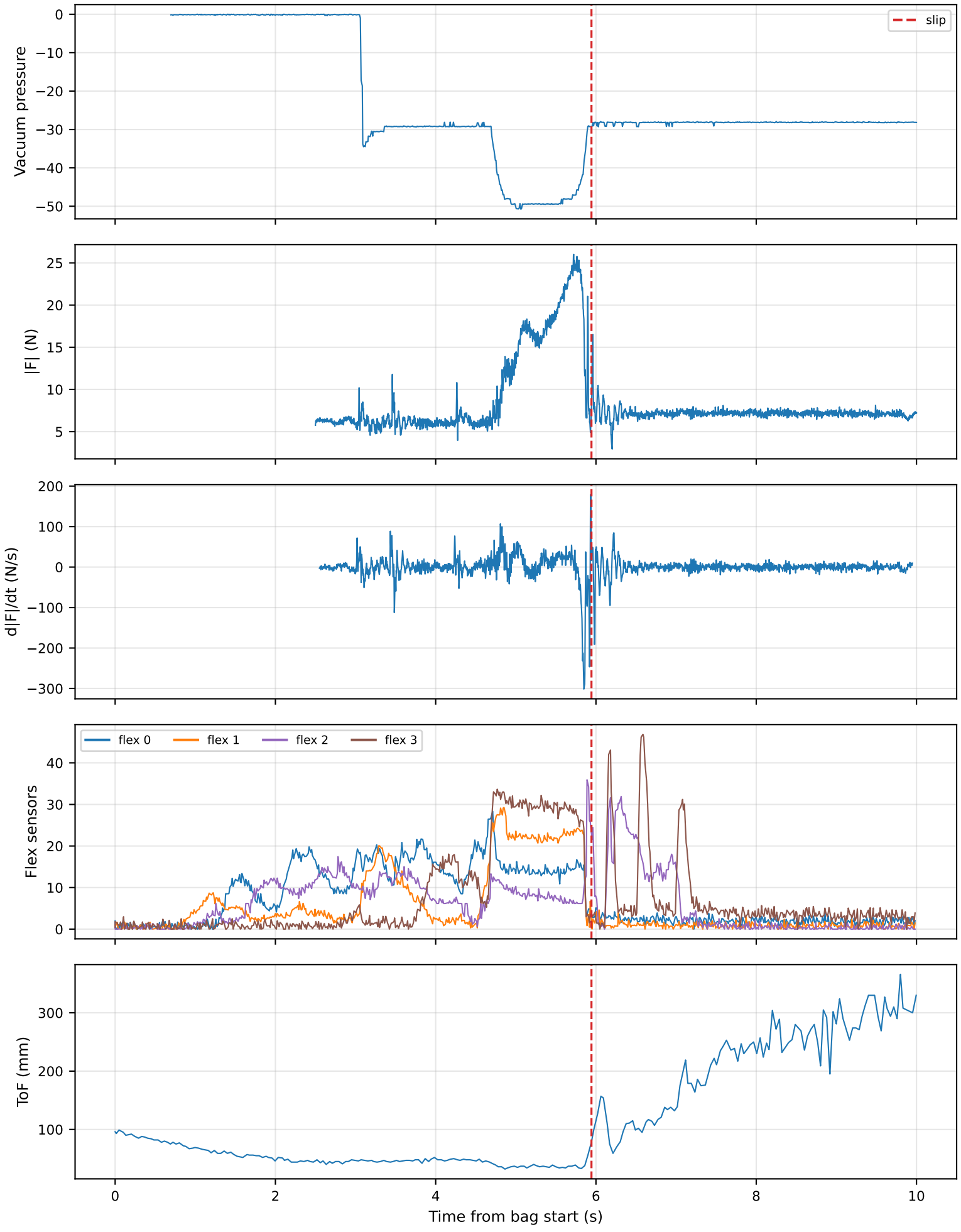

final_approach_and_pick_20261006_143319.db3
SLIP at 5.95 s | 2026-10-06 14:33:21 | batch 133, apple 0 | stiffness_controller
slip
Vacuum pressure
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_143407.db3
PICK at 4.55 s | 2026-10-06 14:34:08 | batch 134, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_143502.db3
SLIP at 6.40 s | 2026-10-06 14:35:04 | batch 135, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

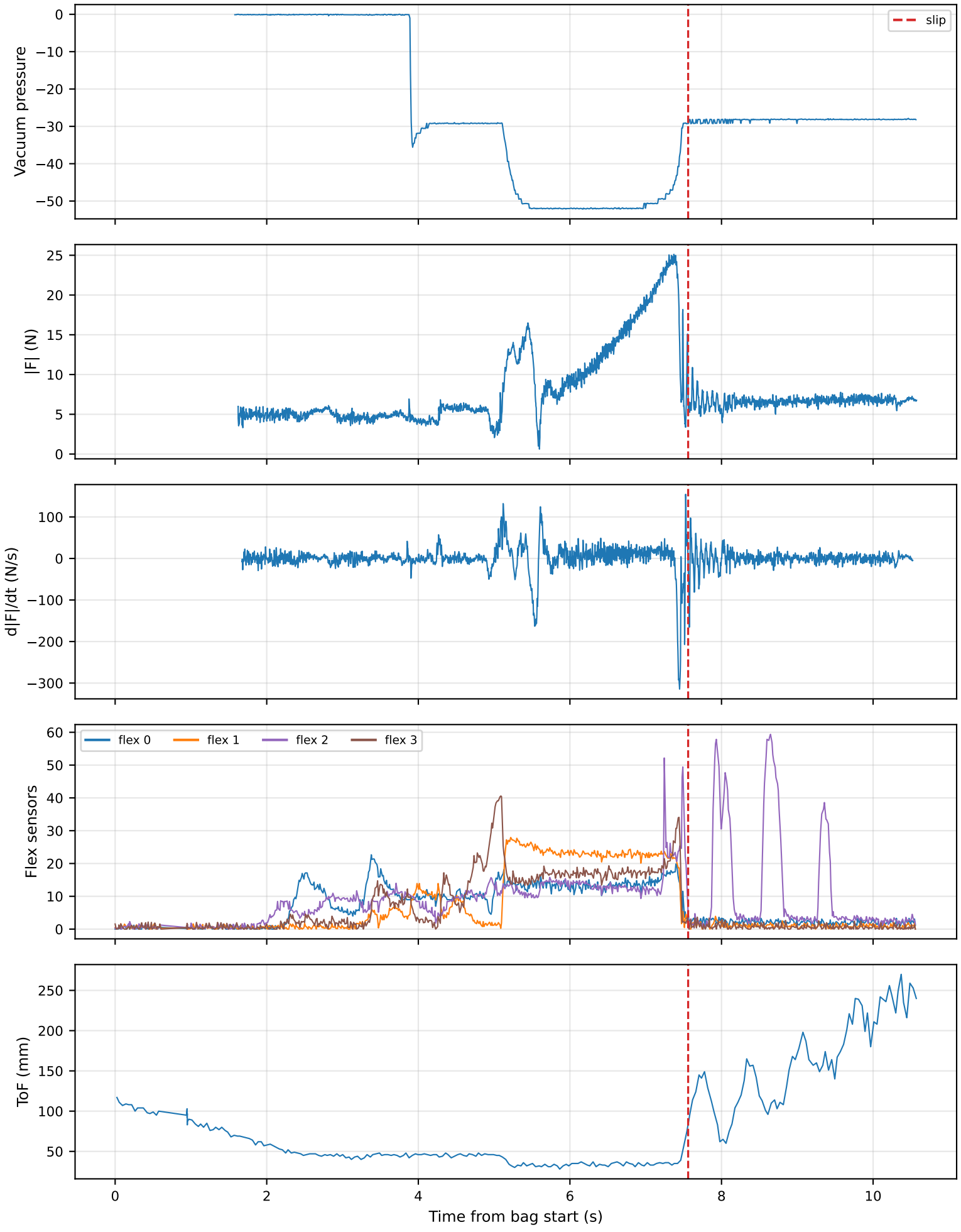
final_approach_and_pick_20261006_143557.db3

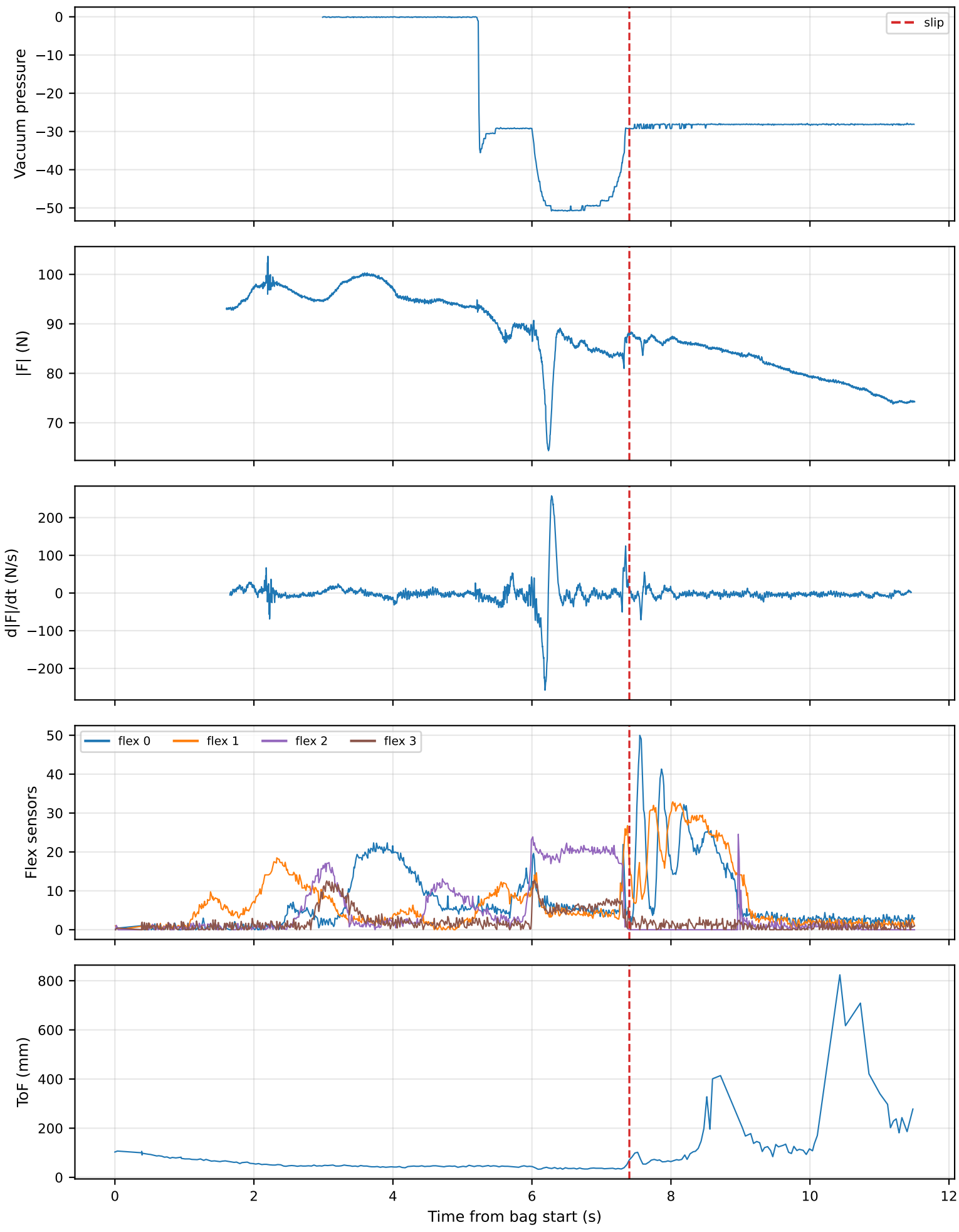
SLIP at 5.94 s | 2026-10-06 14:35:59 | batch 136, apple 0 | stiffness_controller

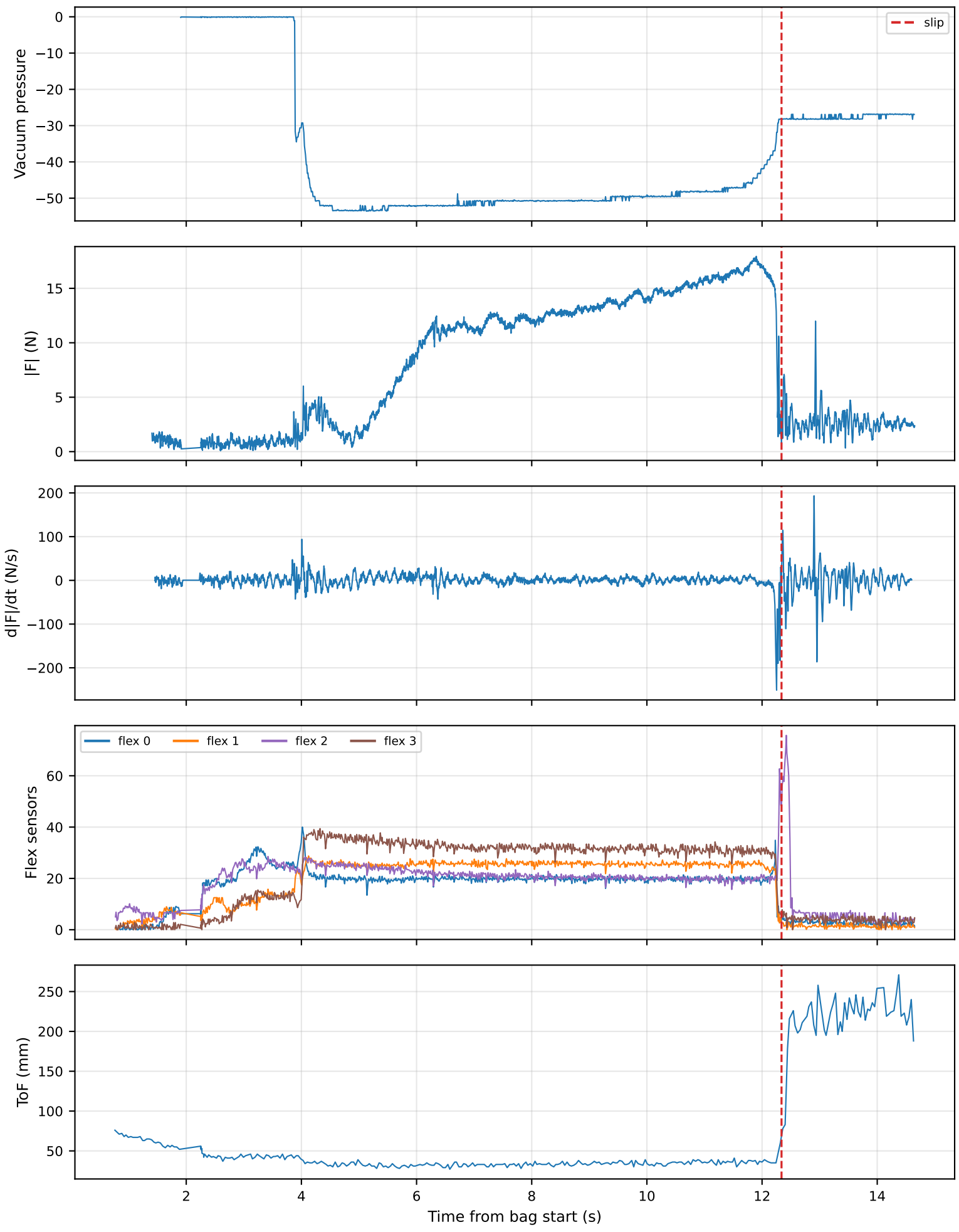


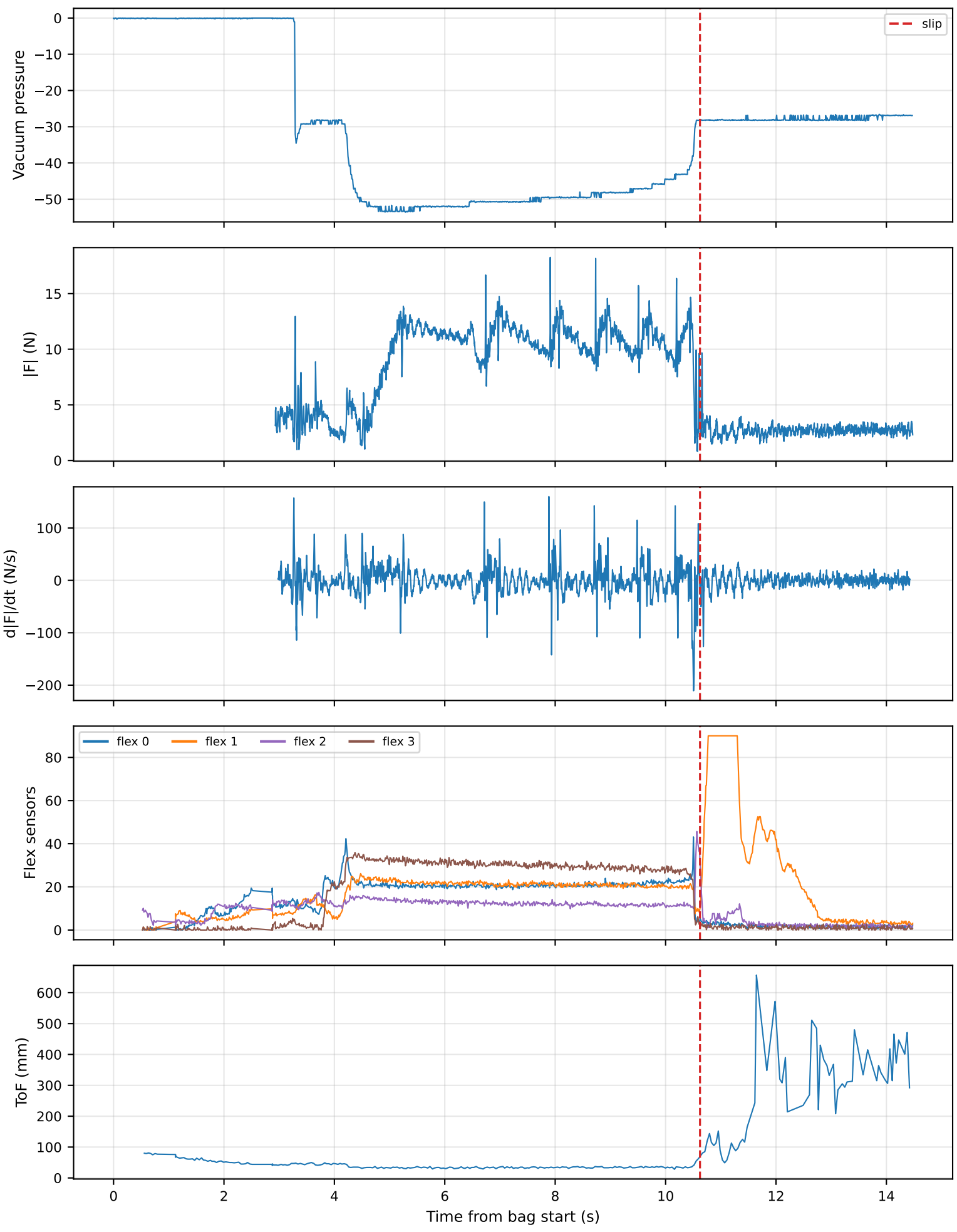
final_approach_and_pick_20261006_143859.db3

SLIP at 7.56 s | 2026-10-06 14:39:00 | batch 138, apple 0 | stiffness_controller



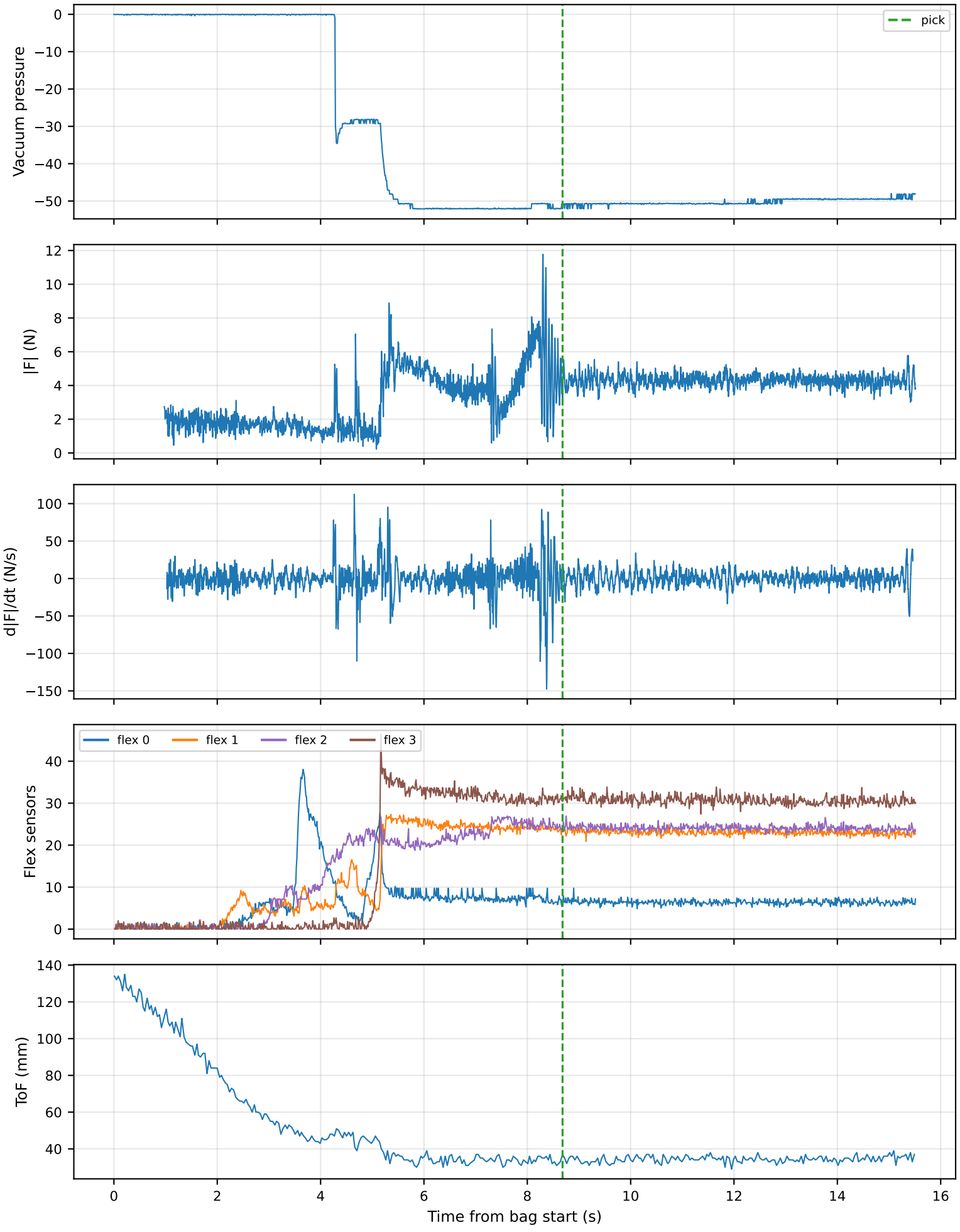

final_approach_and_pick_20261006_144048.db3
SLIP at 7.40 s | 2026-10-06 14:40:49 | batch 139, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

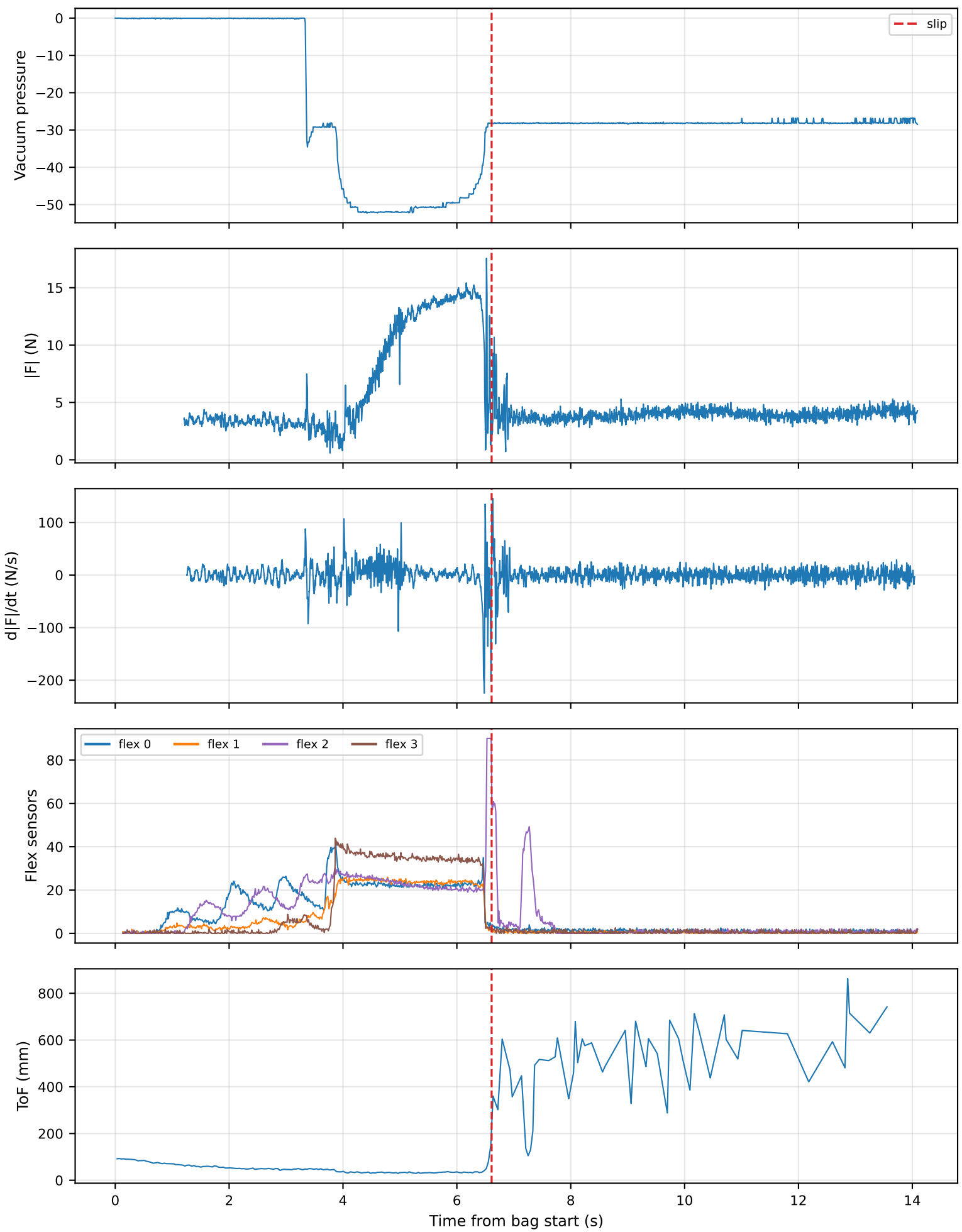

final_approach_and_pick_20261006_151444.db3
SLIP at 12.34 s | 2026-10-06 15:14:45 | batch 140, apple 0 | heuristic_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_152146.db3
SLIP at 10.62 s | 2026-10-06 15:21:47 | batch 143, apple 0 | heuristic_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261006_152303.db3

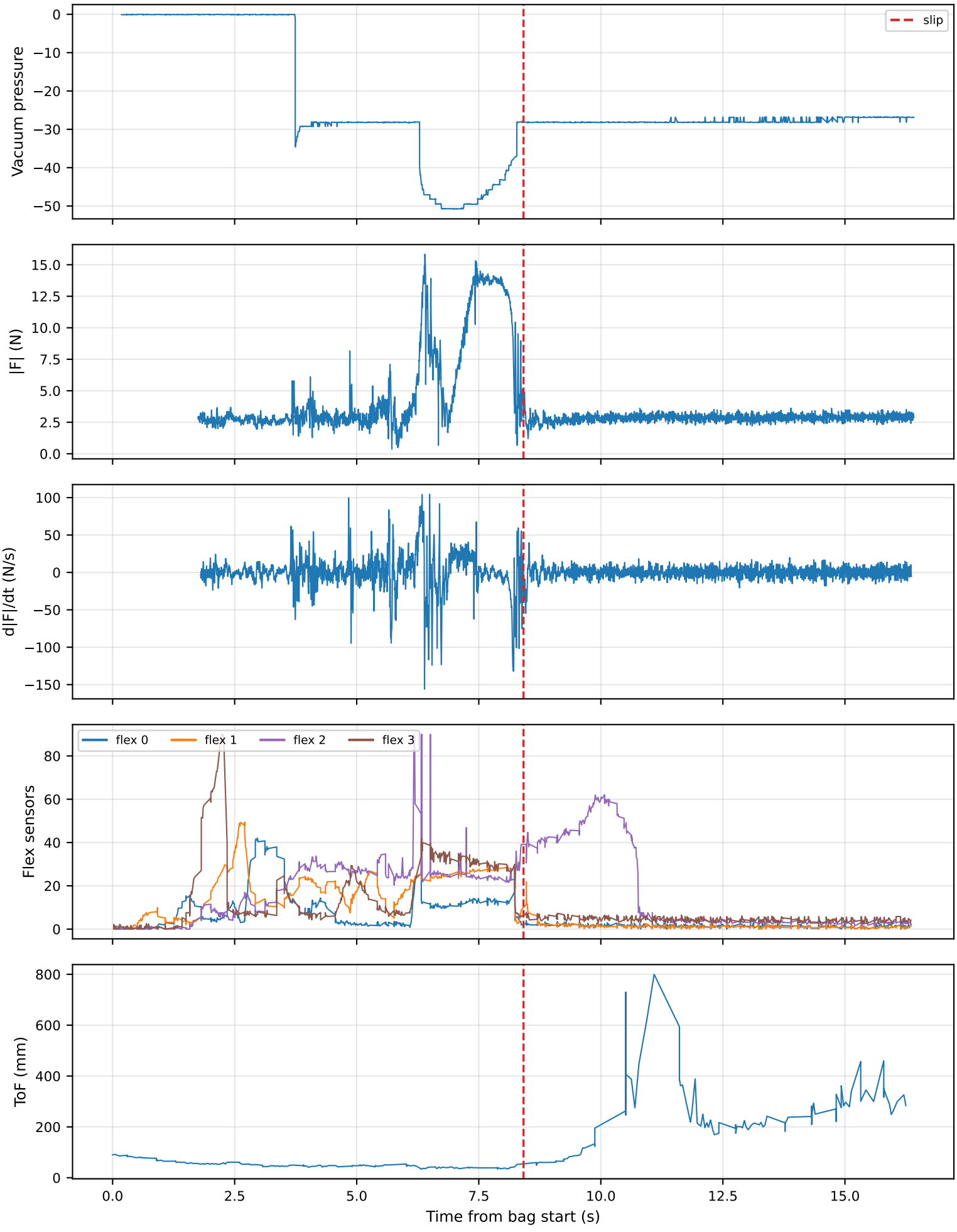
PICK at 8.68 s | 2026-10-06 15:23:04 | batch 144, apple 0 | heuristic_controller

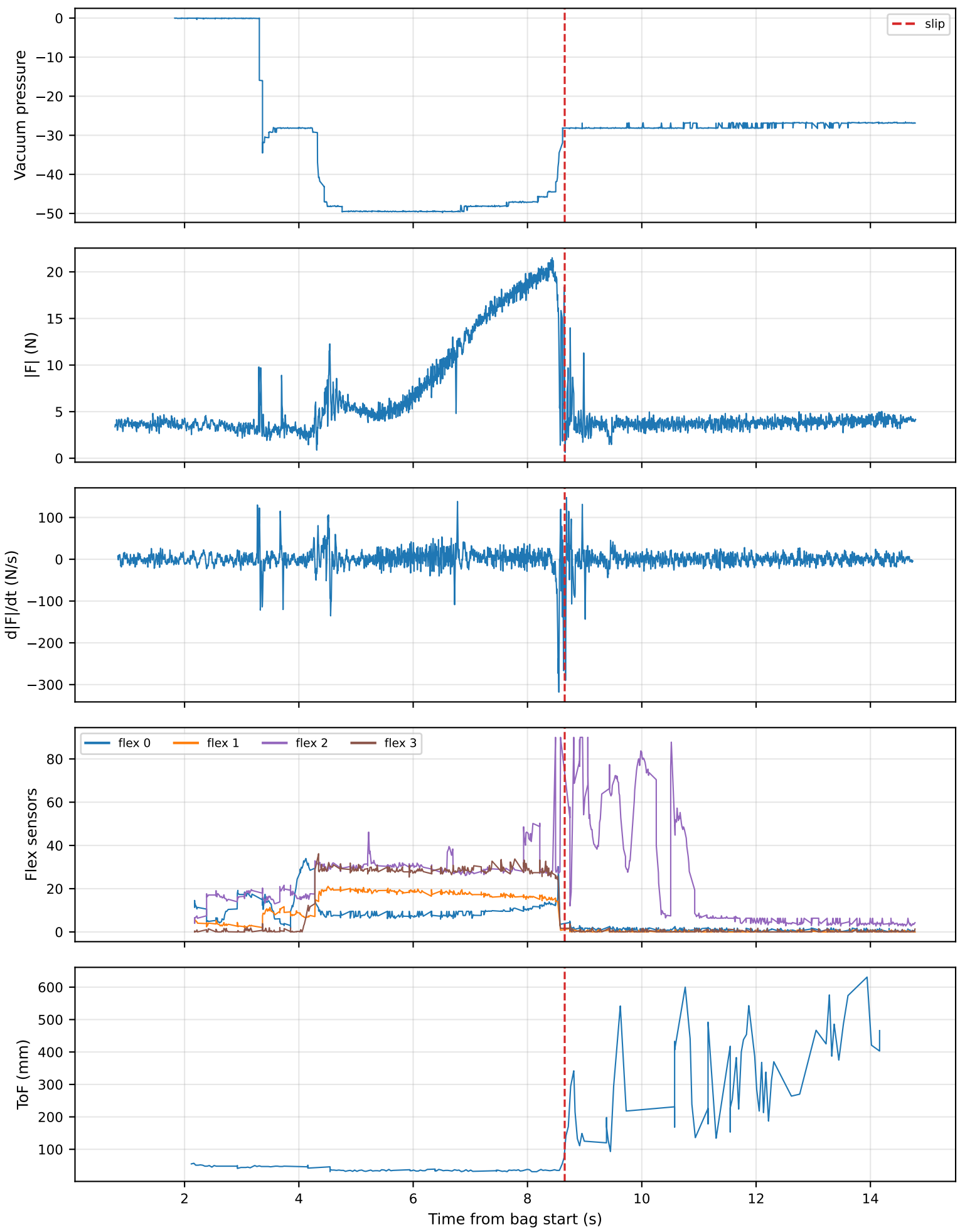


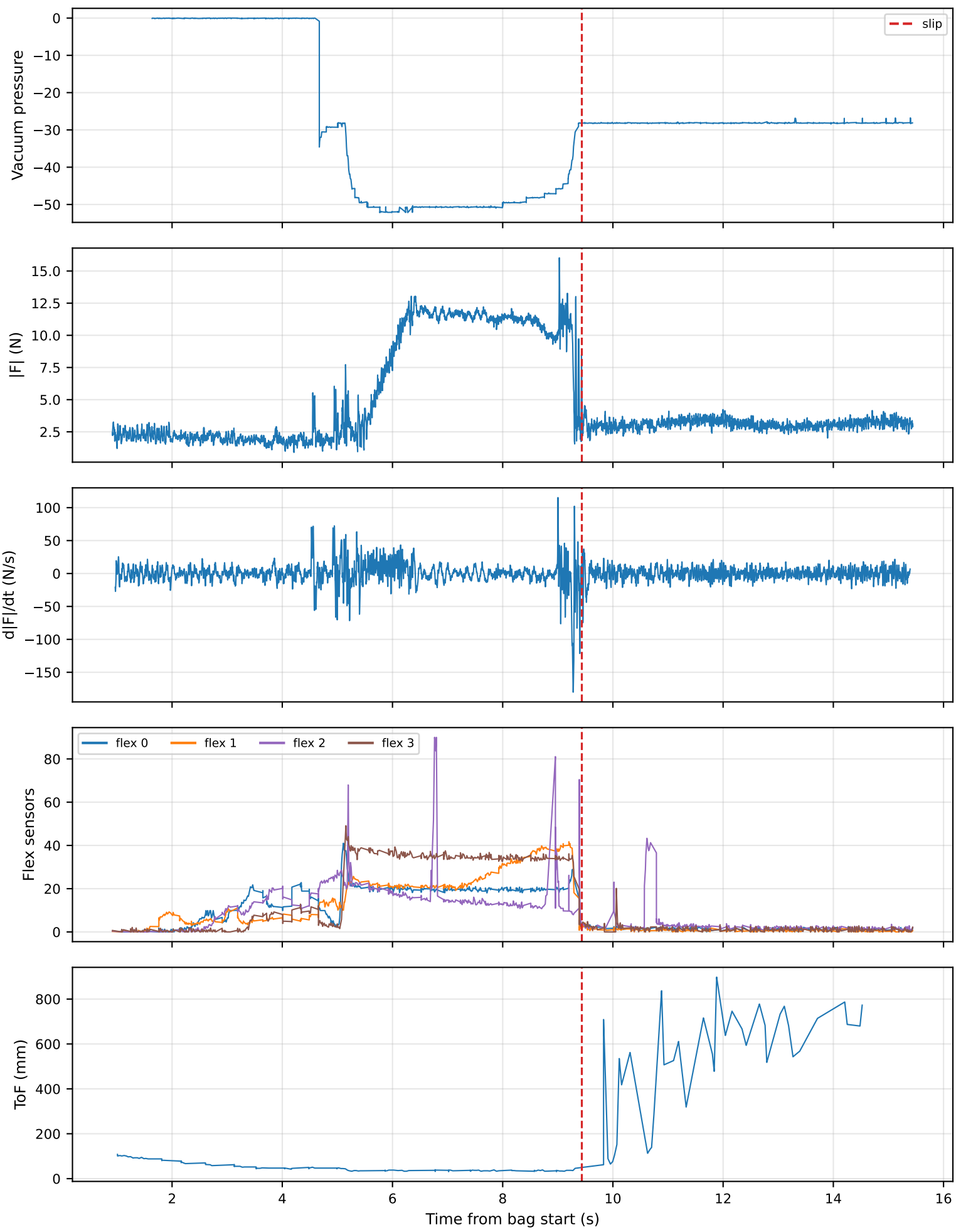

final_approach_and_pick_20261006_152455.db3
SLIP at 6.61 s | 2026-10-06 15:24:57 | batch 145, apple 0 | heuristic_controller
slip
Vacuum pressure
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

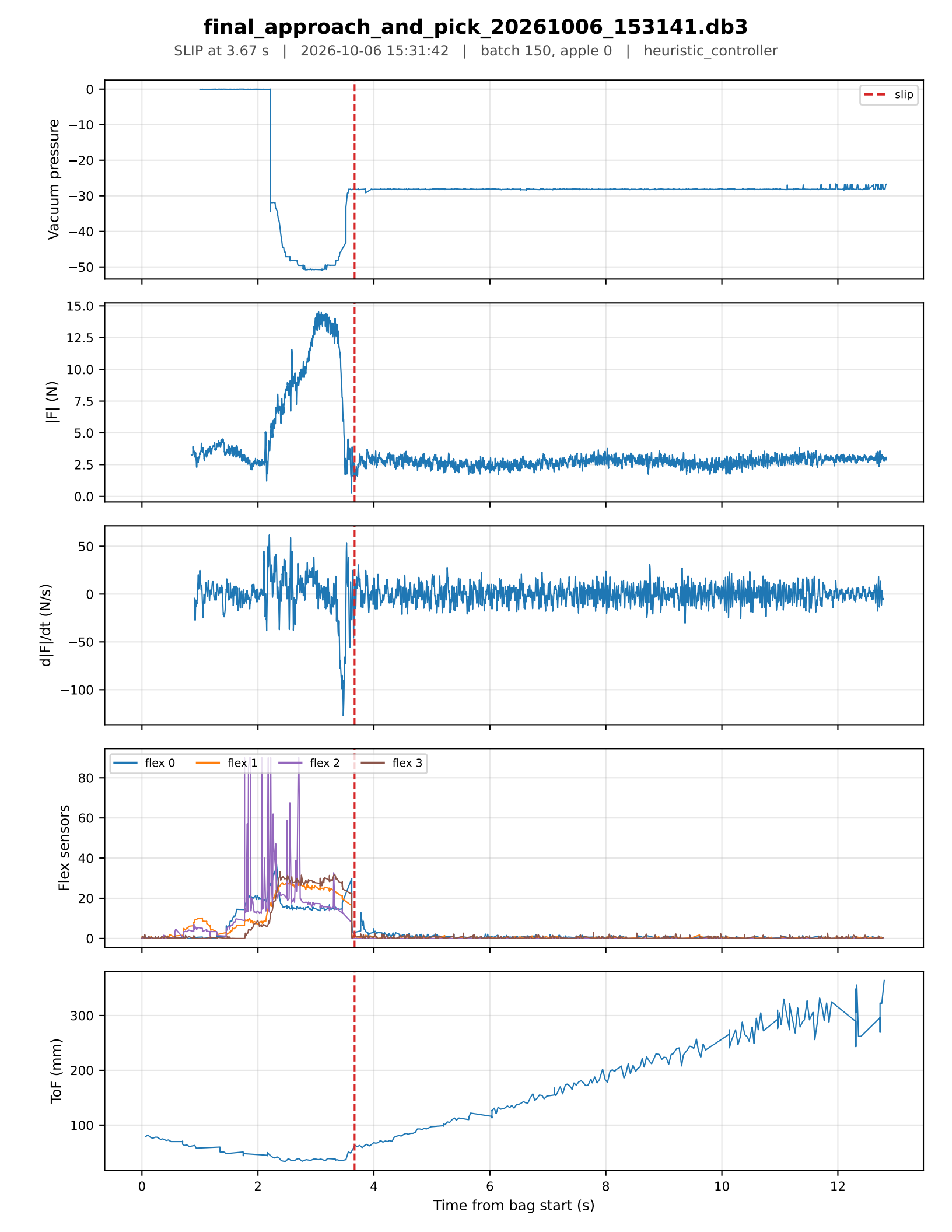
final_approach_and_pick_20261006_152607.db3

SLIP at 8.42 s | 2026-10-06 15:26:08 | batch 146, apple 0 | heuristic_controller



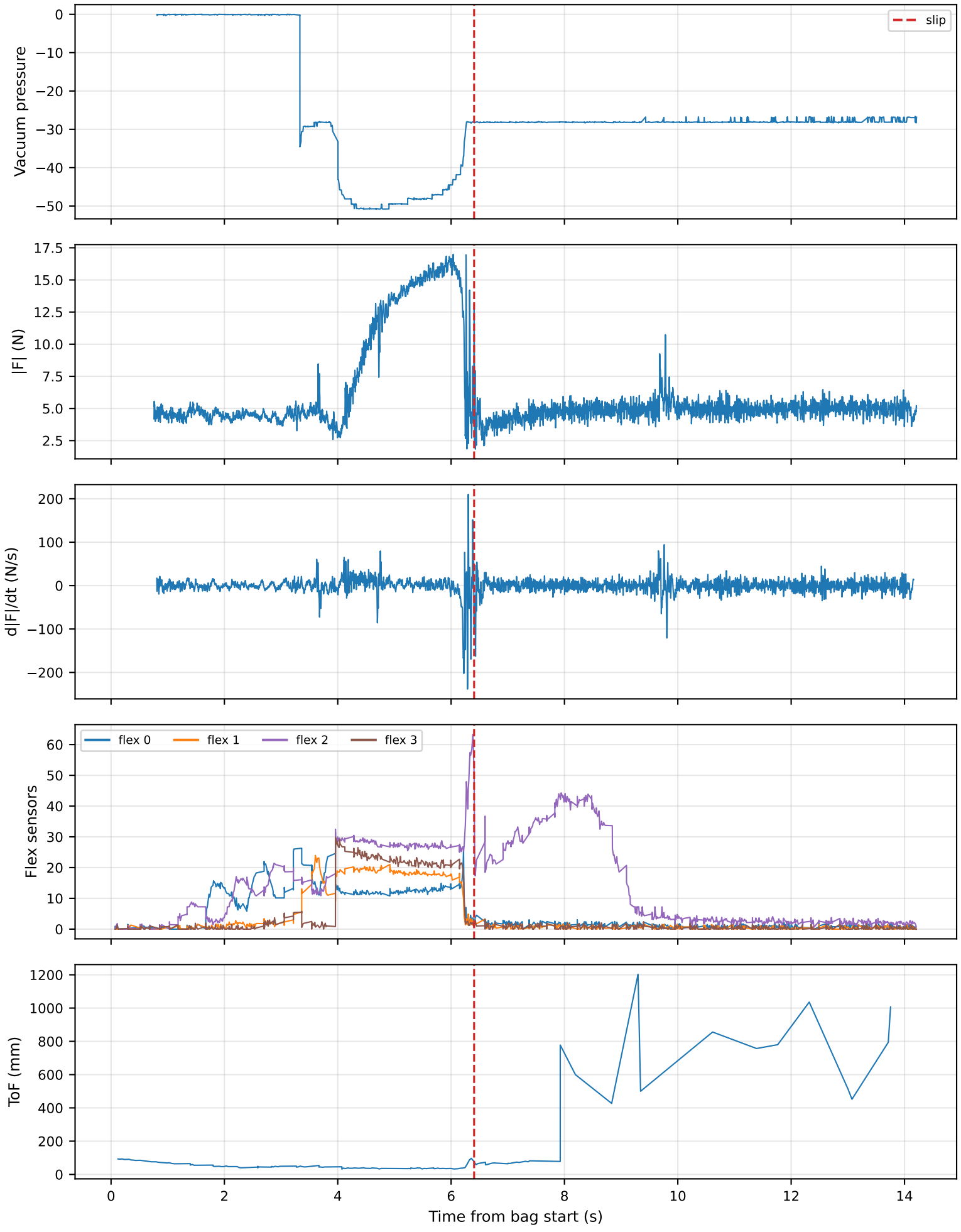

final_approach_and_pick_20261006_152708.db3
SLIP at 8.65 s | 2026-10-06 15:27:10 | batch 147, apple 0 | heuristic_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

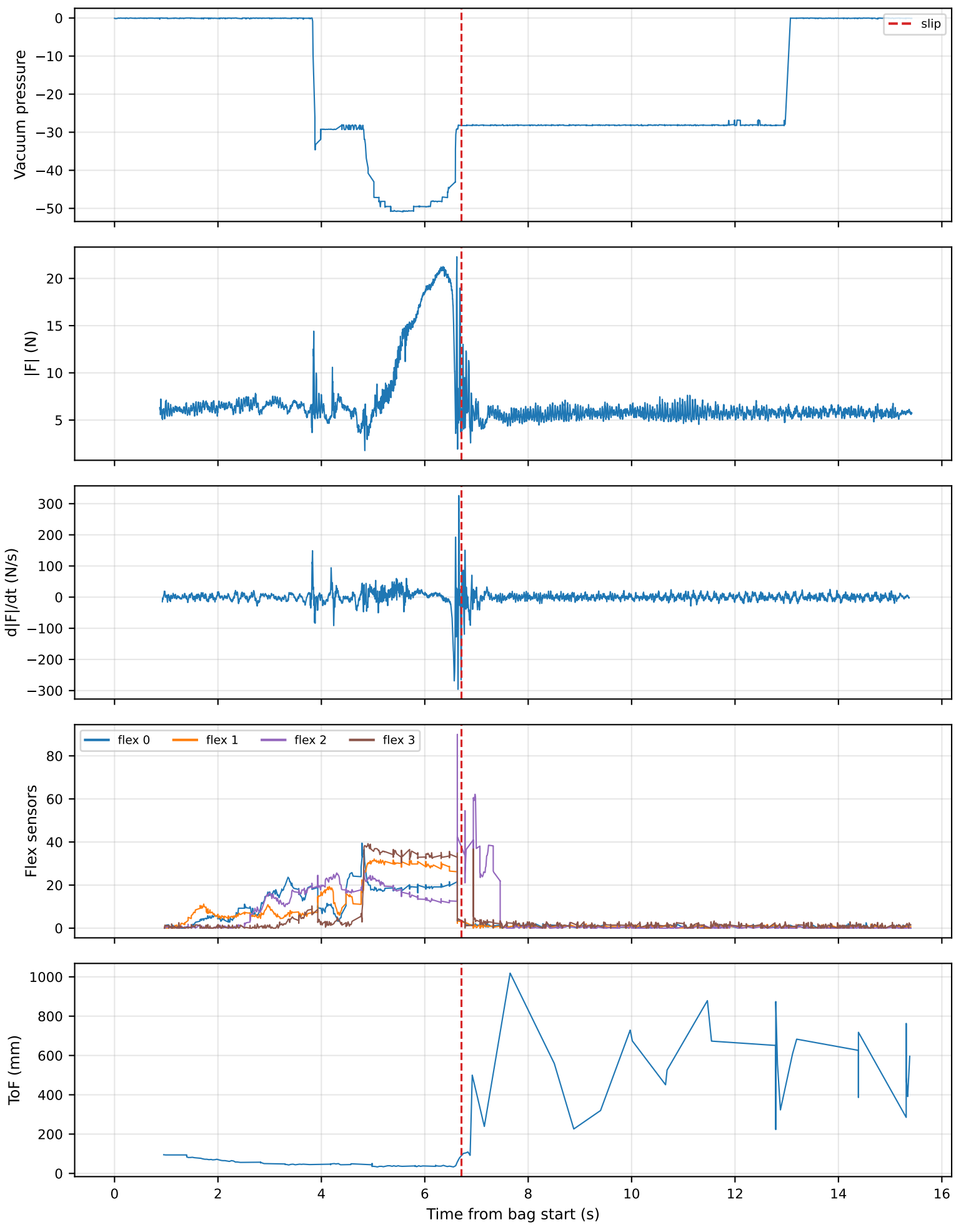

final_approach_and_pick_20261006_153012.db3
SLIP at 9.44 s | 2026-10-06 15:30:13 | batch 149, apple 0 | heuristic_controller
slip
Vacuum pressure
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261006_153141.db3
SLIP at 3.67 s | 2026-10-06 15:31:42 | batch 150, apple 0 | heuristic_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261006_160019.db3

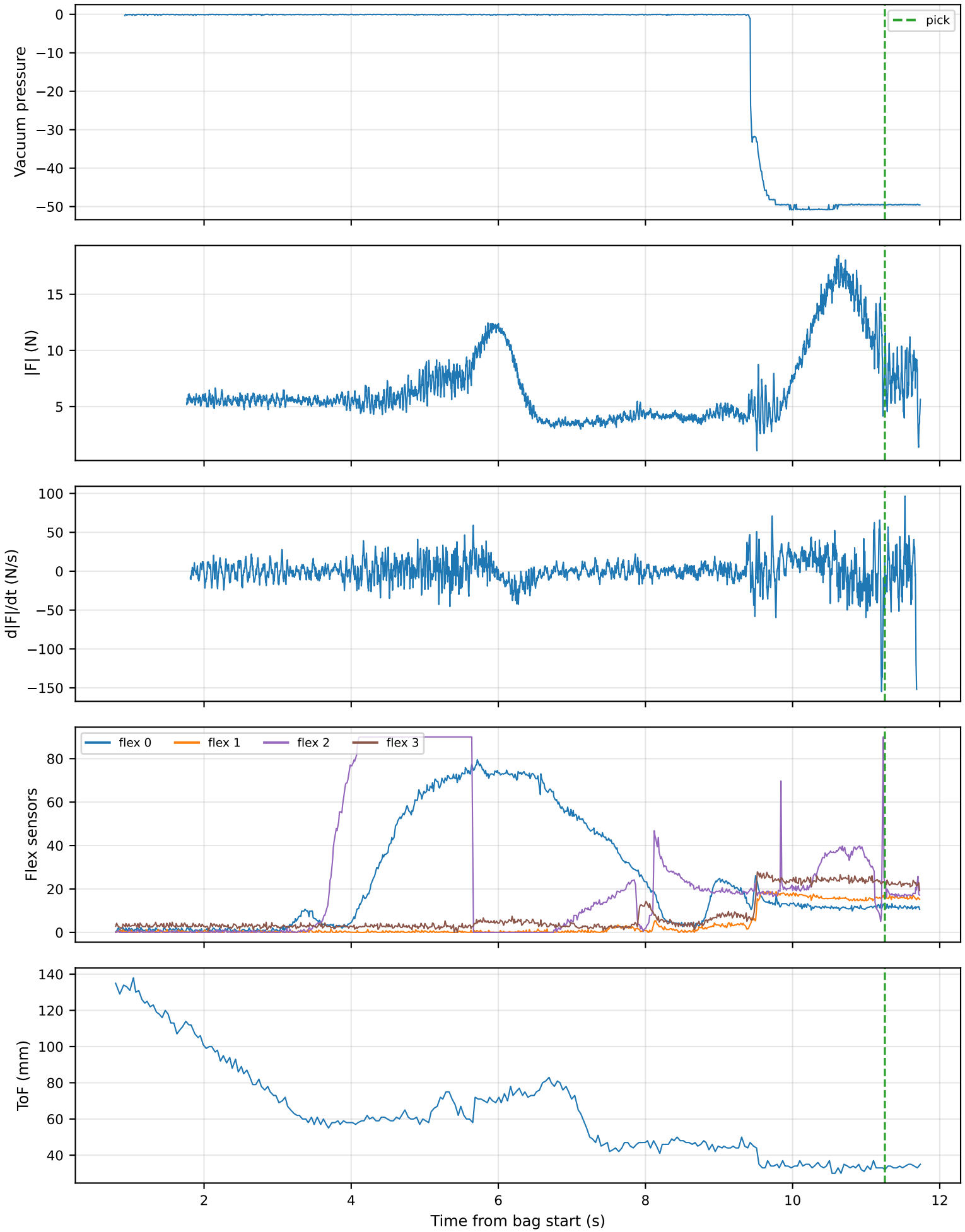
SLIP at 6.41 s | 2026-10-06 16:00:20 | batch 152, apple 0 | heuristic_controller

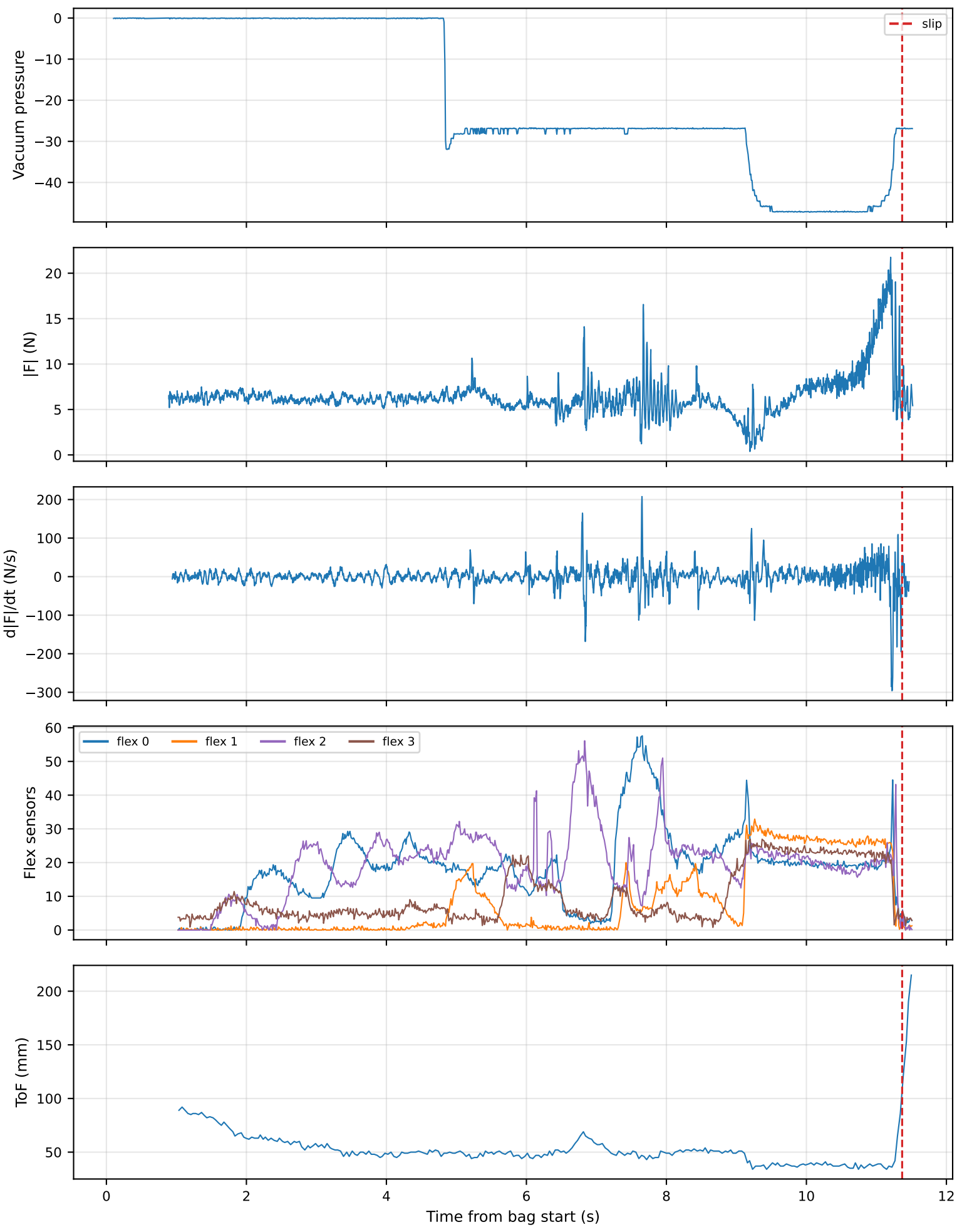


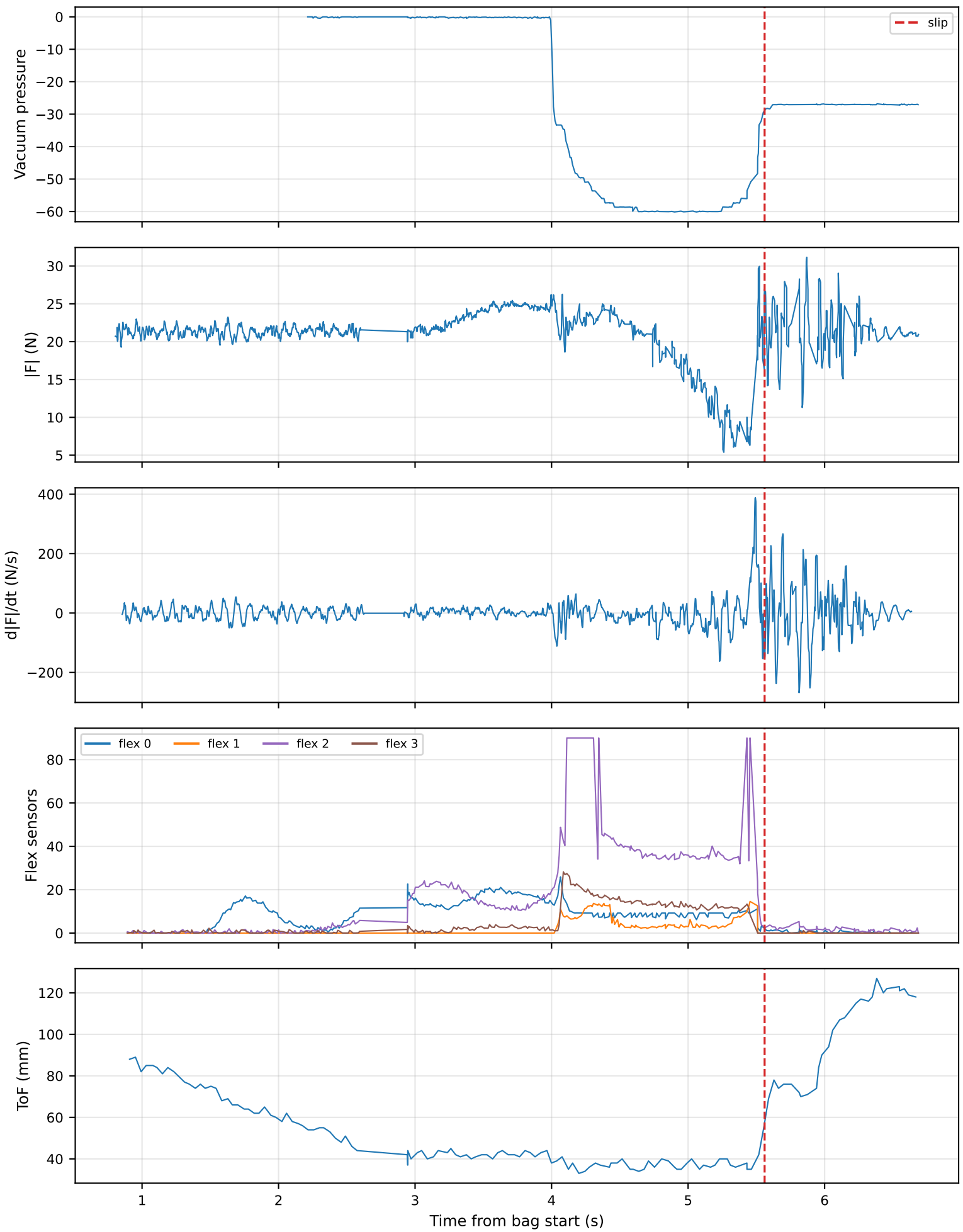

final_approach_and_pick_20261006_160626.db3
SLIP at 6.71 s | 2026-10-06 16:06:27 | batch 154, apple 0 | heuristic_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261006_161320.db3

PICK at 11.25 s | 2026-10-06 16:13:21 | batch 155, apple 5 | pull_twist_controller

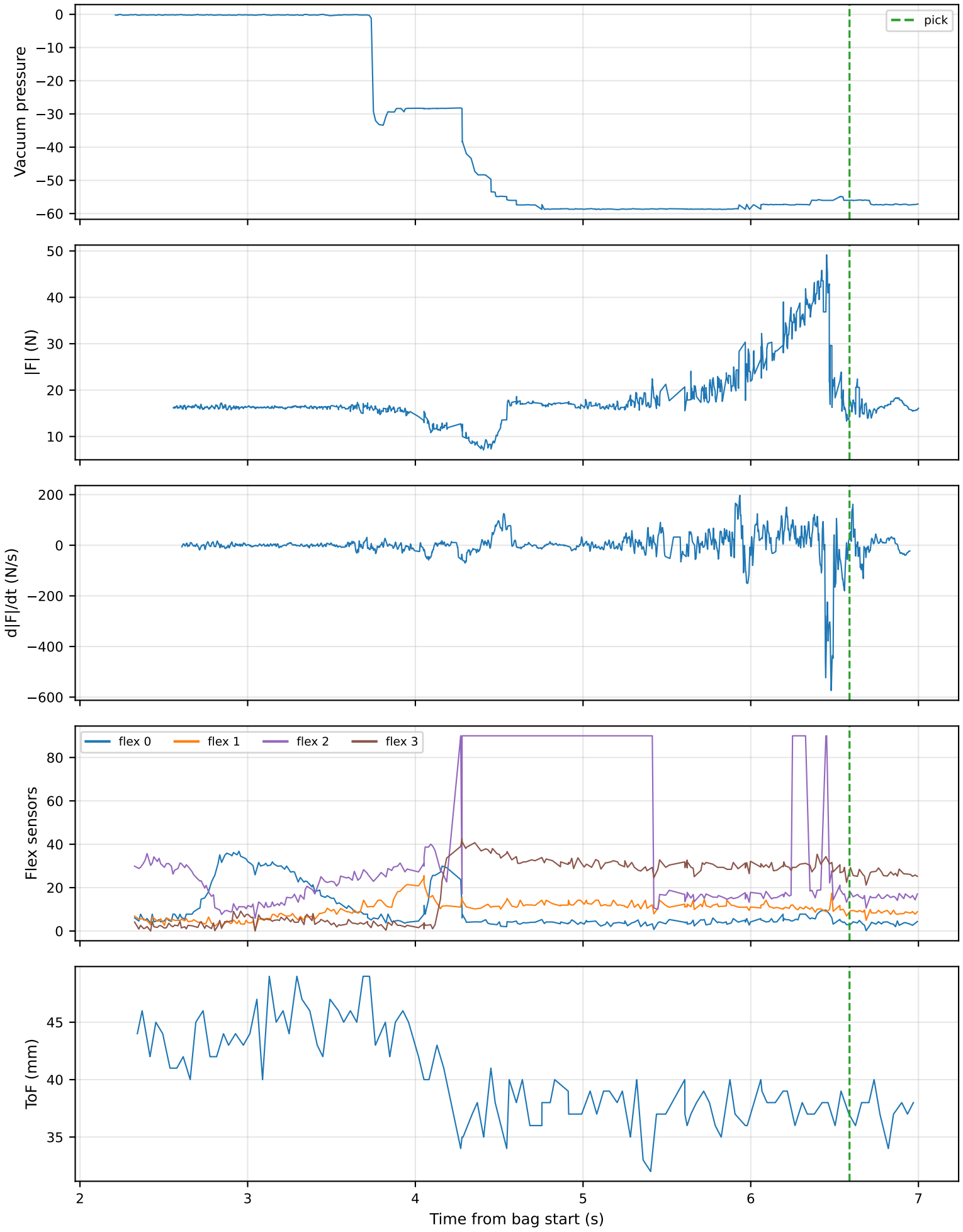


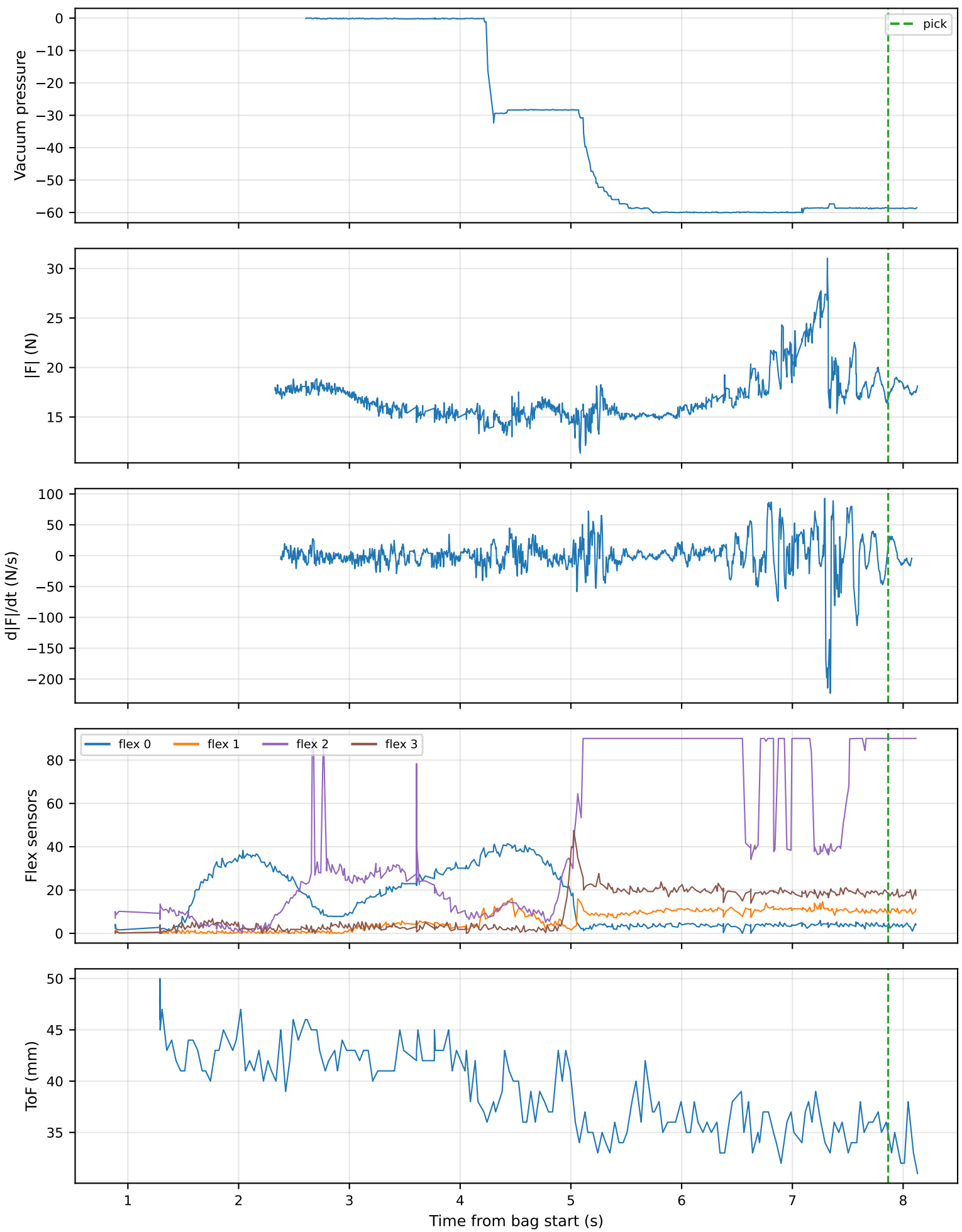

final_approach_and_pick_20261006_162431.db3
SLIP at 11.37 s | 2026-10-06 16:24:32 | batch 39, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

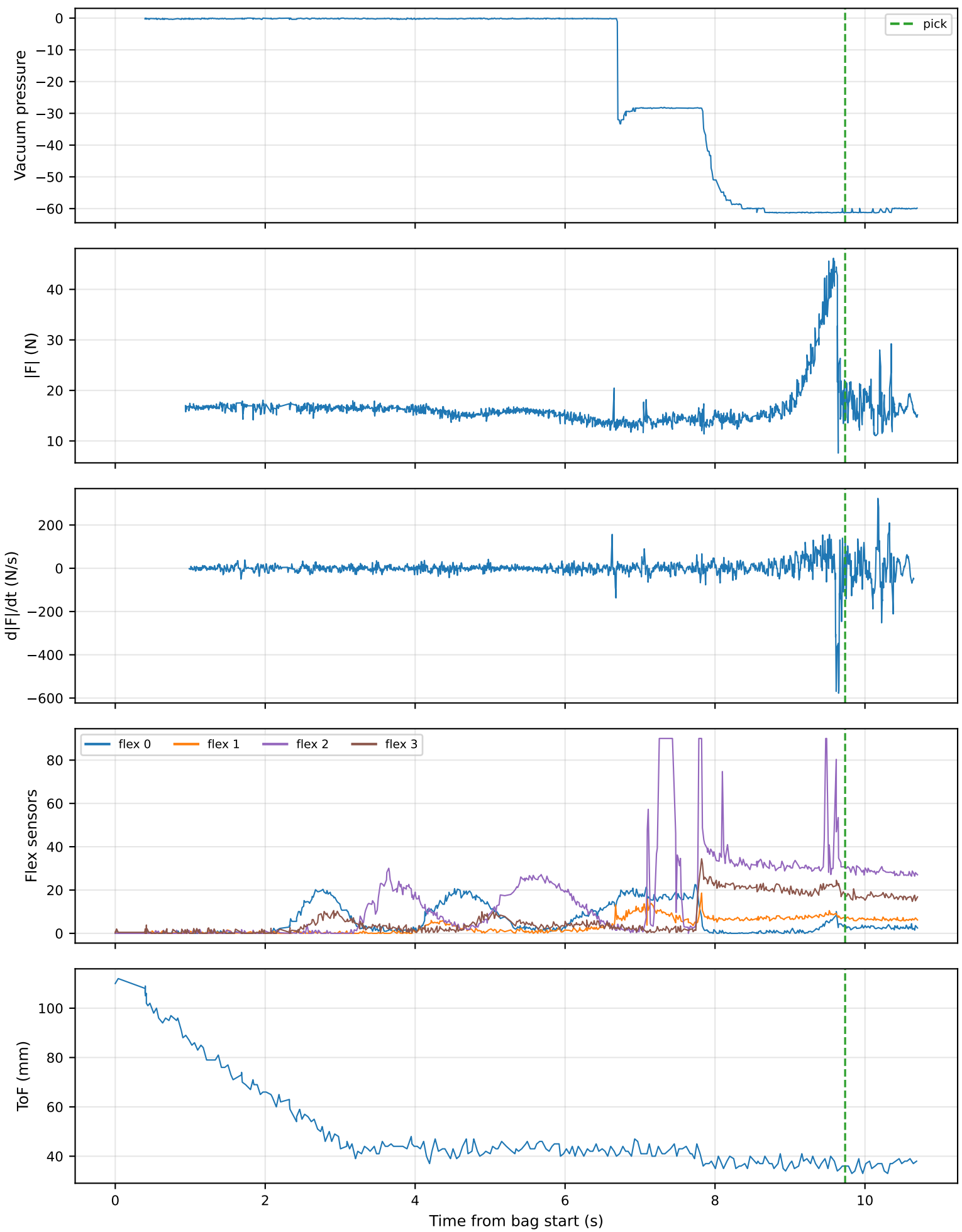

final_approach_and_pick_20261007_102040.db3
SLIP at 5.56 s | 2026-10-07 10:20:43 | batch 46, apple 5 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

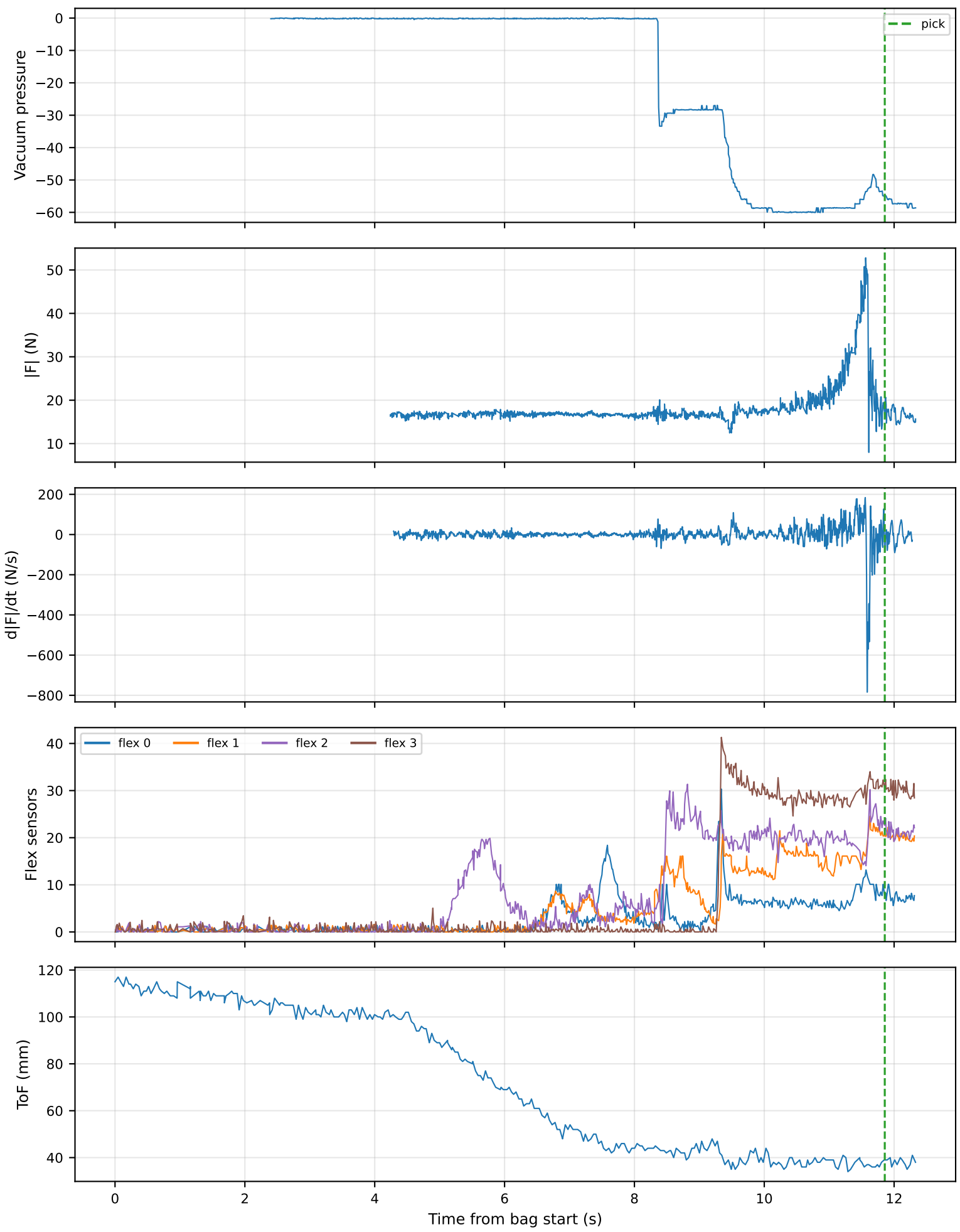
final_approach_and_pick_20261007_102817.db3

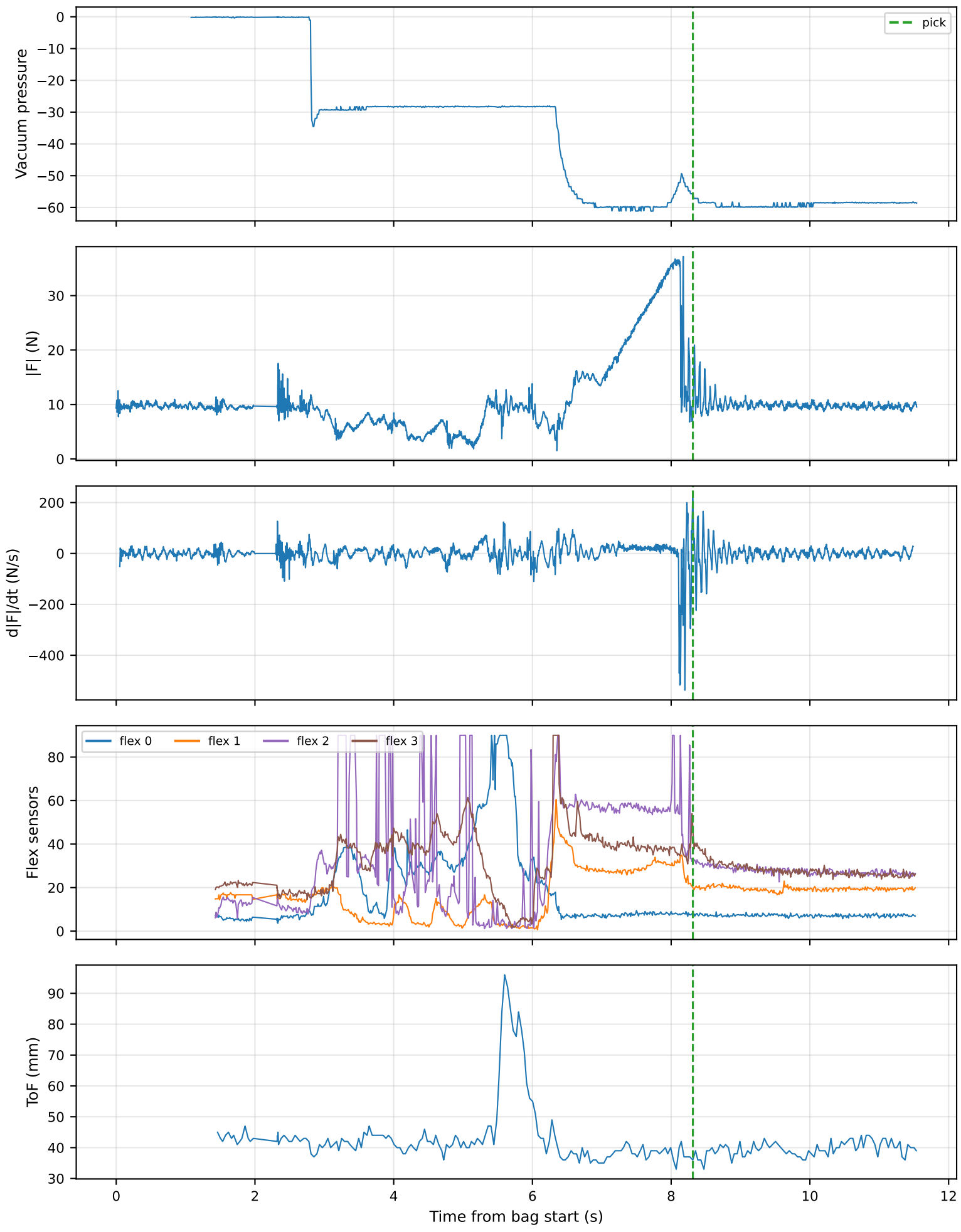
PICK at 6.59 s | 2026-10-07 10:28:22 | batch 48, apple 0 | pull_twist_controller

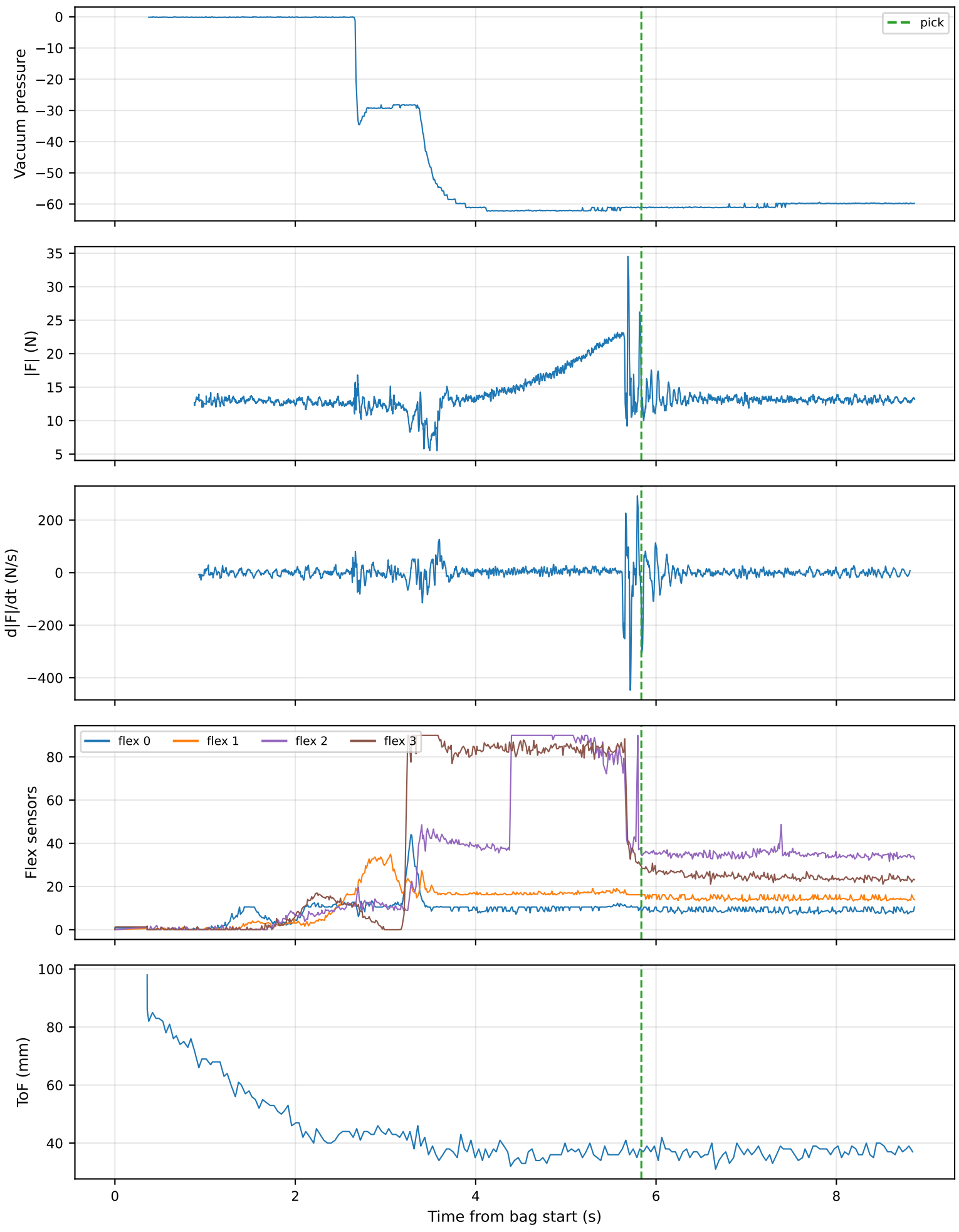


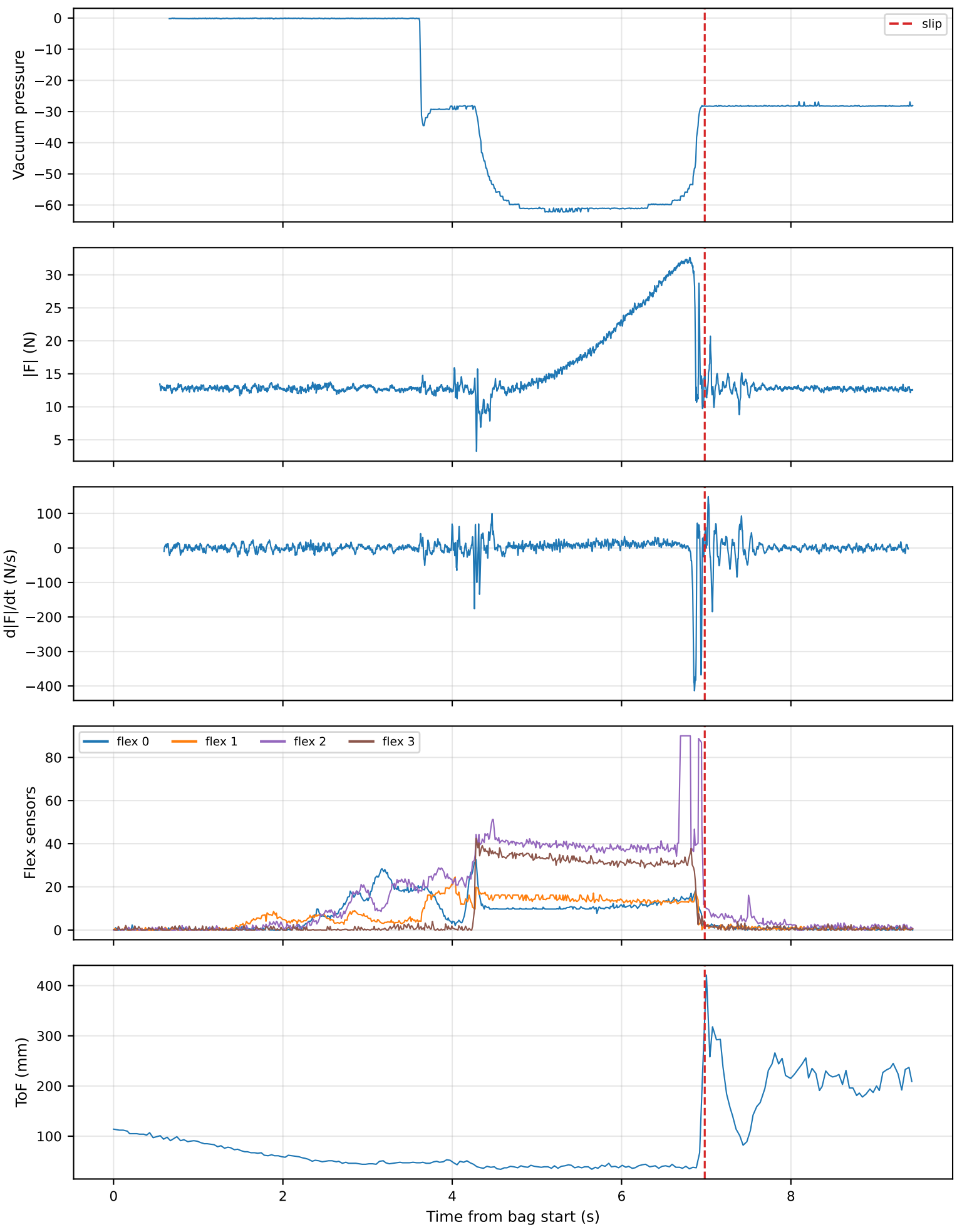

final_approach_and_pick_20261007_103151.db3
PICK at 7.87 s | 2026-10-07 10:31:55 | batch 49, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

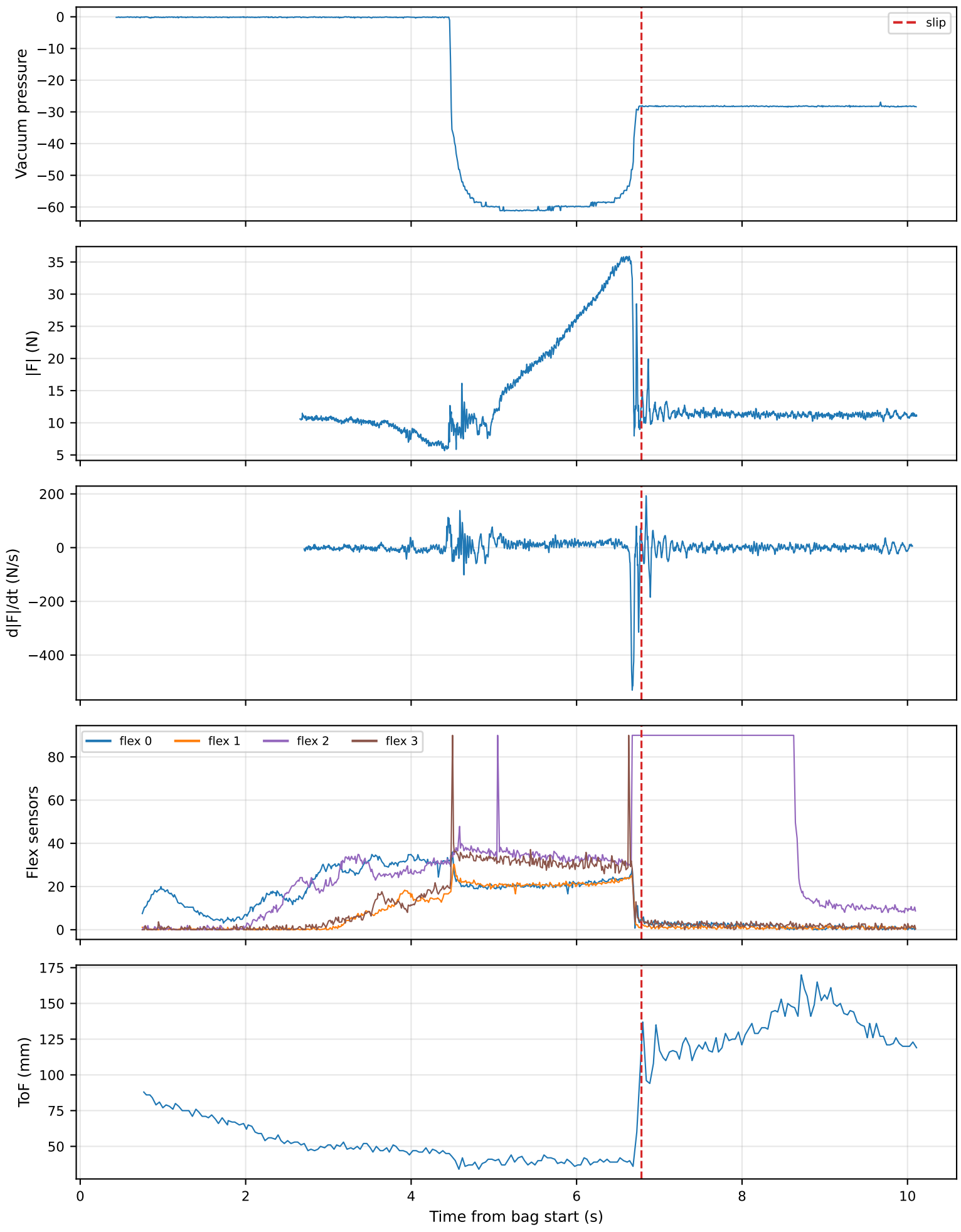

final_approach_and_pick_20261007_103348.db3
PICK at 9.74 s | 2026-10-07 10:33:51 | batch 51, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

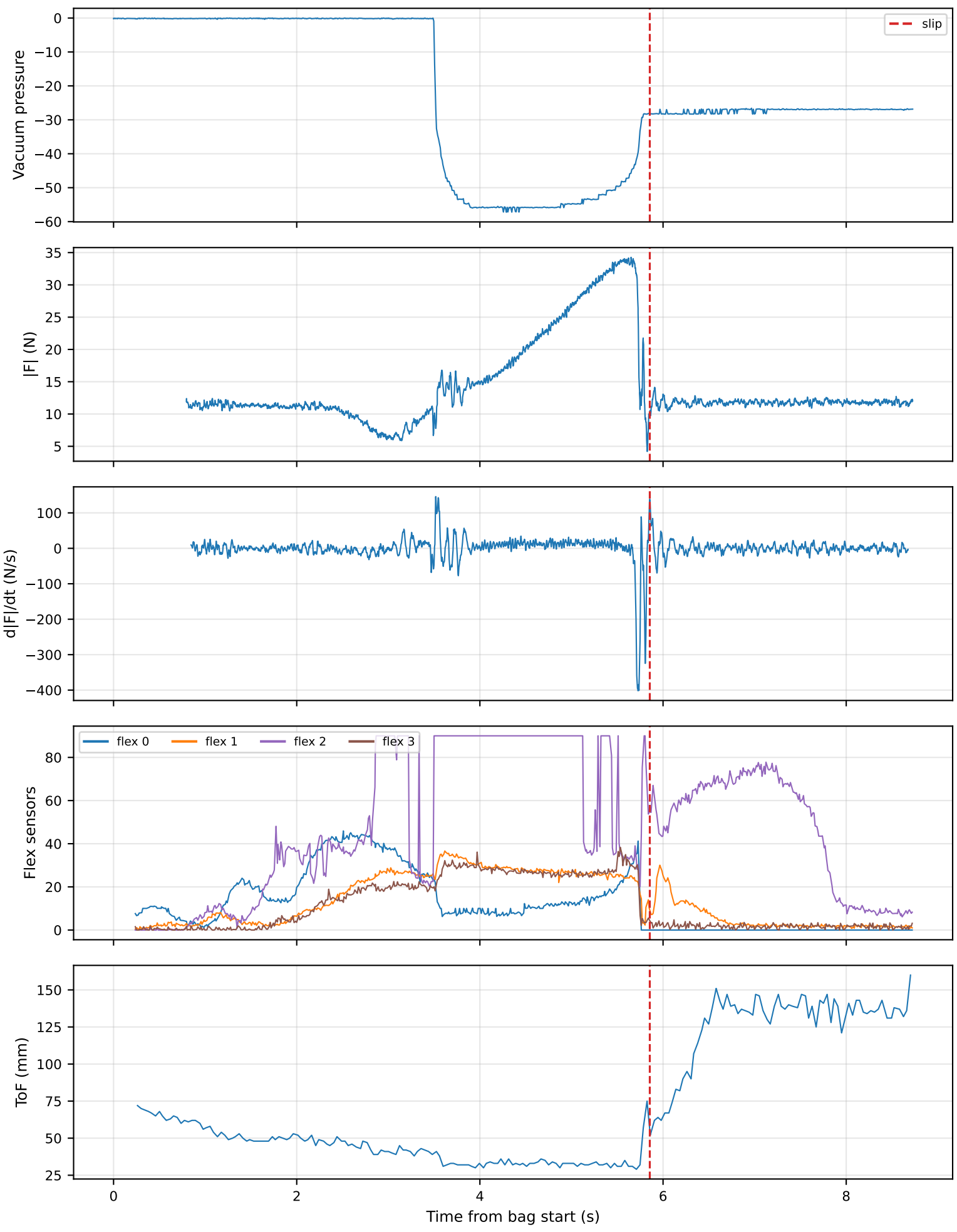

final_approach_and_pick_20261007_103444.db3
PICK at 11.86 s | 2026-10-07 10:34:45 | batch 52, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

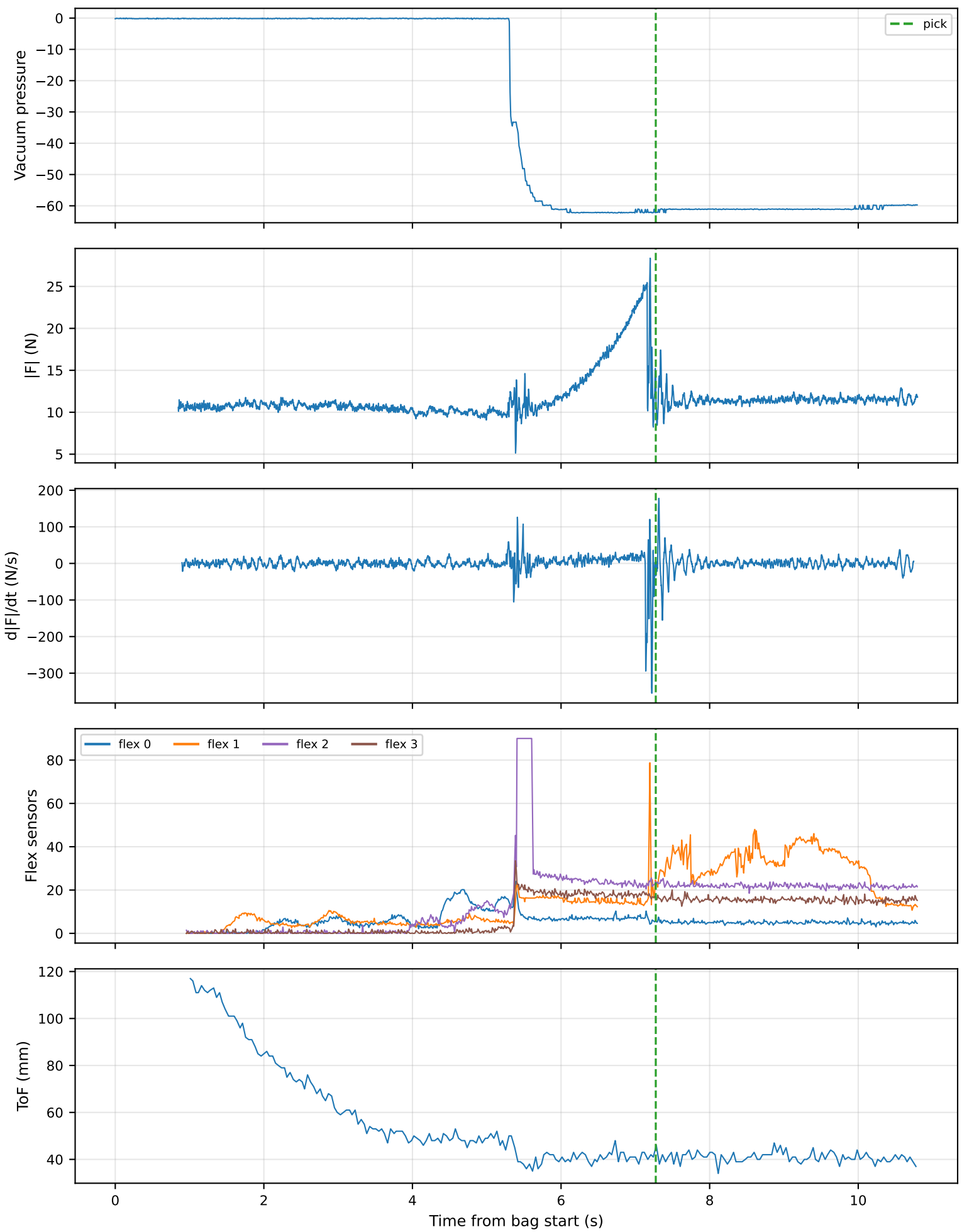

final_approach_and_pick_20261007_115418.db3
PICK at 8.31 s | 2026-10-07 11:54:20 | batch 55, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_115631.db3
PICK at 5.84 s | 2026-10-07 11:56:32 | batch 56, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_115755.db3
SLIP at 6.98 s | 2026-10-07 11:57:56 | batch 57, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

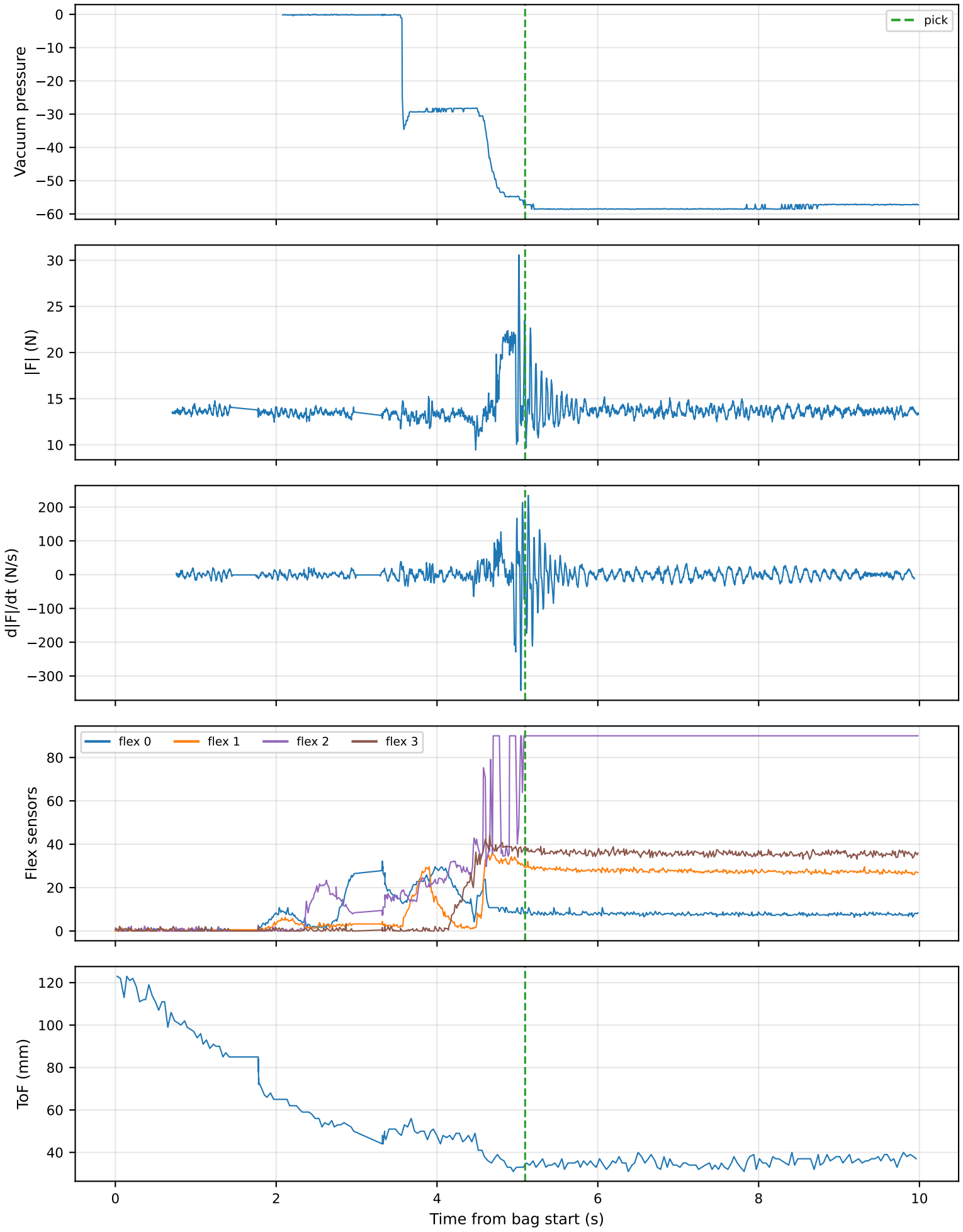

final_approach_and_pick_20261007_115916.db3
SLIP at 6.78 s | 2026-10-07 11:59:18 | batch 58, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

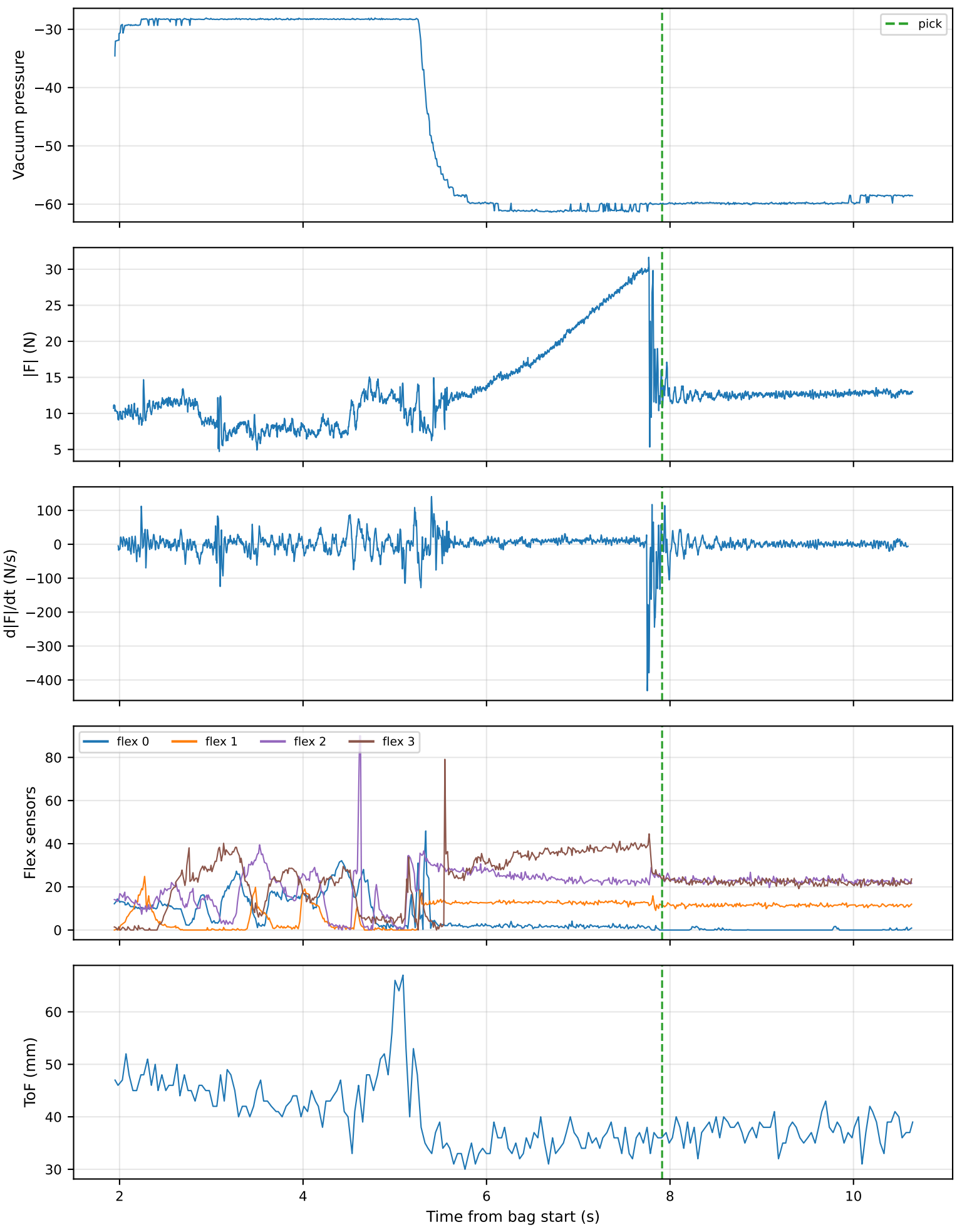

final_approach_and_pick_20261007_120047.db3
SLIP at 5.86 s | 2026-10-07 12:00:48 | batch 59, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

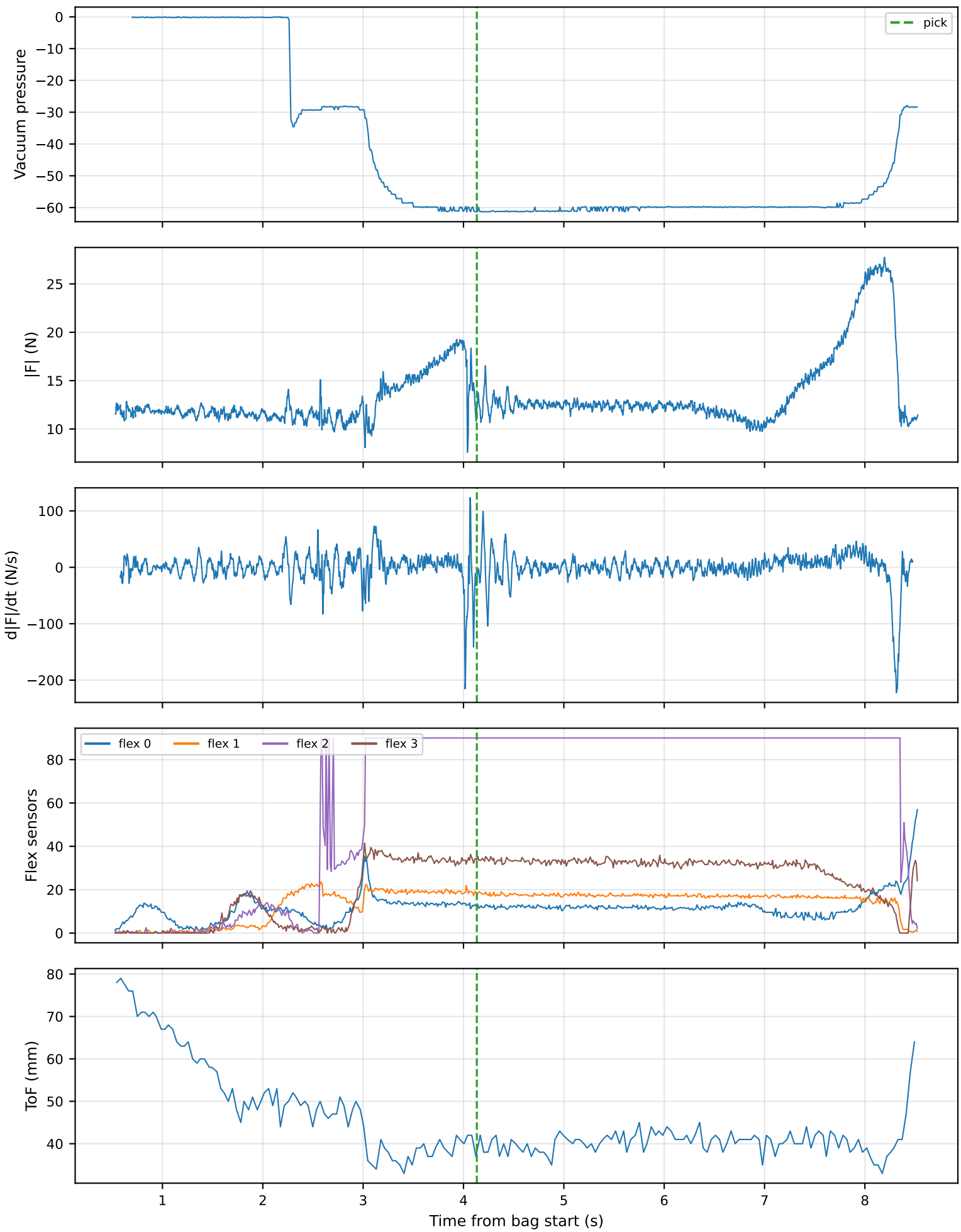

final_approach_and_pick_20261007_120204.db3
PICK at 7.27 s | 2026-10-07 12:02:05 | batch 60, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

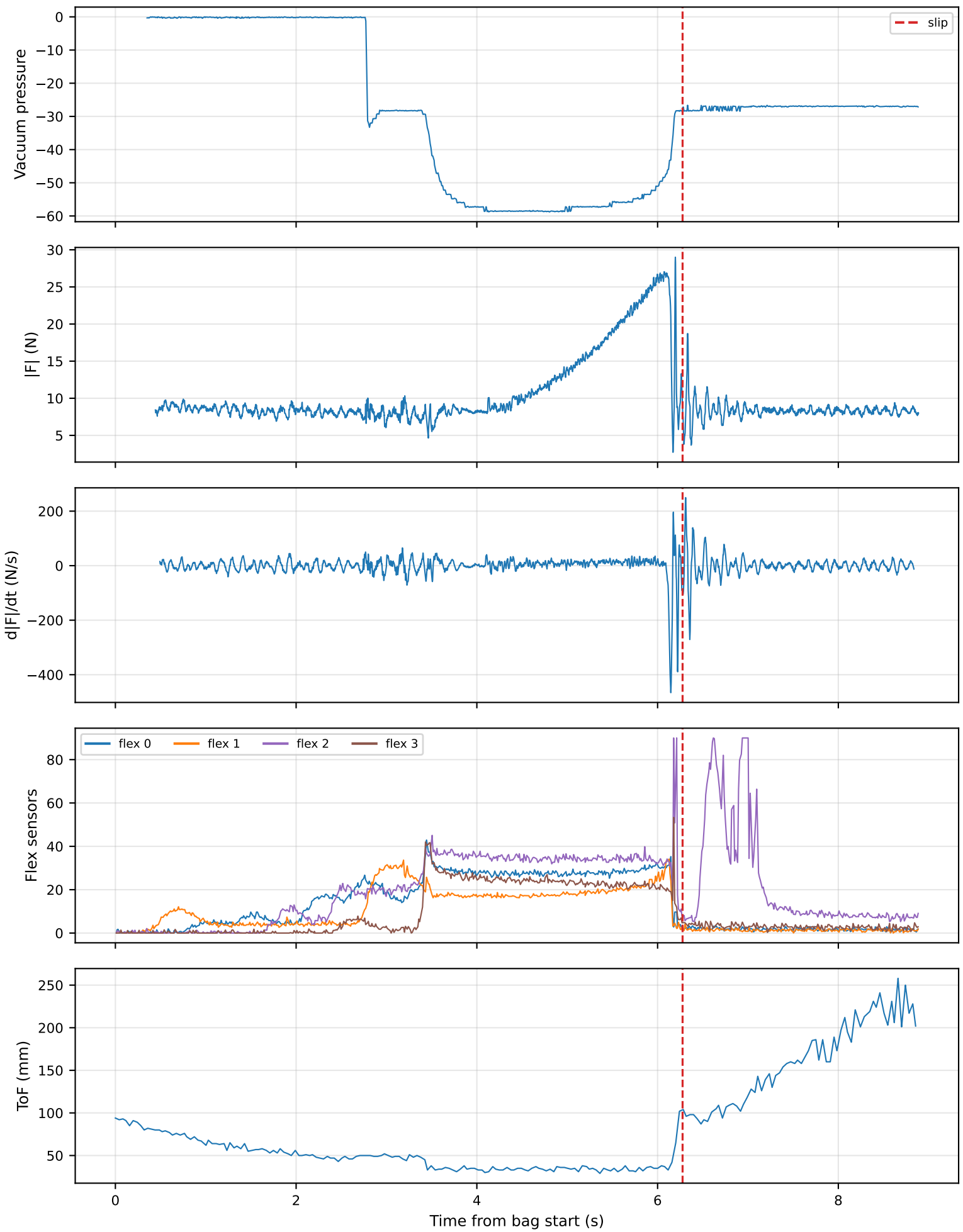
final_approach_and_pick_20261007_120415.db3

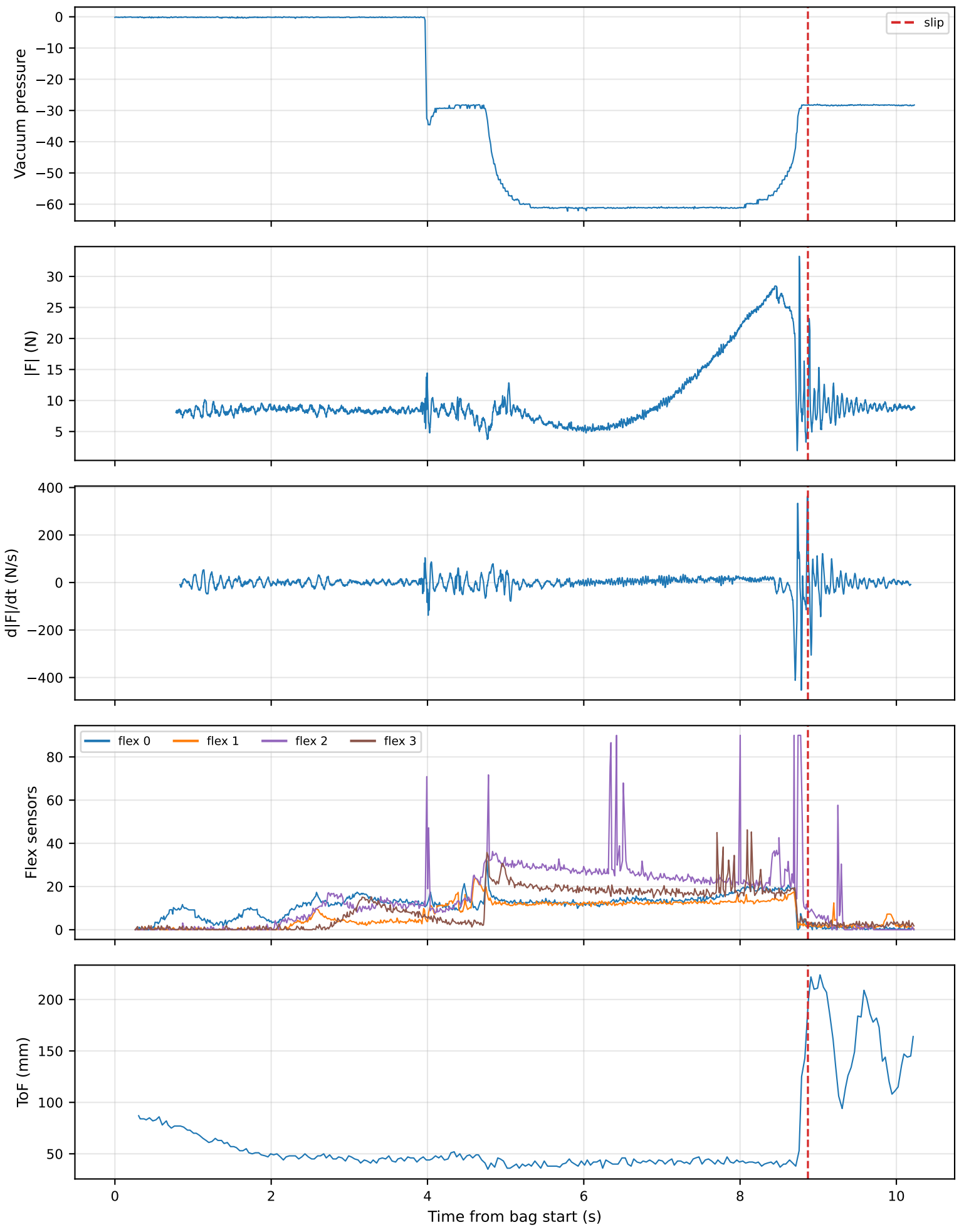
PICK at 5.10 s | 2026-10-07 12:04:16 | batch 61, apple 0 | stiffness_controller

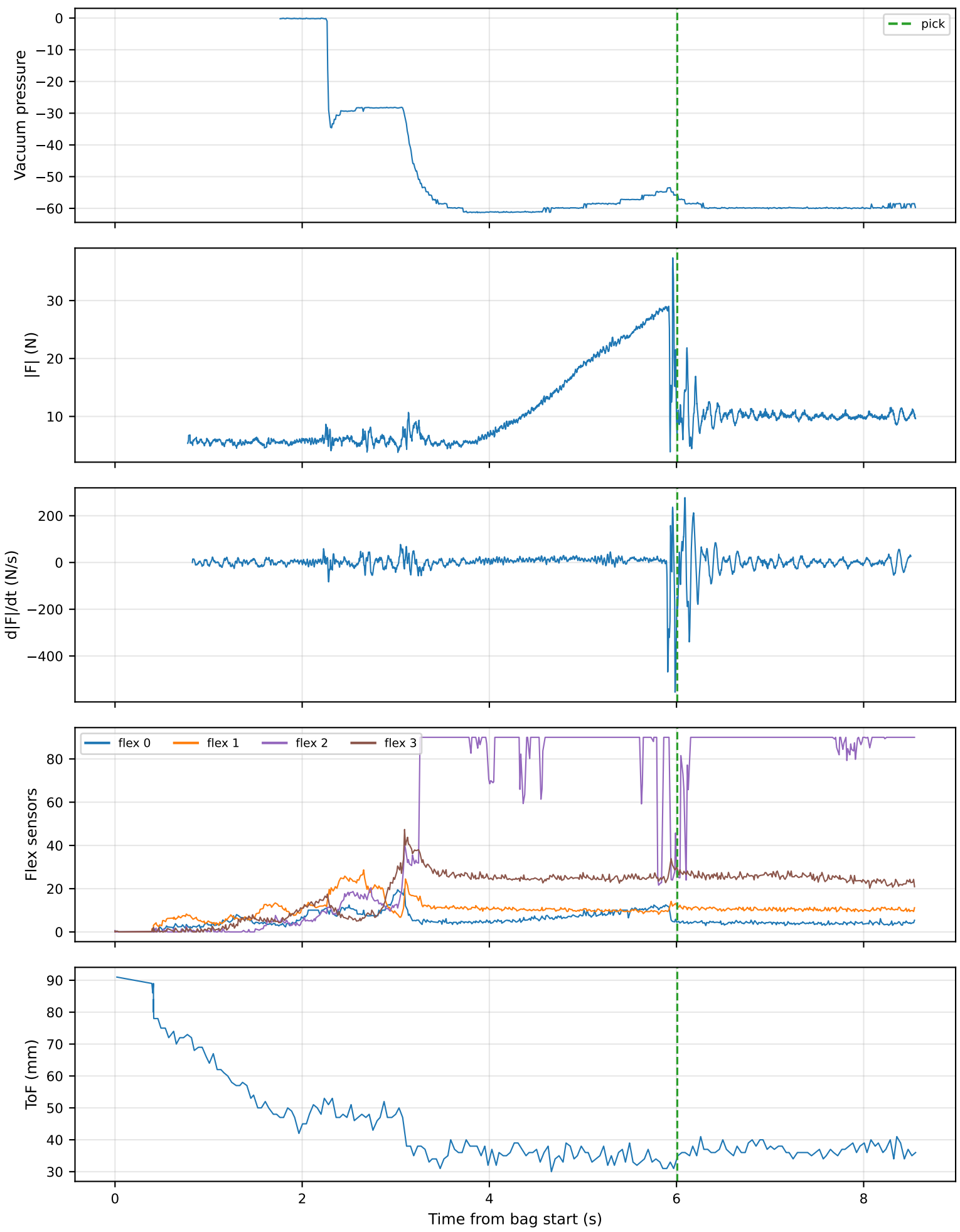


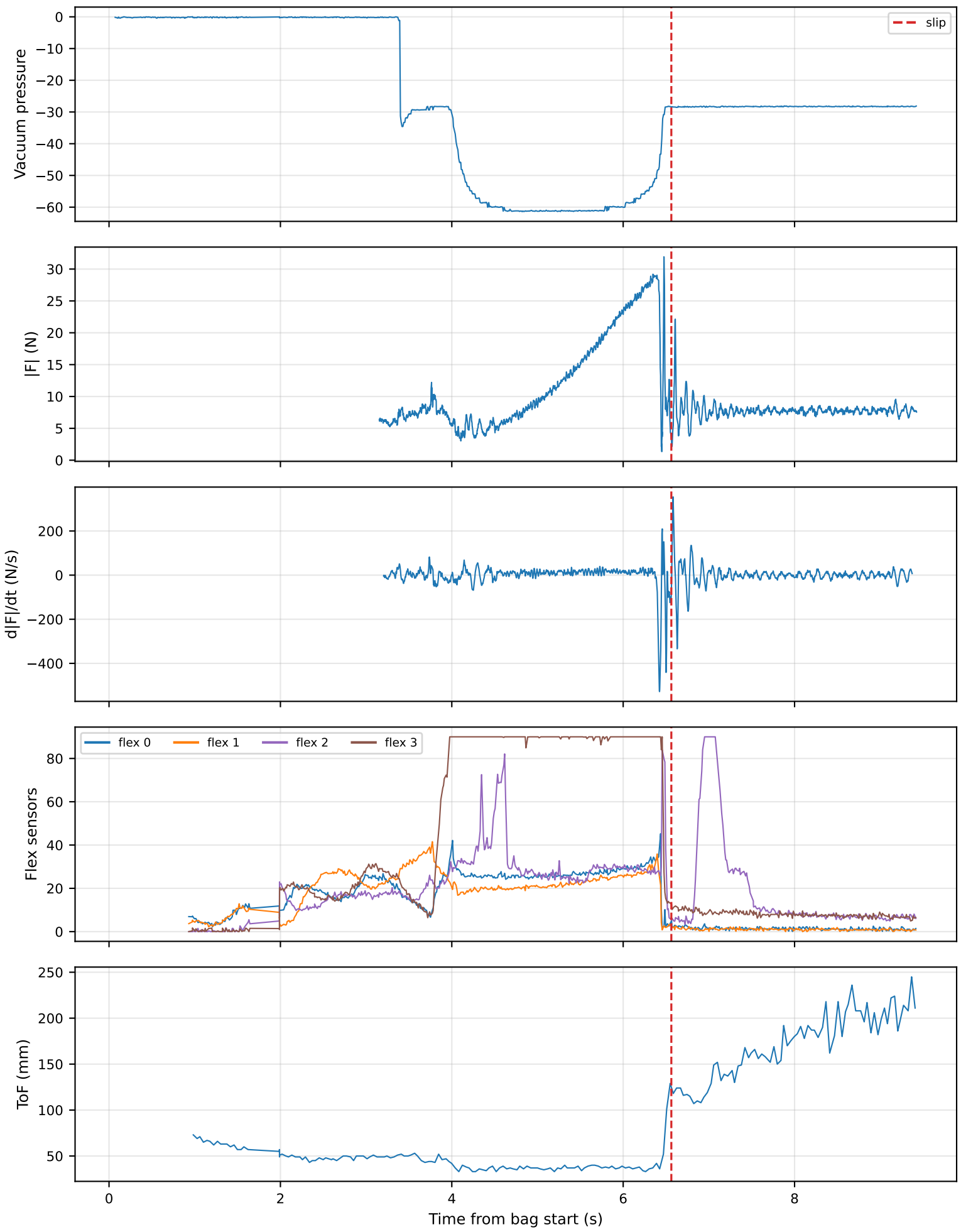

final_approach_and_pick_20261007_120536.db3
PICK at 7.91 s | 2026-10-07 12:05:37 | batch 62, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

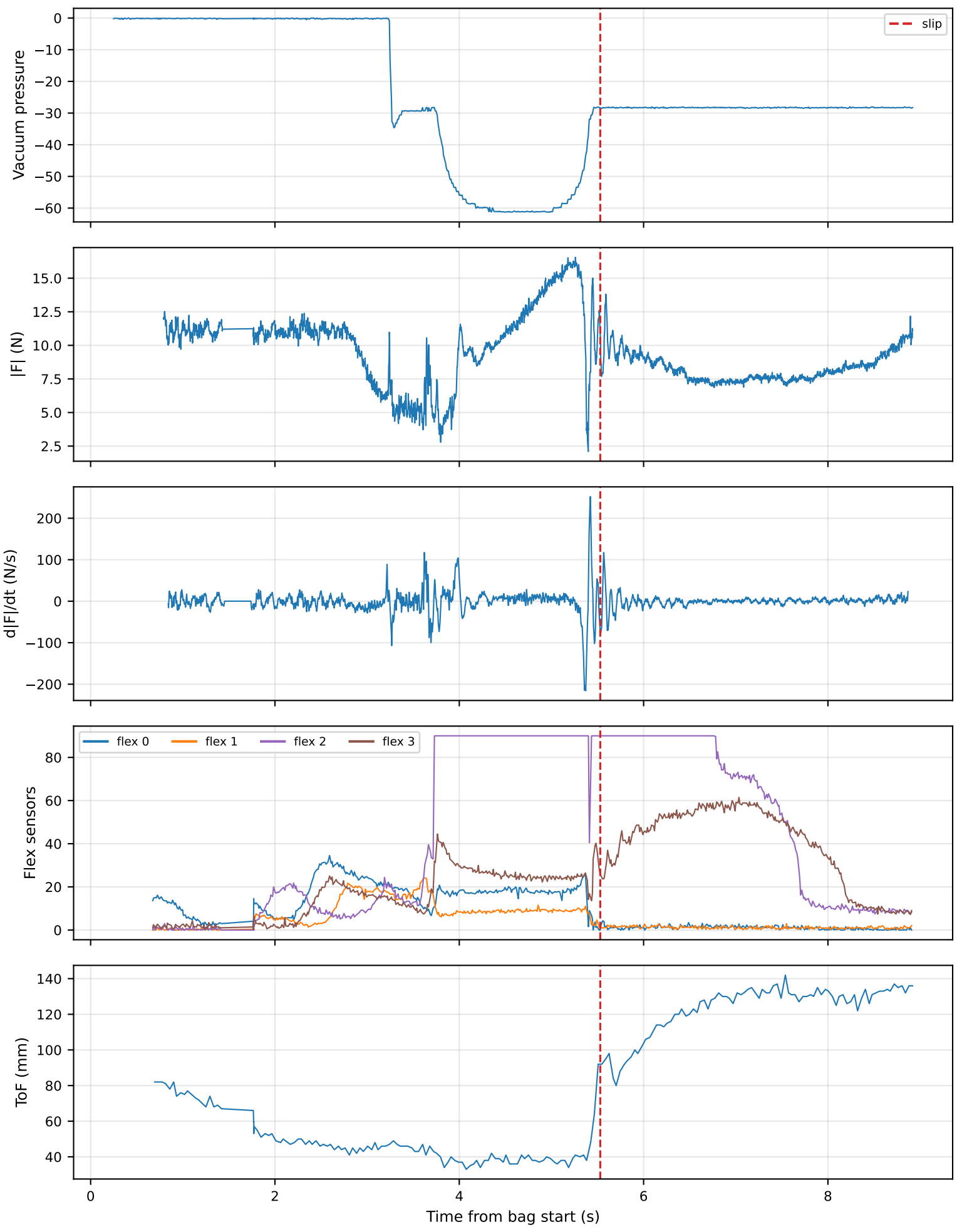

final_approach_and_pick_20261007_120701.db3
PICK at 4.13 s | 2026-10-07 12:07:02 | batch 63, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_121352.db3
SLIP at 6.28 s | 2026-10-07 12:13:53 | batch 65, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_121530.db3
SLIP at 8.87 s | 2026-10-07 12:15:31 | batch 66, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

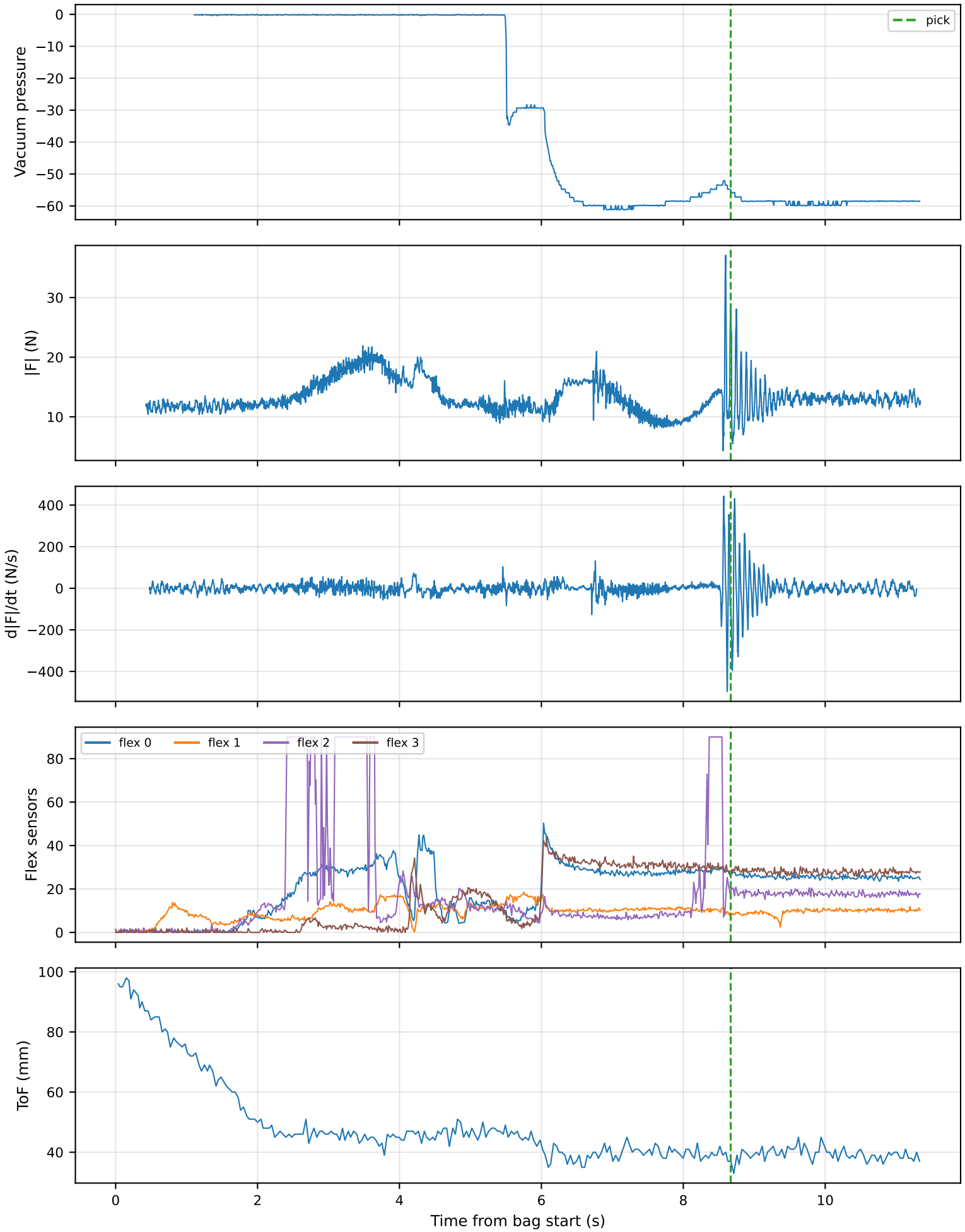

final_approach_and_pick_20261007_121720.db3
PICK at 6.01 s | 2026-10-07 12:17:21 | batch 67, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

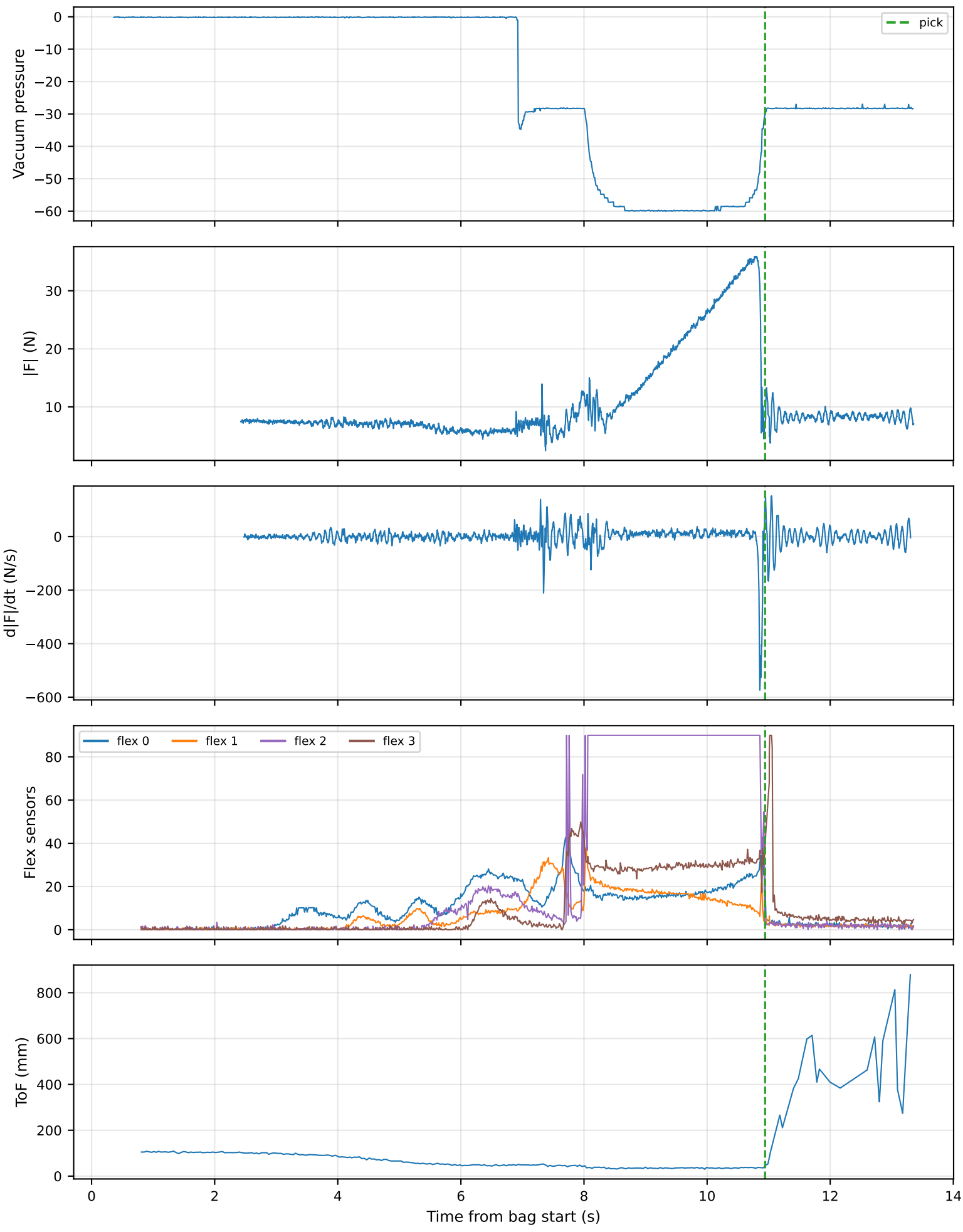

final_approach_and_pick_20261007_121907.db3
SLIP at 6.56 s | 2026-10-07 12:19:08 | batch 68, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

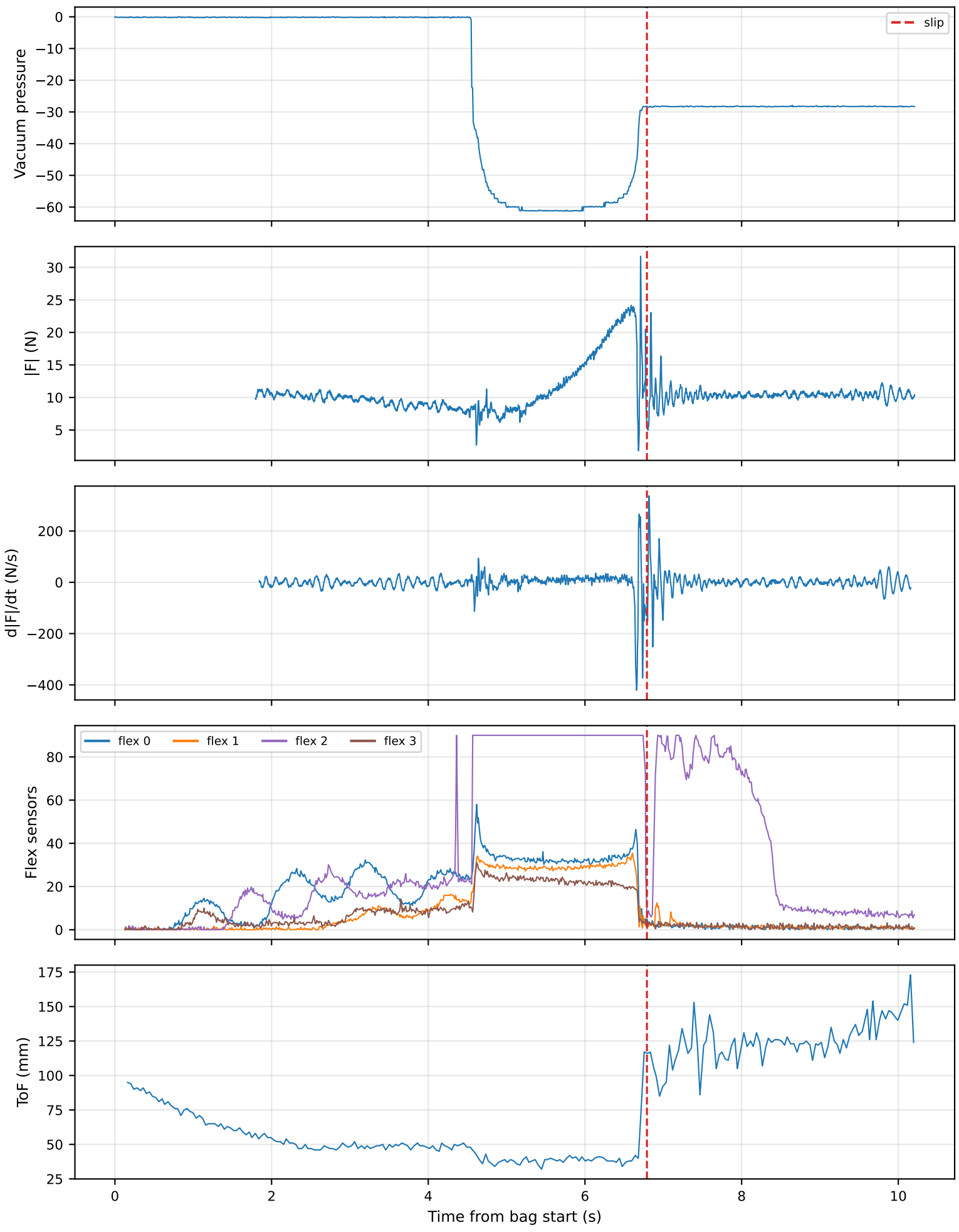

final_approach_and_pick_20261007_122337.db3
SLIP at 5.53 s | 2026-10-07 12:23:38 | batch 56, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

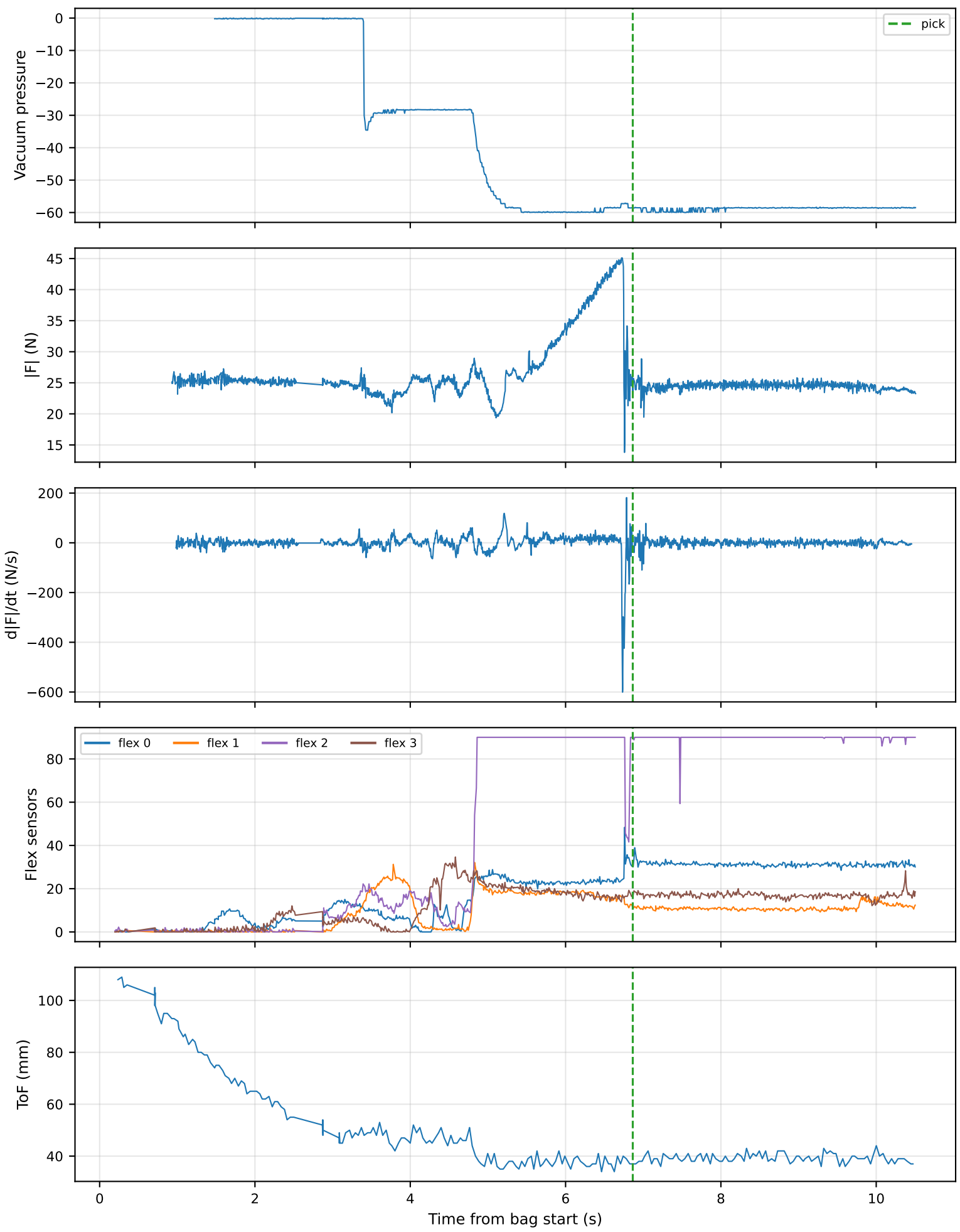
final_approach_and_pick_20261007_122619.db3

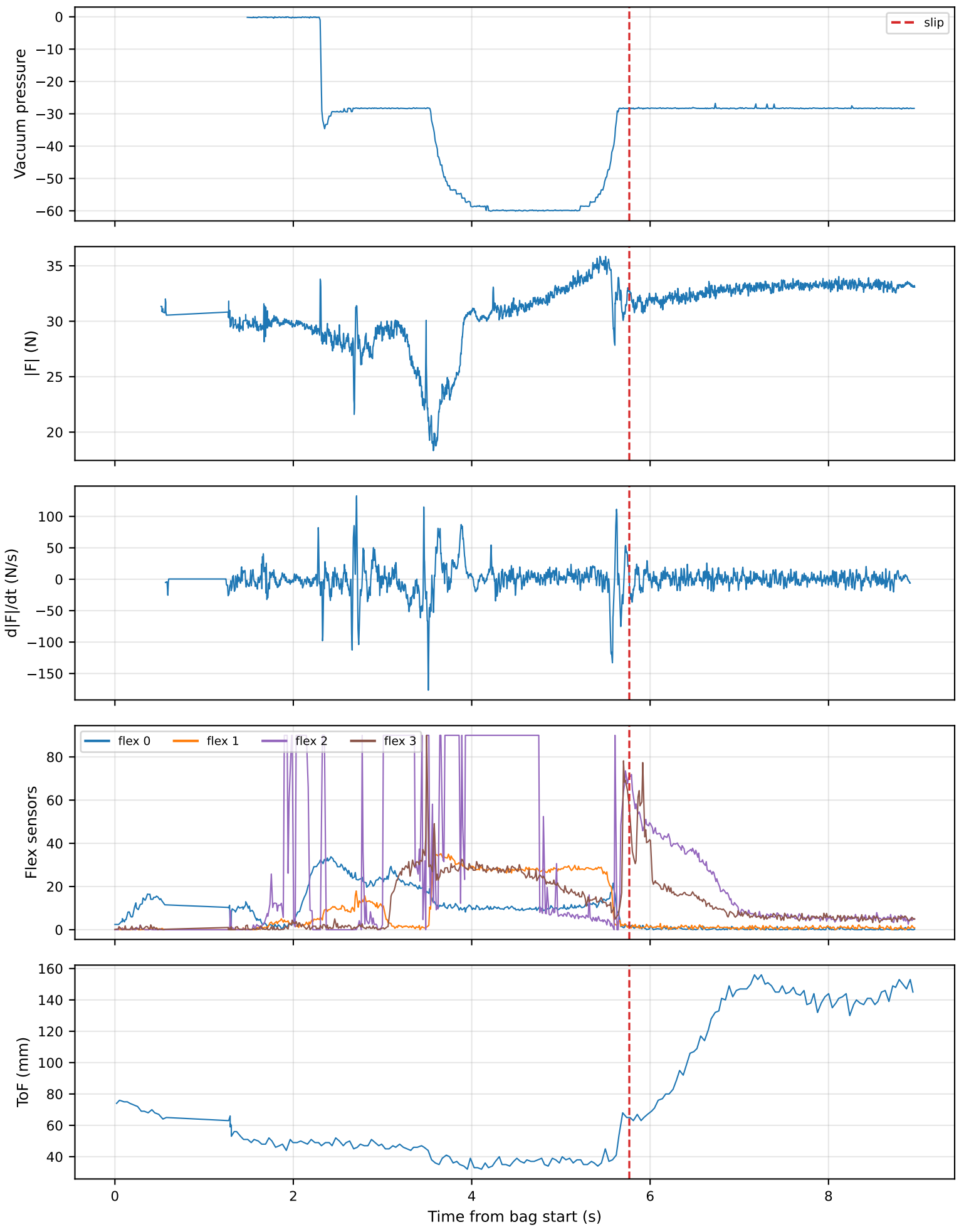
PICK at 8.67 s | 2026-10-07 12:26:22 | batch 70, apple 0 | stiffness_controller

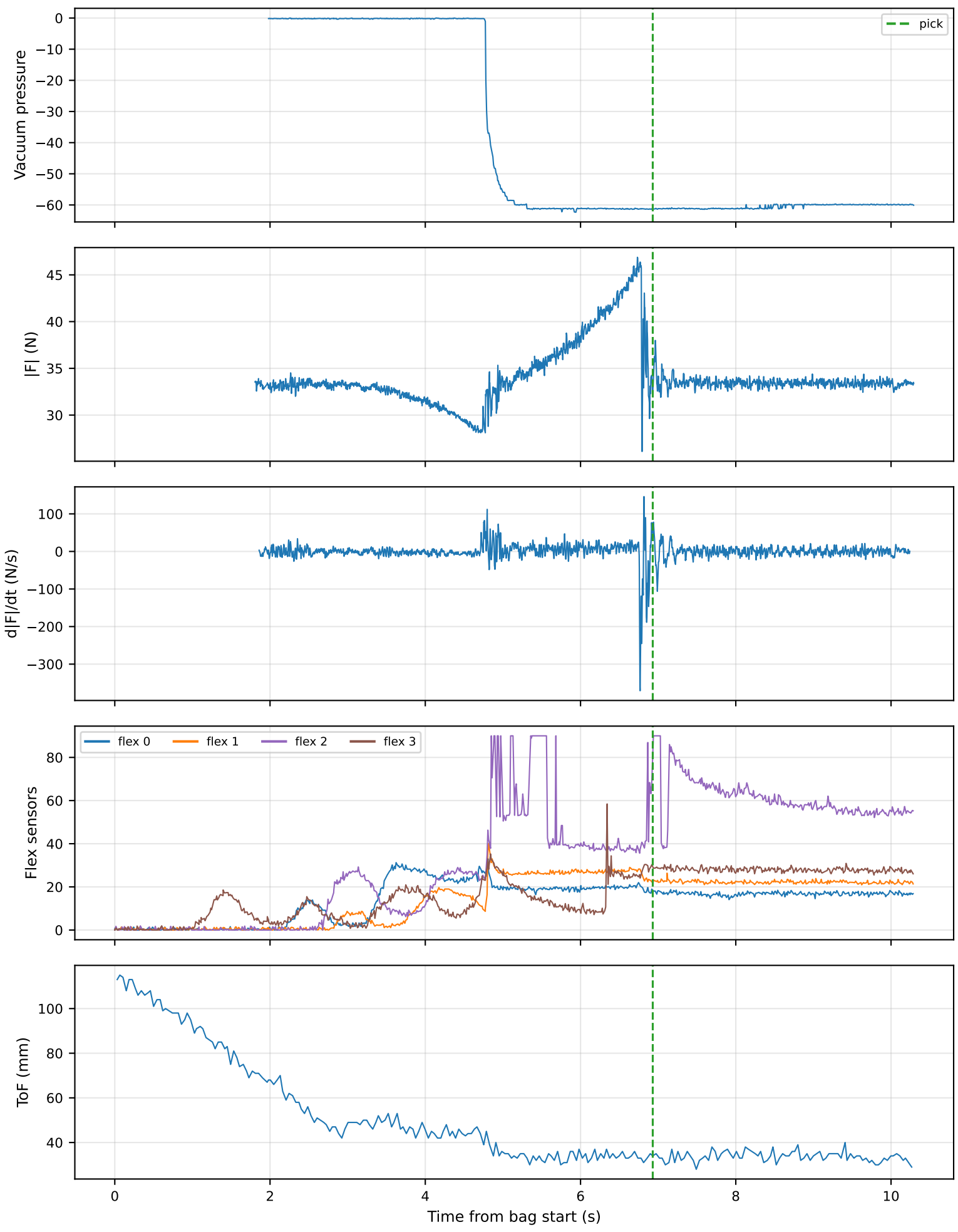



final_approach_and_pick_20261007_123828.db3
PICK at 10.94 s | 2026-10-07 12:38:29 | batch 71, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_123957.db3
SLIP at 6.79 s | 2026-10-07 12:39:58 | batch 72, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

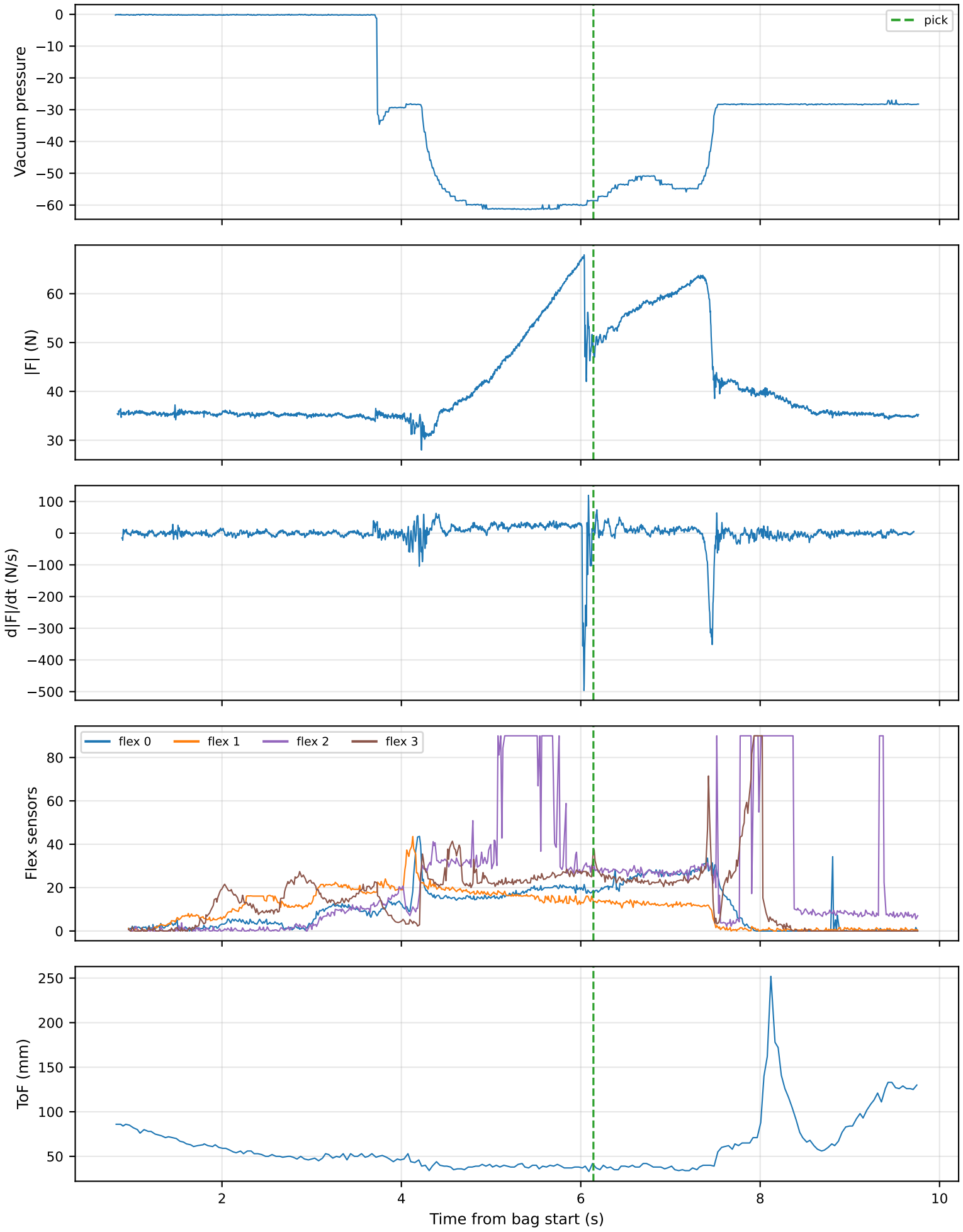

final_approach_and_pick_20261007_124257.db3
PICK at 6.87 s | 2026-10-07 12:42:58 | batch 73, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

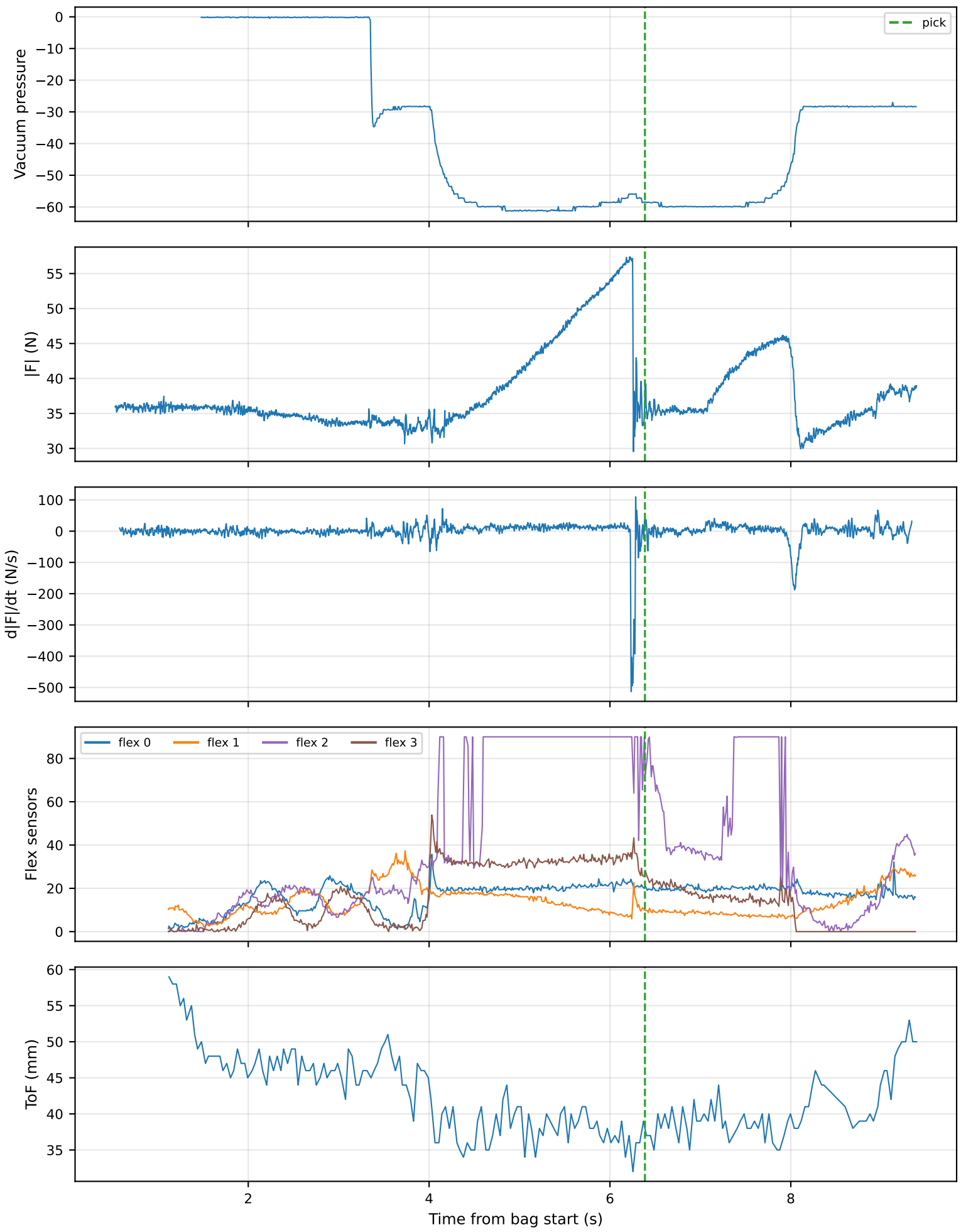

final_approach_and_pick_20261007_124600.db3
SLIP at 5.77 s | 2026-10-07 12:46:02 | batch 74, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

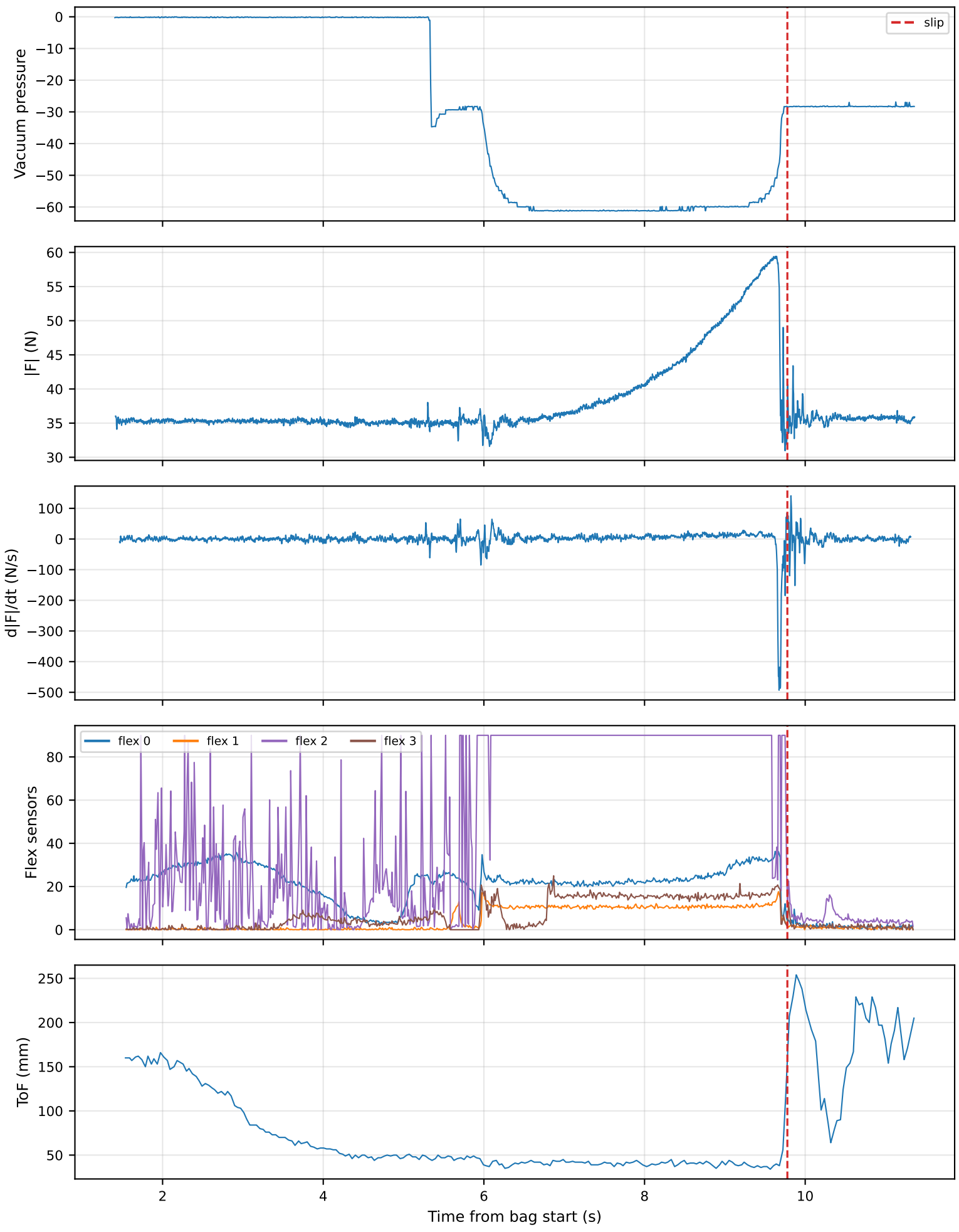

final_approach_and_pick_20261007_124706.db3
PICK at 6.93 s | 2026-10-07 12:47:07 | batch 75, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261007_124905.db3

PICK at 6.14 s | 2026-10-07 12:49:06 | batch 76, apple 0 | stiffness_controller

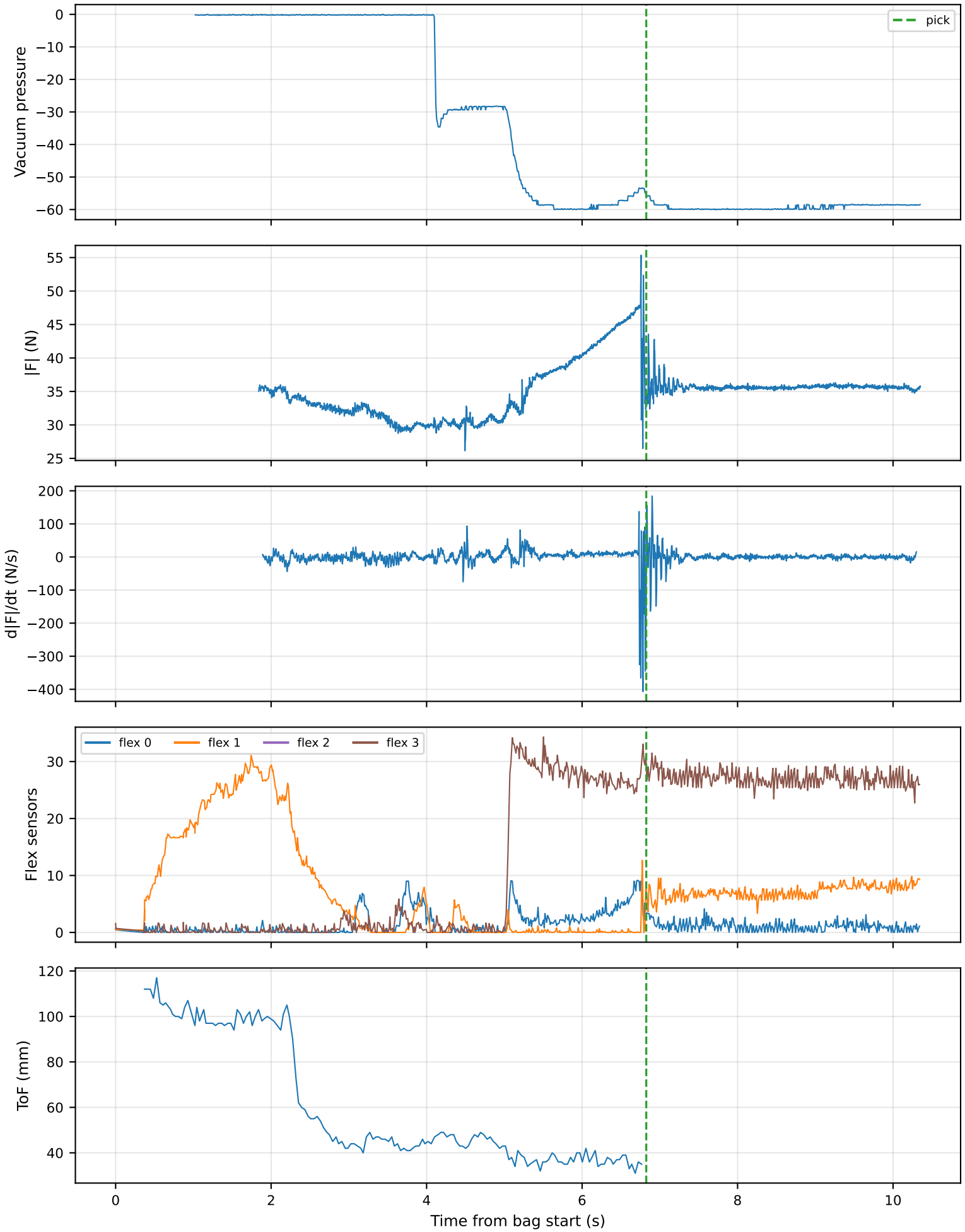


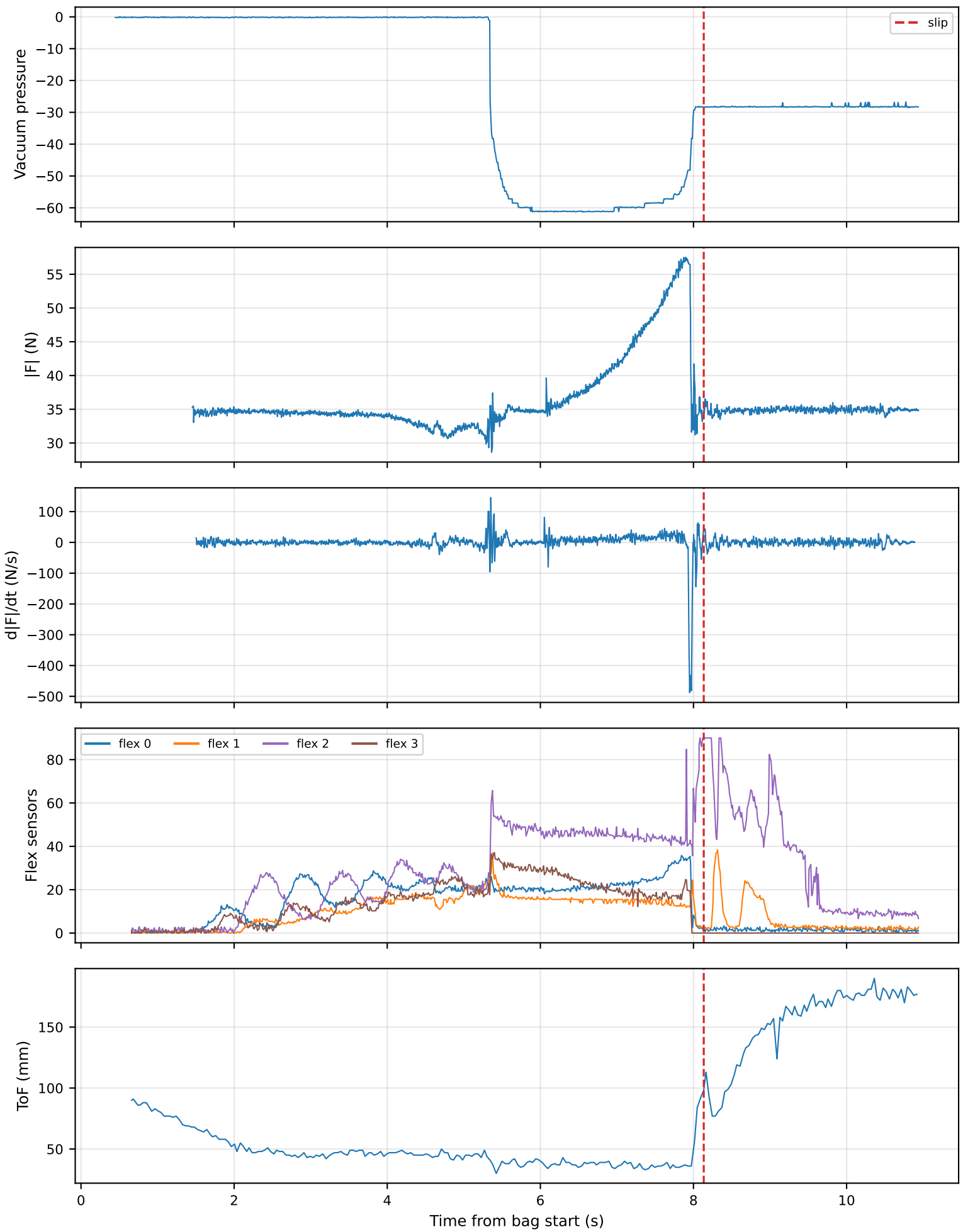

final_approach_and_pick_20261007_125025.db3
PICK at 6.39 s | 2026-10-07 12:50:26 | batch 77, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

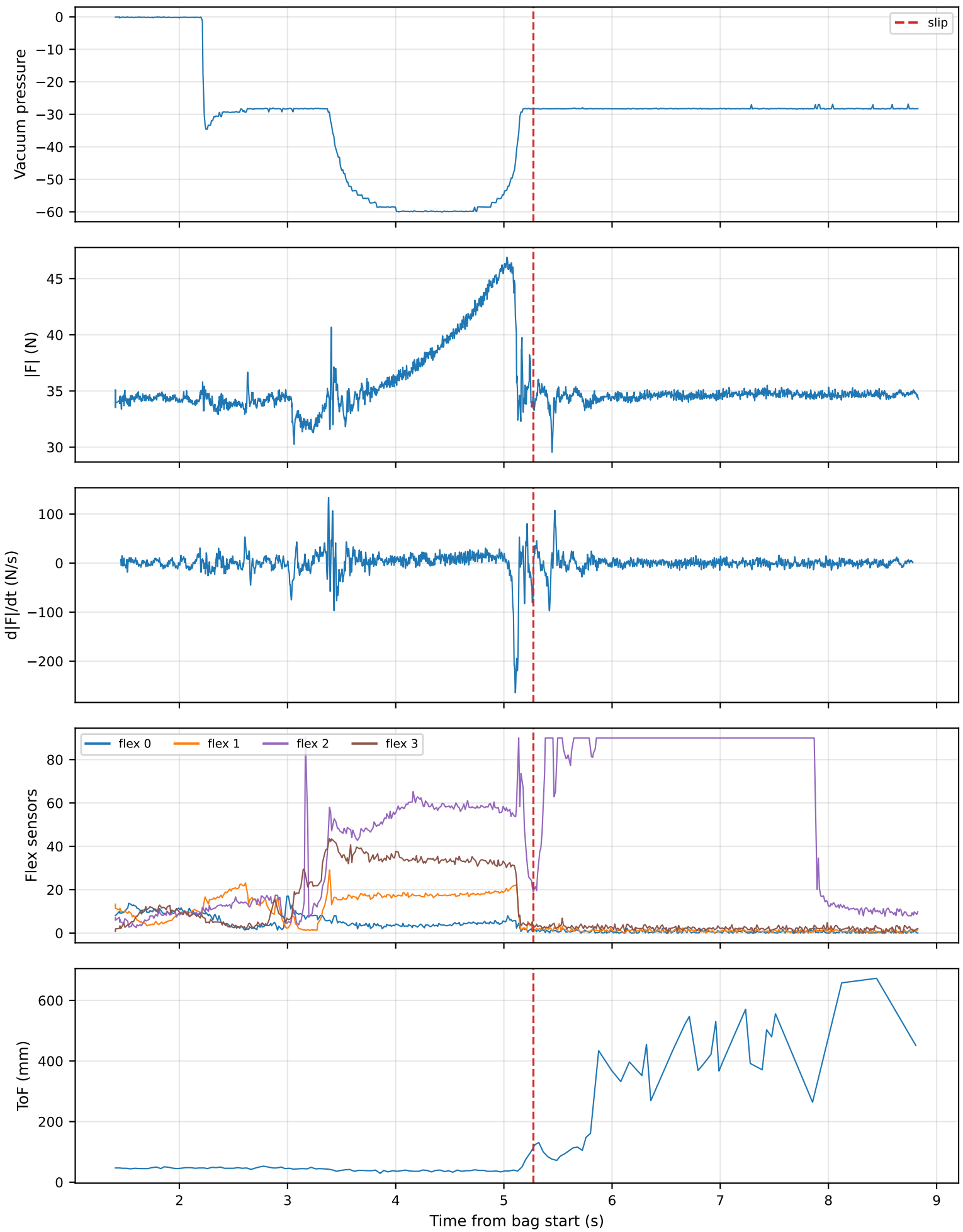

final_approach_and_pick_20261007_125333.db3
SLIP at 9.78 s | 2026-10-07 12:53:35 | batch 78, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

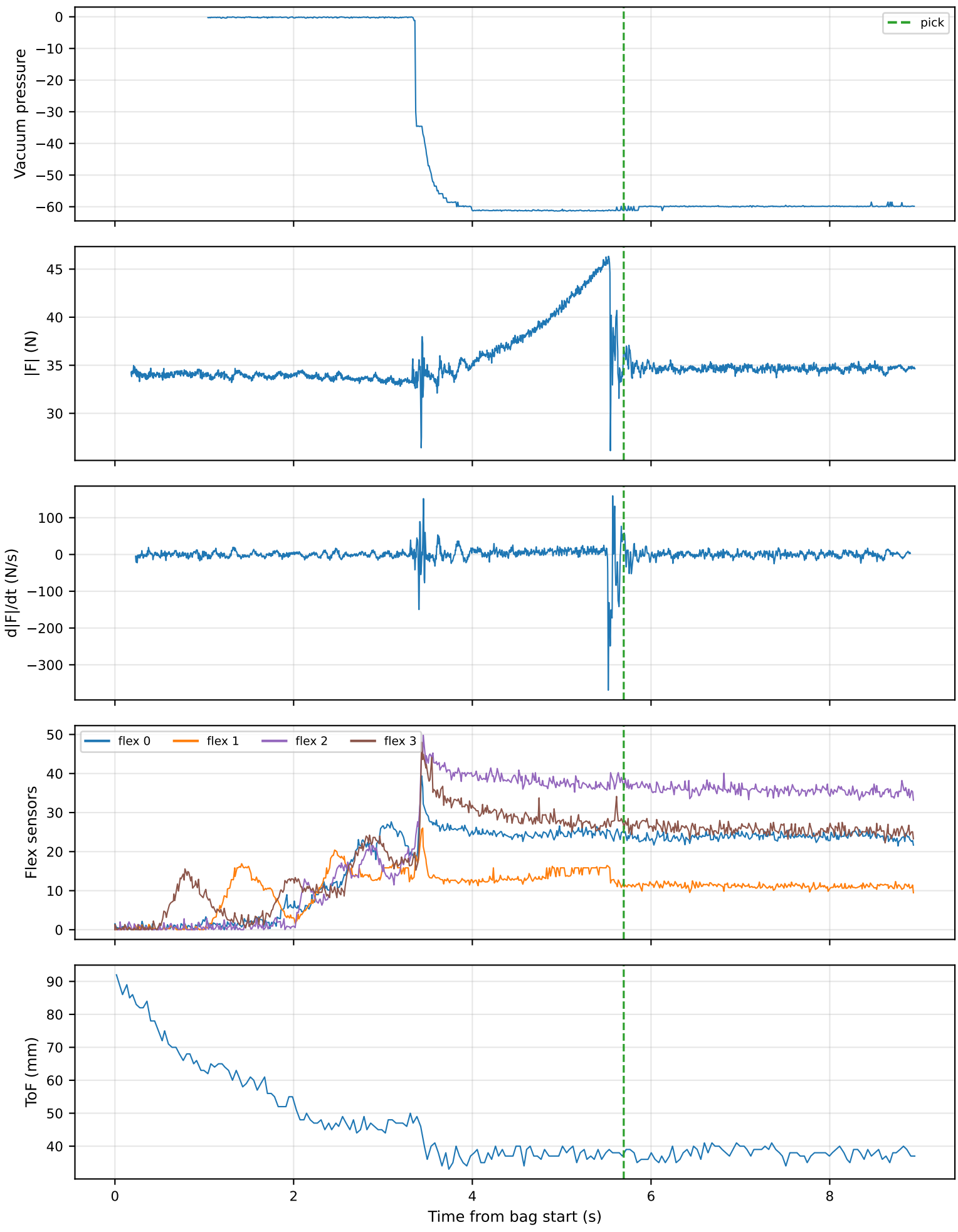
final_approach_and_pick_20261007_125441.db3

PICK at 6.83 s | 2026-10-07 12:54:42 | batch 60, apple 0 | stiffness_controller



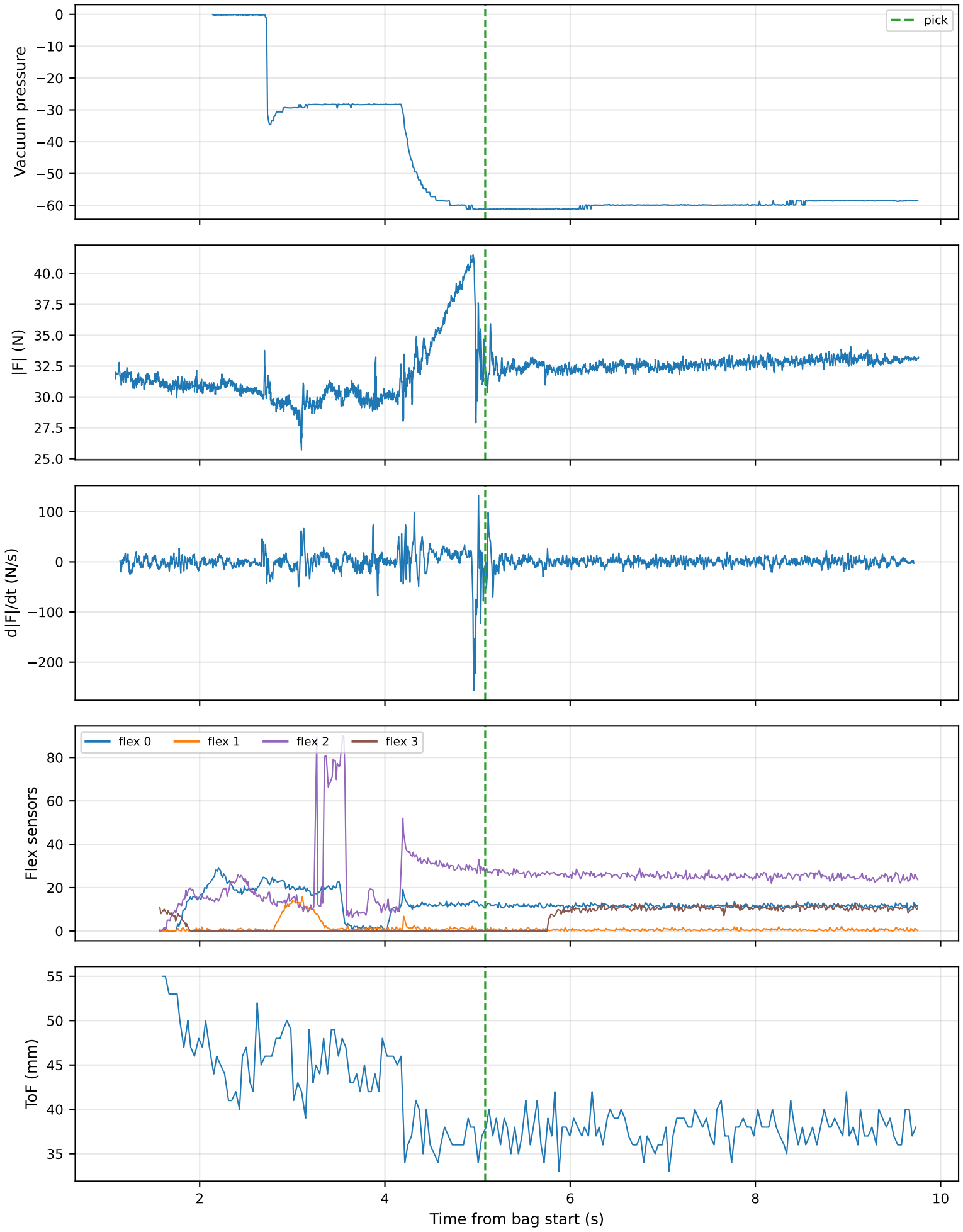

final_approach_and_pick_20261007_130536.db3
SLIP at 8.13 s | 2026-10-07 13:05:38 | batch 79, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

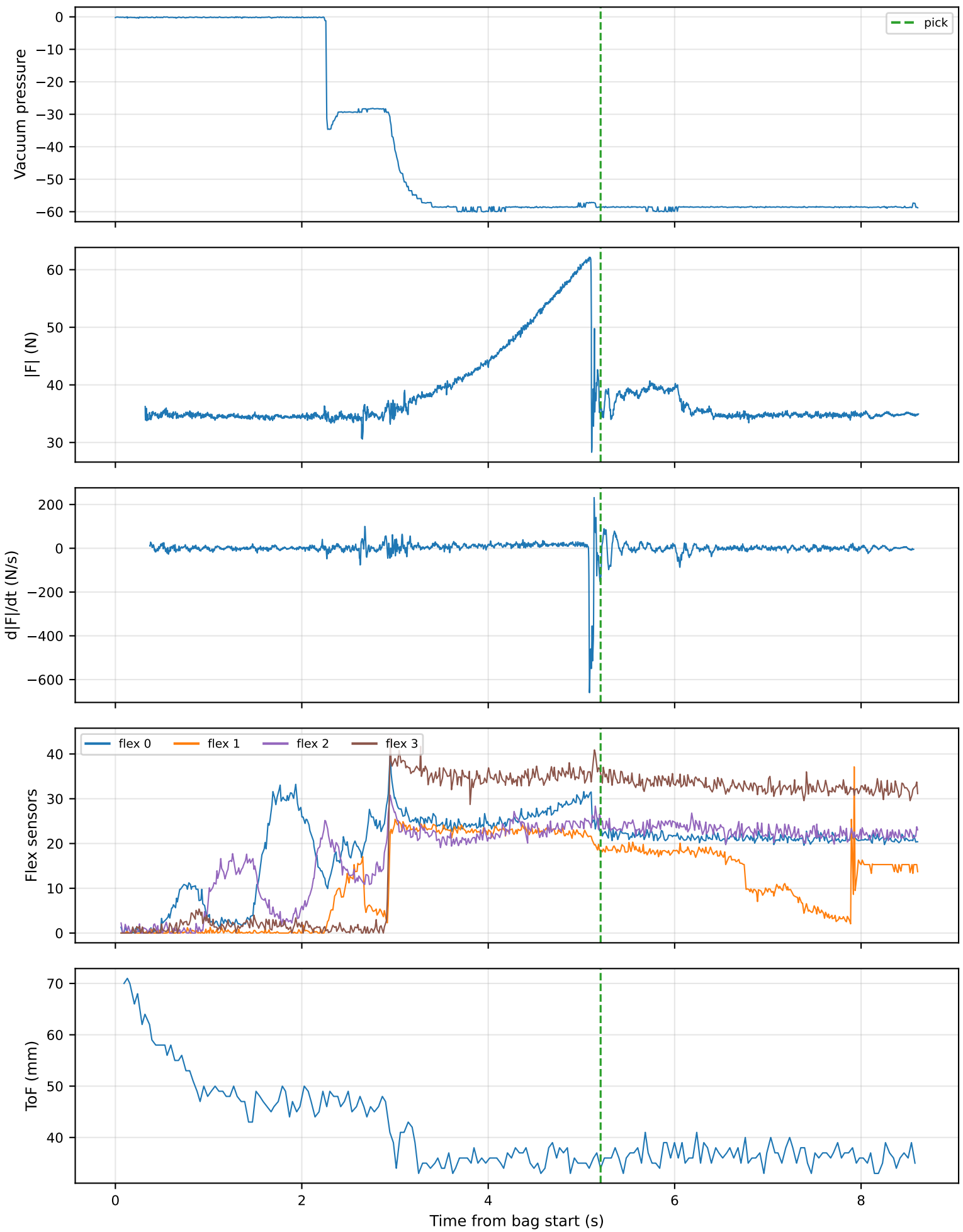

final_approach_and_pick_20261007_130717.db3
SLIP at 5.27 s | 2026-10-07 13:07:18 | batch 80, apple 0 | stiffness_controller
slip
Vacuum pressure
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

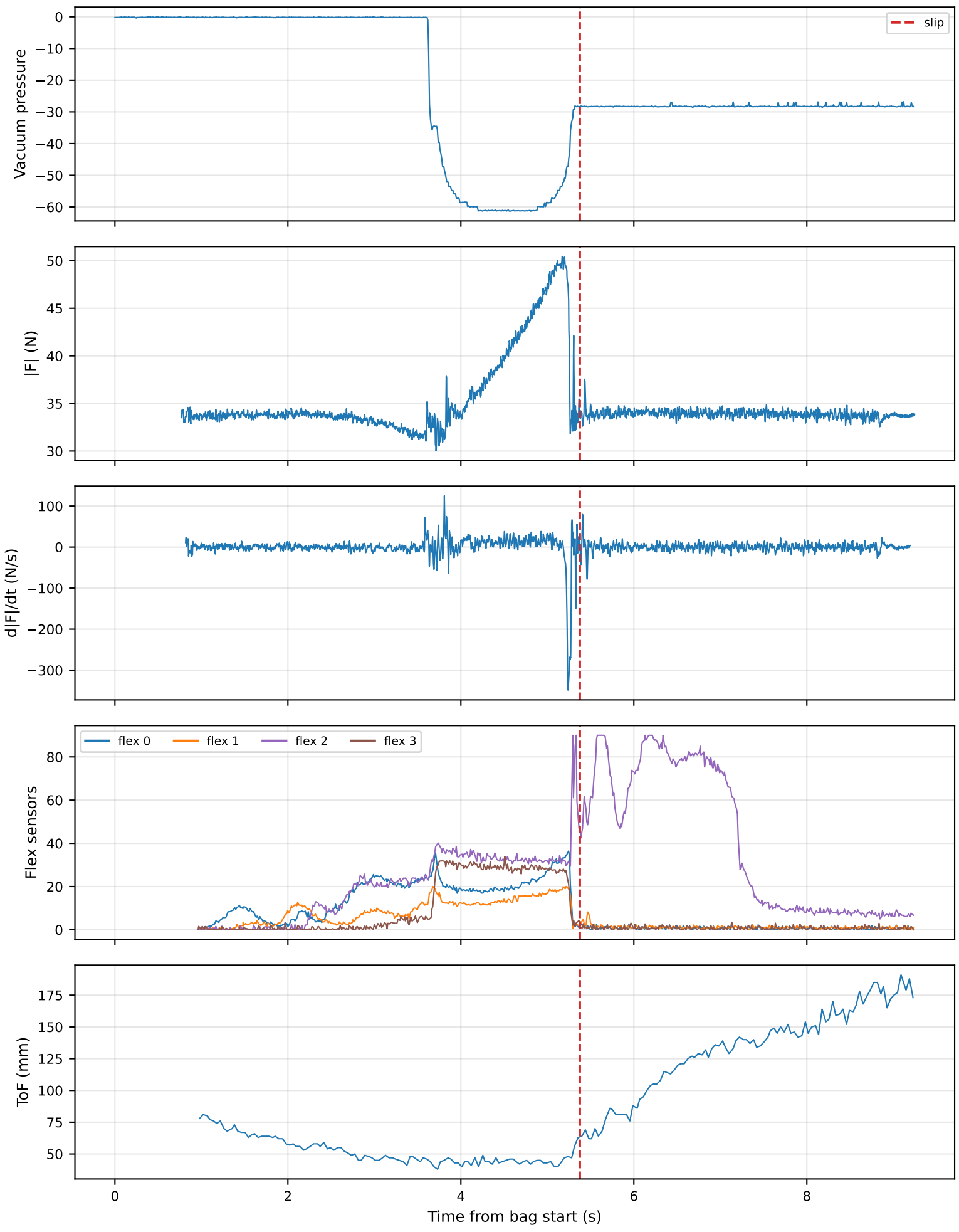

final_approach_and_pick_20261007_131319.db3
PICK at 5.69 s | 2026-10-07 13:13:21 | batch 82, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261007_131523.db3

PICK at 5.08 s | 2026-10-07 13:15:24 | batch 83, apple 0 | stiffness_controller

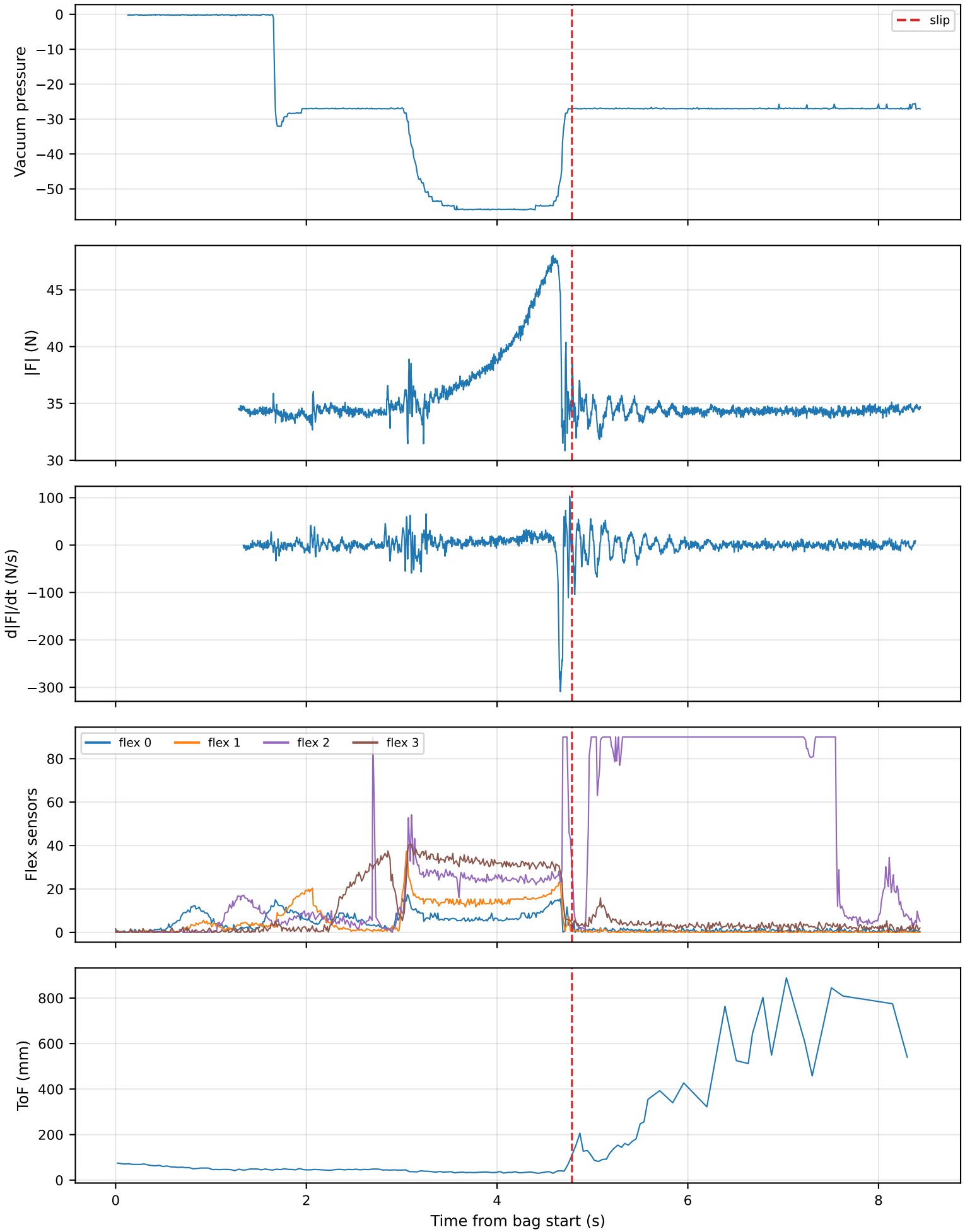


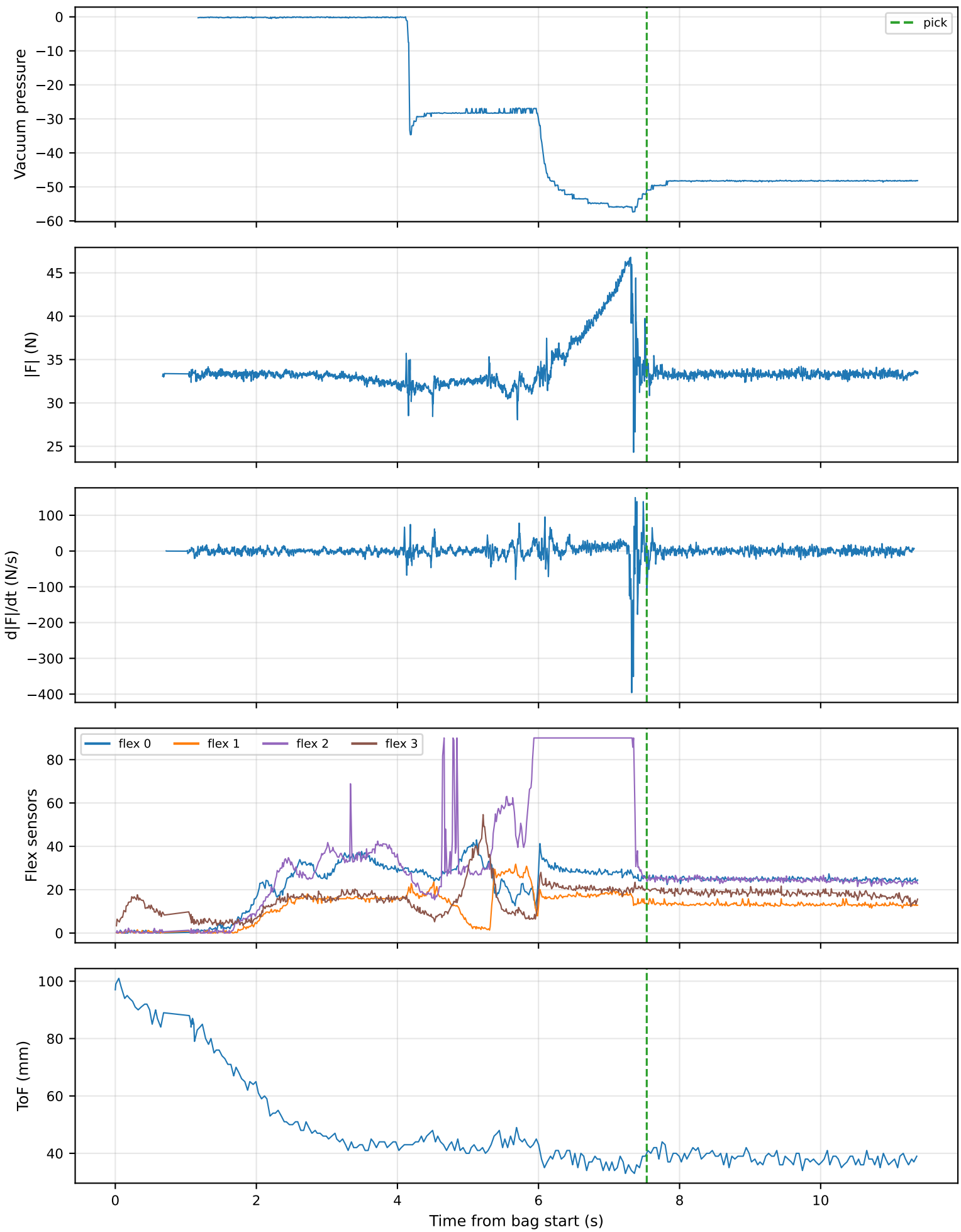

final_approach_and_pick_20261007_131805.db3
PICK at 5.21 s | 2026-10-07 13:18:07 | batch 84, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

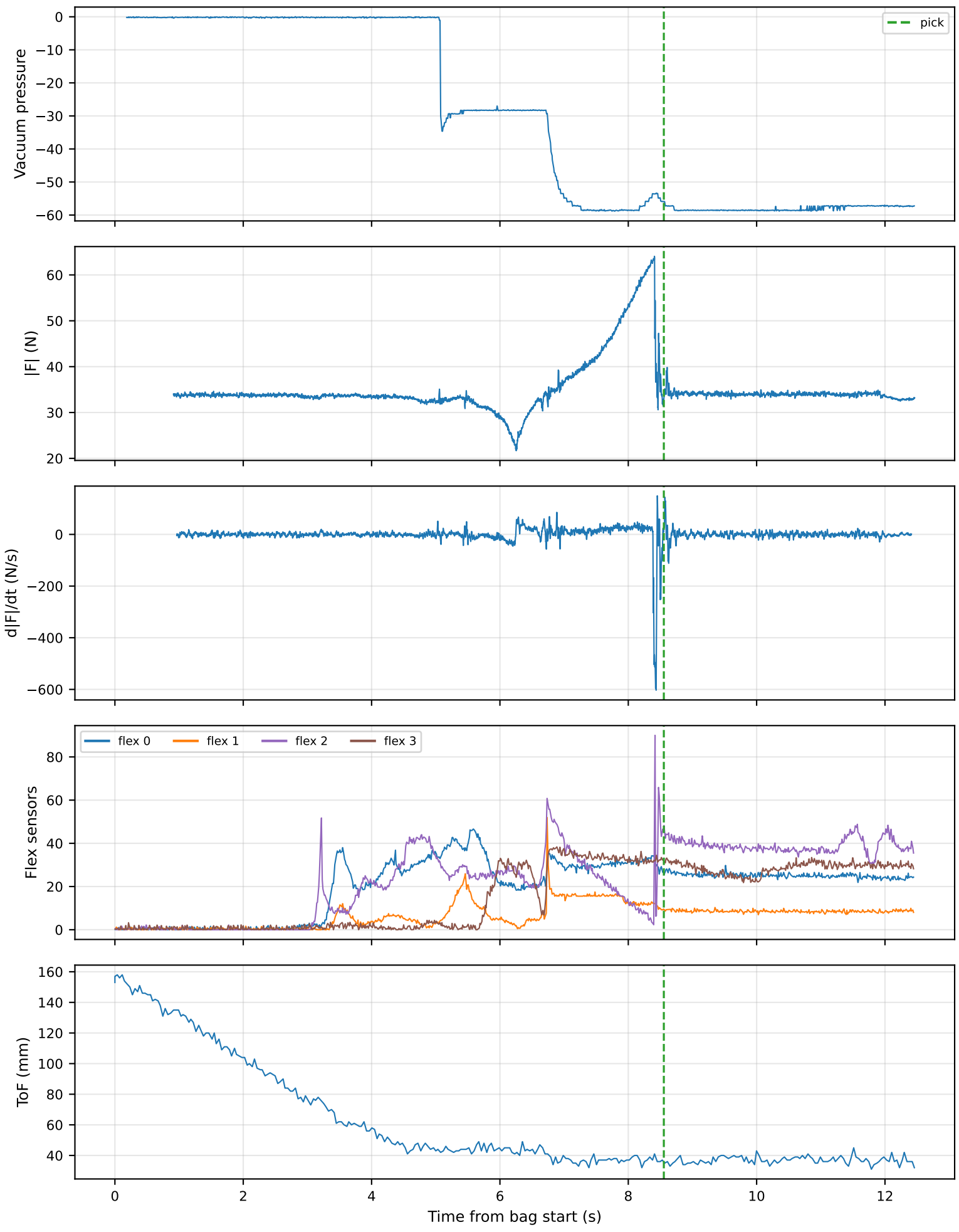

final_approach_and_pick_20261007_131908.db3
SLIP at 5.38 s | 2026-10-07 13:19:09 | batch 85, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

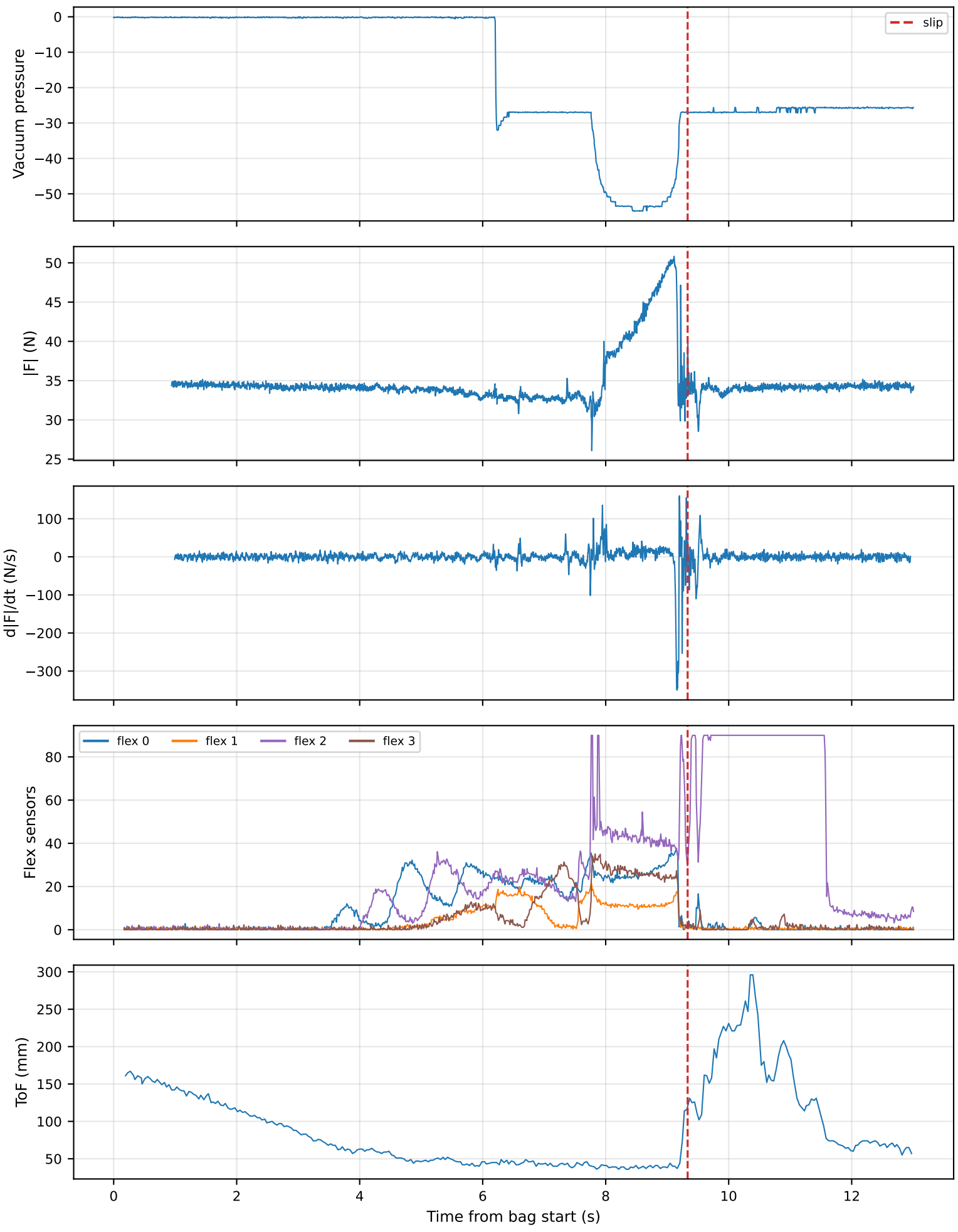
final_approach_and_pick_20261007_132011.db3

SLIP at 4.79 s | 2026-10-07 13:20:13 | batch 86, apple 0 | stiffness_controller



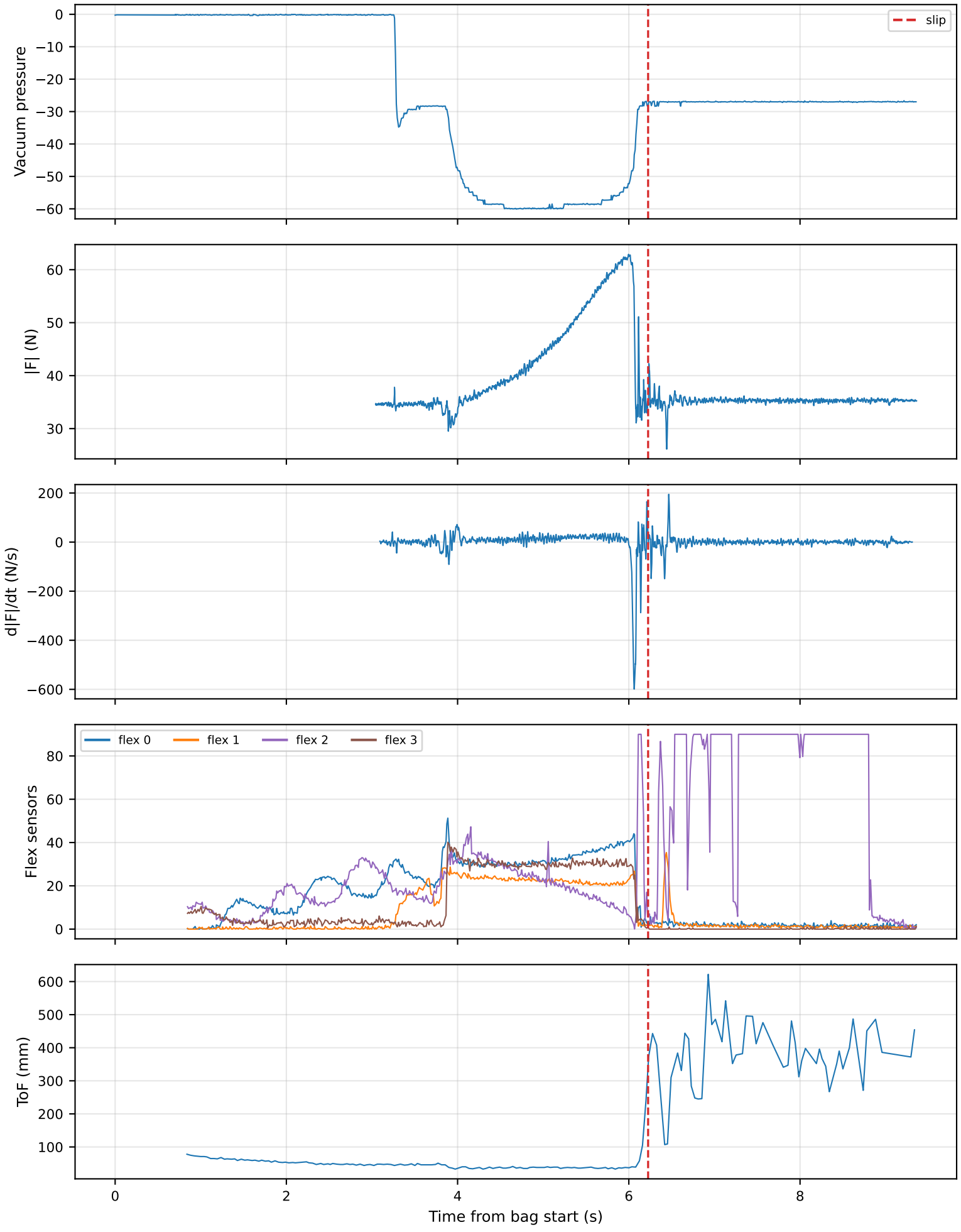

final_approach_and_pick_20261007_132110.db3
PICK at 7.53 s | 2026-10-07 13:21:11 | batch 87, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

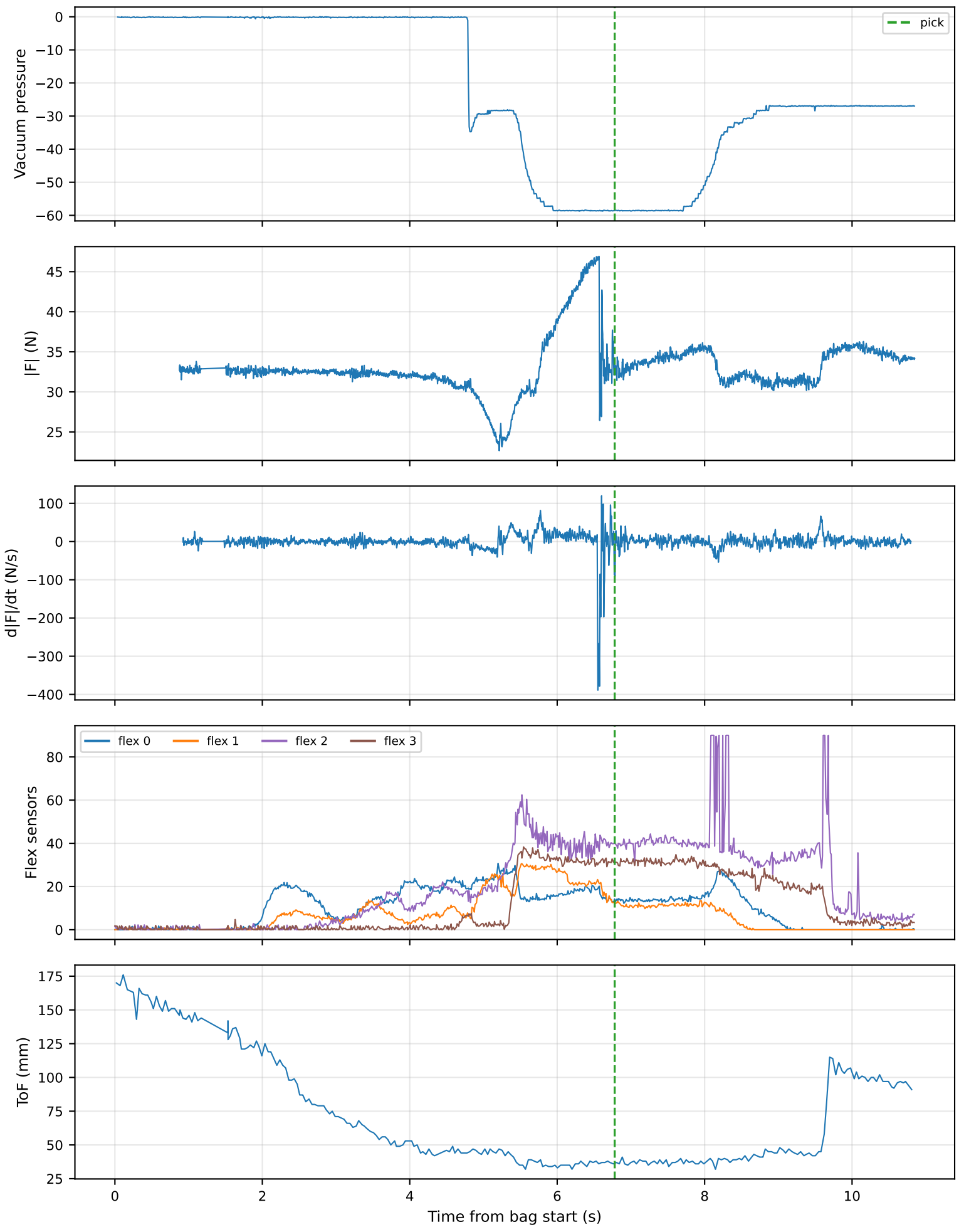

final_approach_and_pick_20261007_132207.db3
PICK at 8.55 s | 2026-10-07 13:22:08 | batch 88, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

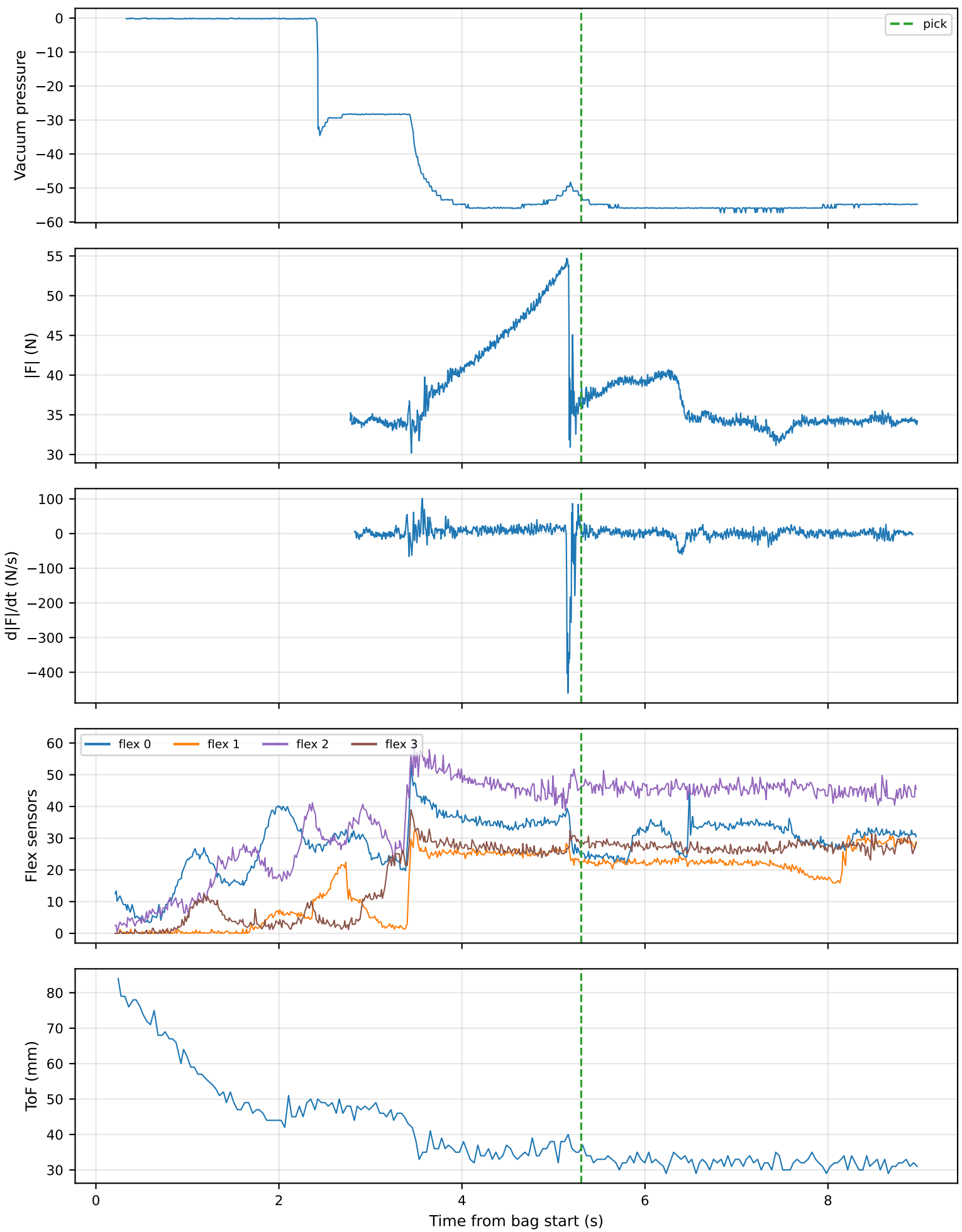

final_approach_and_pick_20261007_132319.db3
SLIP at 9.33 s | 2026-10-07 13:23:21 | batch 89, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261007_132424.db3

SLIP at 6.23 s | 2026-10-07 13:24:25 | batch 90, apple 0 | stiffness_controller

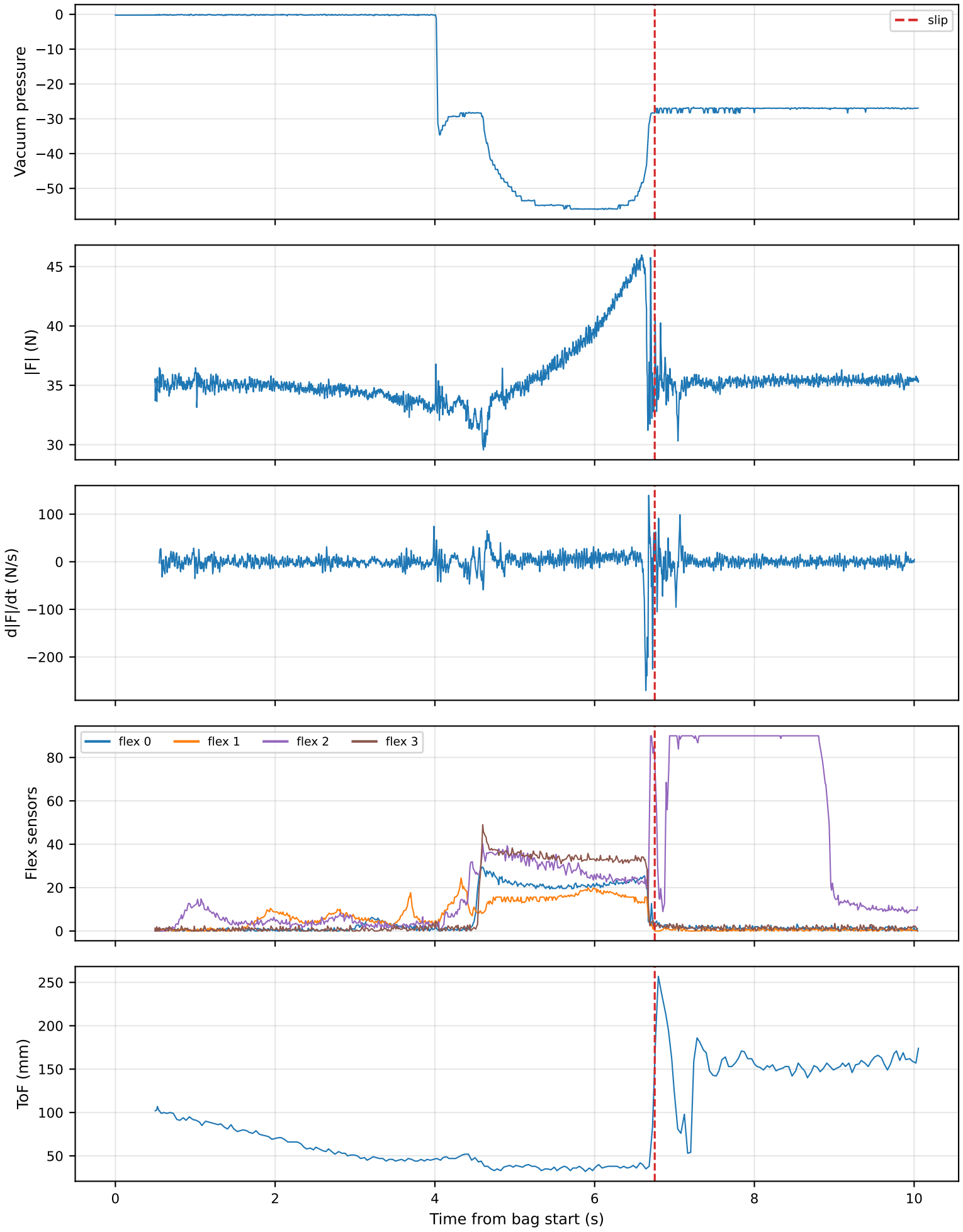


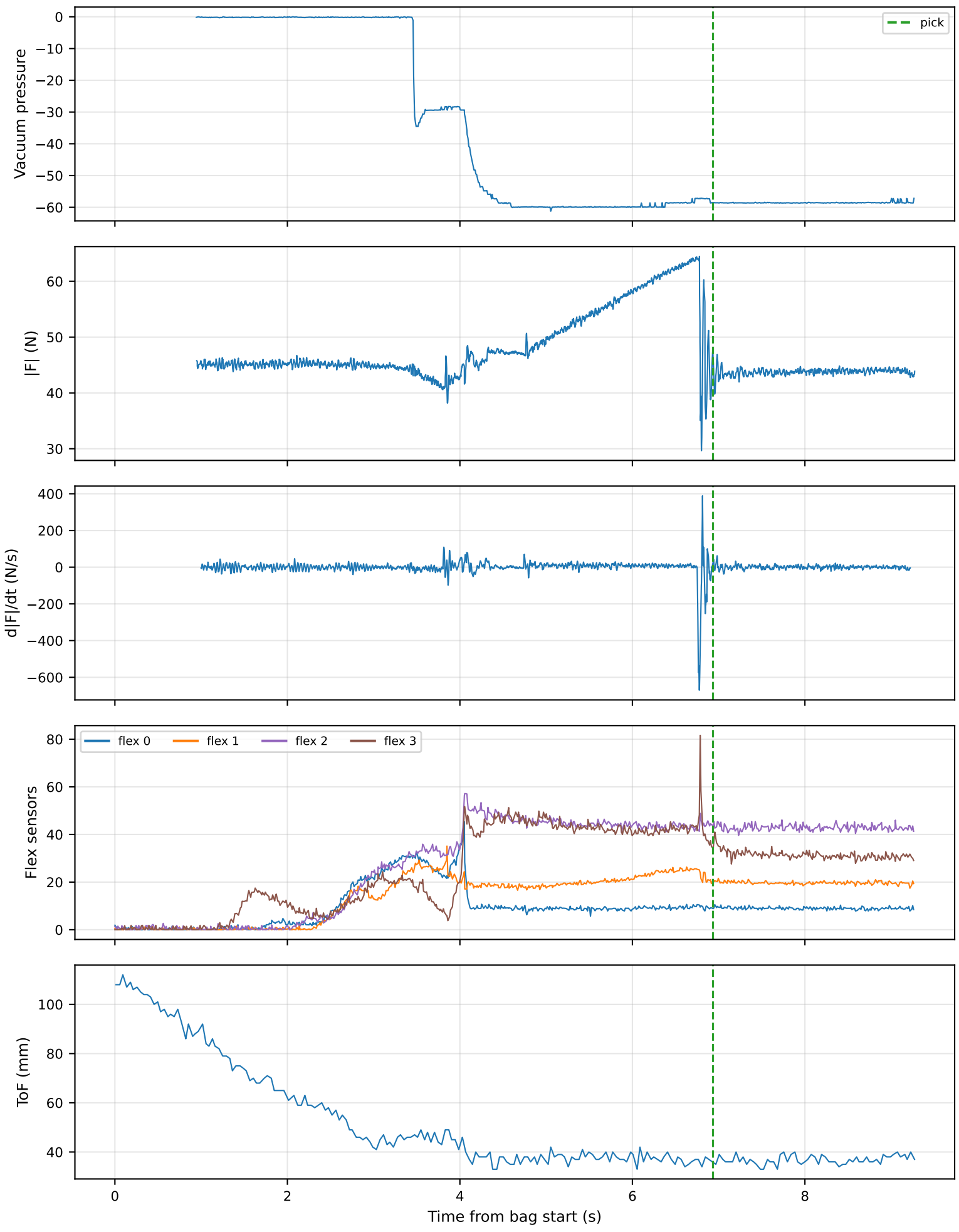

final_approach_and_pick_20261007_132528.db3
PICK at 6.78 s | 2026-10-07 13:25:29 | batch 91, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

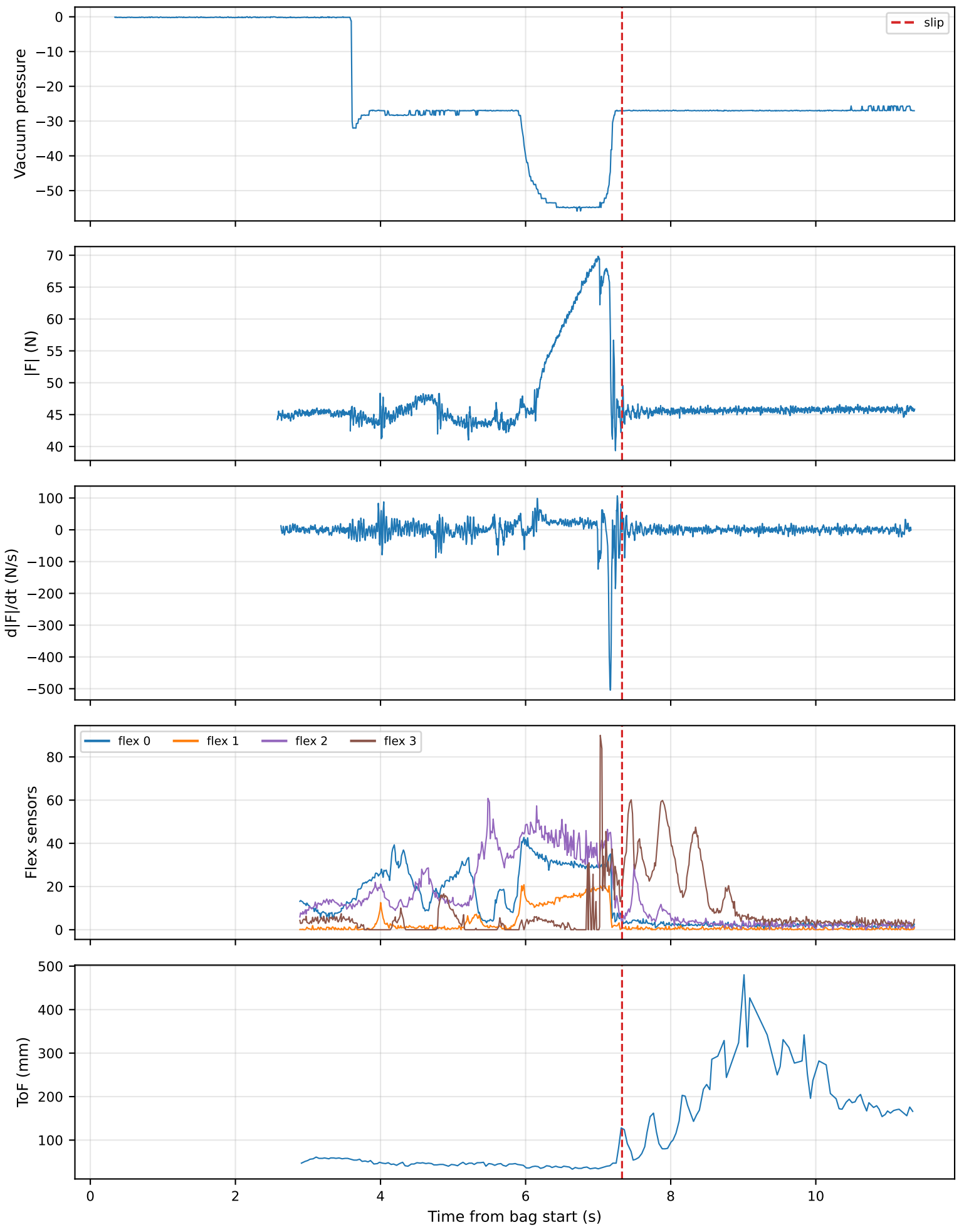

final_approach_and_pick_20261007_132657.db3
PICK at 5.30 s | 2026-10-07 13:26:59 | batch 92, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

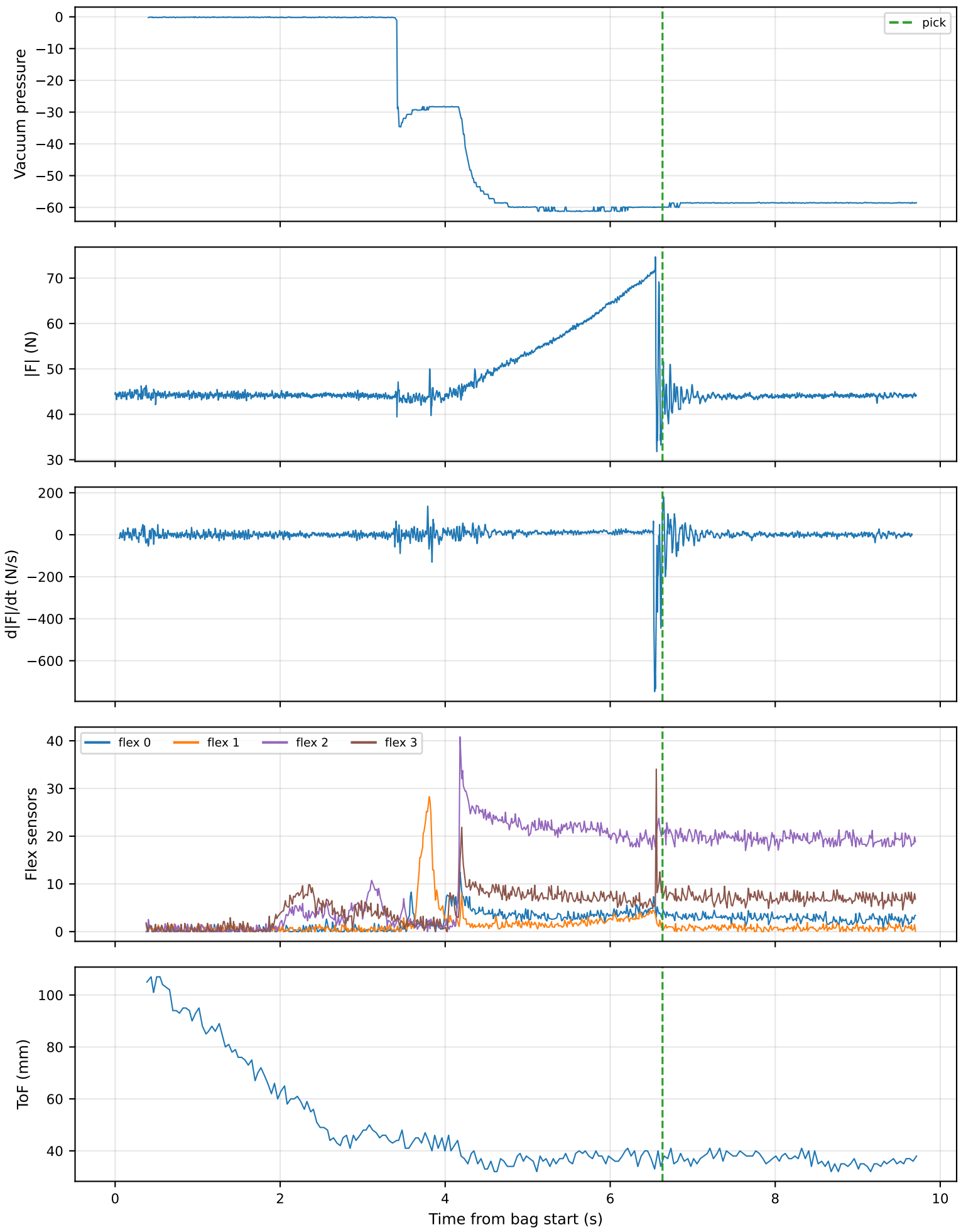
final_approach_and_pick_20261007_132813.db3

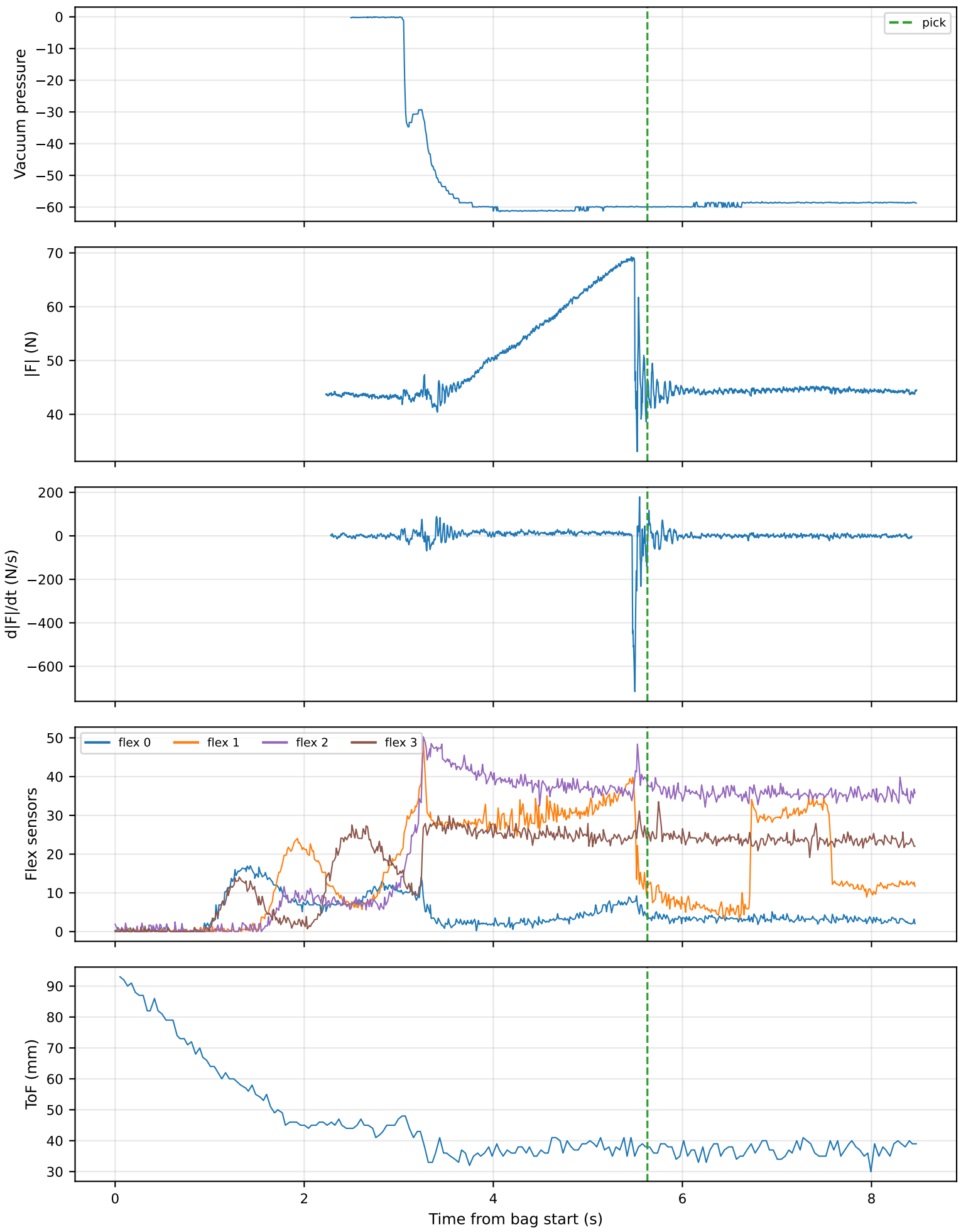
SLIP at 6.75 s | 2026-10-07 13:28:15 | batch 93, apple 0 | stiffness_controller

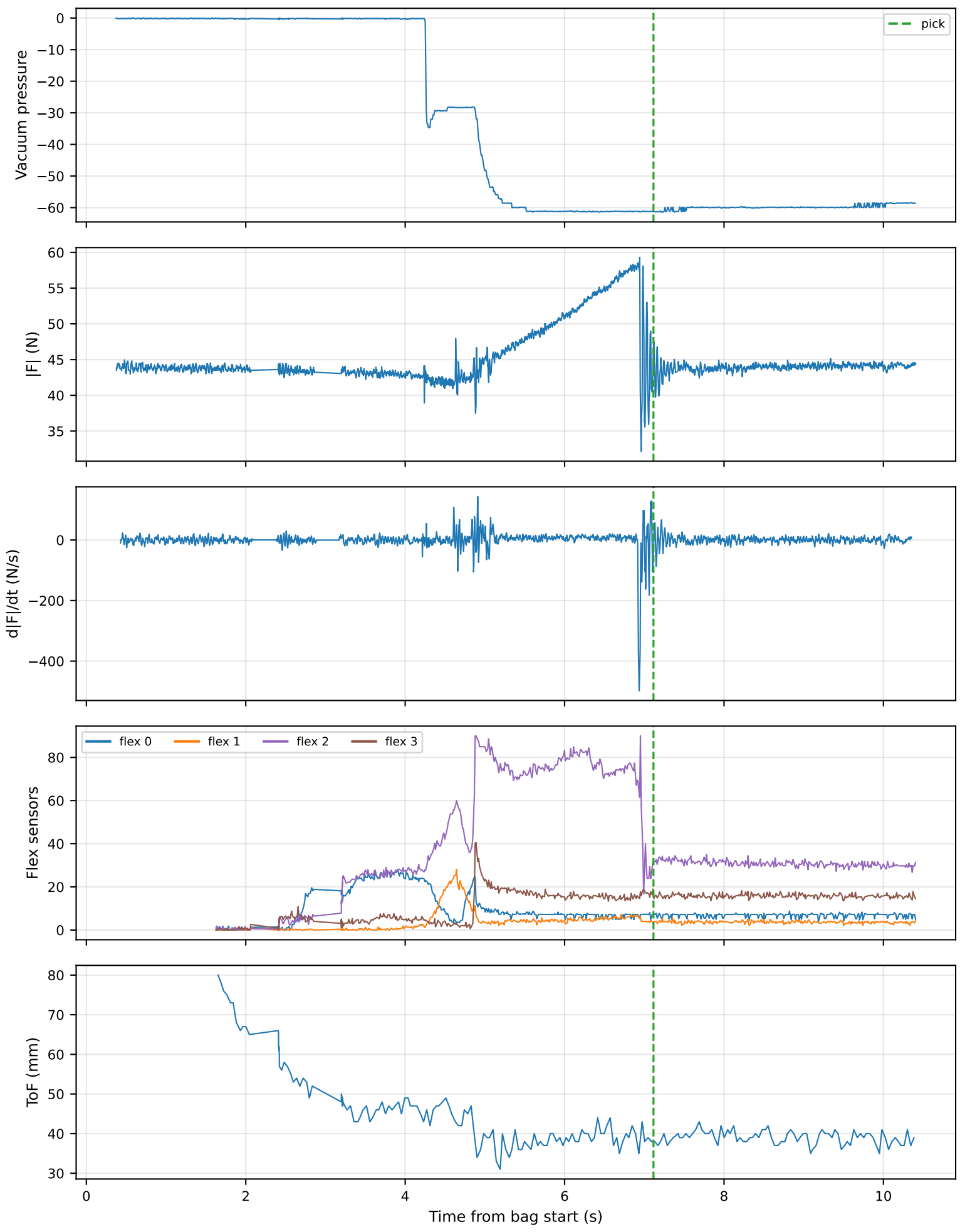


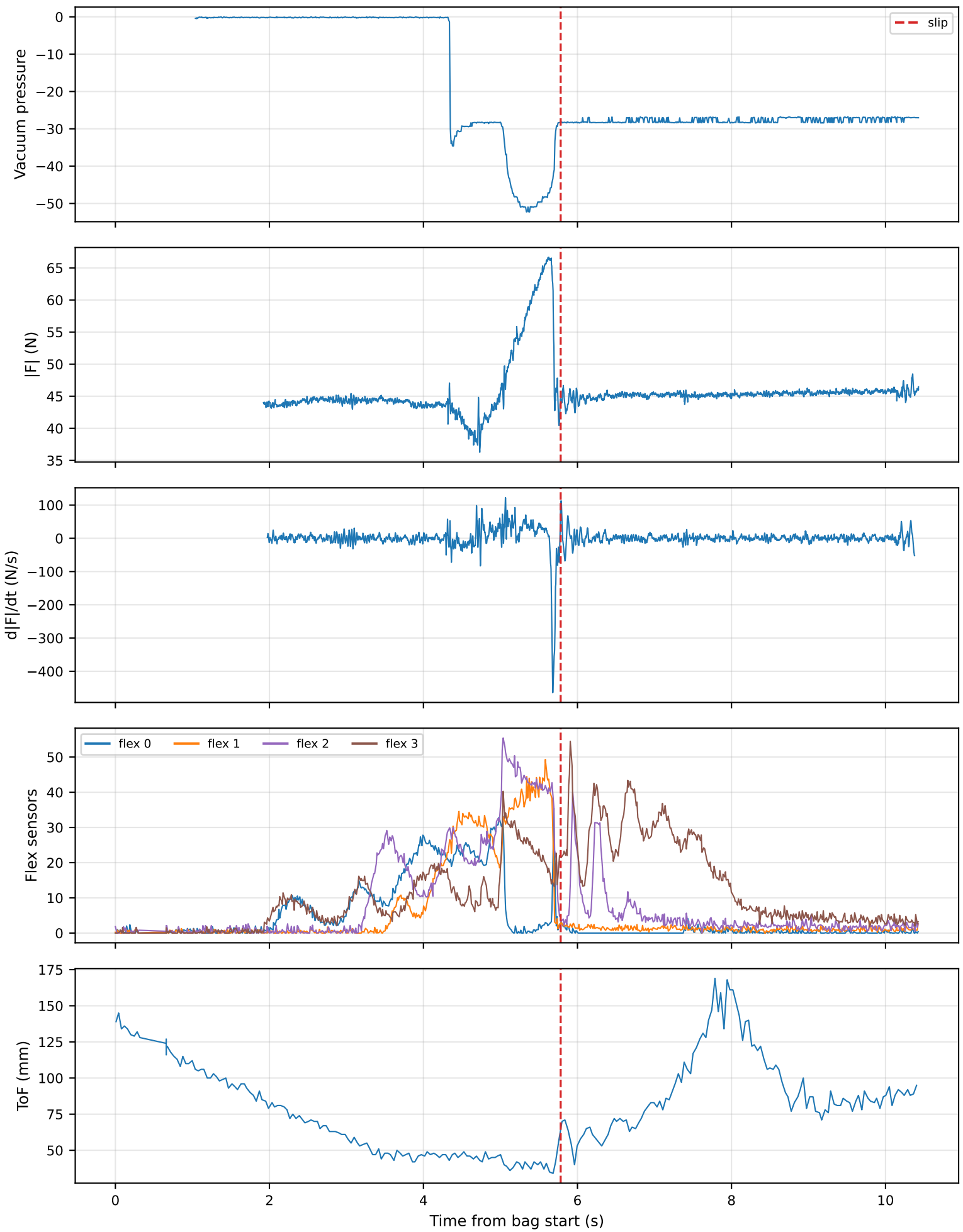

final_approach_and_pick_20261007_133327.db3
PICK at 6.93 s | 2026-10-07 13:33:28 | batch 95, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

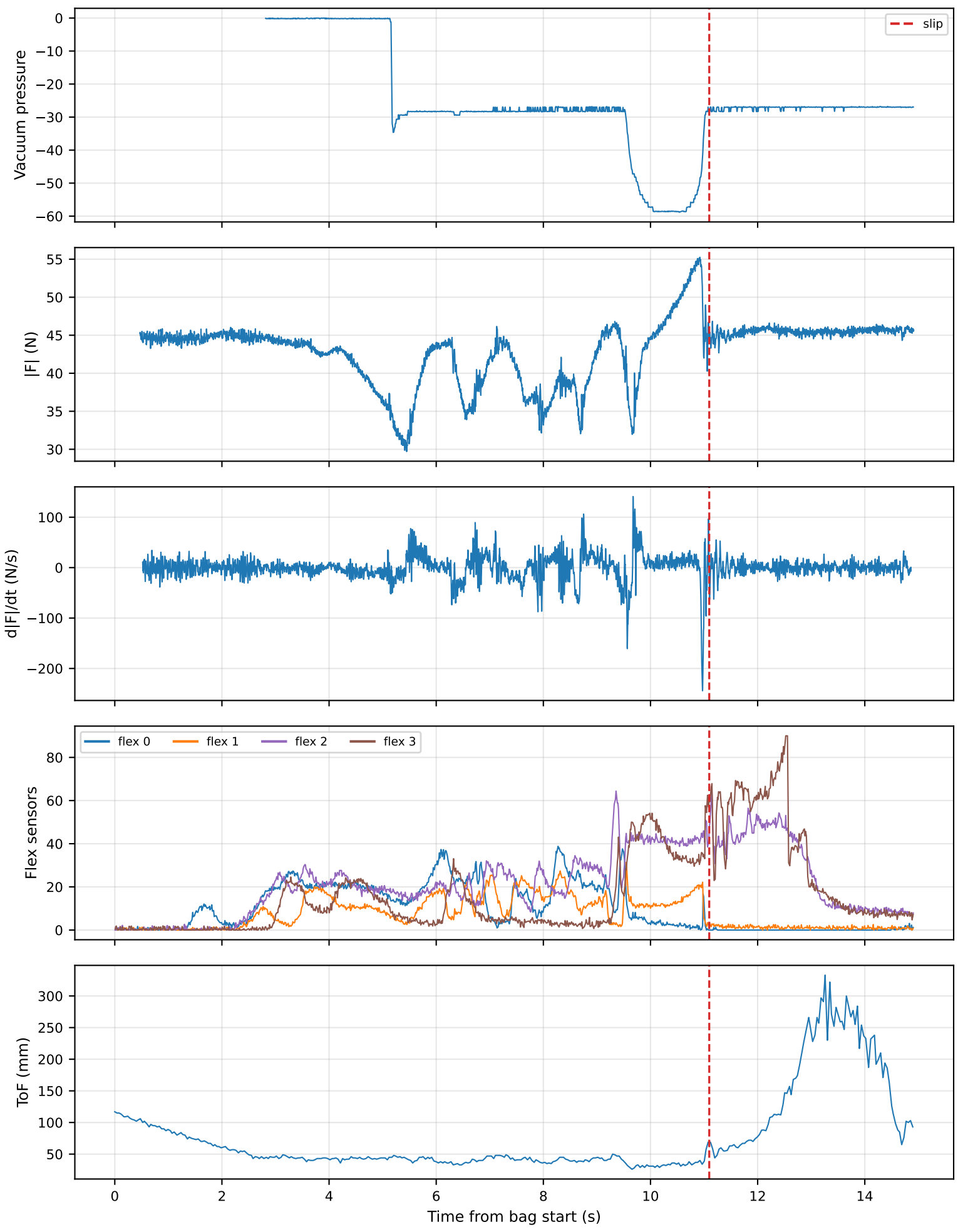

final_approach_and_pick_20261007_133551.db3
SLIP at 7.33 s | 2026-10-07 13:35:52 | batch 97, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_133707.db3
PICK at 6.63 s | 2026-10-07 13:37:08 | batch 98, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_133842.db3
PICK at 5.63 s | 2026-10-07 13:38:44 | batch 99, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

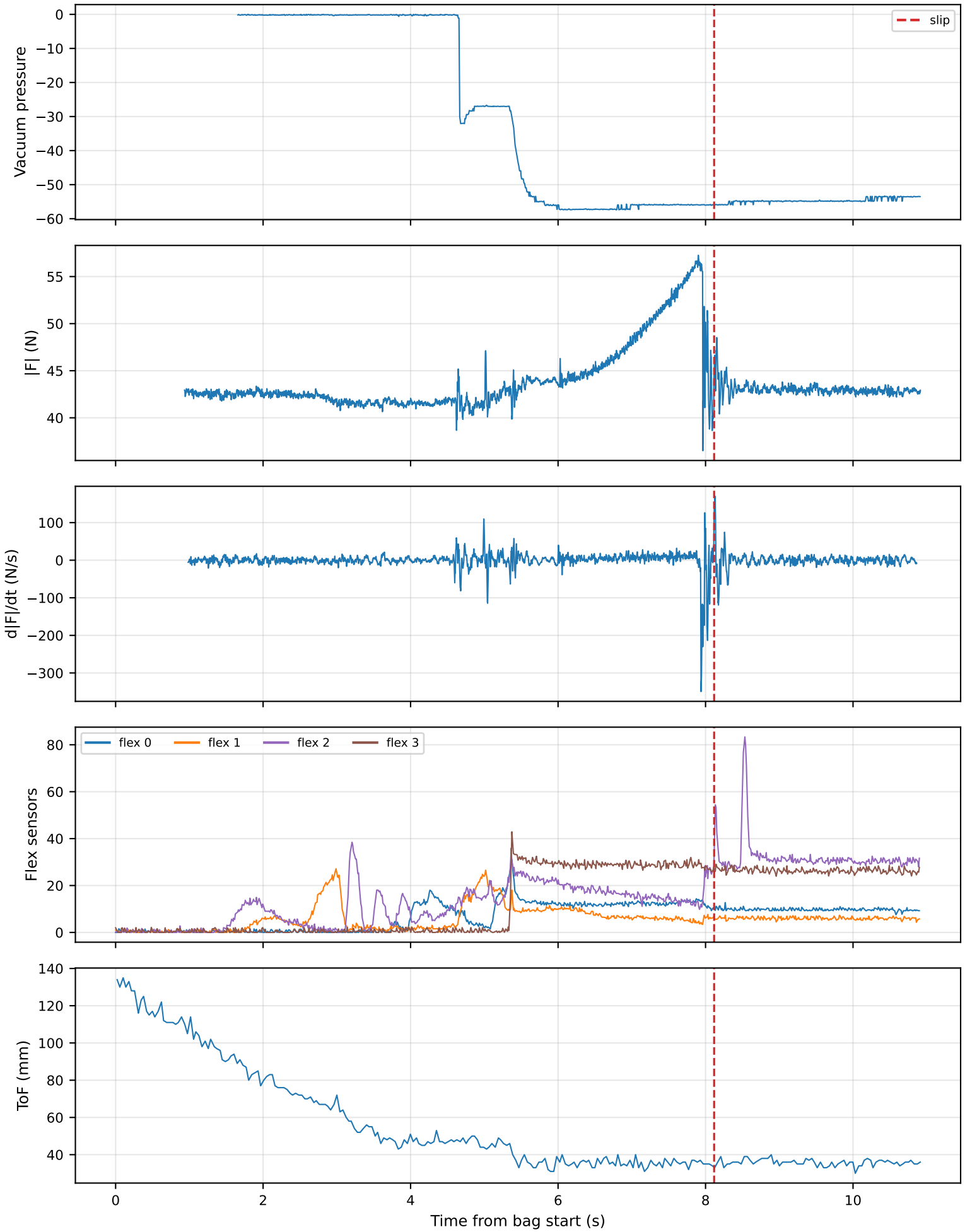

final_approach_and_pick_20261007_134119.db3
PICK at 7.12 s | 2026-10-07 13:41:20 | batch 100, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

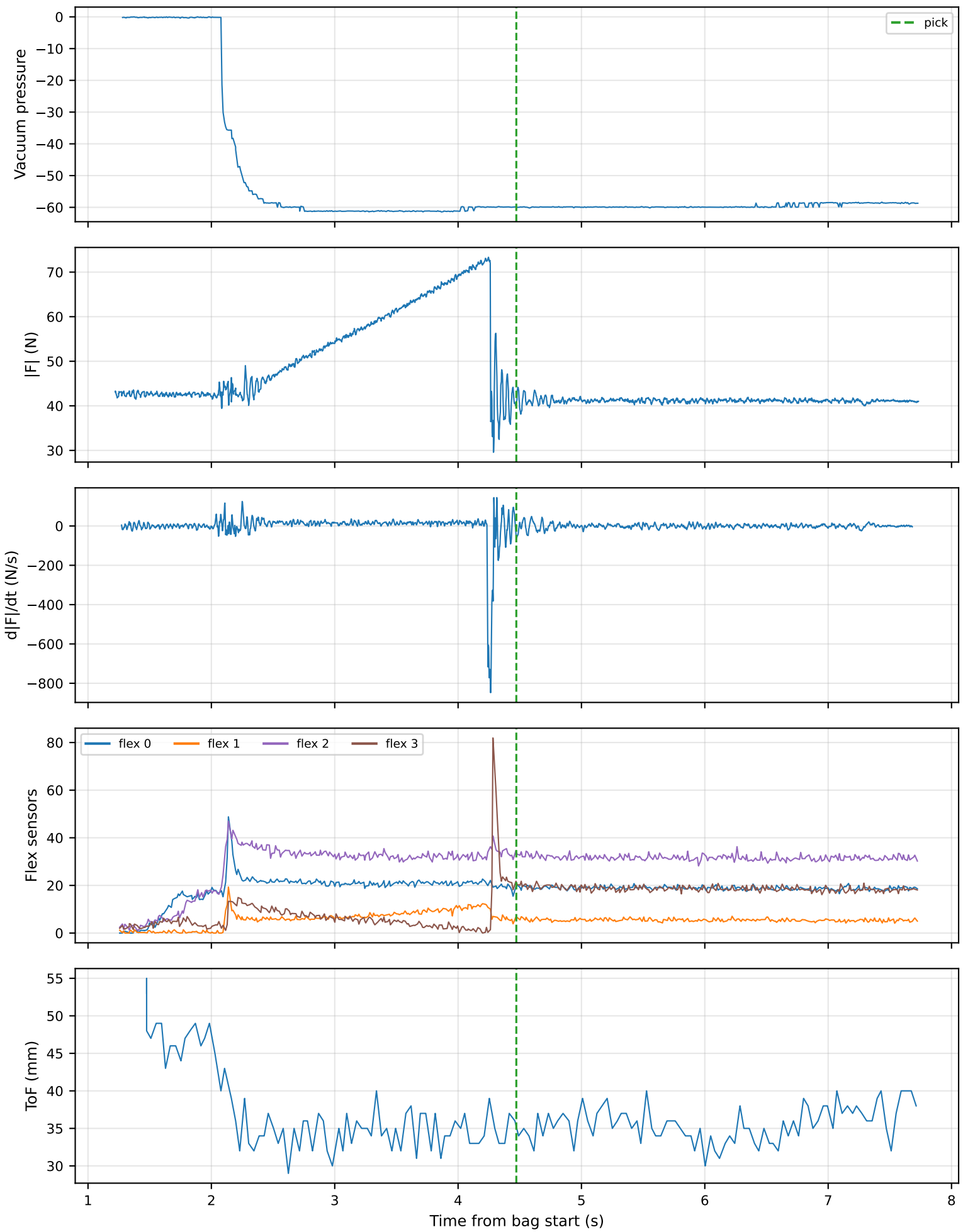

final_approach_and_pick_20261007_134228.db3
SLIP at 5.78 s | 2026-10-07 13:42:29 | batch 101, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

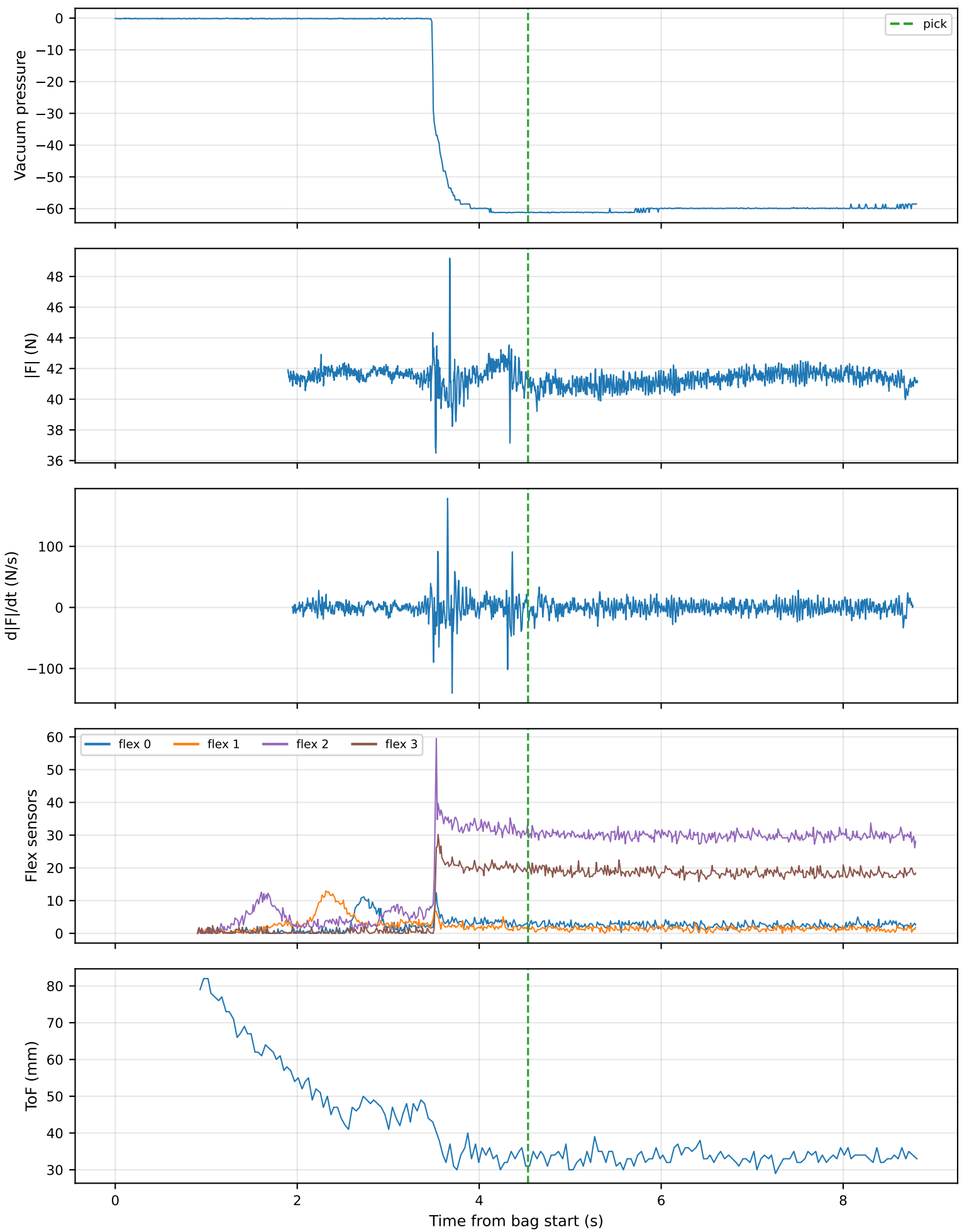

final_approach_and_pick_20261007_134330.db3
SLIP at 11.10 s | 2026-10-07 13:43:31 | batch 102, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

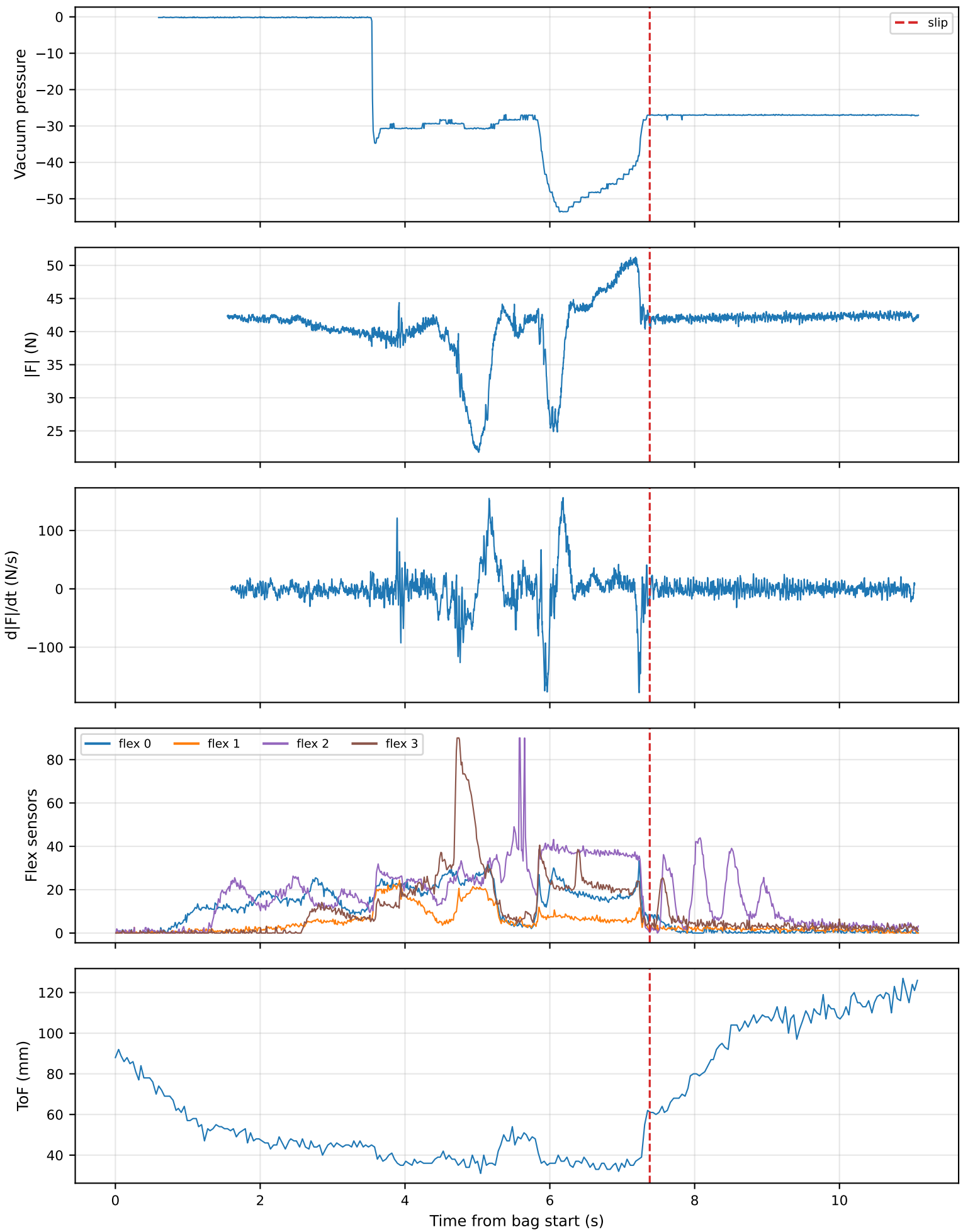
final_approach_and_pick_20261007_135129.db3

SLIP at 8.12 s | 2026-10-07 13:51:29 | batch 104, apple 0 | stiffness_controller



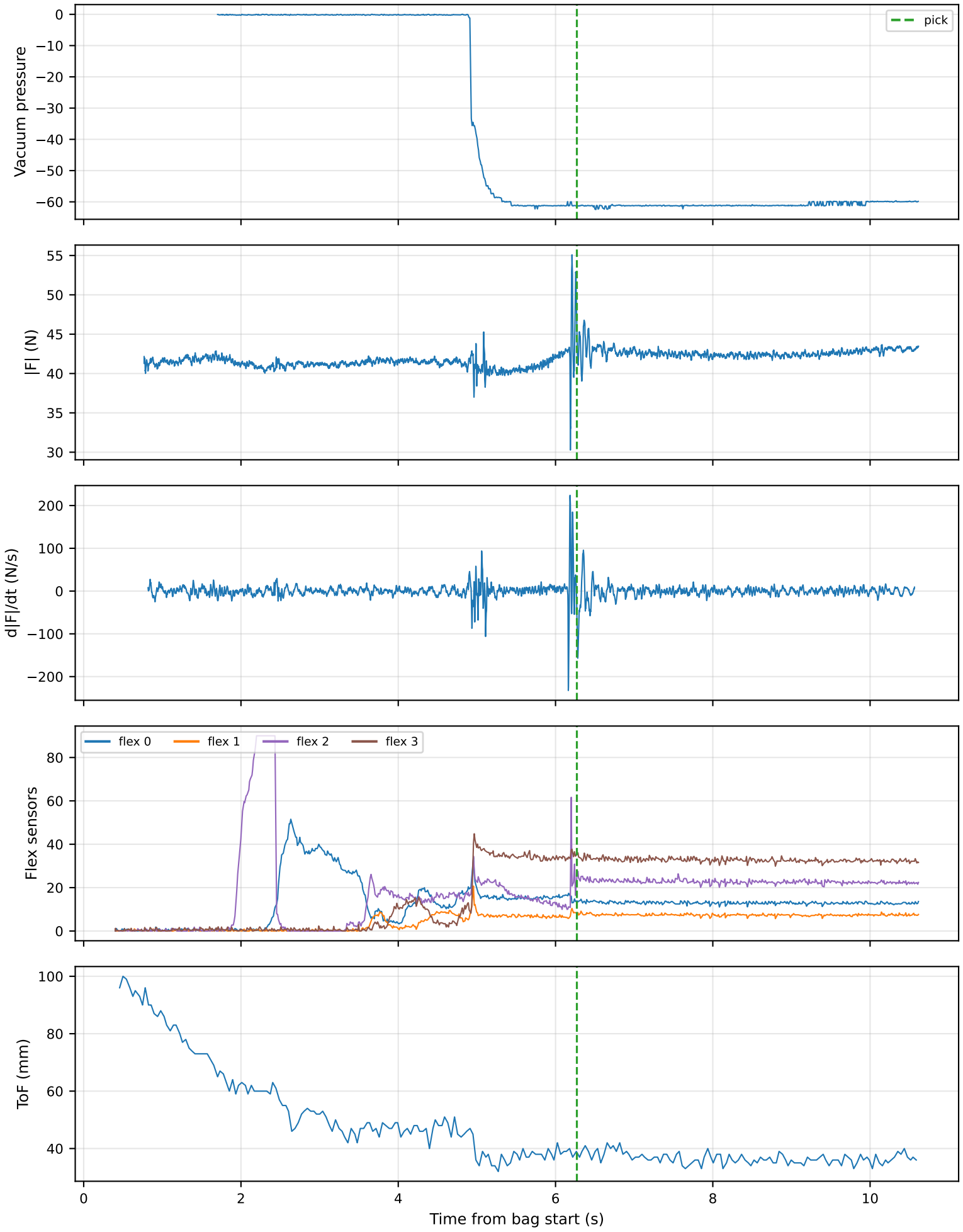

final_approach_and_pick_20261007_135248.db3
PICK at 4.47 s | 2026-10-07 13:52:50 | batch 105, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_135343.db3
PICK at 4.54 s | 2026-10-07 13:53:44 | batch 106, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_135852.db3
SLIP at 7.38 s | 2026-10-07 13:58:53 | batch 107, apple 0 | stiffness_controller
slip
Vacuum pressure
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

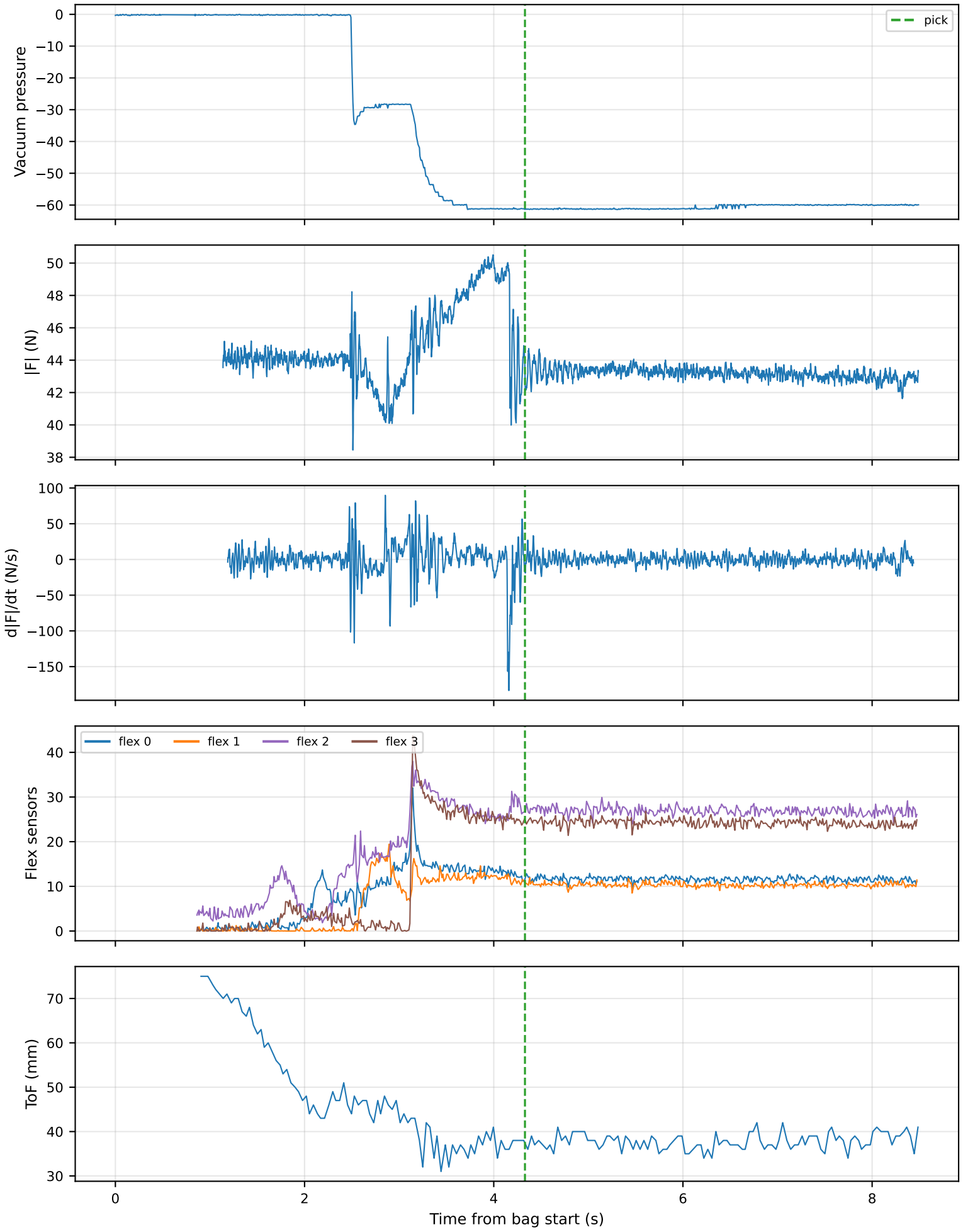
final_approach_and_pick_20261007_135945.db3

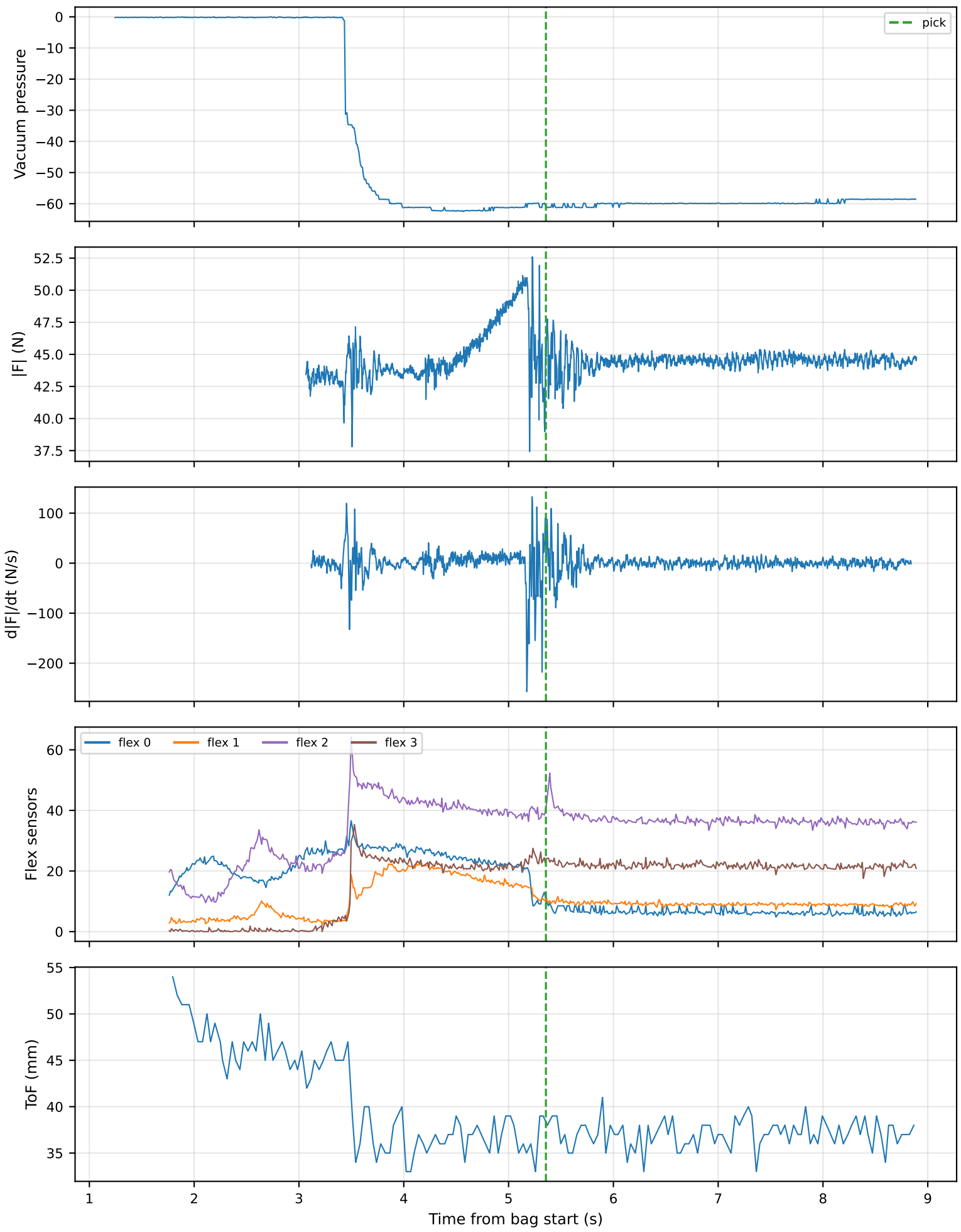
PICK at 6.27 s | 2026-10-07 13:59:46 | batch 108, apple 0 | stiffness_controller

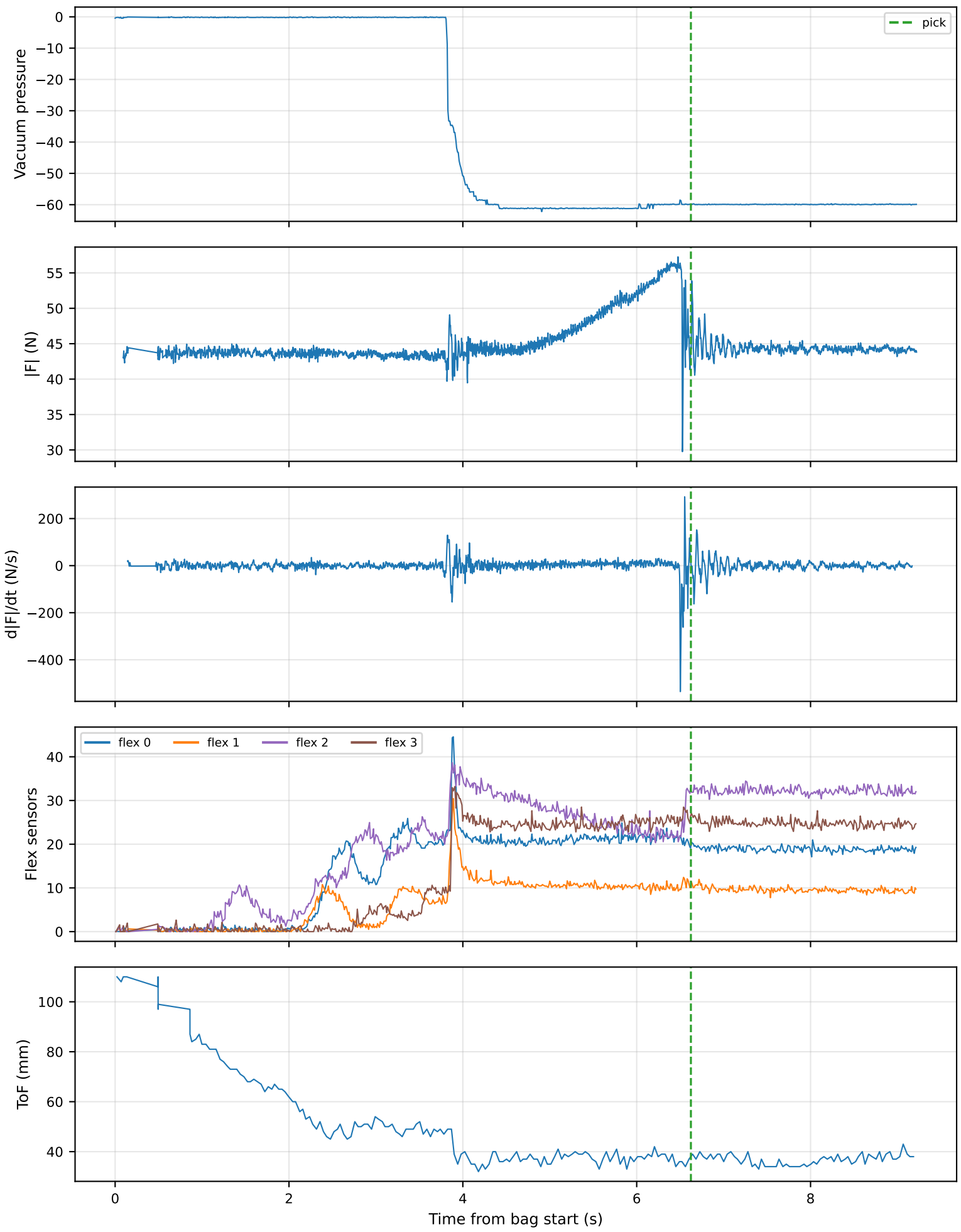


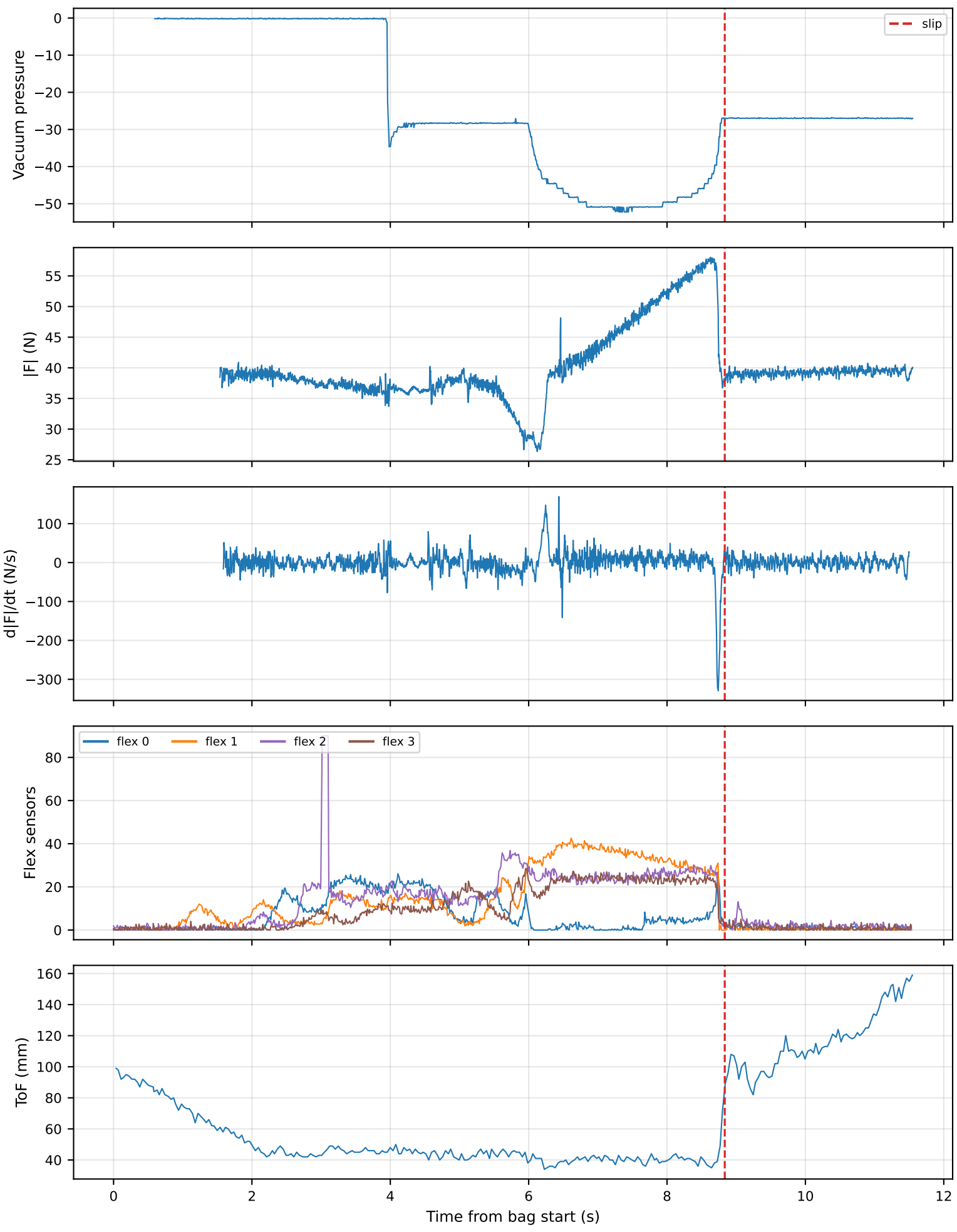
final_approach_and_pick_20261007_140109.db3

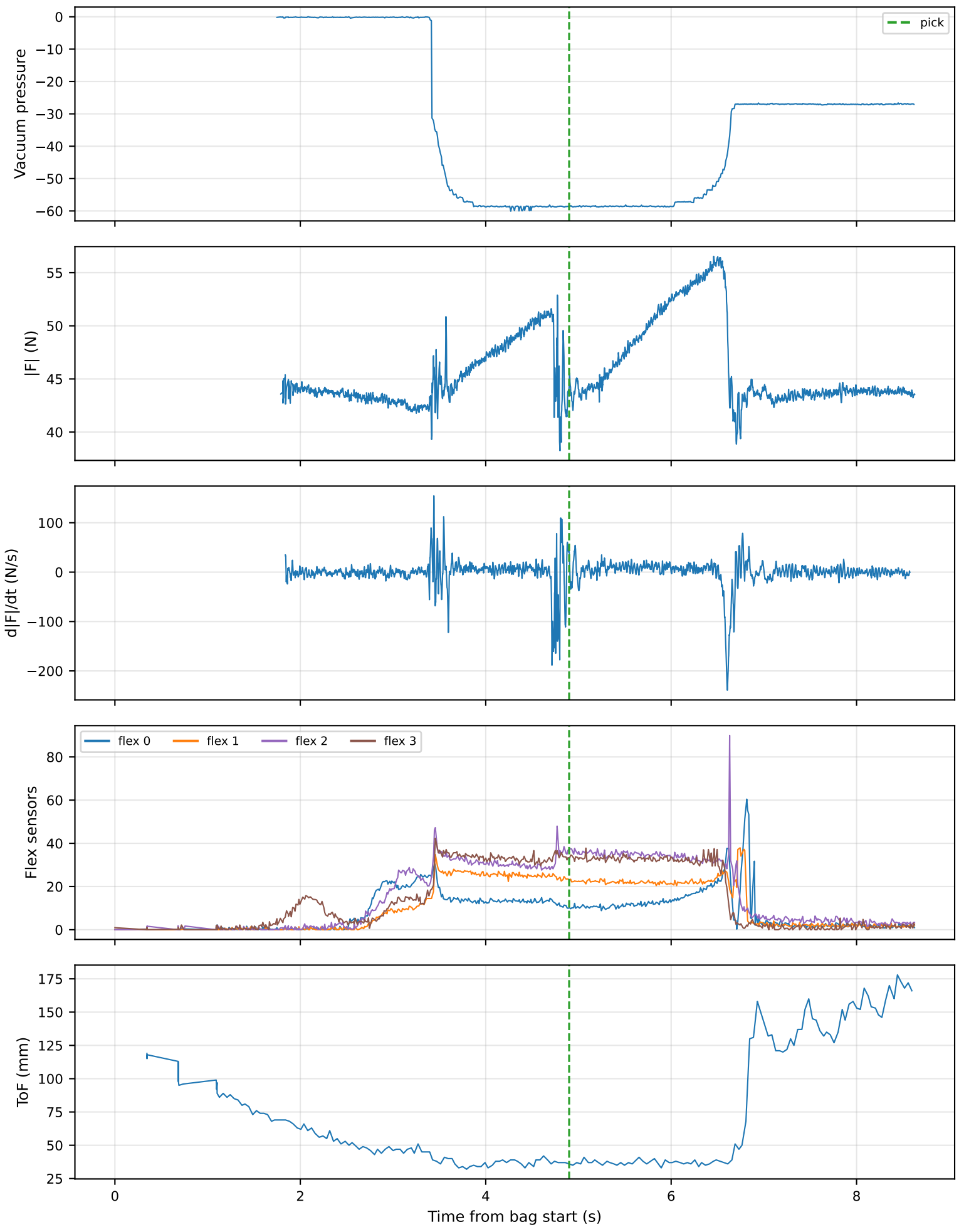
PICK at 4.33 s | 2026-10-07 14:01:10 | batch 109, apple 0 | stiffness_controller

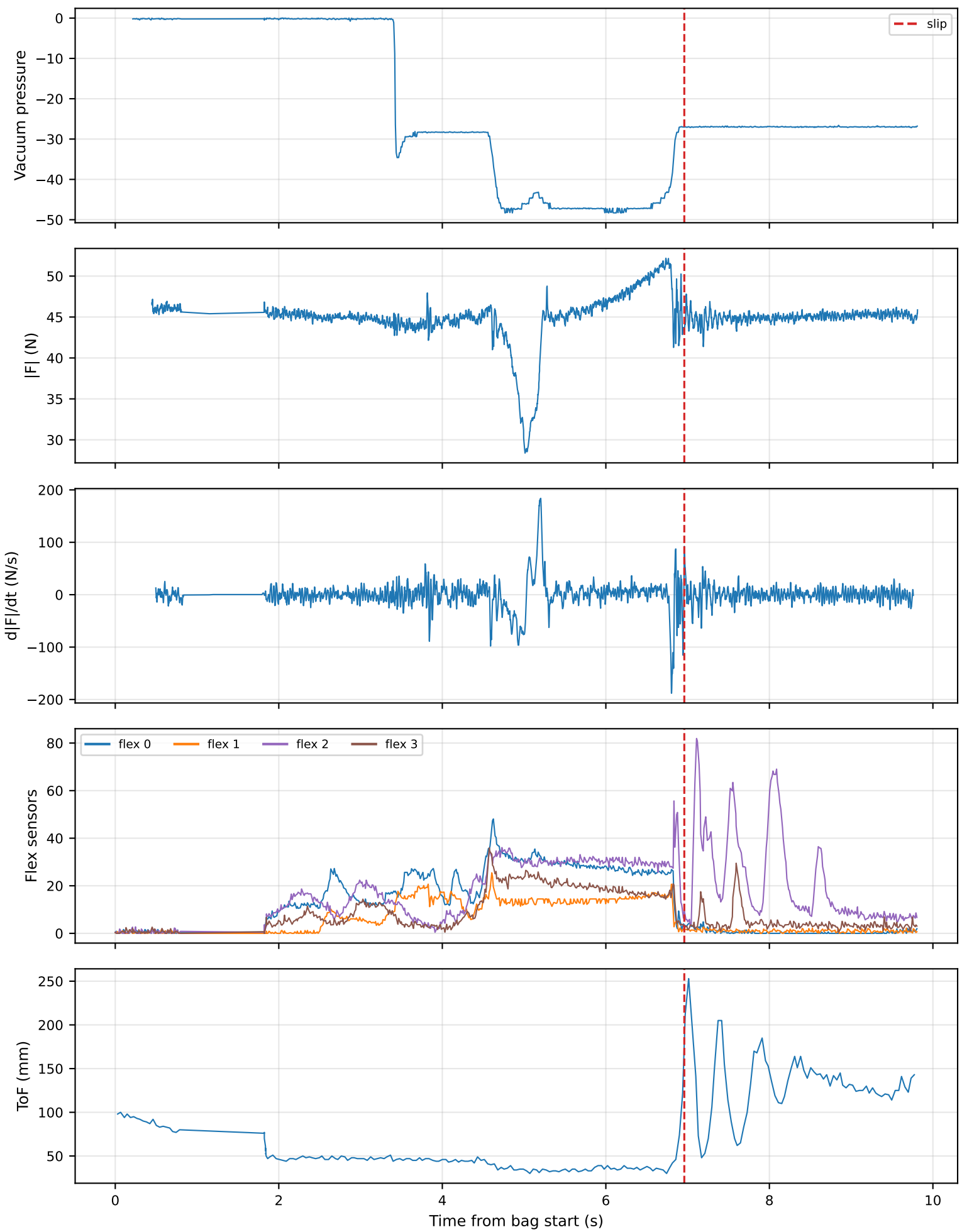


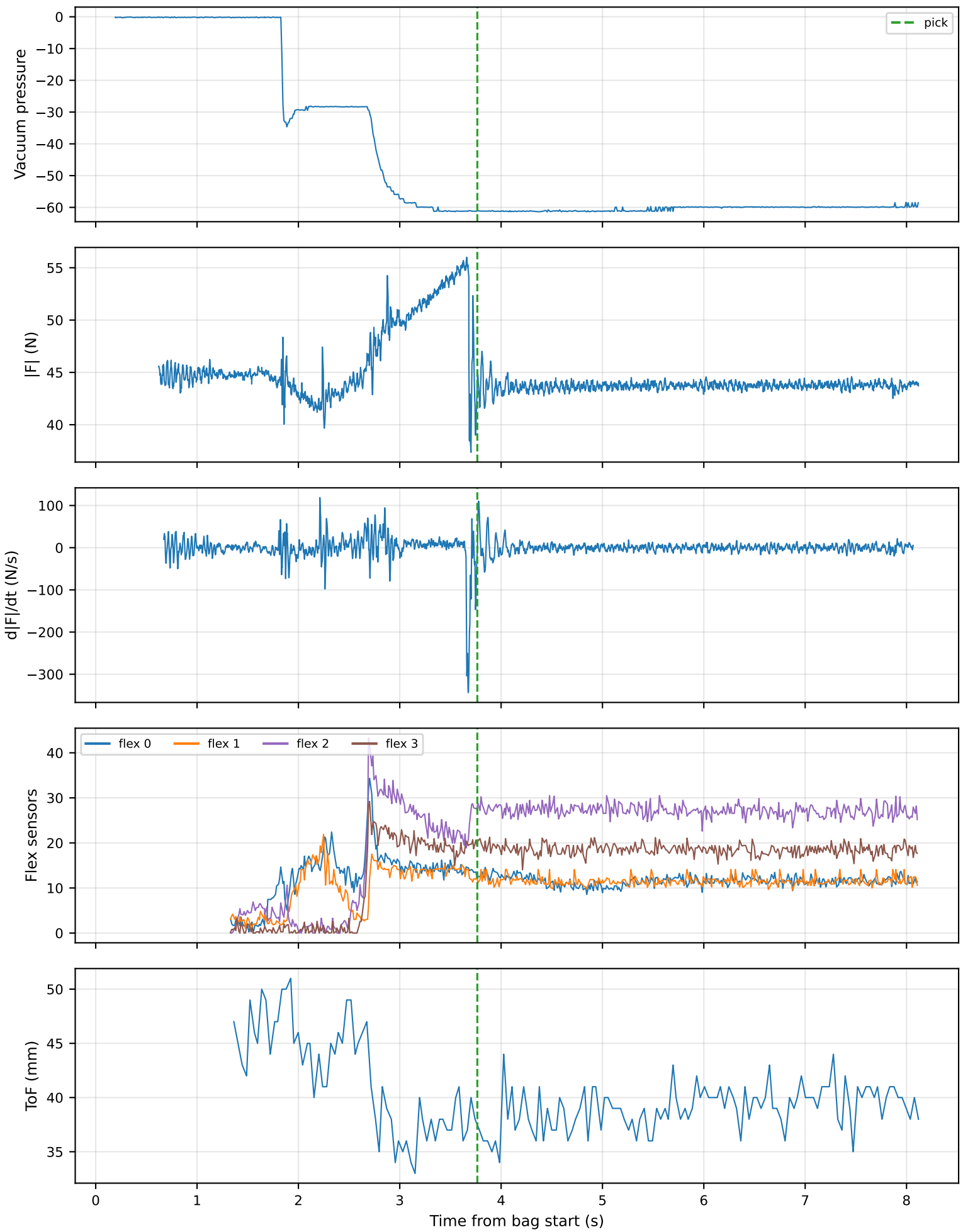

final_approach_and_pick_20261007_140309.db3
PICK at 5.36 s | 2026-10-07 14:03:10 | batch 110, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

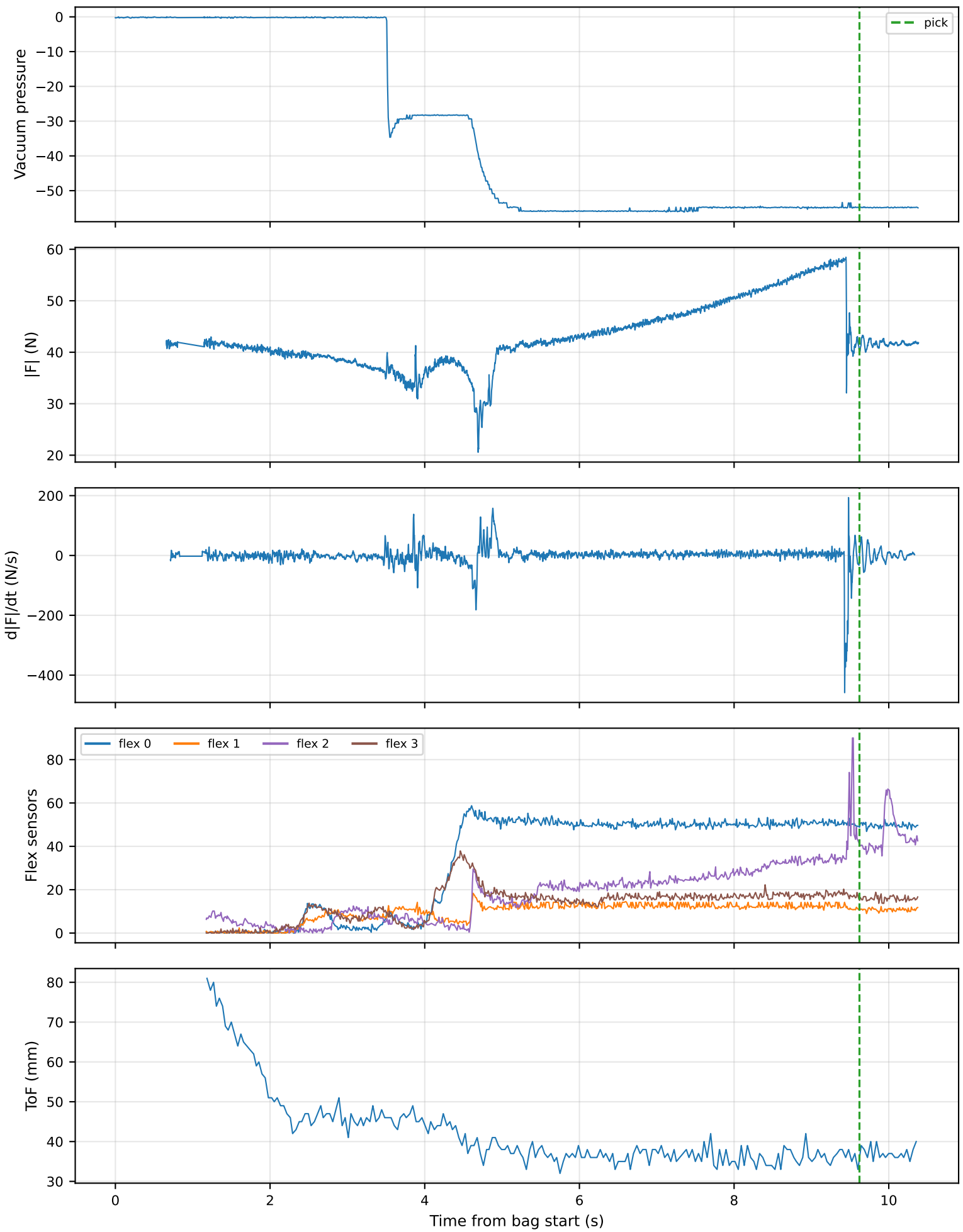

final_approach_and_pick_20261007_140409.db3
PICK at 6.62 s | 2026-10-07 14:04:10 | batch 111, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_140511.db3
SLIP at 8.83 s | 2026-10-07 14:05:12 | batch 112, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_140604.db3
PICK at 4.90 s | 2026-10-07 14:06:05 | batch 71, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

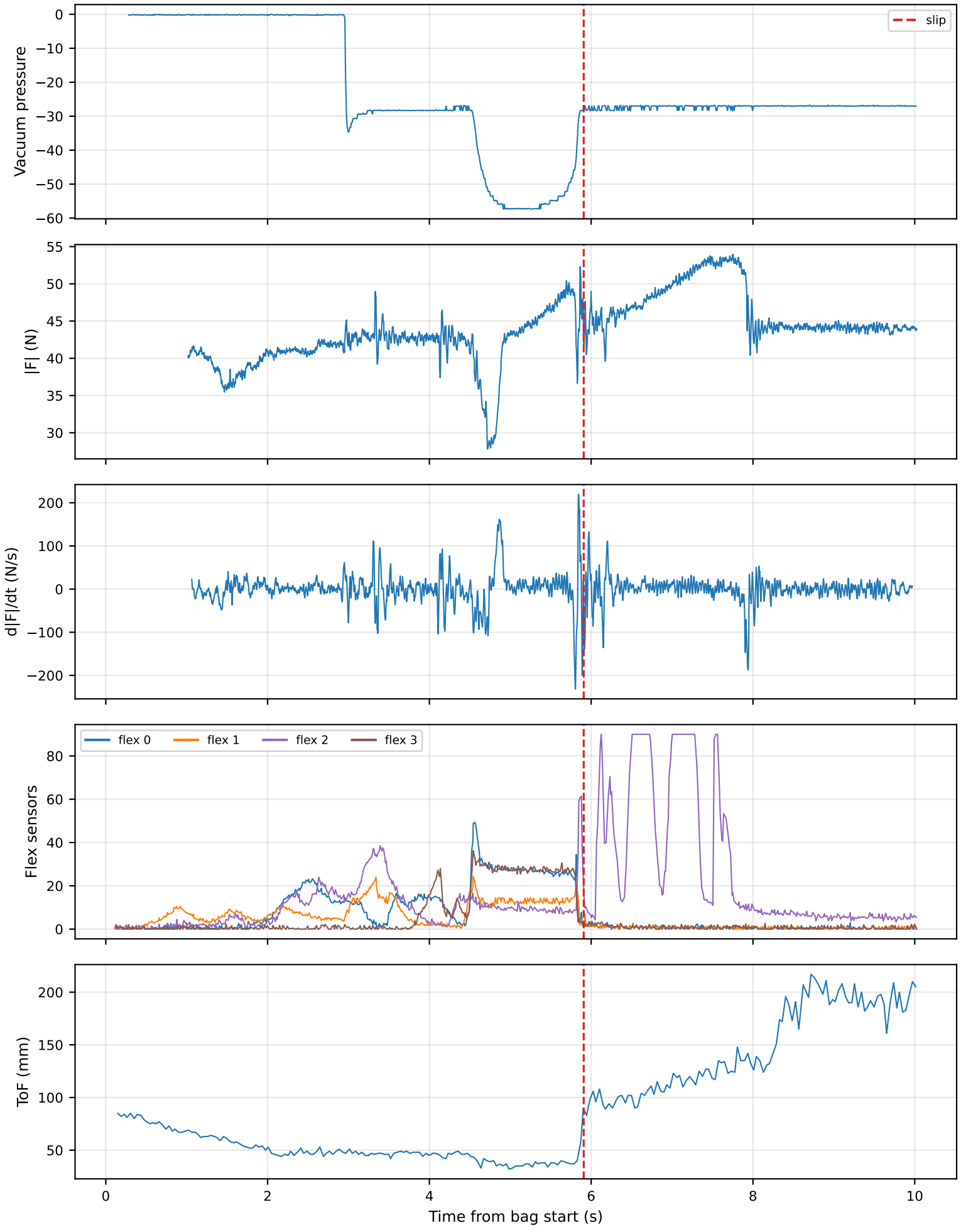

final_approach_and_pick_20261007_140756.db3
SLIP at 6.96 s | 2026-10-07 14:07:57 | batch 113, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

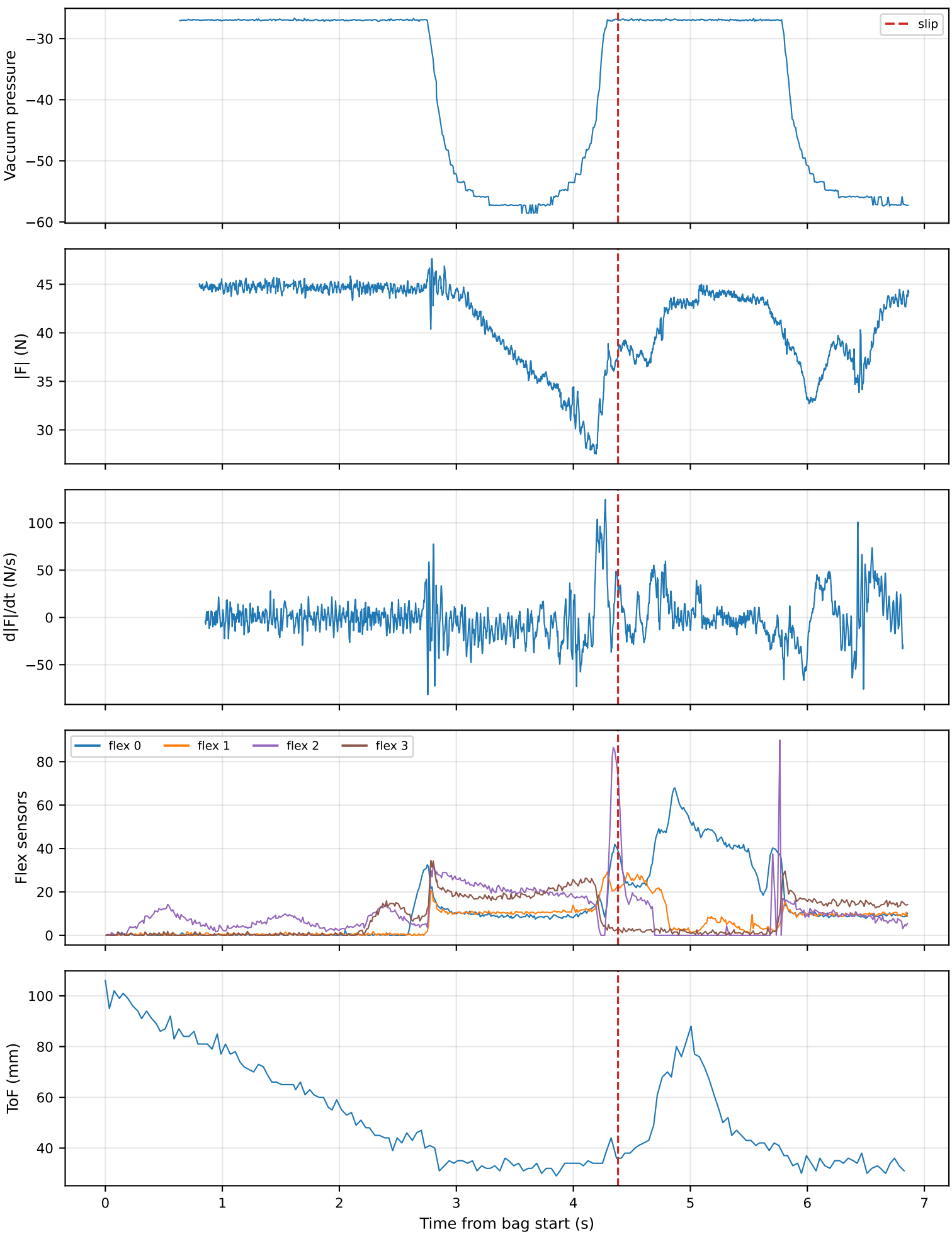

final_approach_and_pick_20261007_140947.db3
PICK at 3.77 s | 2026-10-07 14:09:48 | batch 114, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

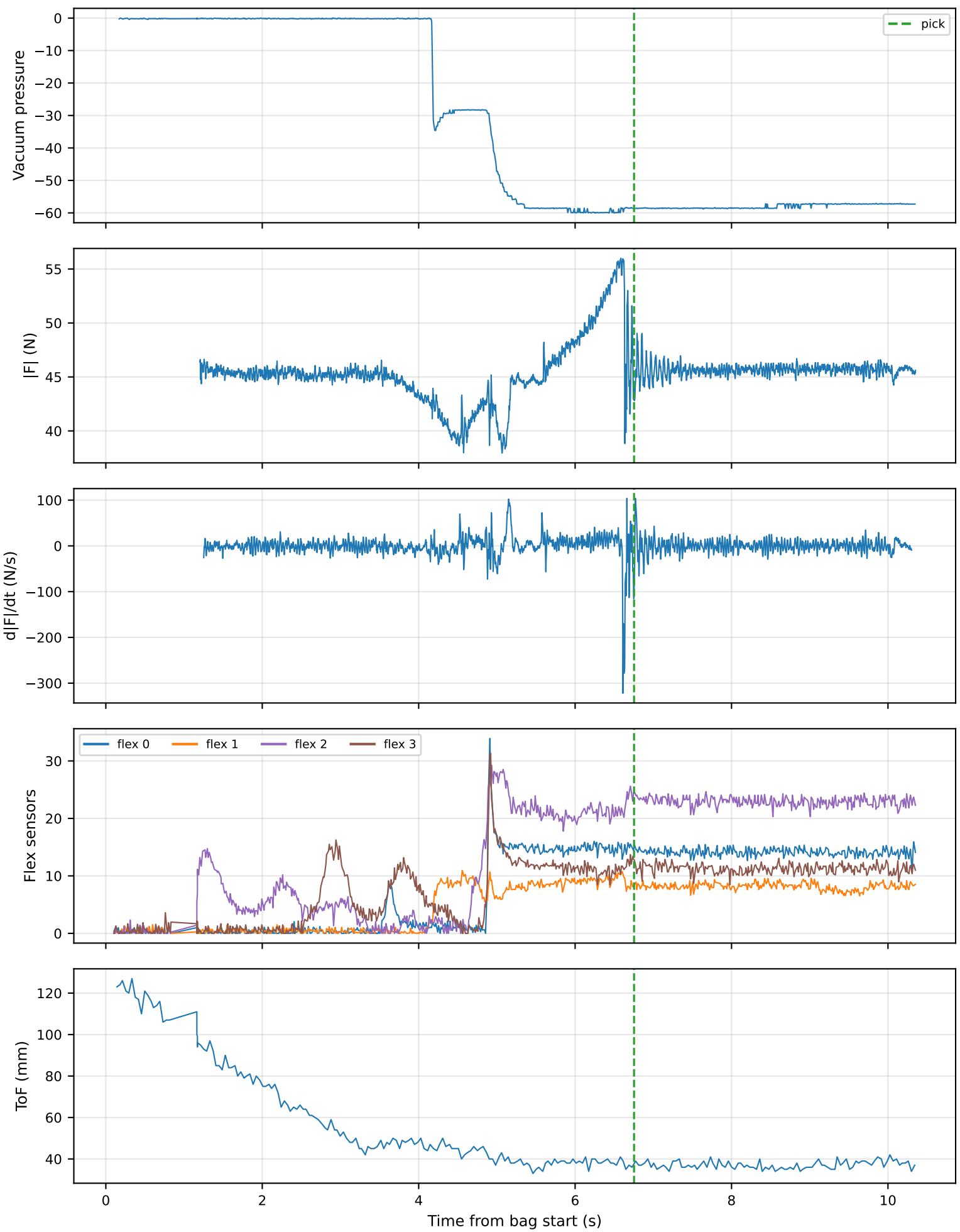

final_approach_and_pick_20261007_141046.db3
PICK at 9.62 s | 2026-10-07 14:10:47 | batch 115, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

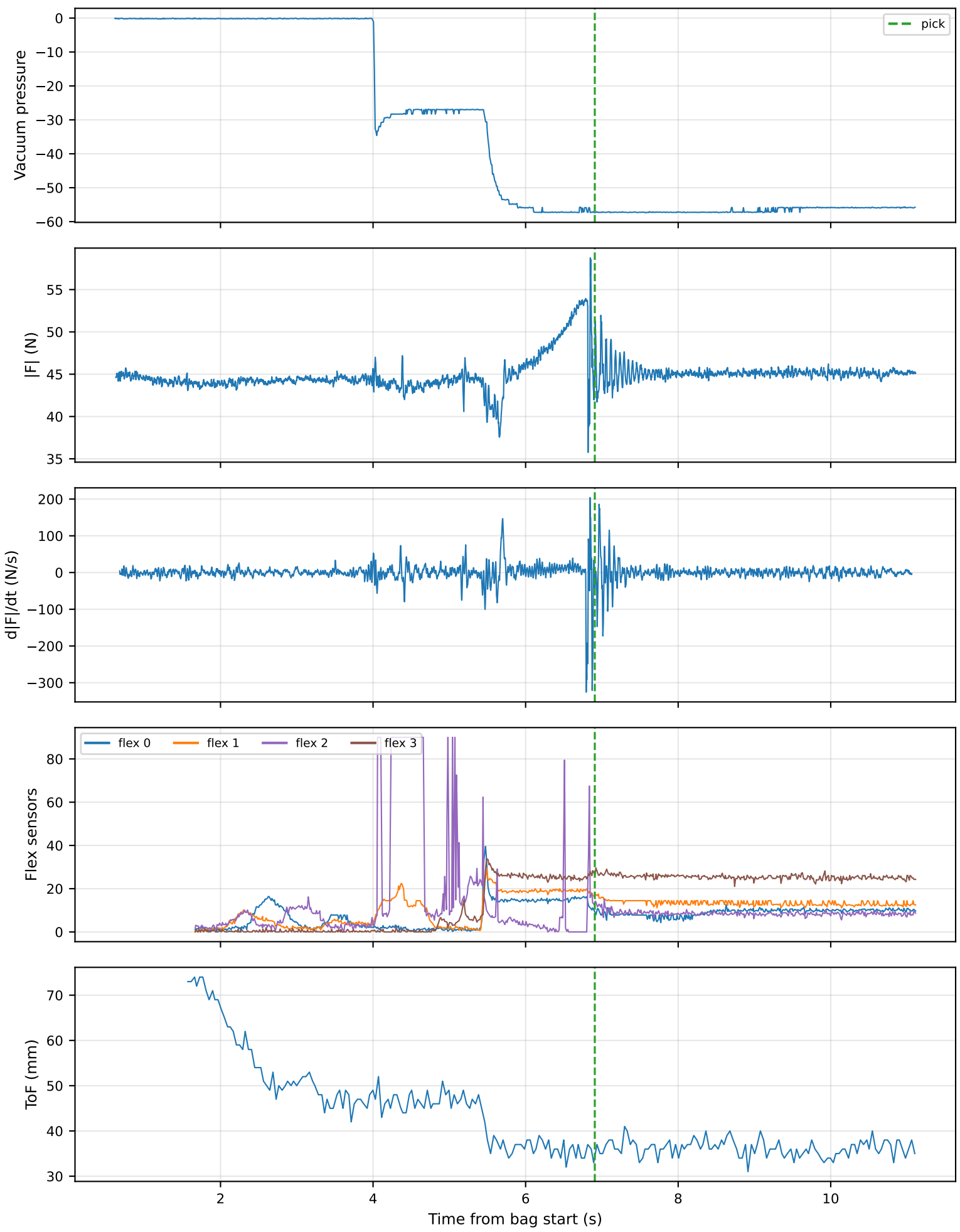
final_approach_and_pick_20261007_141137.db3

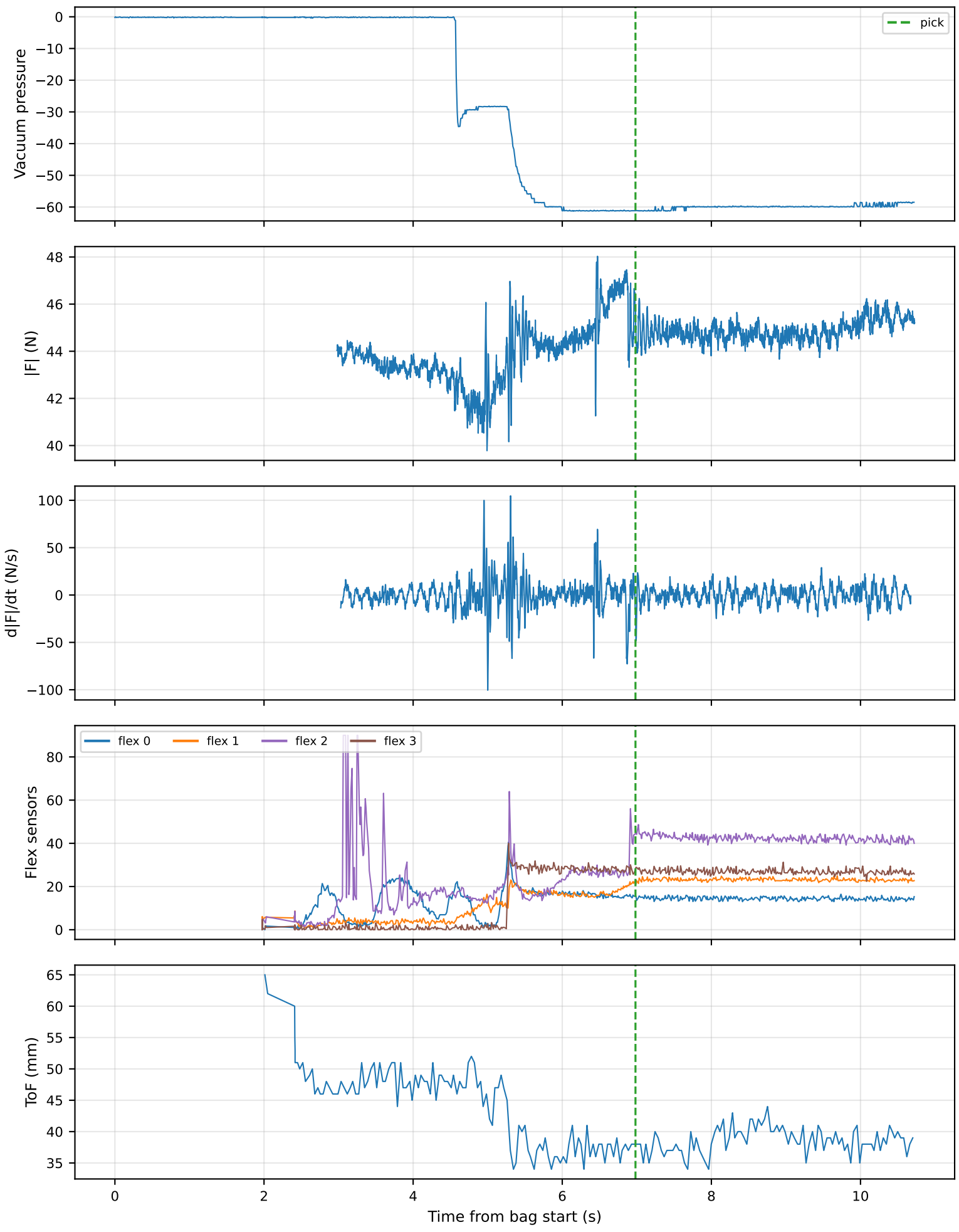
SLIP at 5.91 s | 2026-10-07 14:11:39 | batch 116, apple 0 | stiffness_controller

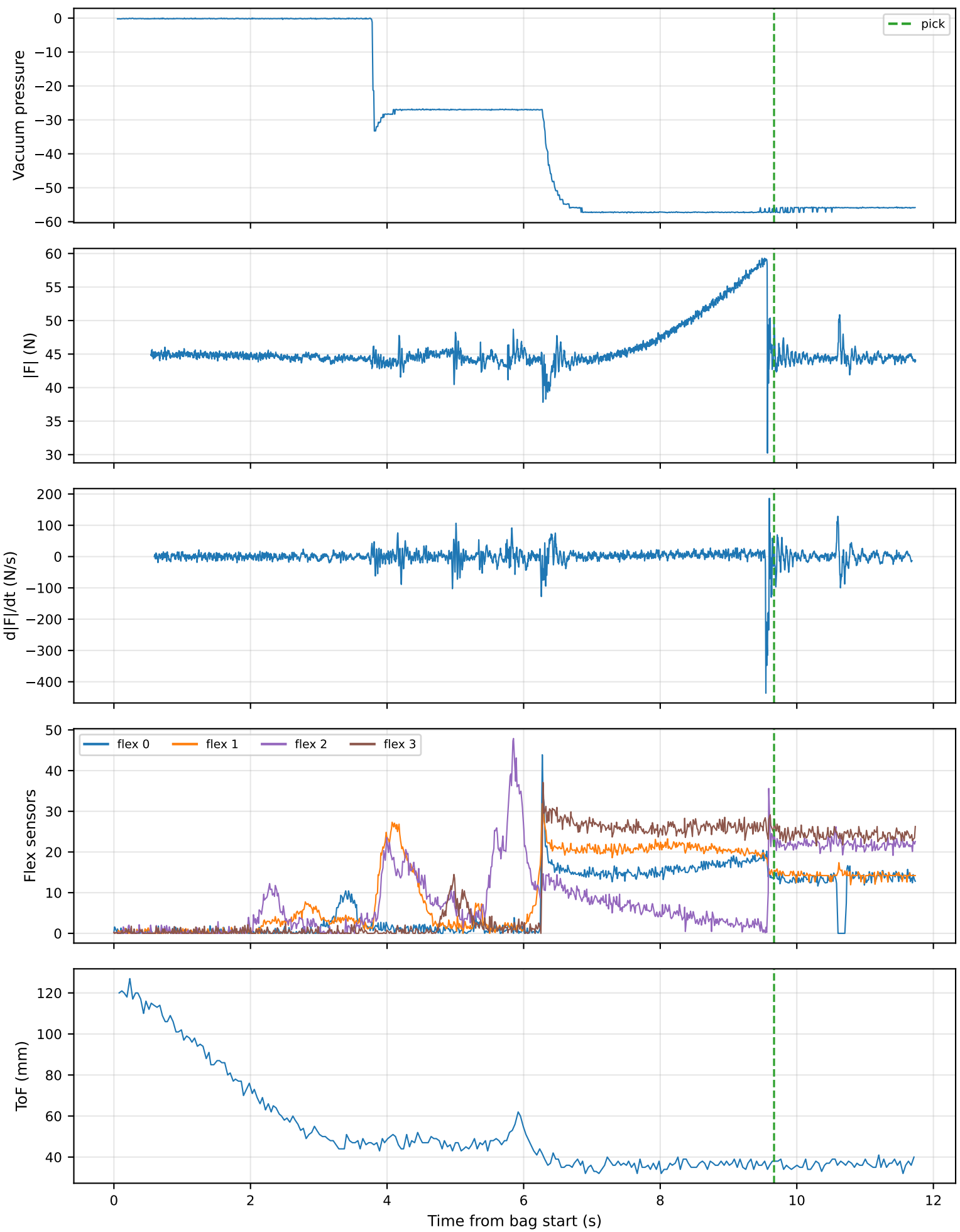


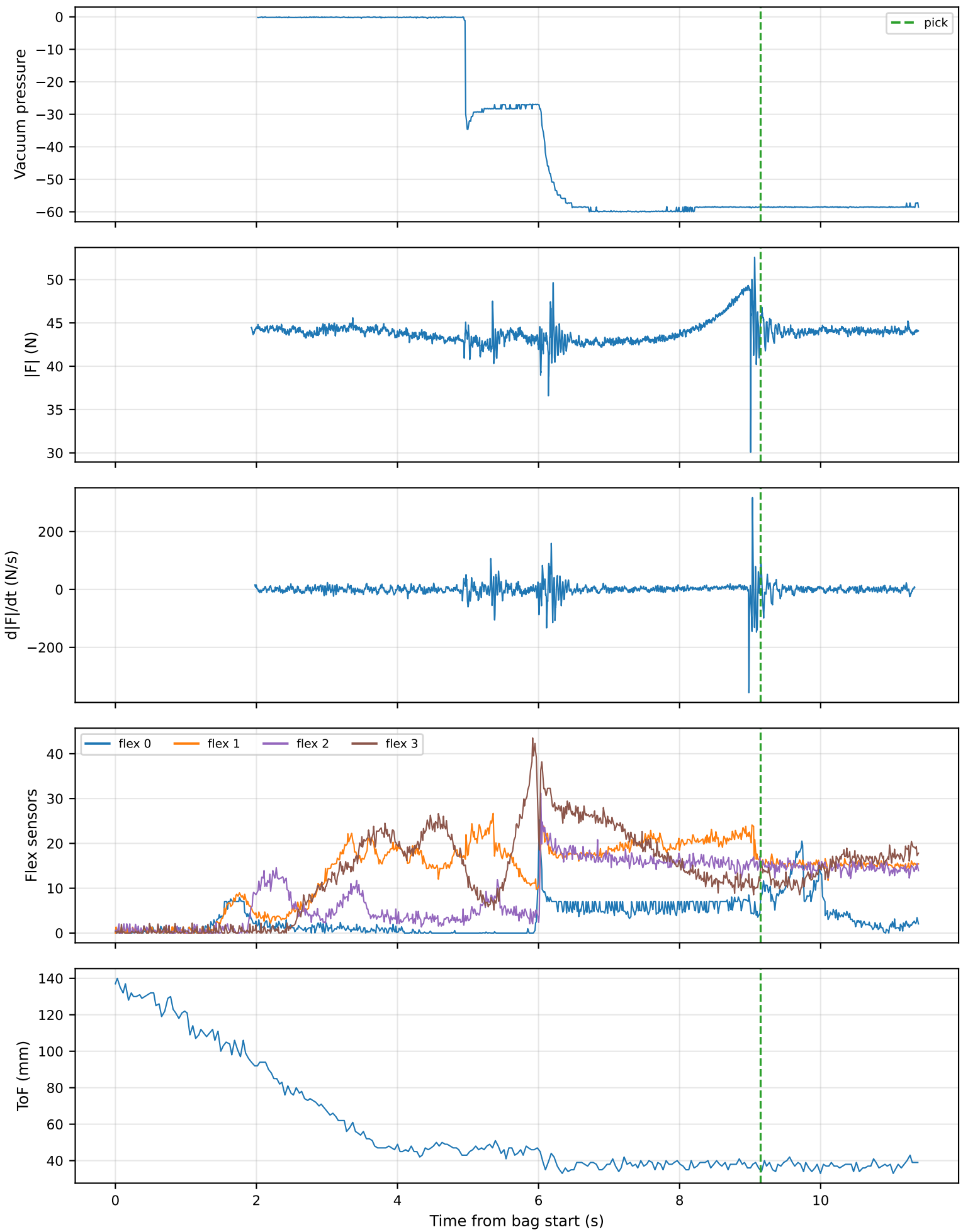

final_approach_and_pick_20261007_141358.db3
SLIP at 4.38 s | 2026-10-07 14:13:59 | batch 74, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

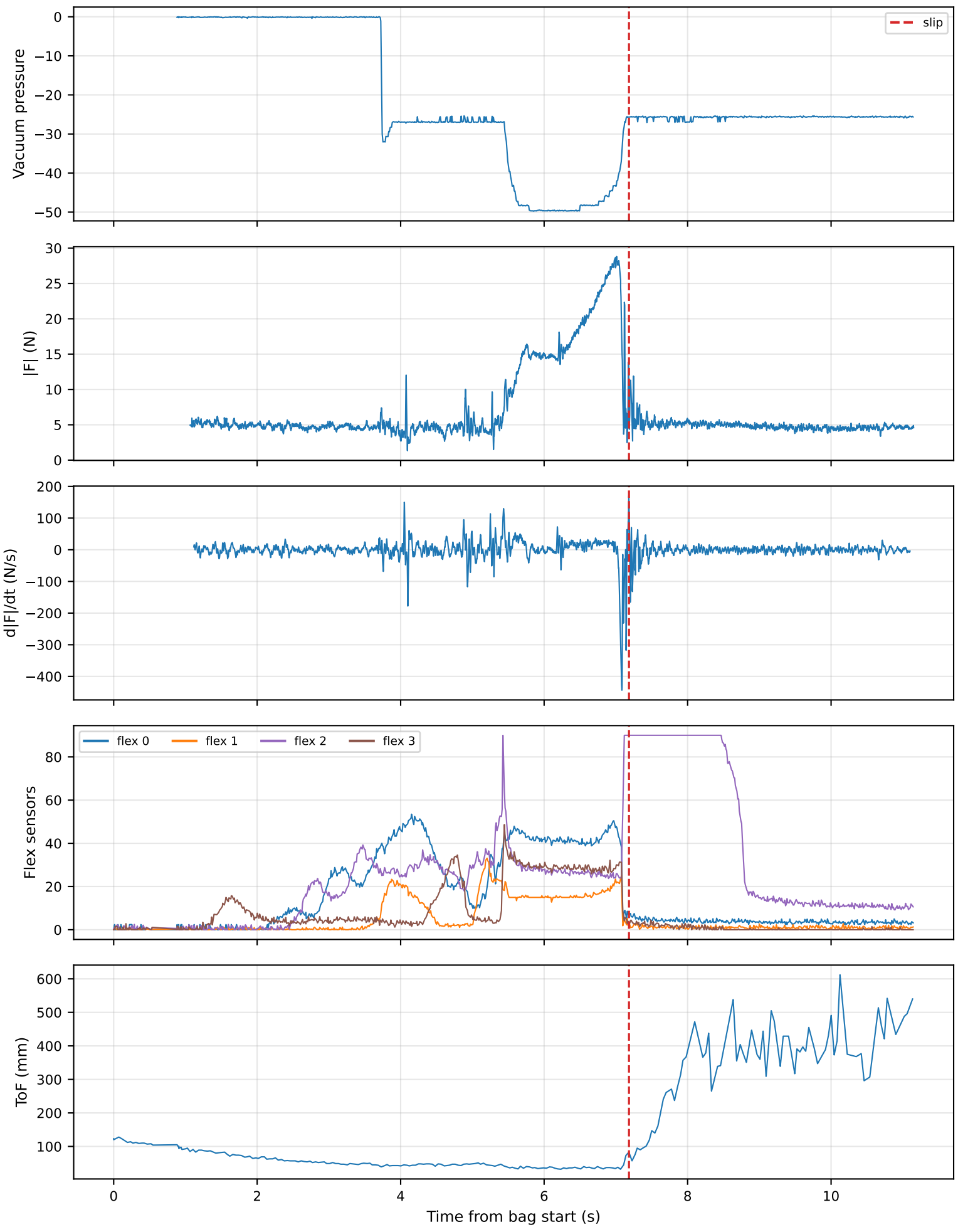

final_approach_and_pick_20261007_141520.db3
PICK at 6.75 s | 2026-10-07 14:15:21 | batch 117, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

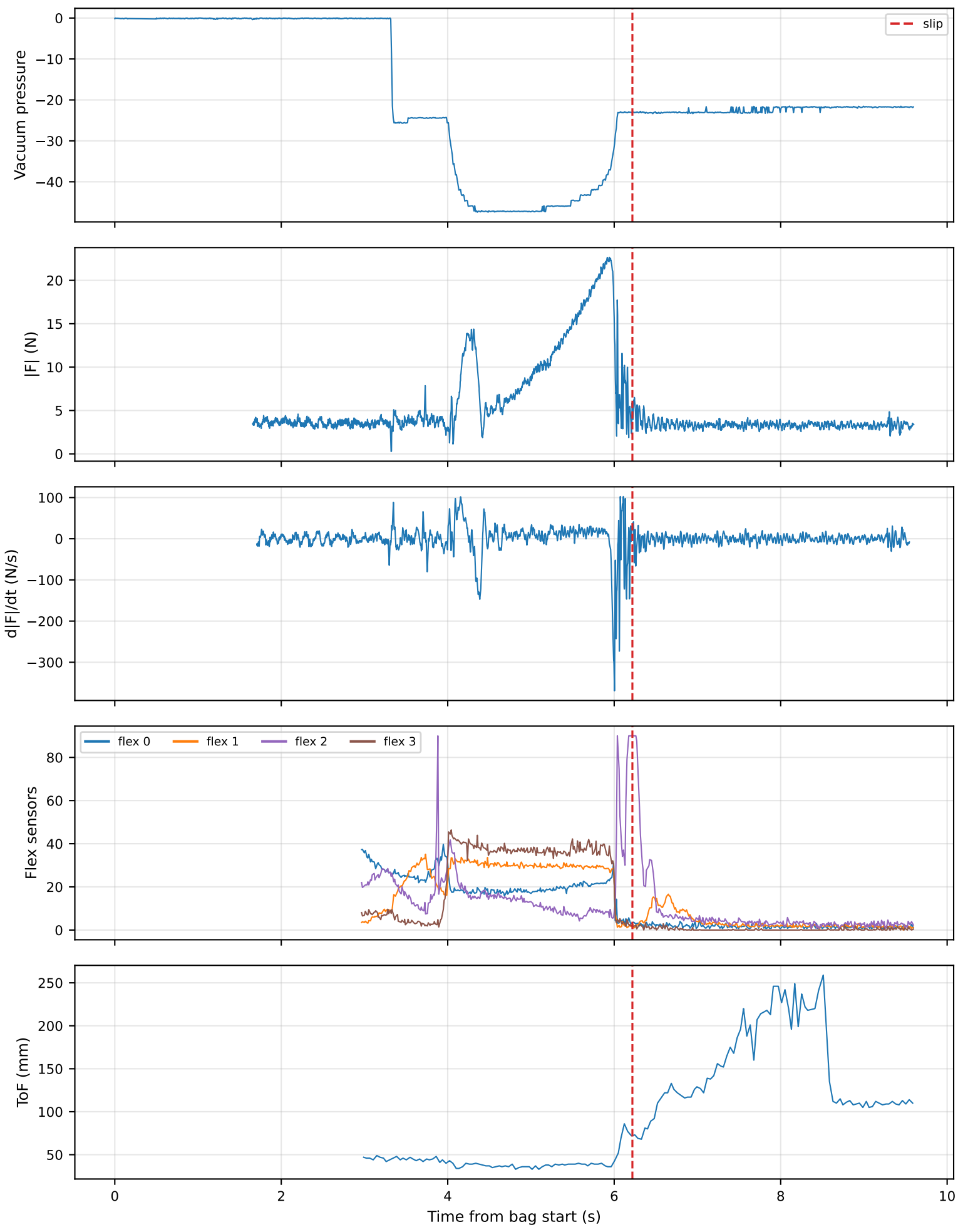

final_approach_and_pick_20261007_141632.db3
PICK at 6.91 s | 2026-10-07 14:16:34 | batch 118, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

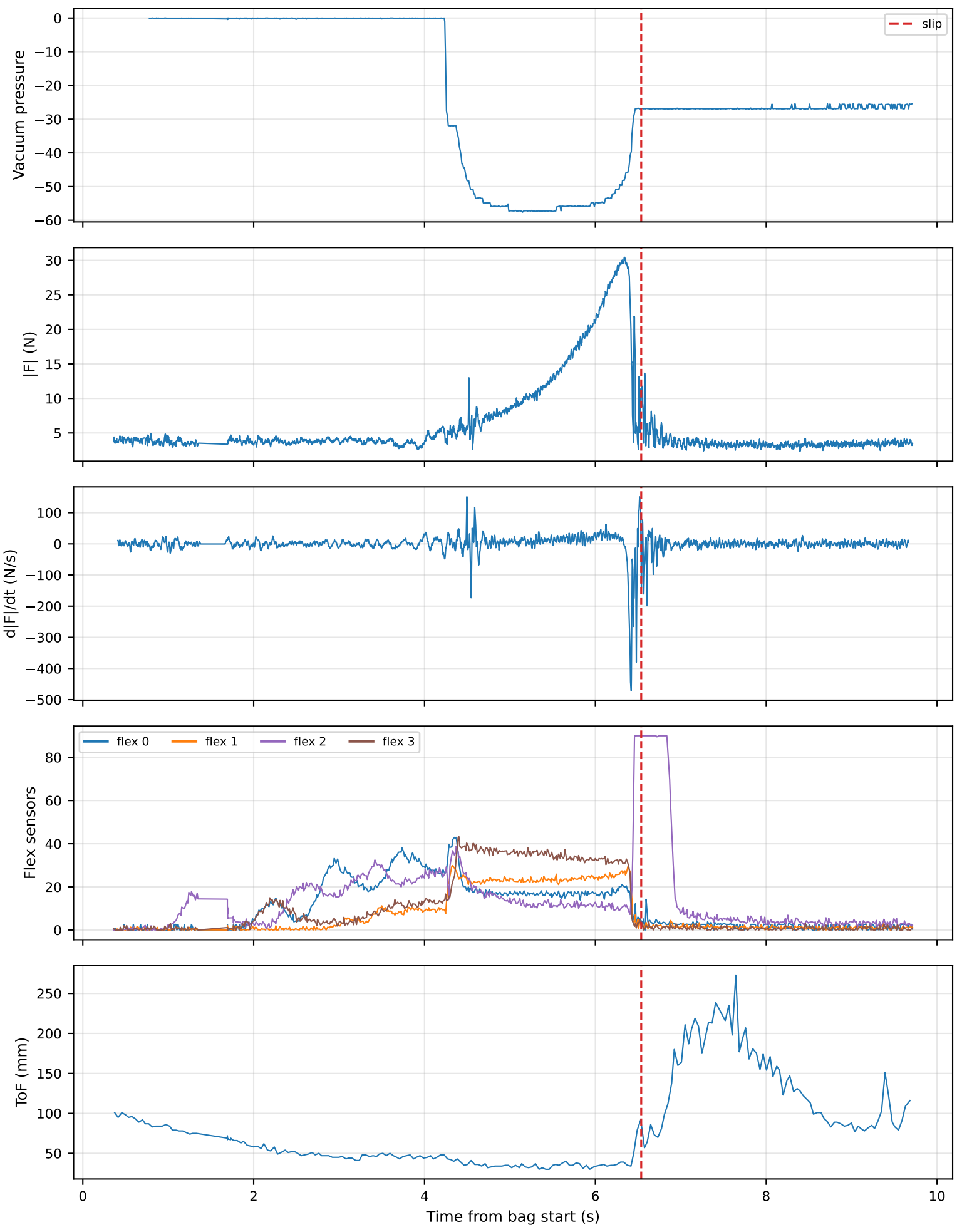

final_approach_and_pick_20261007_141759.db3
PICK at 6.98 s | 2026-10-07 14:18:00 | batch 119, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_141851.db3
PICK at 9.67 s | 2026-10-07 14:18:52 | batch 120, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_141945.db3
PICK at 9.15 s | 2026-10-07 14:19:45 | batch 121, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

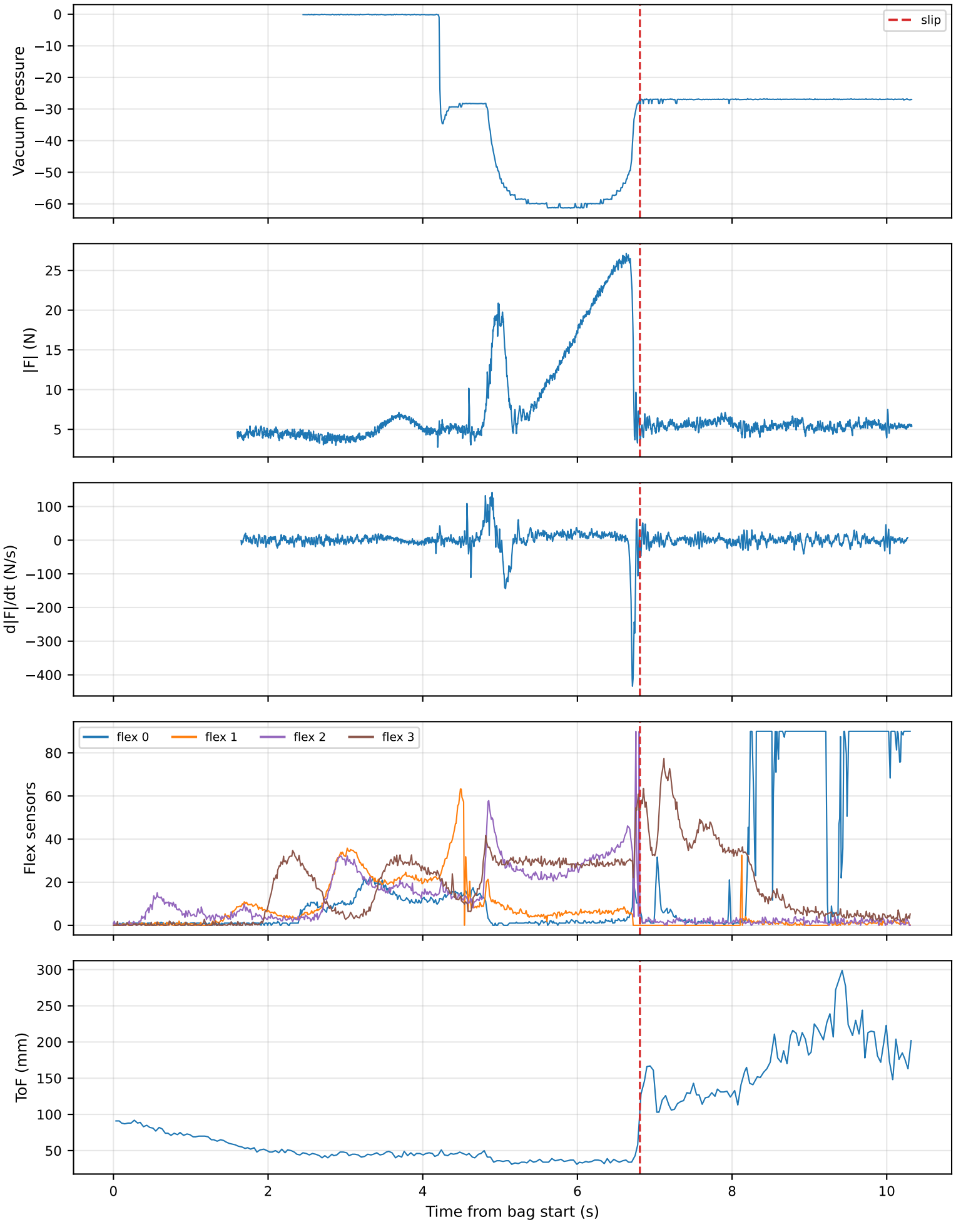

final_approach_and_pick_20261007_152258.db3
SLIP at 7.18 s | 2026-10-07 15:22:59 | batch 122, apple 0 | stiffness_controller
slip
Vacuum pressure
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

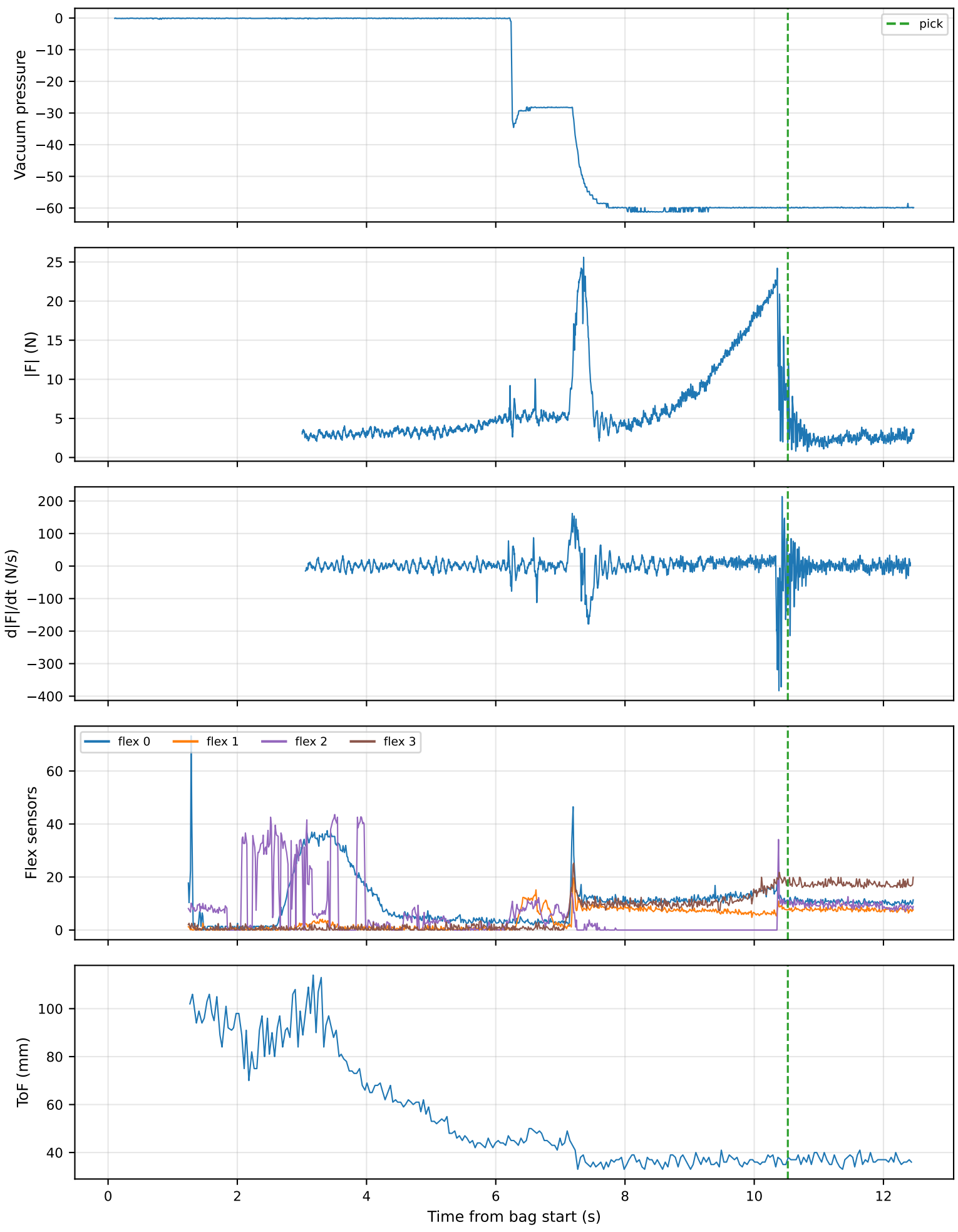

final_approach_and_pick_20261007_152501.db3
SLIP at 6.22 s | 2026-10-07 15:25:02 | batch 123, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

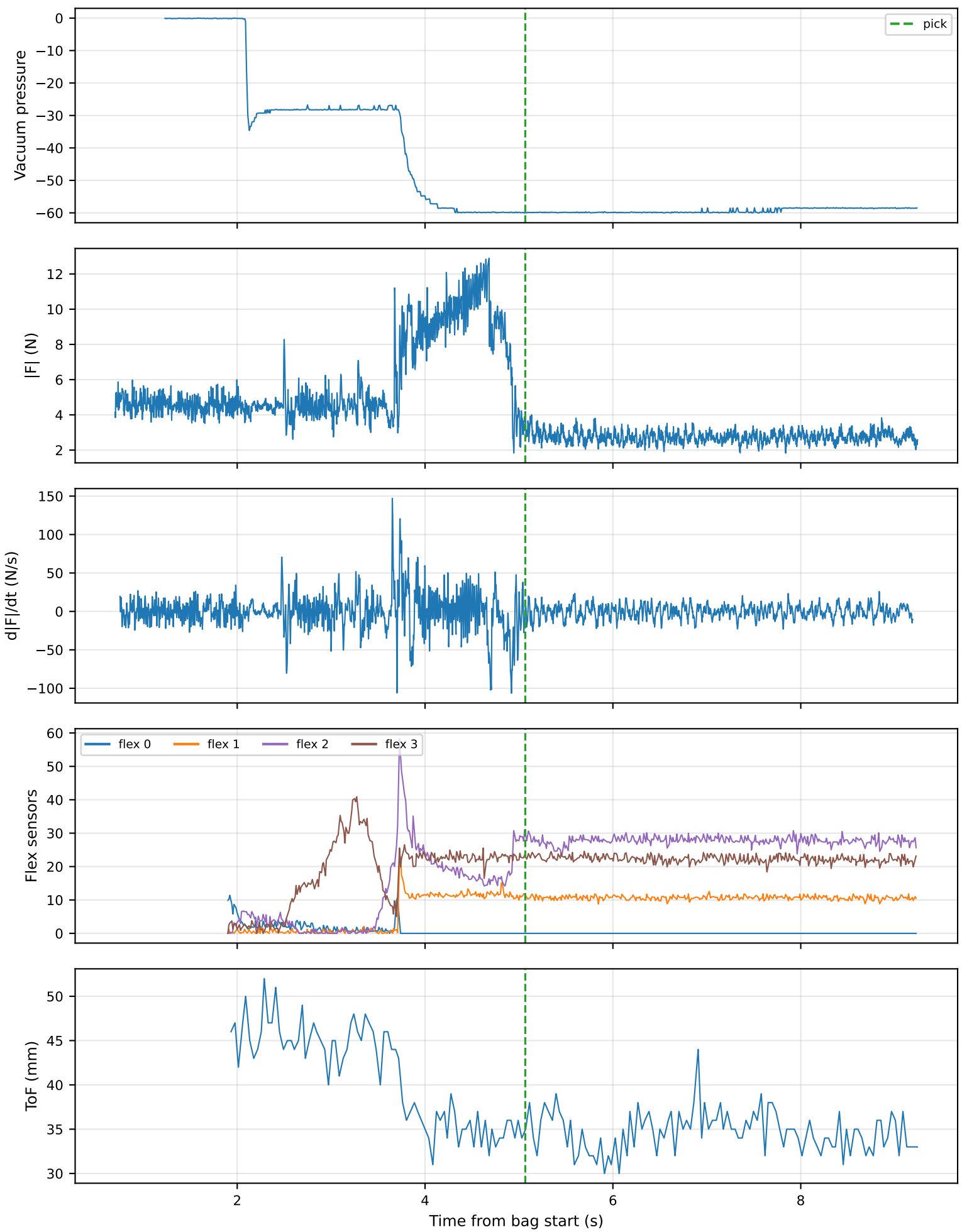

final_approach_and_pick_20261007_153034.db3
SLIP at 6.54 s | 2026-10-07 15:30:35 | batch 126, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

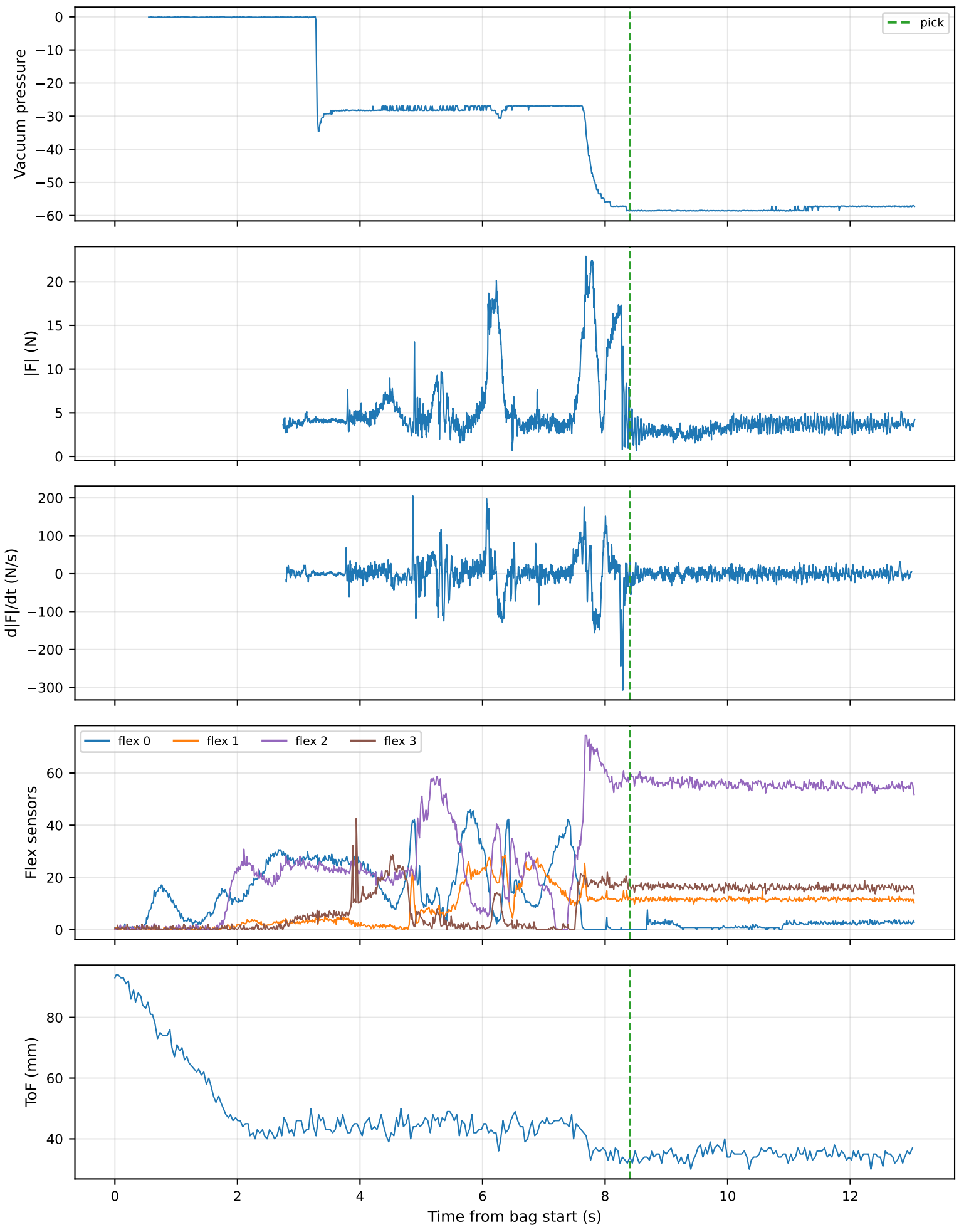
final_approach_and_pick_20261007_153258.db3

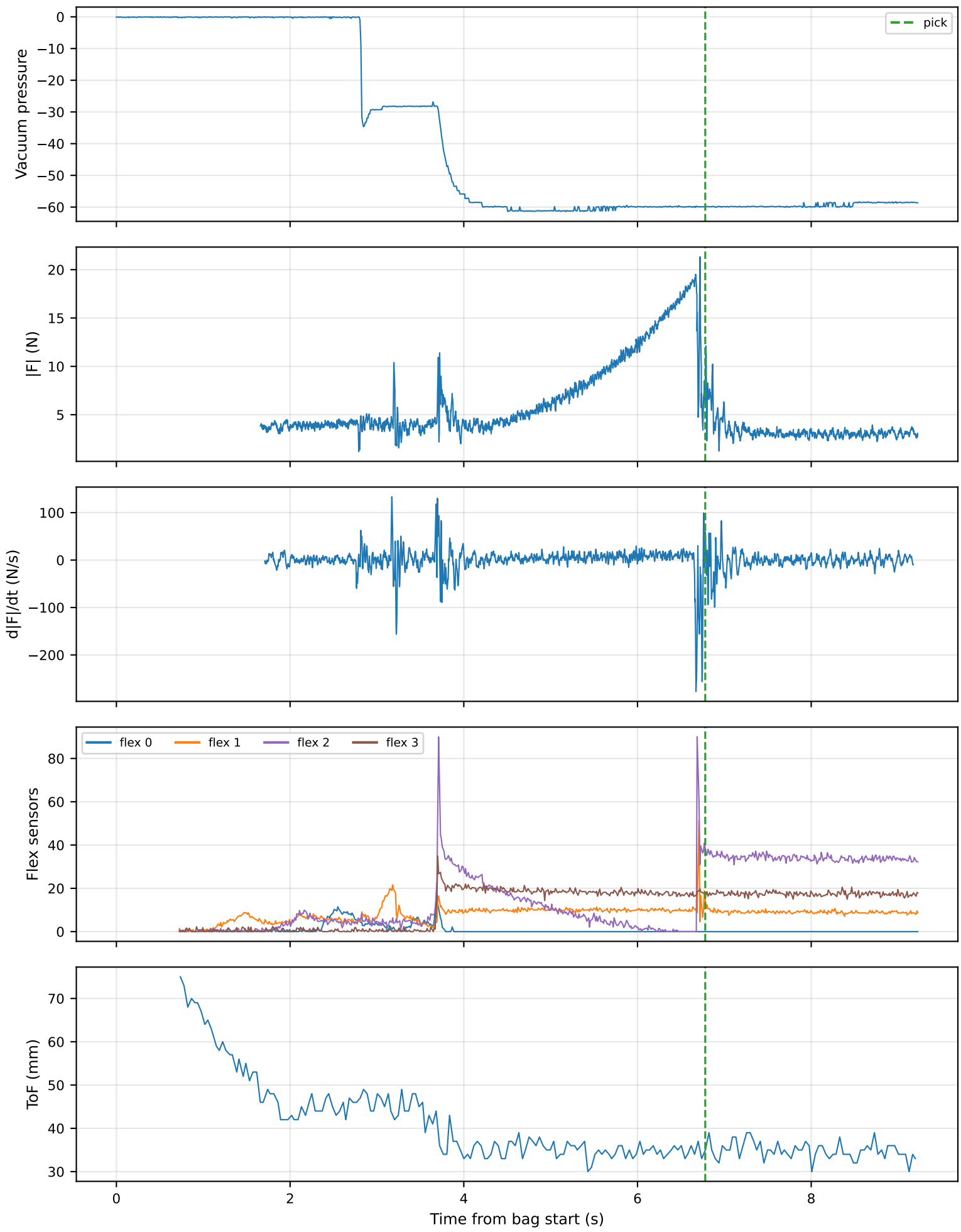
SLIP at 6.81 s | 2026-10-07 15:32:59 | batch 128, apple 0 | stiffness_controller

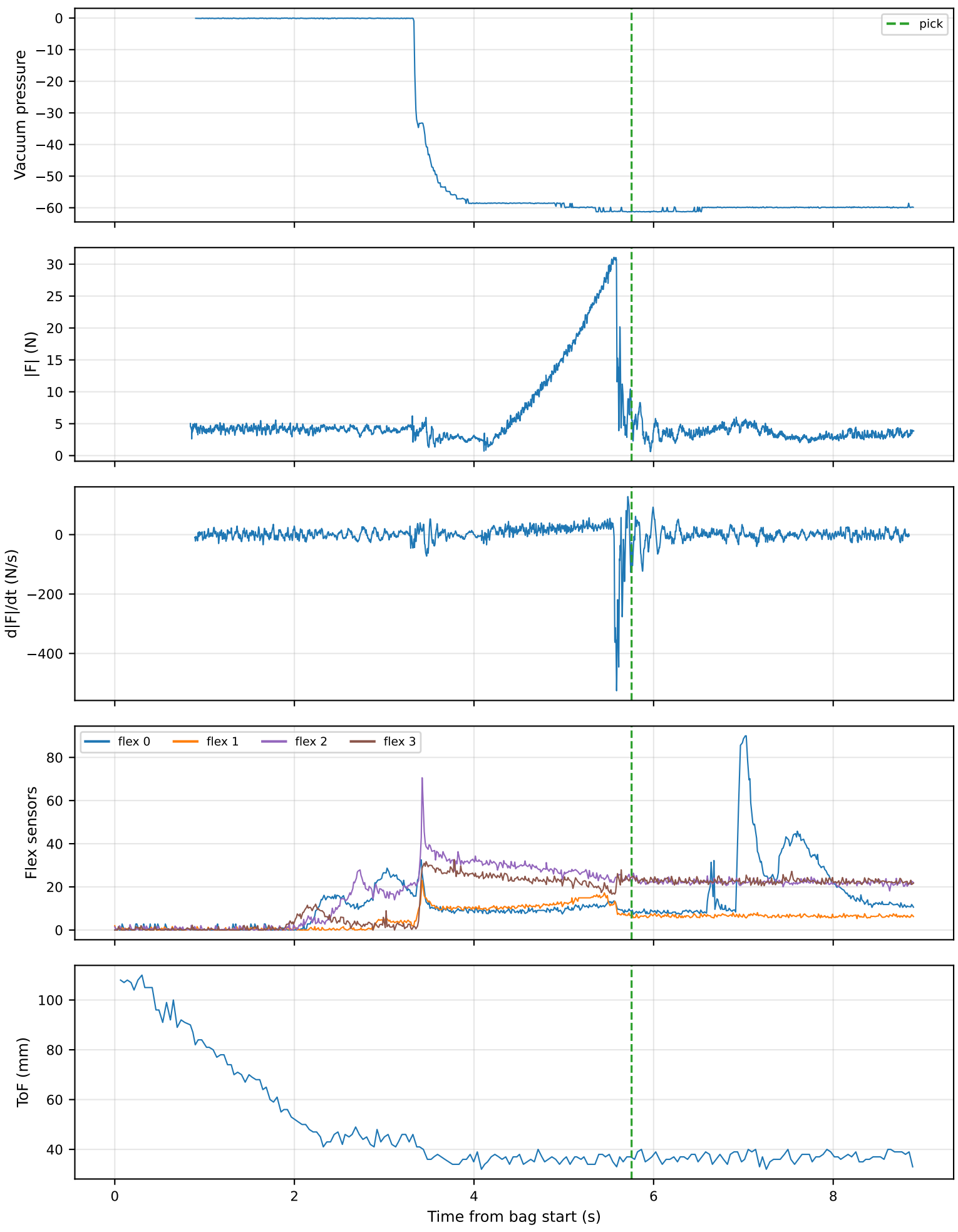



final_approach_and_pick_20261007_153356.db3
PICK at 10.52 s | 2026-10-07 15:33:57 | batch 129, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_153455.db3
PICK at 5.07 s | 2026-10-07 15:34:57 | batch 130, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

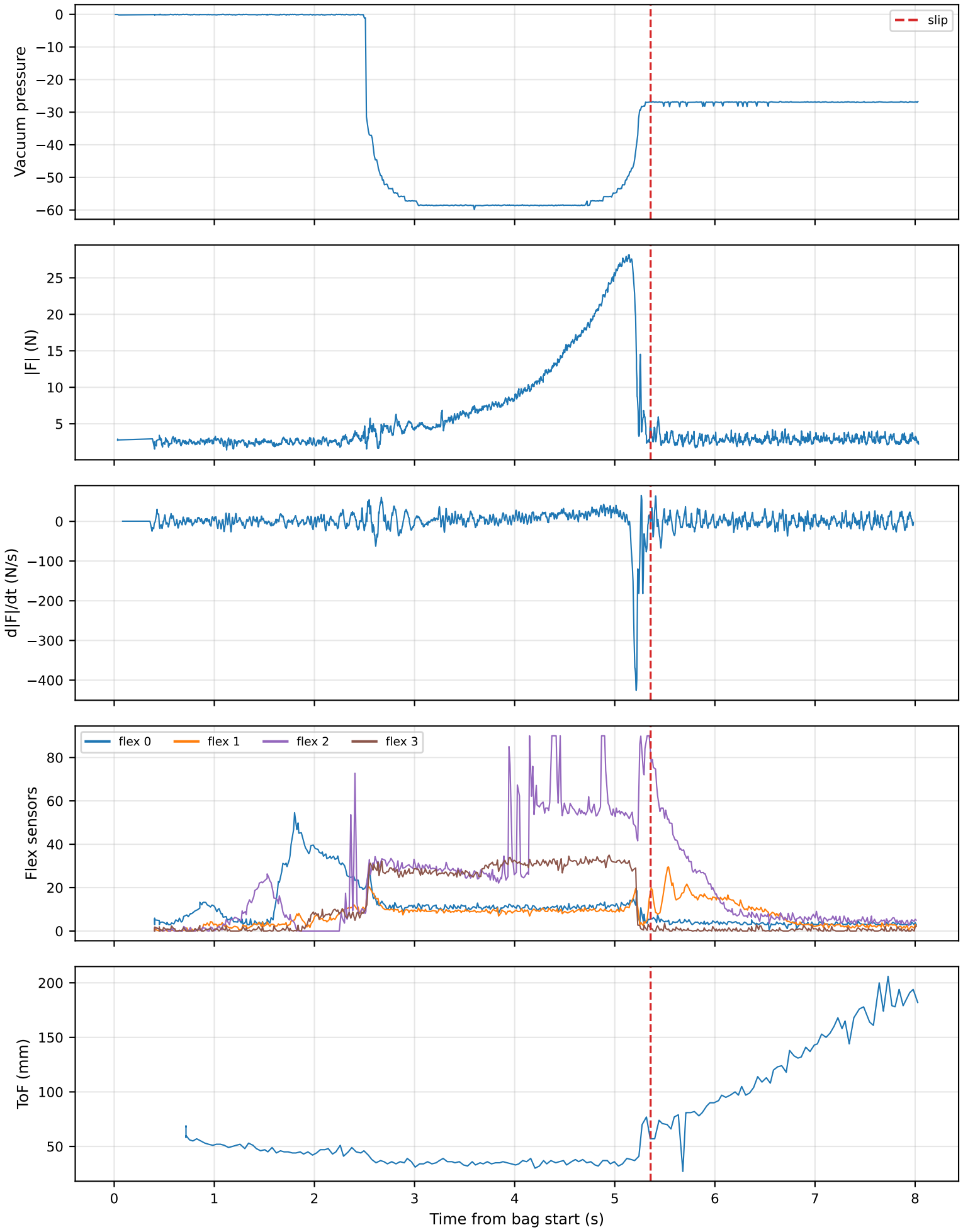

final_approach_and_pick_20261007_153609.db3
PICK at 8.40 s | 2026-10-07 15:36:09 | batch 131, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

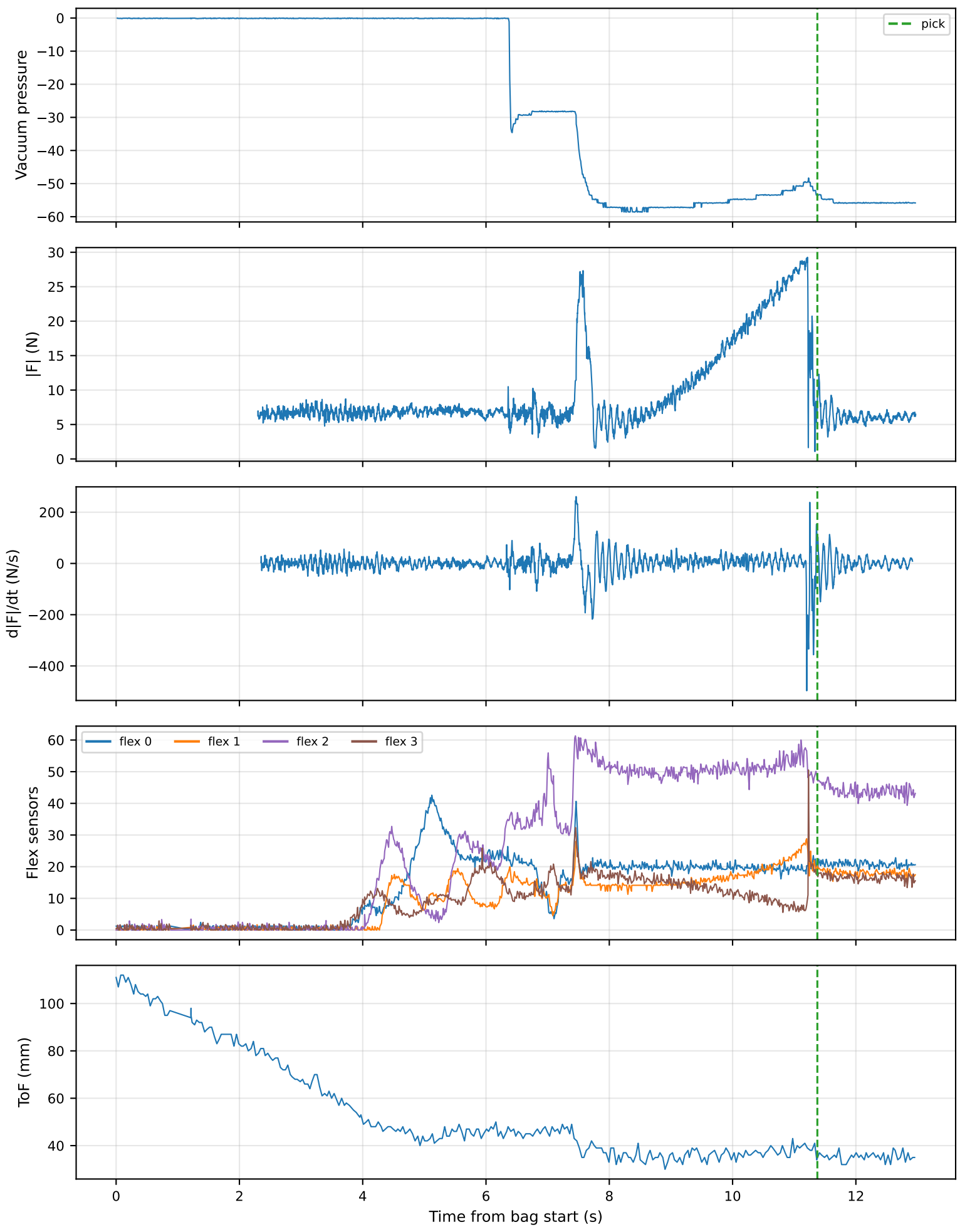

final_approach_and_pick_20261007_153812.db3
PICK at 6.78 s | 2026-10-07 15:38:13 | batch 132, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_154039.db3
PICK at 5.75 s | 2026-10-07 15:40:40 | batch 133, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261007_154702.db3

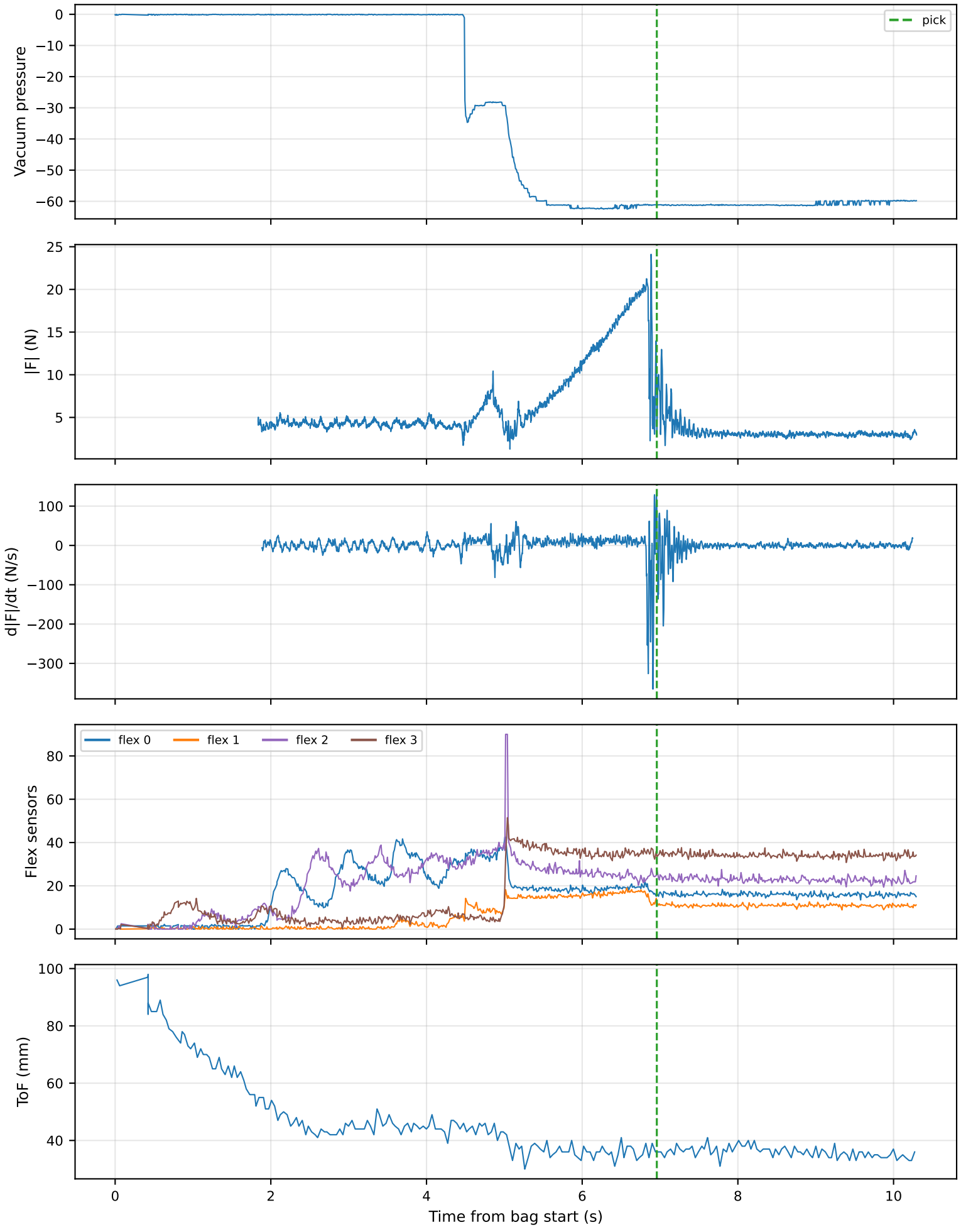
SLIP at 5.36 s | 2026-10-07 15:47:03 | batch 134, apple 0 | stiffness_controller




final_approach_and_pick_20261007_154814.db3
PICK at 11.38 s | 2026-10-07 15:48:15 | batch 135, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

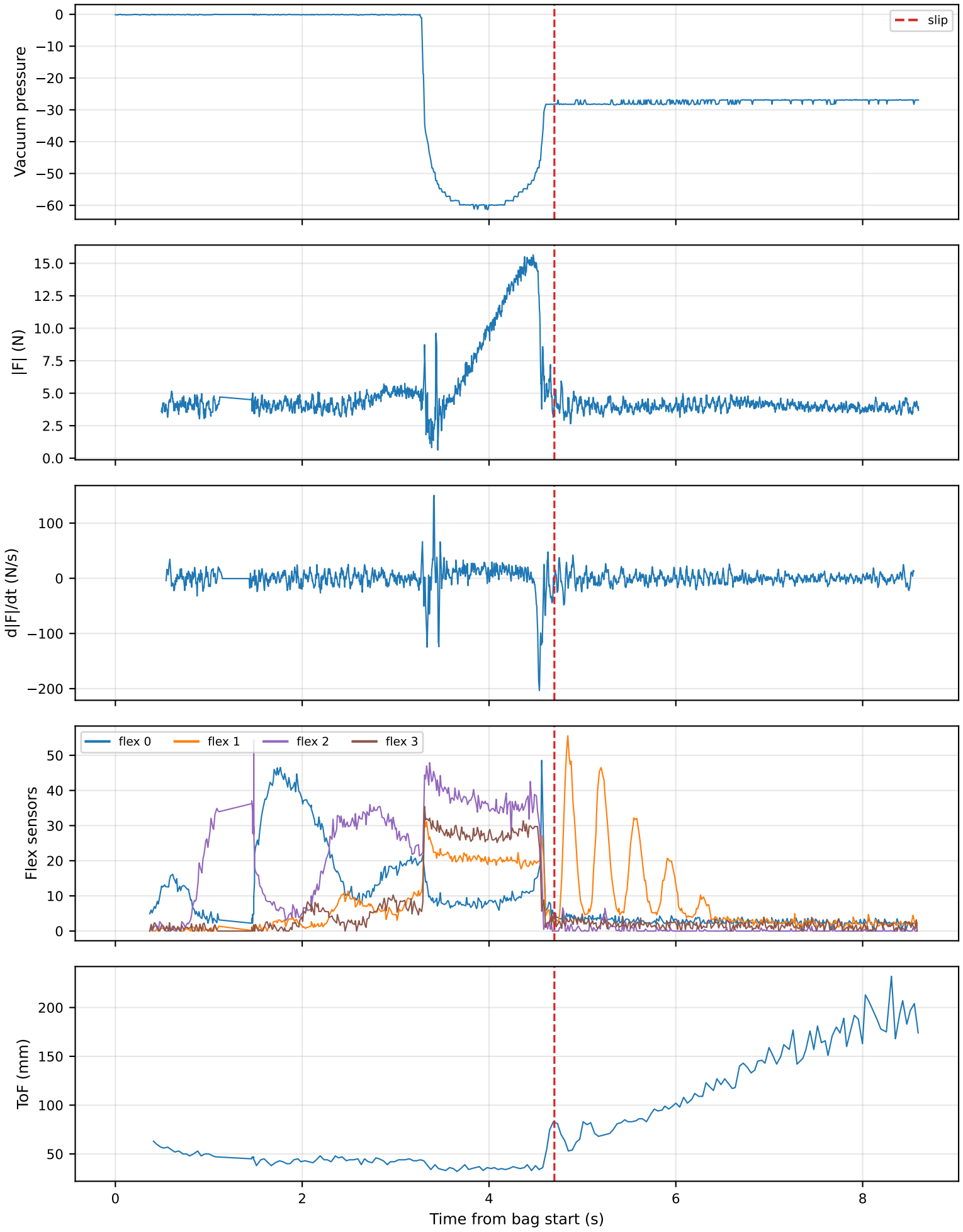
final_approach_and_pick_20261007_154928.db3

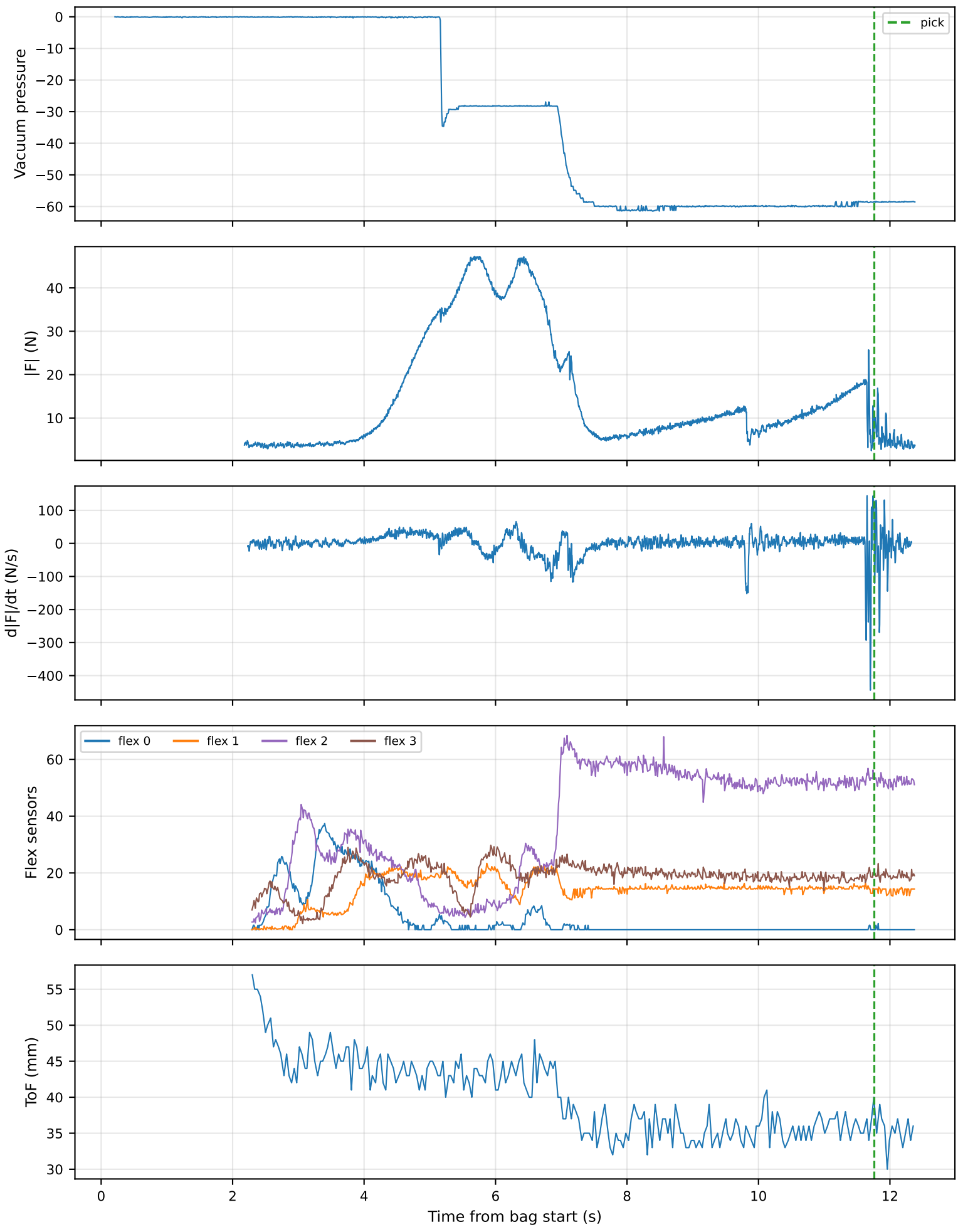
PICK at 6.96 s | 2026-10-07 15:49:29 | batch 136, apple 0 | stiffness_controller

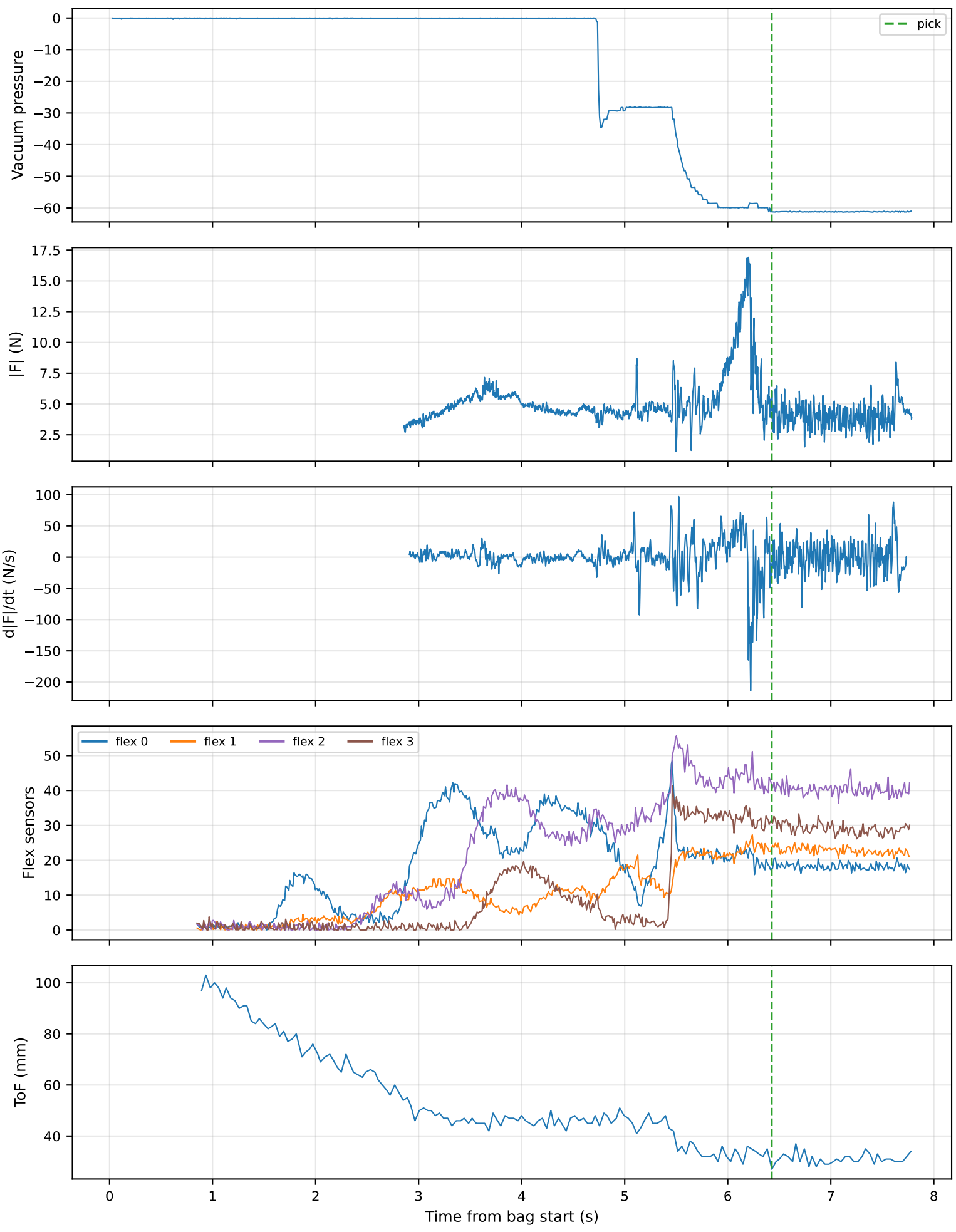


final_approach_and_pick_20261007_155051.db3

SLIP at 4.70 s | 2026-10-07 15:50:53 | batch 137, apple 0 | stiffness_controller

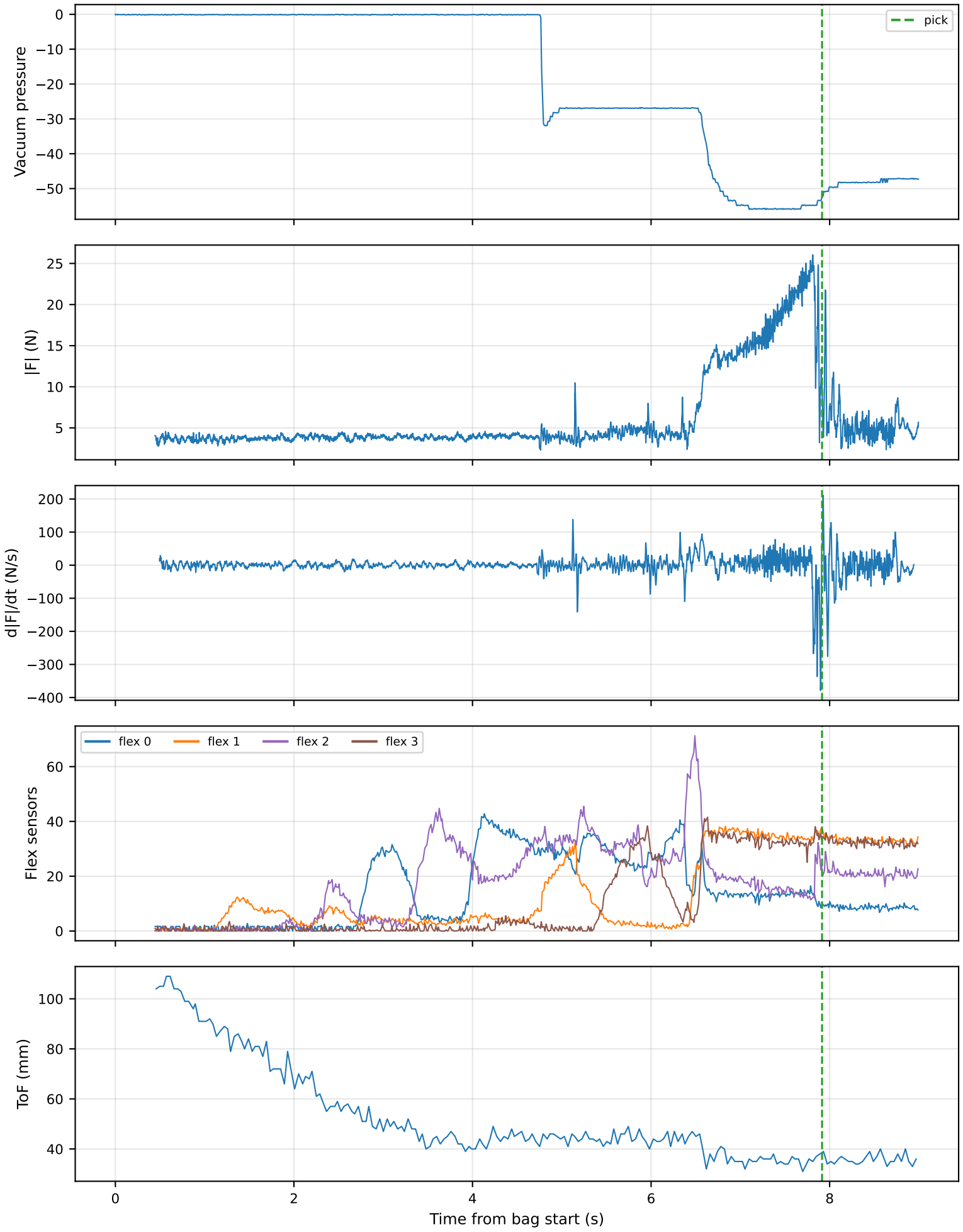


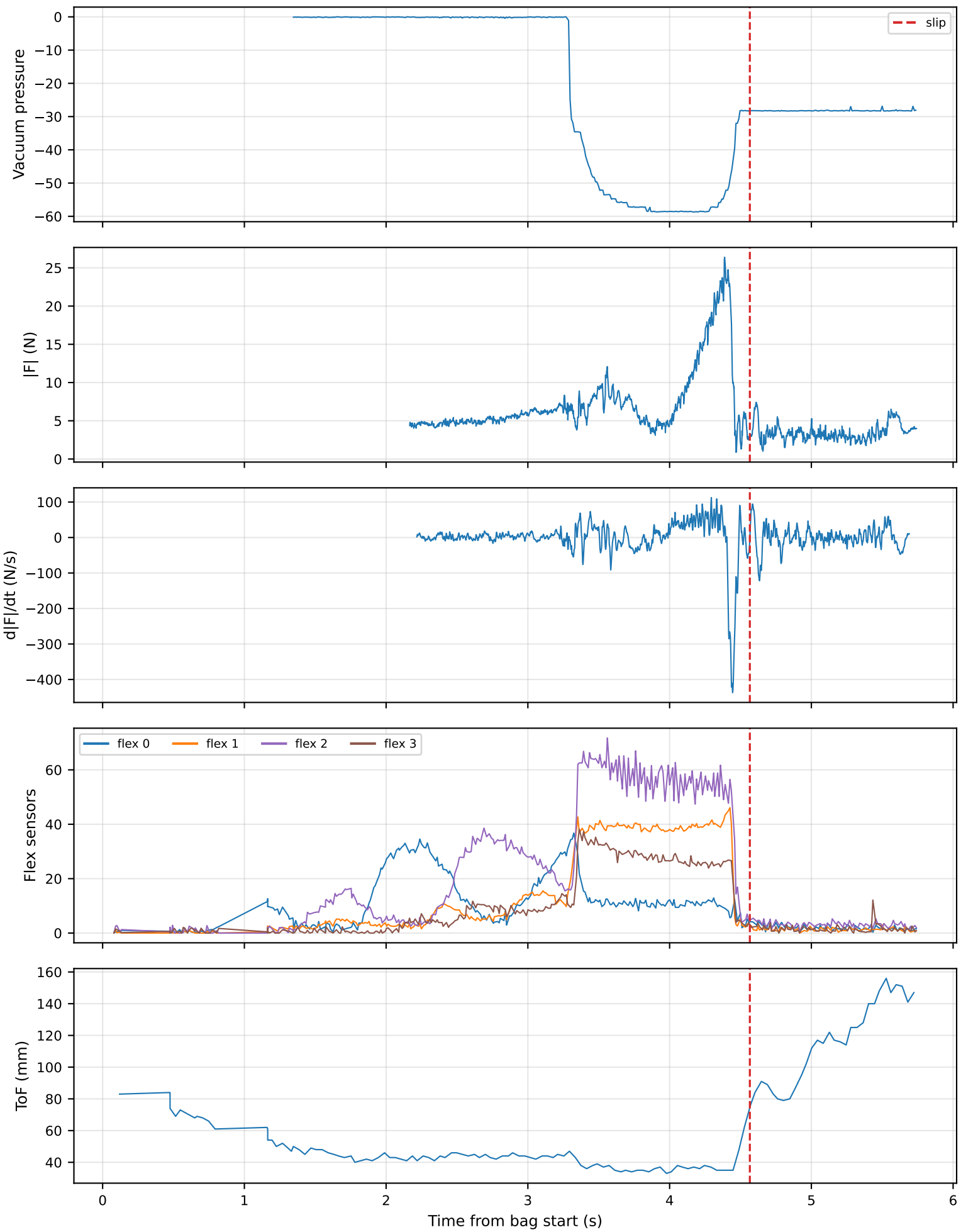

final_approach_and_pick_20261007_155244.db3
PICK at 11.76 s | 2026-10-07 15:52:45 | batch 138, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

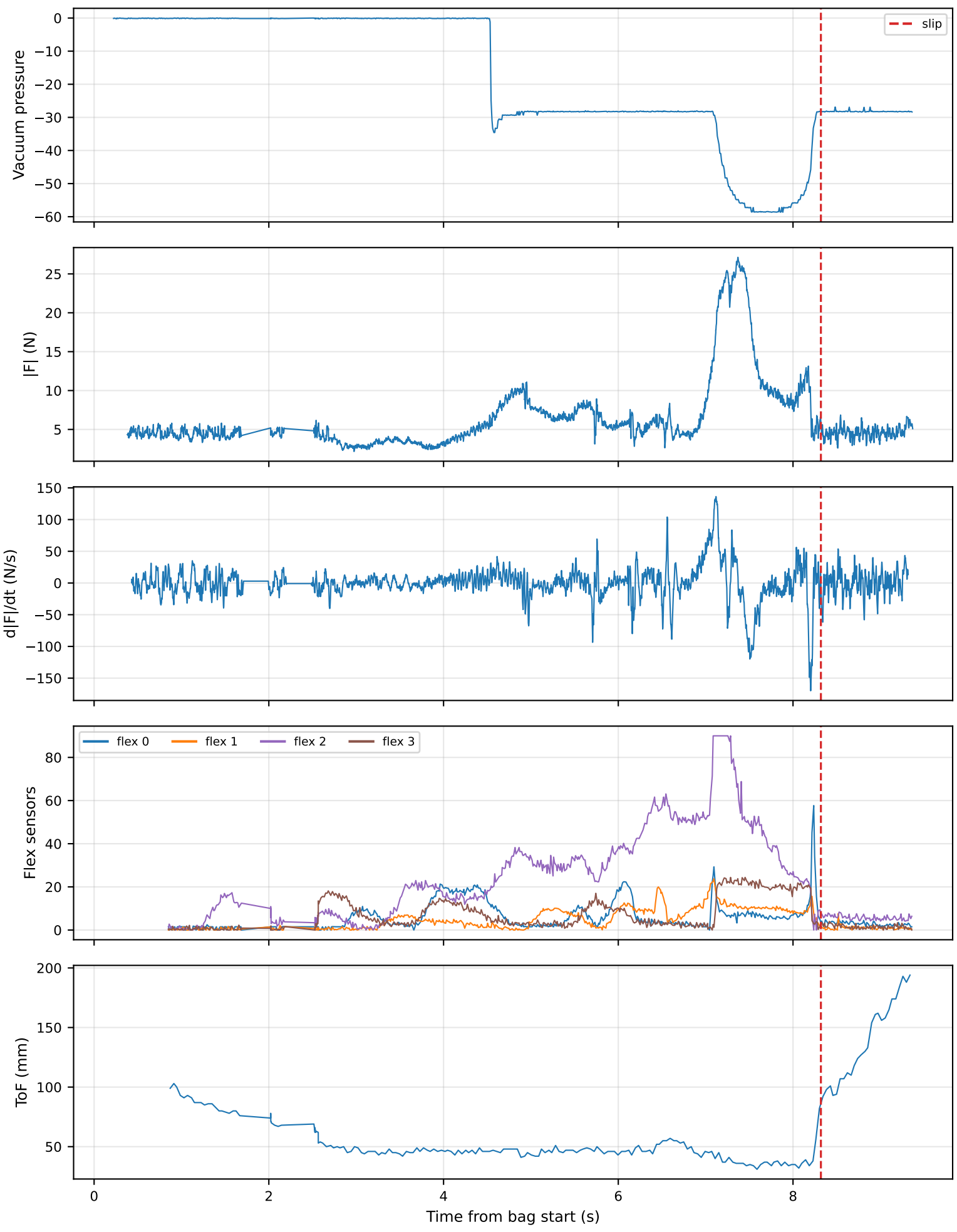

final_approach_and_pick_20261007_160059.db3
PICK at 6.43 s | 2026-10-07 16:01:00 | batch 139, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261007_160231.db3

PICK at 7.92 s | 2026-10-07 16:02:31 | batch 140, apple 0 | pull_twist_controller

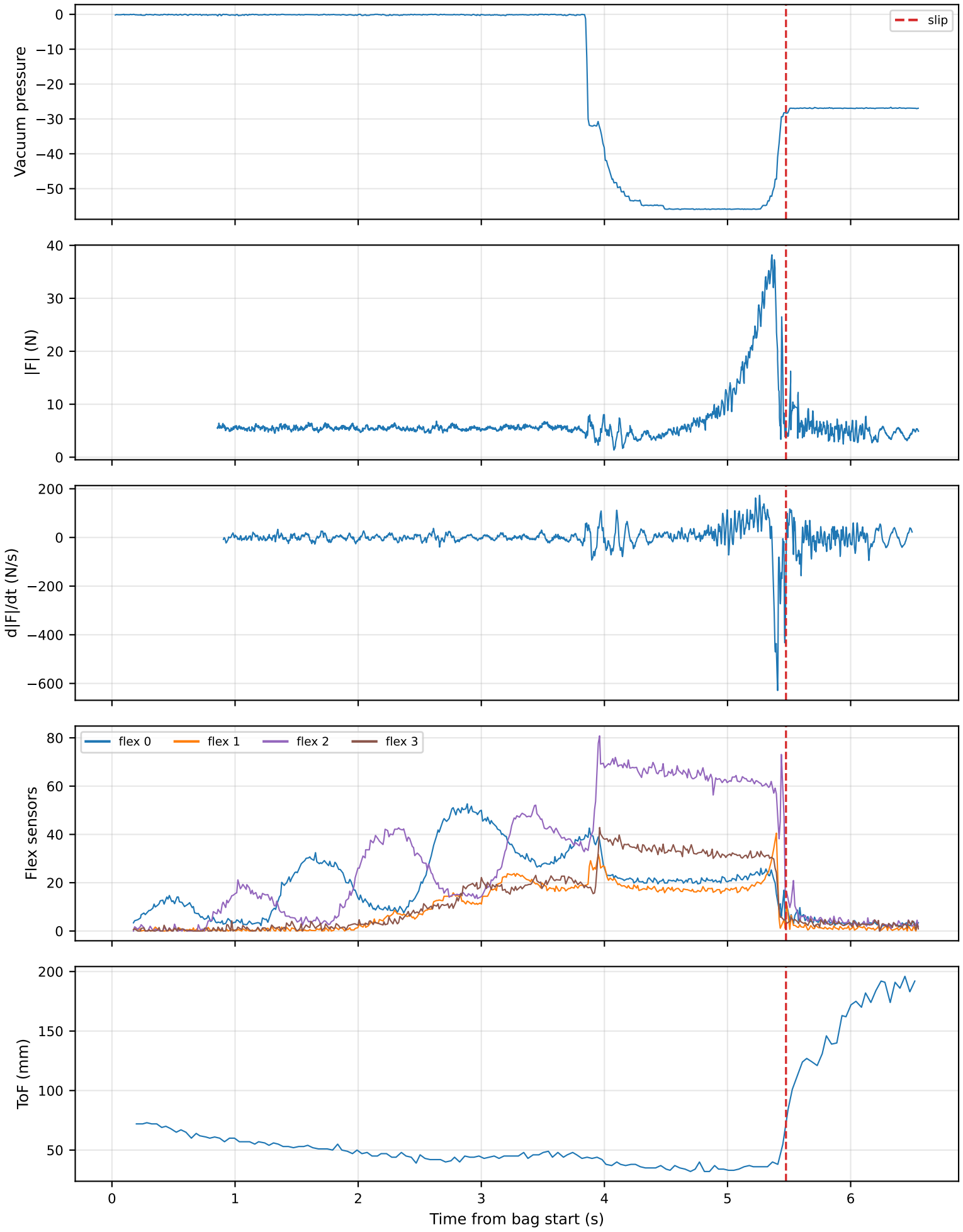


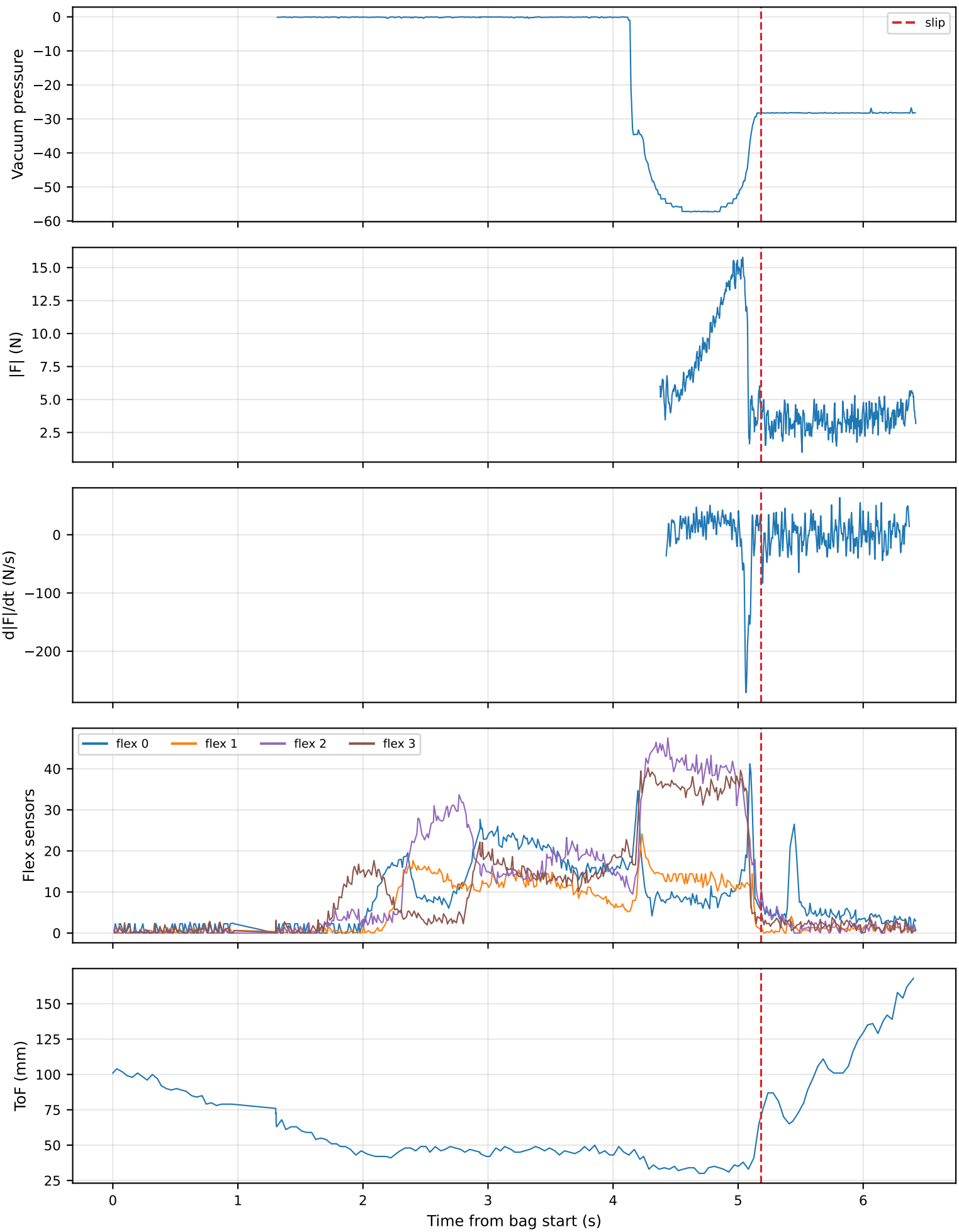

final_approach_and_pick_20261007_160442.db3
SLIP at 4.57 s | 2026-10-07 16:04:43 | batch 141, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_160735.db3
SLIP at 8.32 s | 2026-10-07 16:07:36 | batch 142, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261007_160844.db3

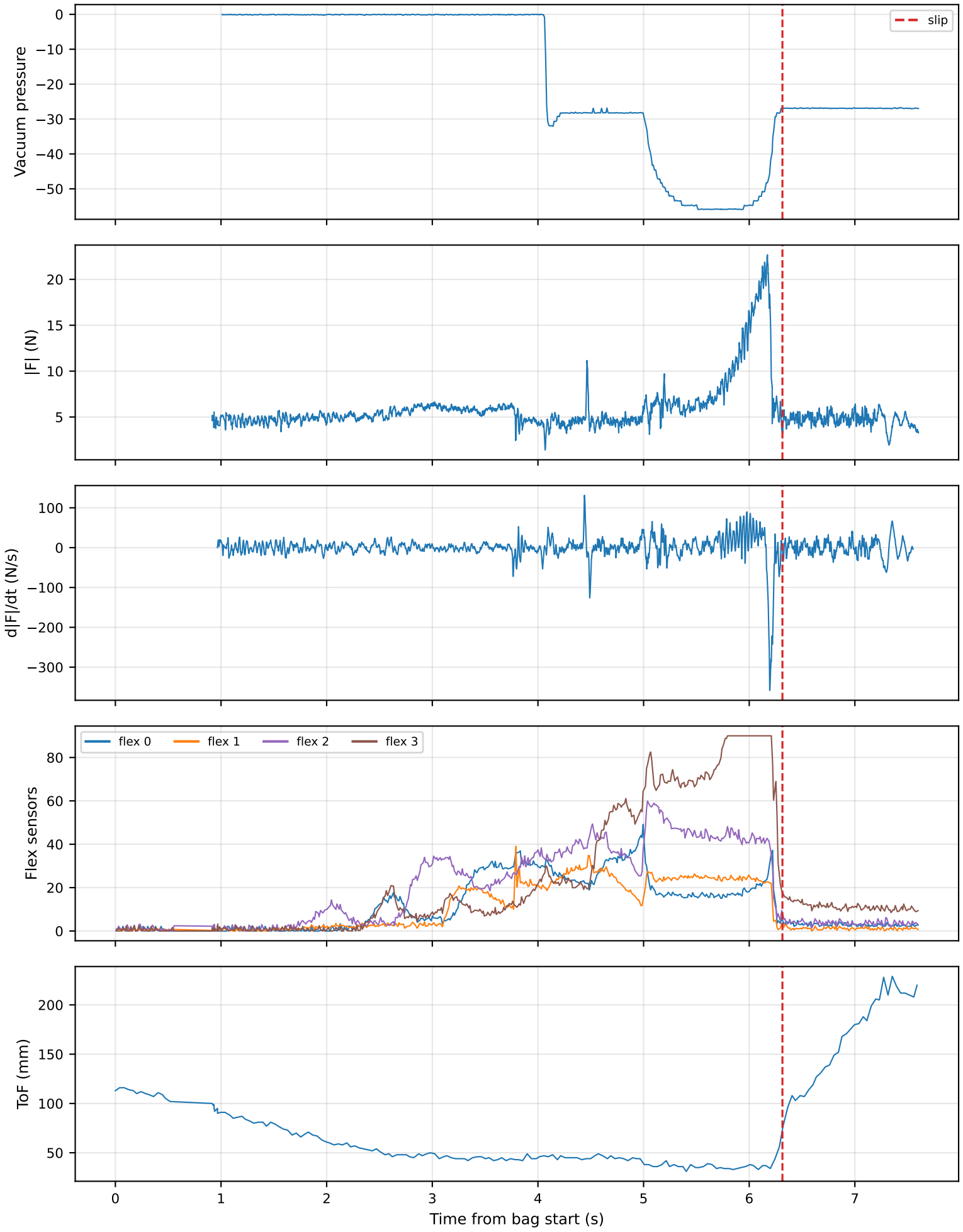
SLIP at 5.47 s | 2026-10-07 16:08:46 | batch 143, apple 0 | pull_twist_controller

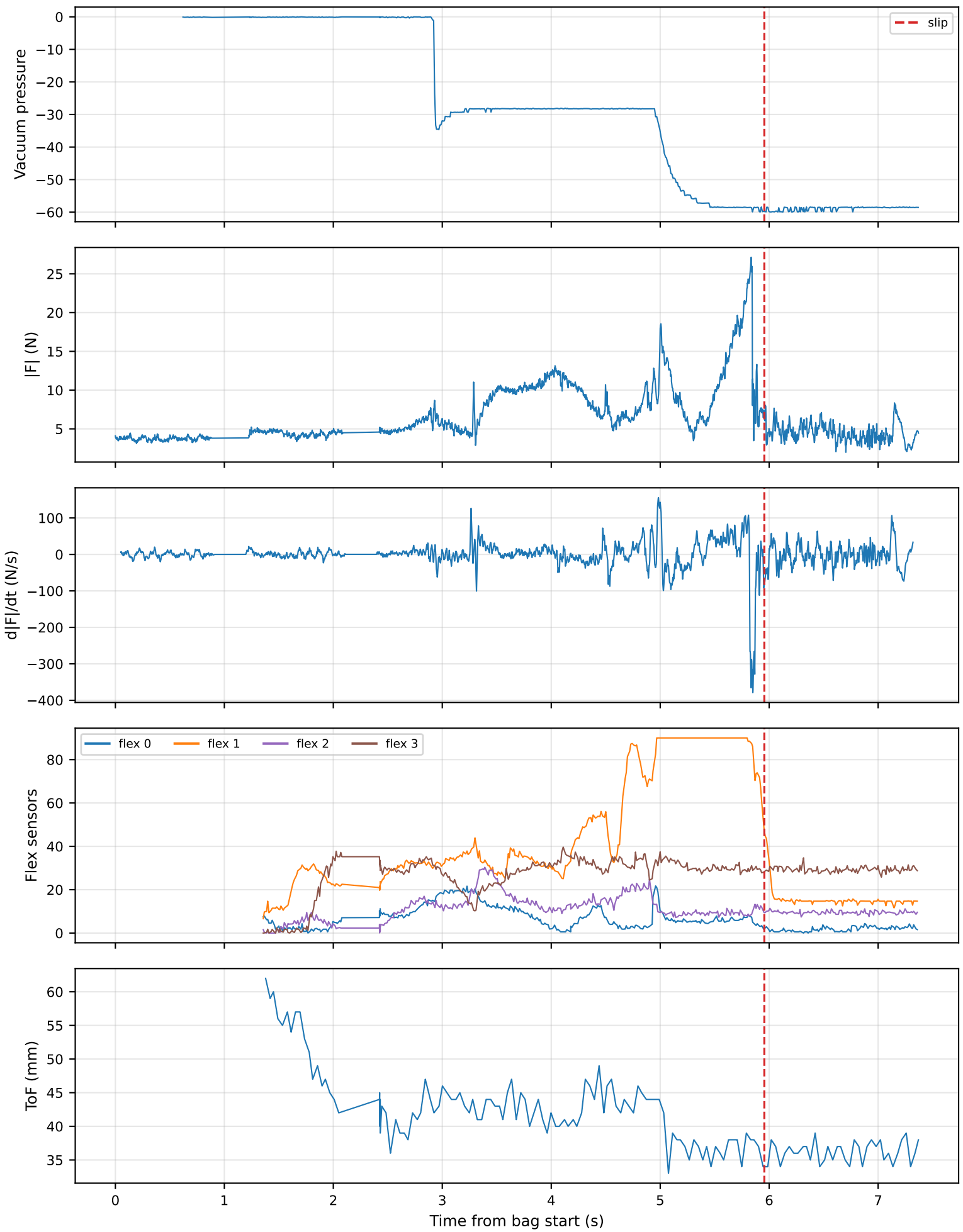


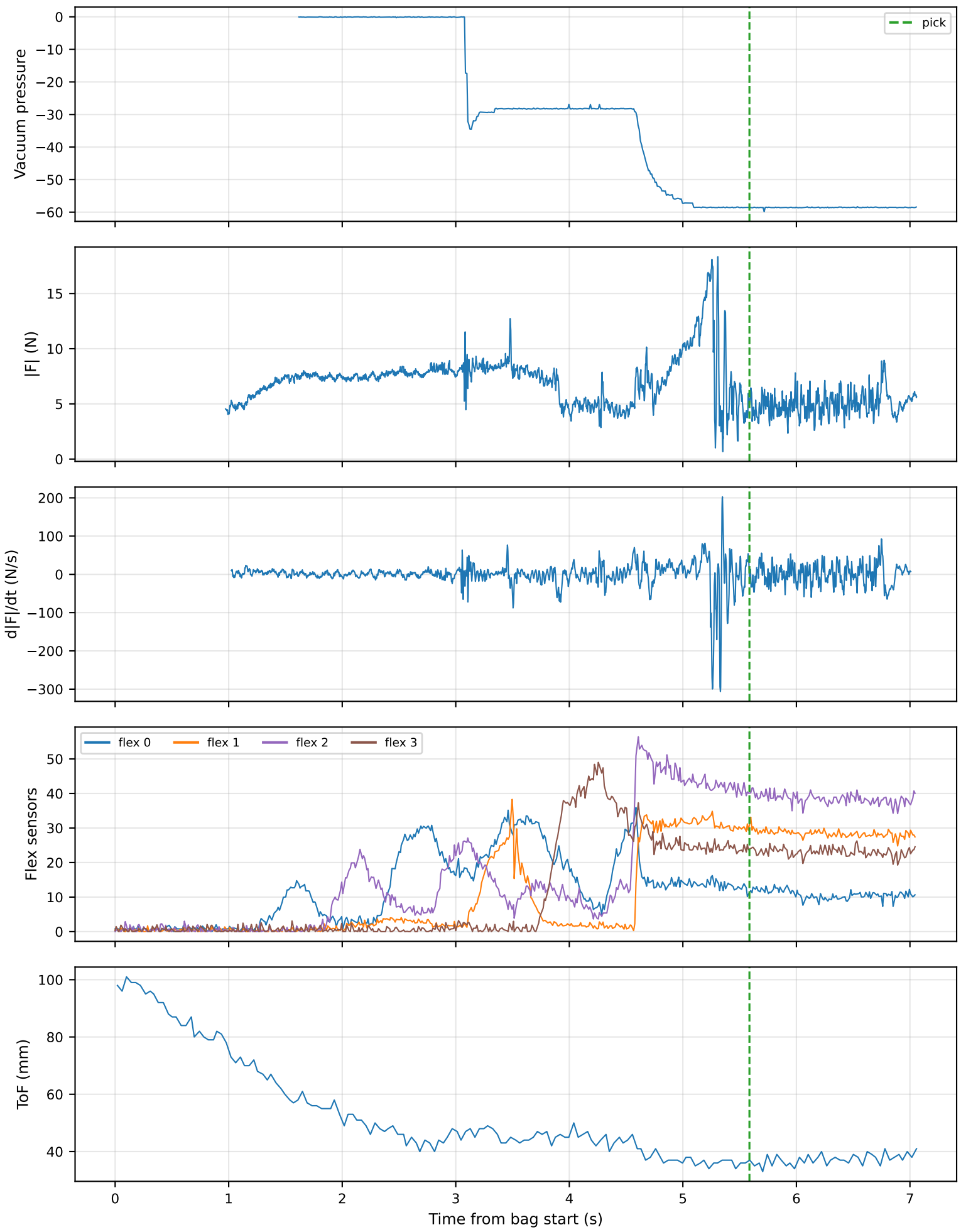

final_approach_and_pick_20261007_161007.db3
SLIP at 5.18 s | 2026-10-07 16:10:08 | batch 144, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

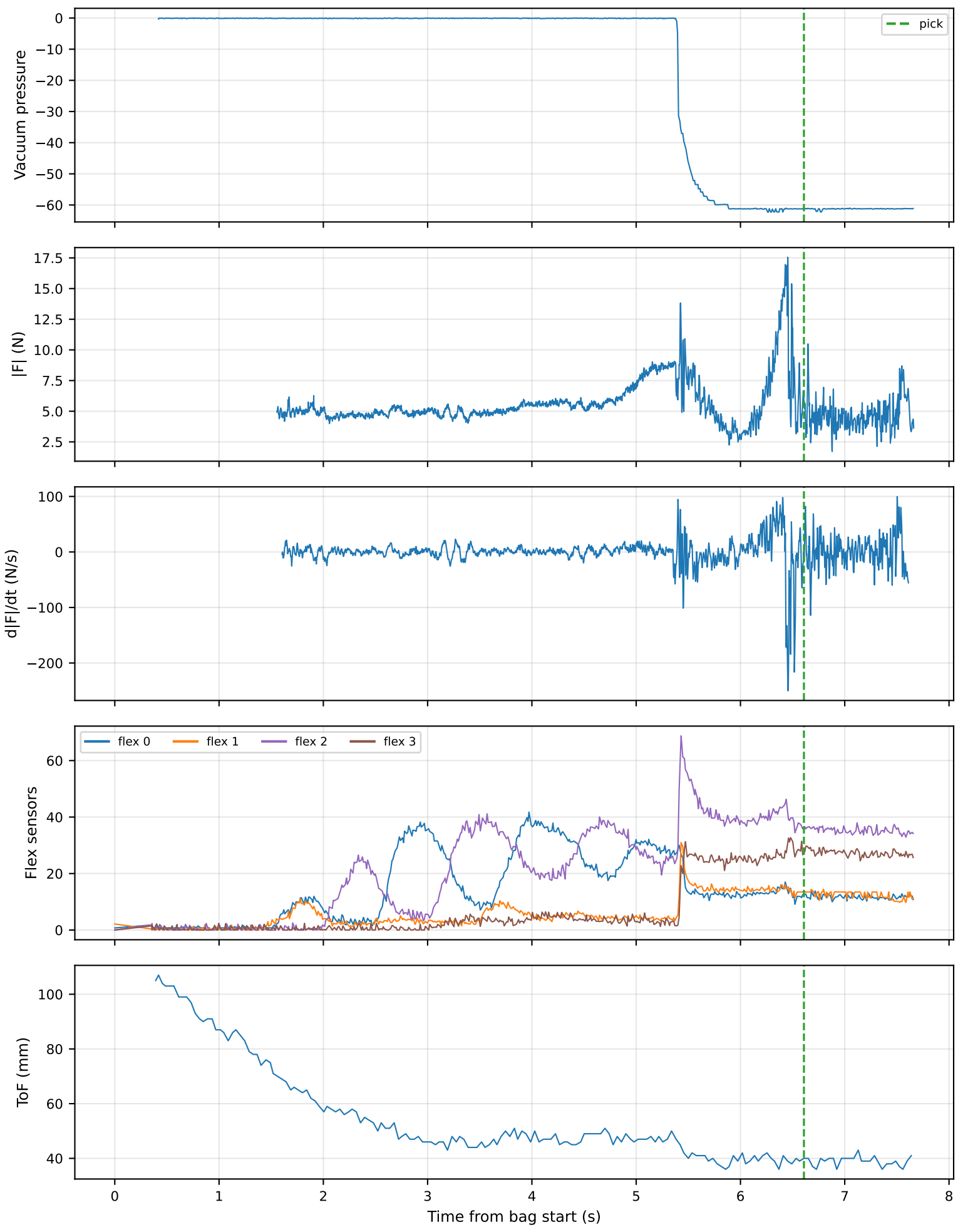
final_approach_and_pick_20261007_161325.db3

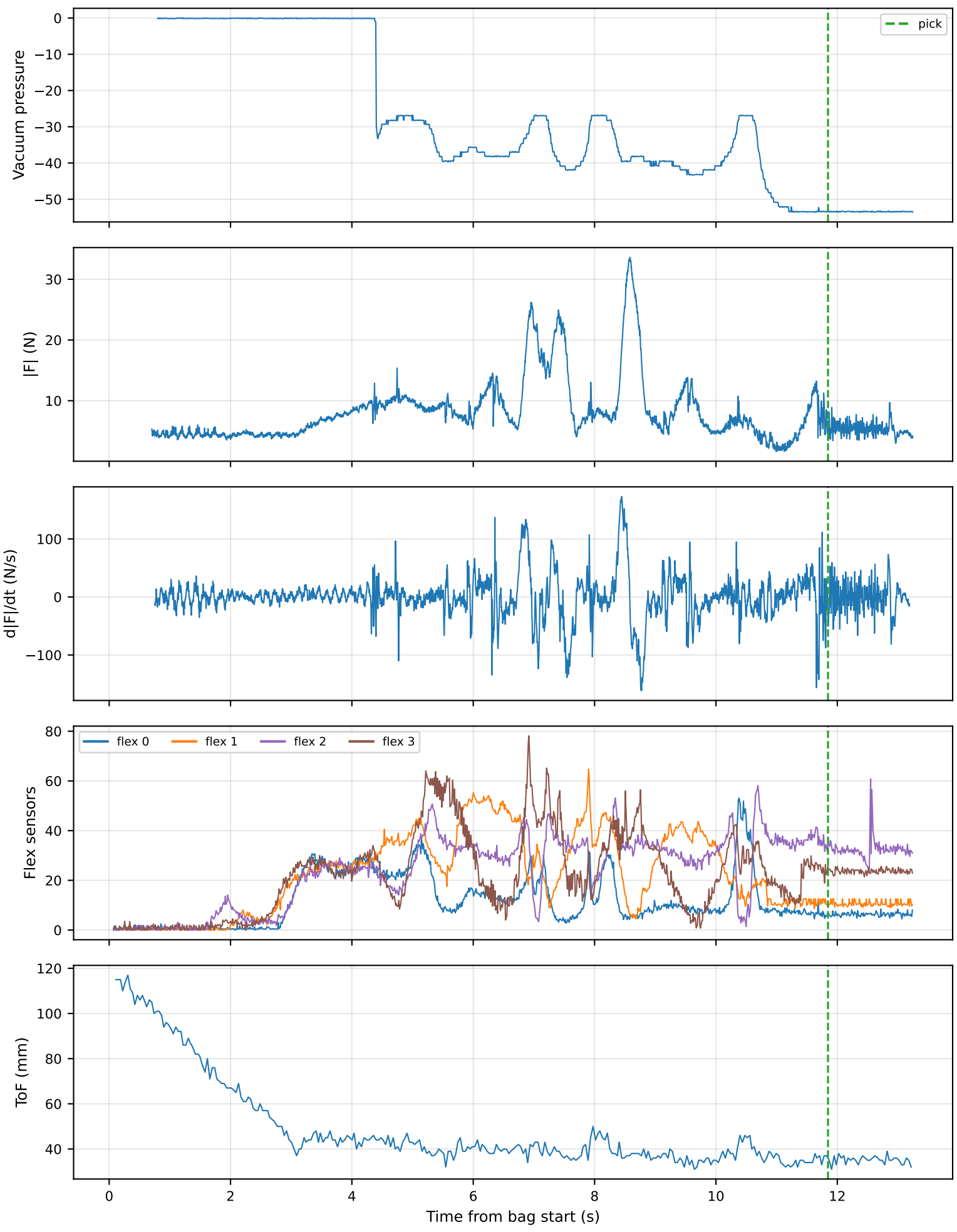
SLIP at 6.31 s | 2026-10-07 16:13:26 | batch 145, apple 0 | pull_twist_controller

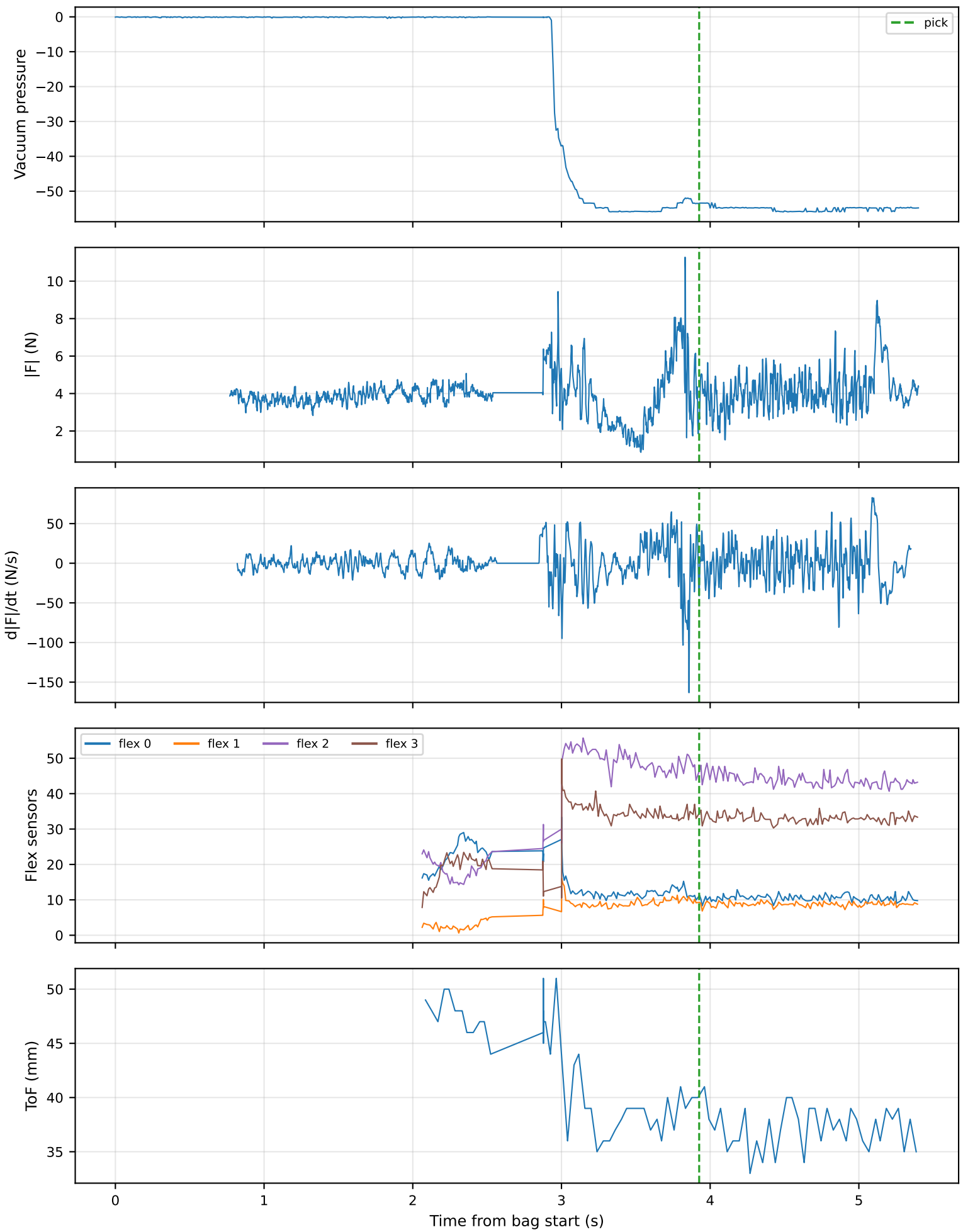


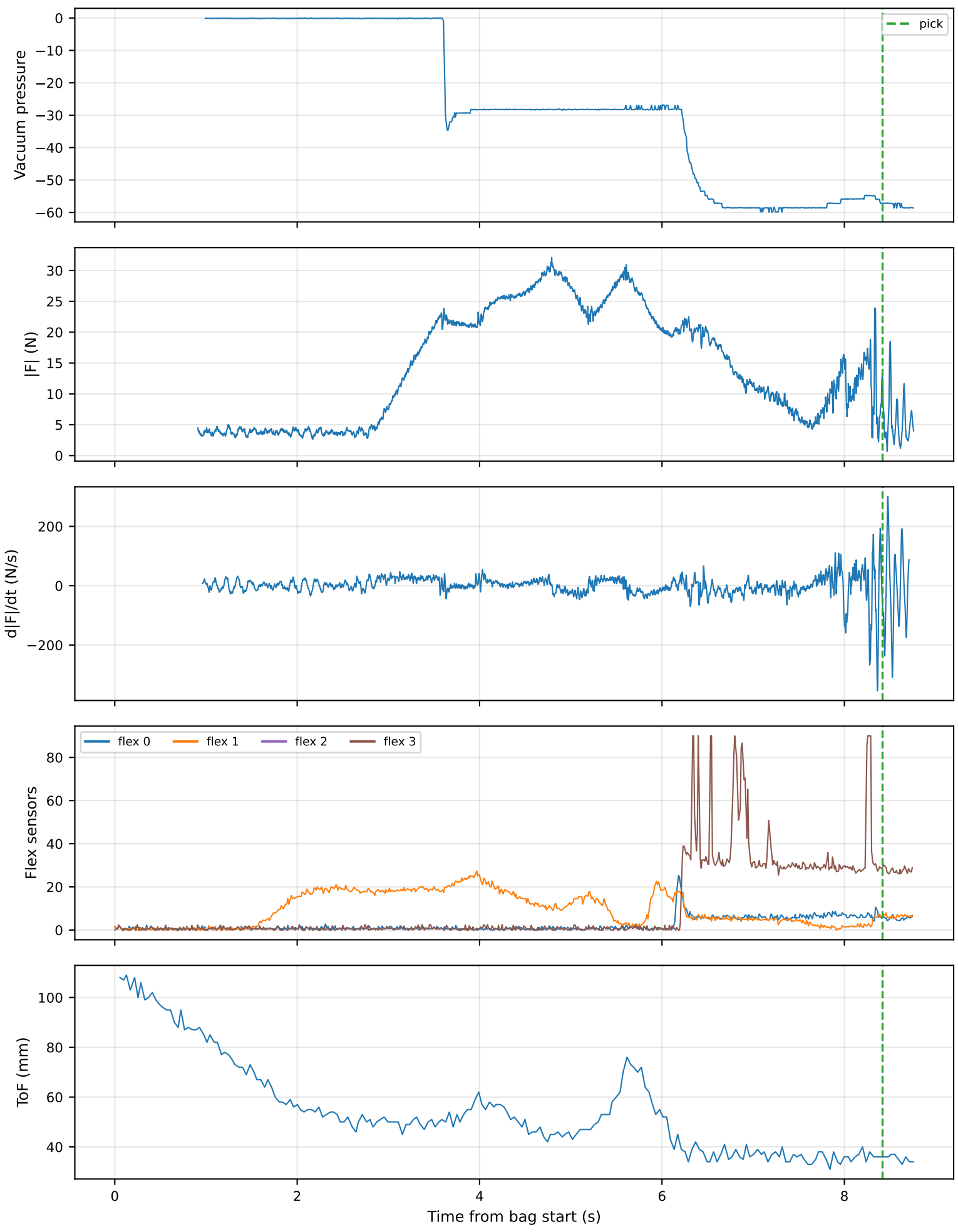

final_approach_and_pick_20261007_161425.db3
SLIP at 5.96 s | 2026-10-07 16:14:27 | batch 146, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

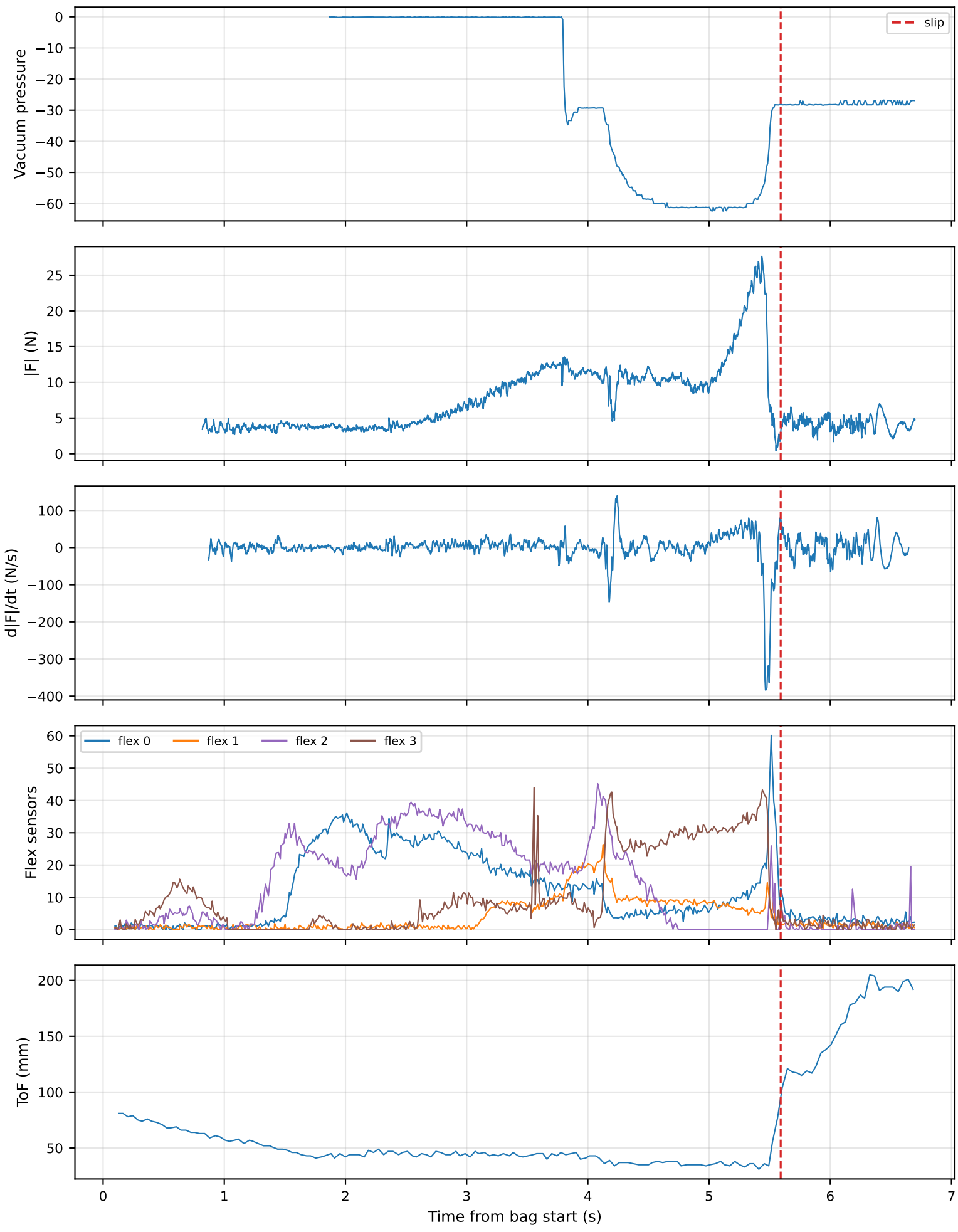

final_approach_and_pick_20261007_161516.db3
PICK at 5.59 s | 2026-10-07 16:15:17 | batch 147, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_161602.db3
PICK at 6.61 s | 2026-10-07 16:16:03 | batch 148, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_161654.db3
PICK at 11.85 s | 2026-10-07 16:16:55 | batch 149, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

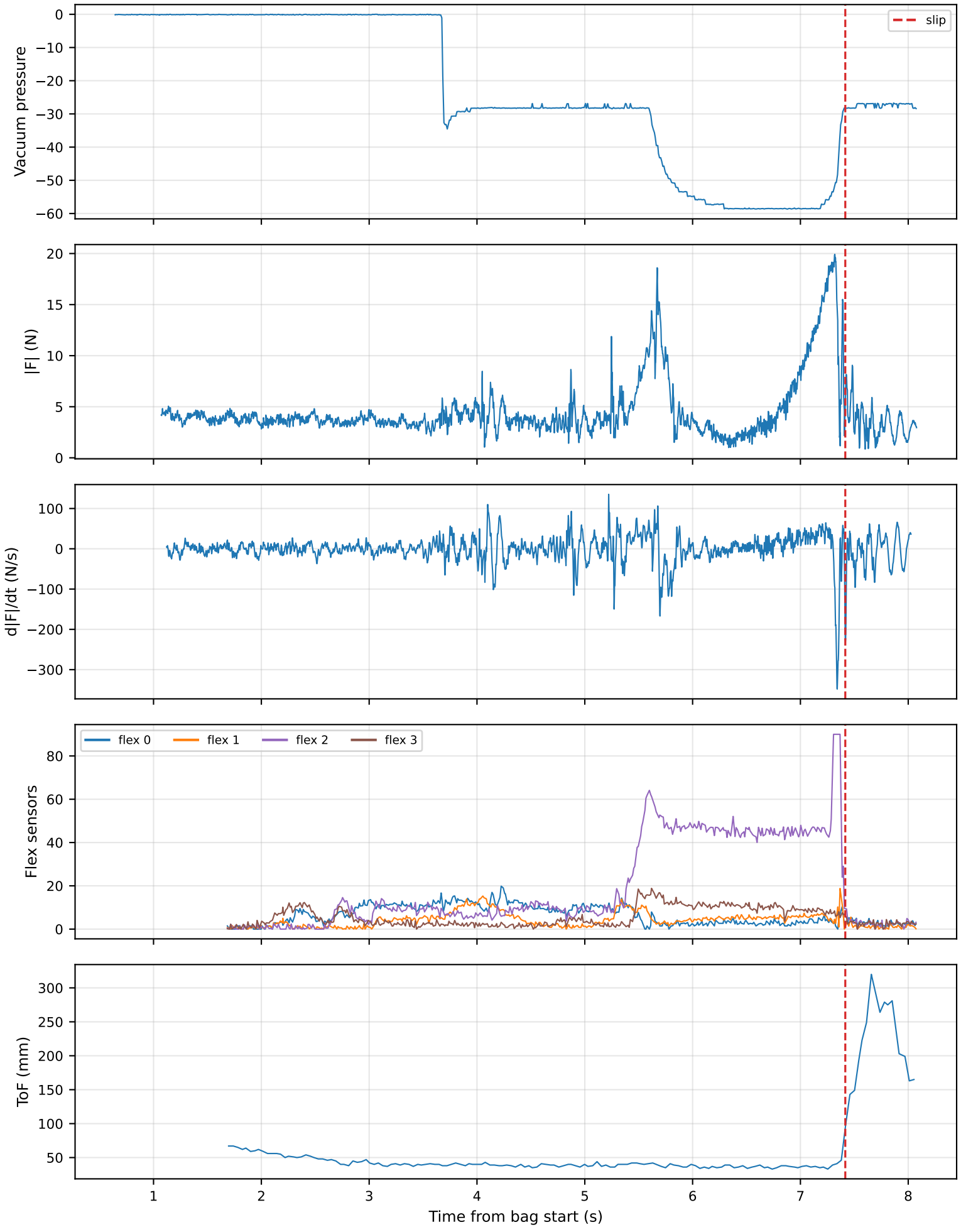

final_approach_and_pick_20261007_161801.db3
PICK at 3.93 s | 2026-10-07 16:18:02 | batch 150, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_162149.db3
PICK at 8.42 s | 2026-10-07 16:21:50 | batch 151, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_162422.db3
SLIP at 5.59 s | 2026-10-07 16:24:23 | batch 152, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

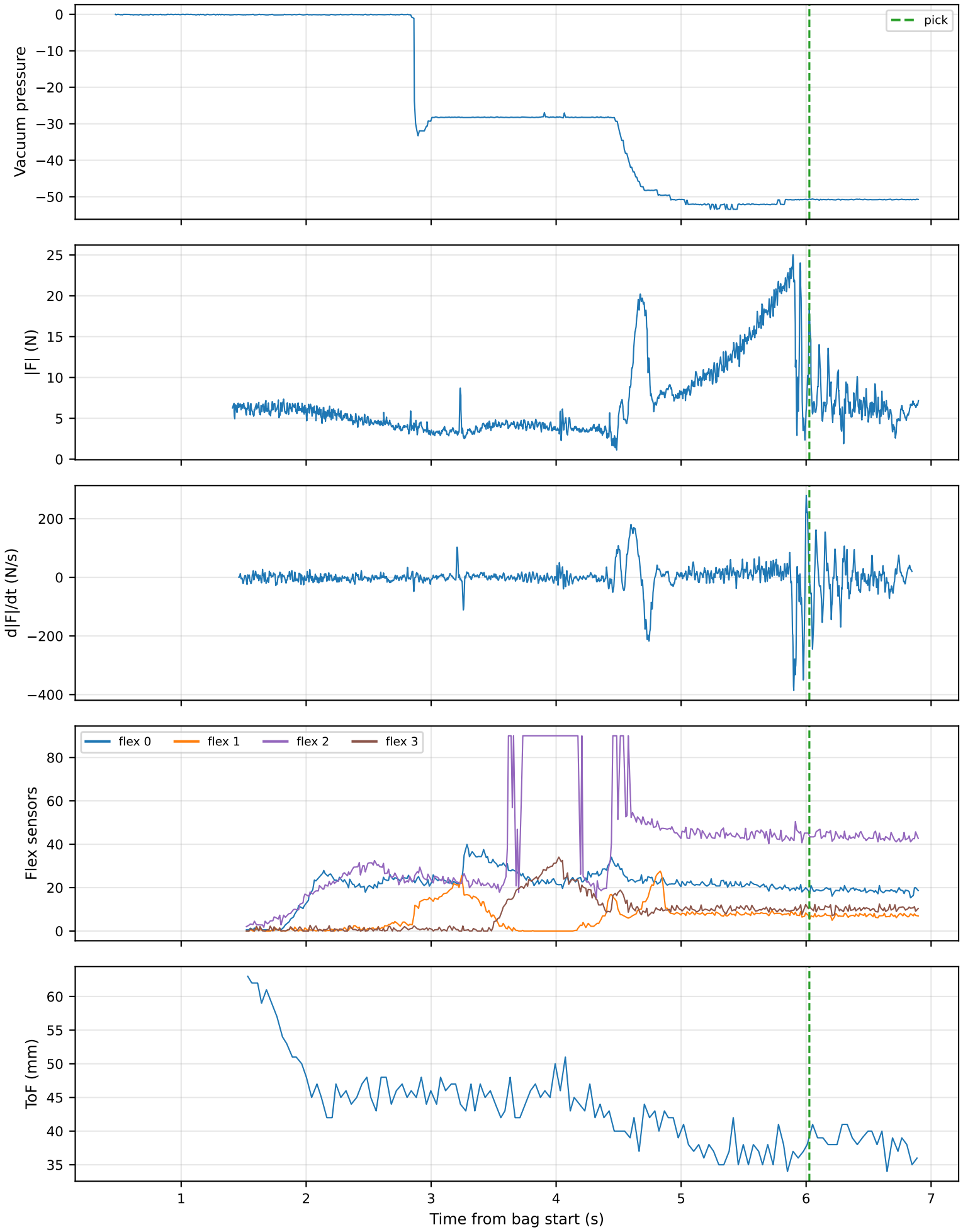
final_approach_and_pick_20261007_163312.db3

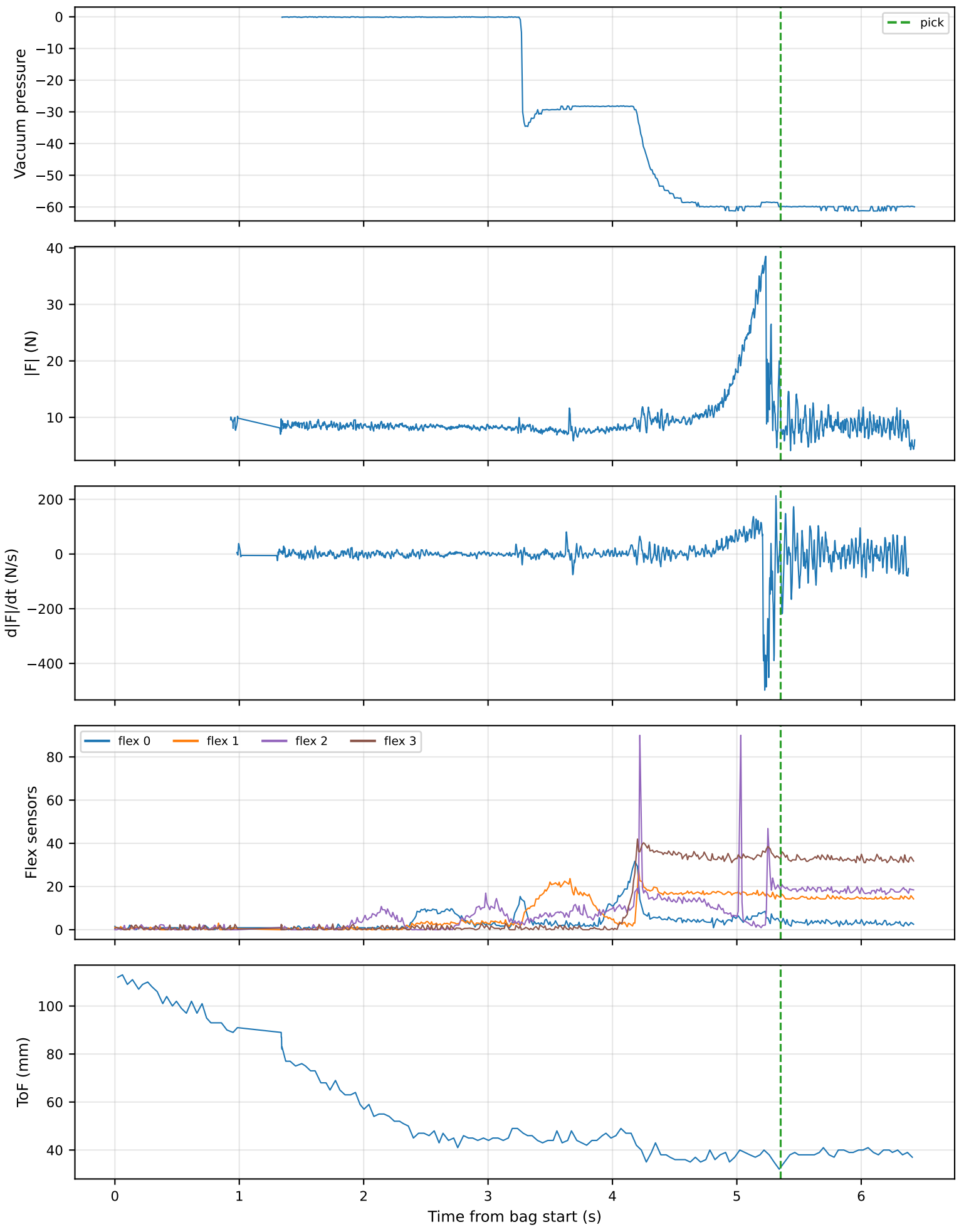
SLIP at 7.42 s | 2026-10-07 16:33:13 | batch 153, apple 0 | pull_twist_controller

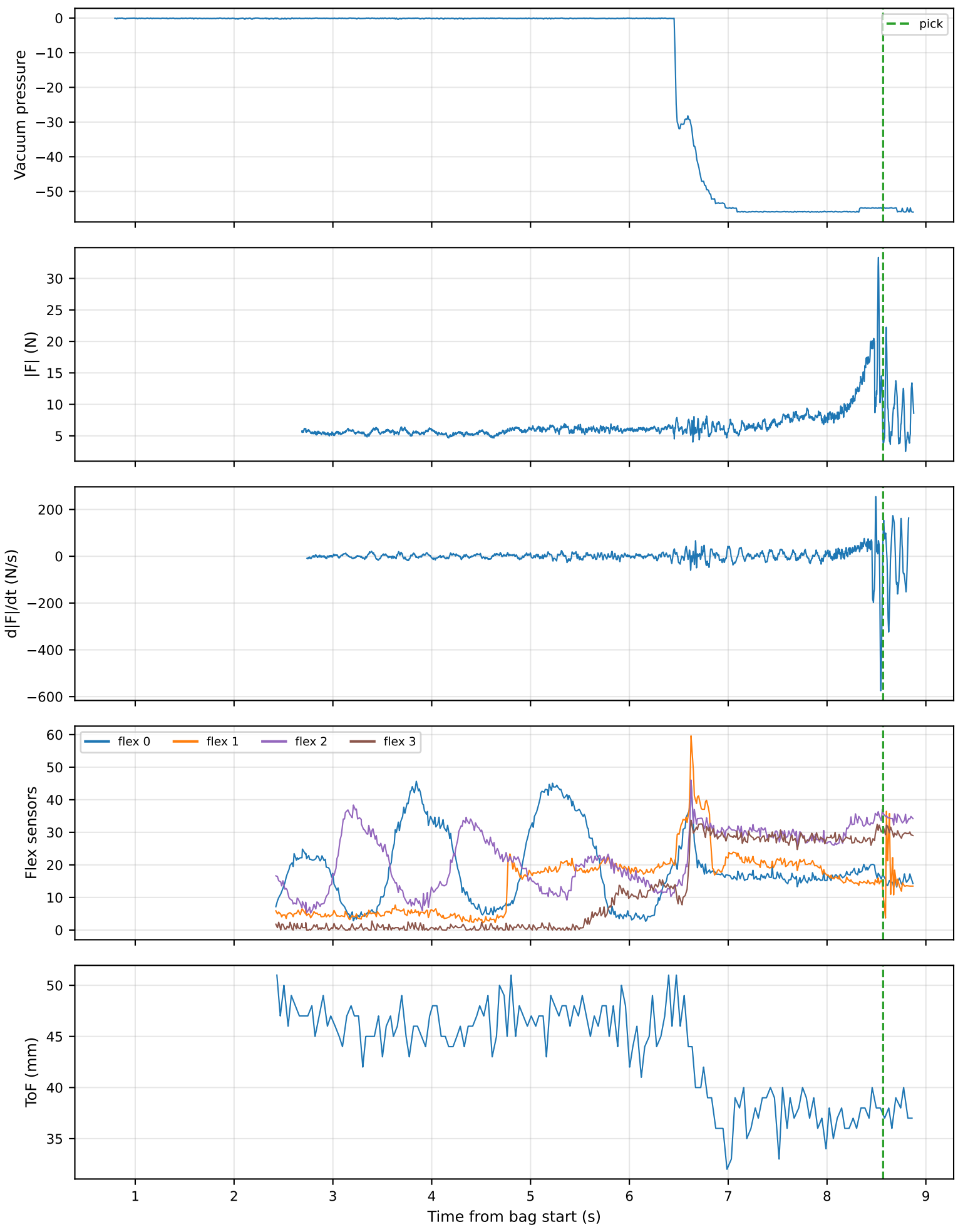


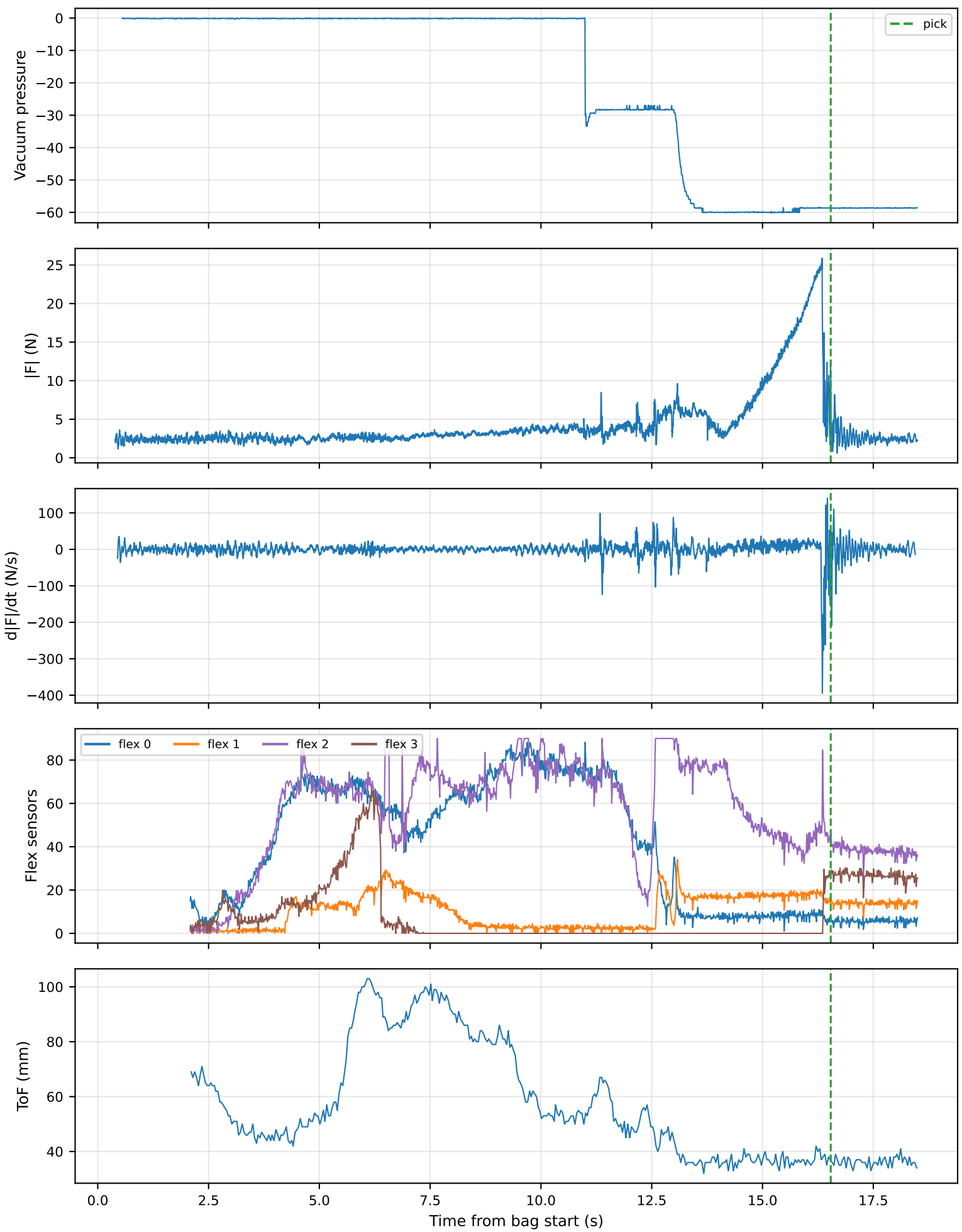
final_approach_and_pick_20261007_163835.db3

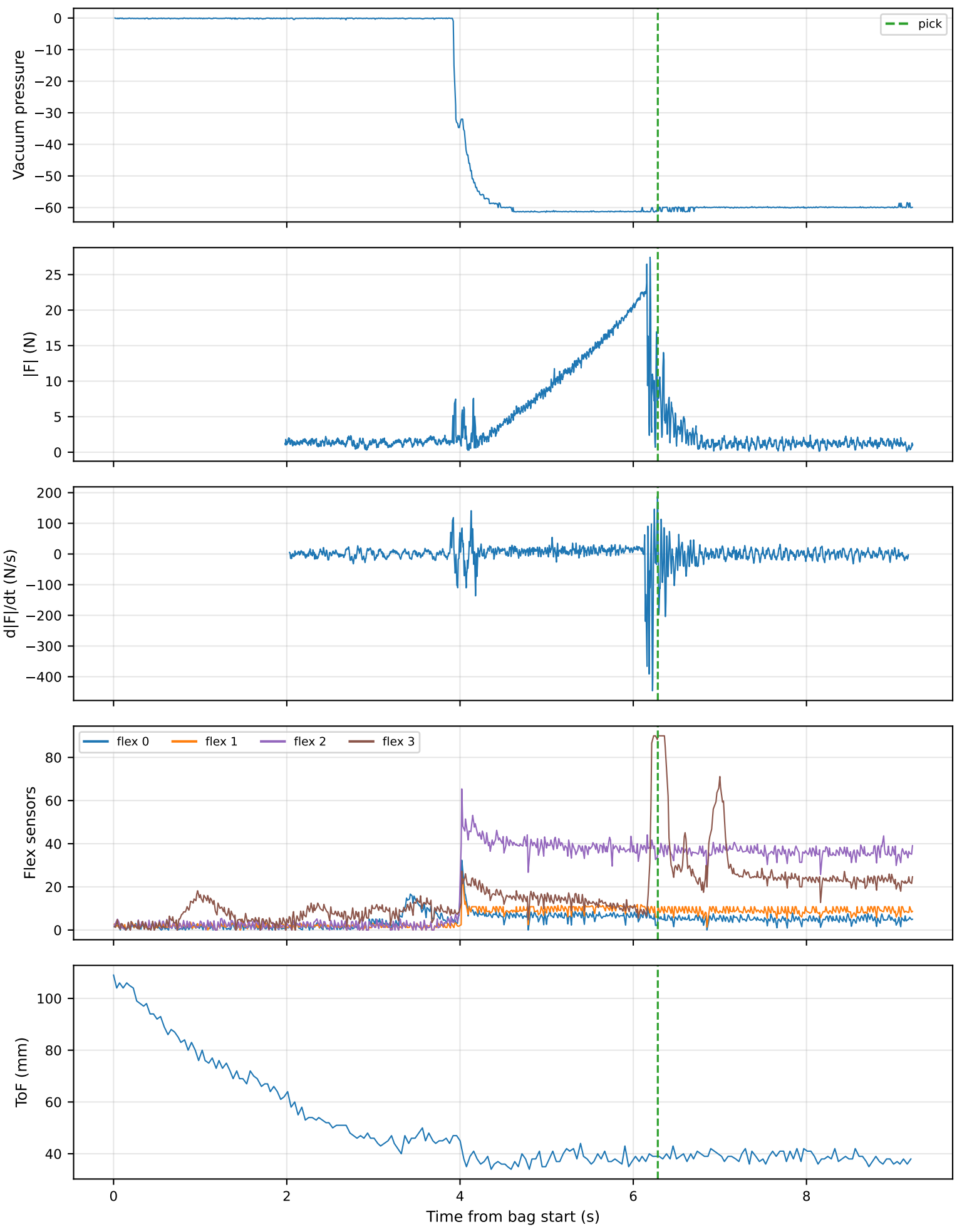
PICK at 6.03 s | 2026-10-07 16:38:36 | batch 155, apple 0 | pull_twist_controller

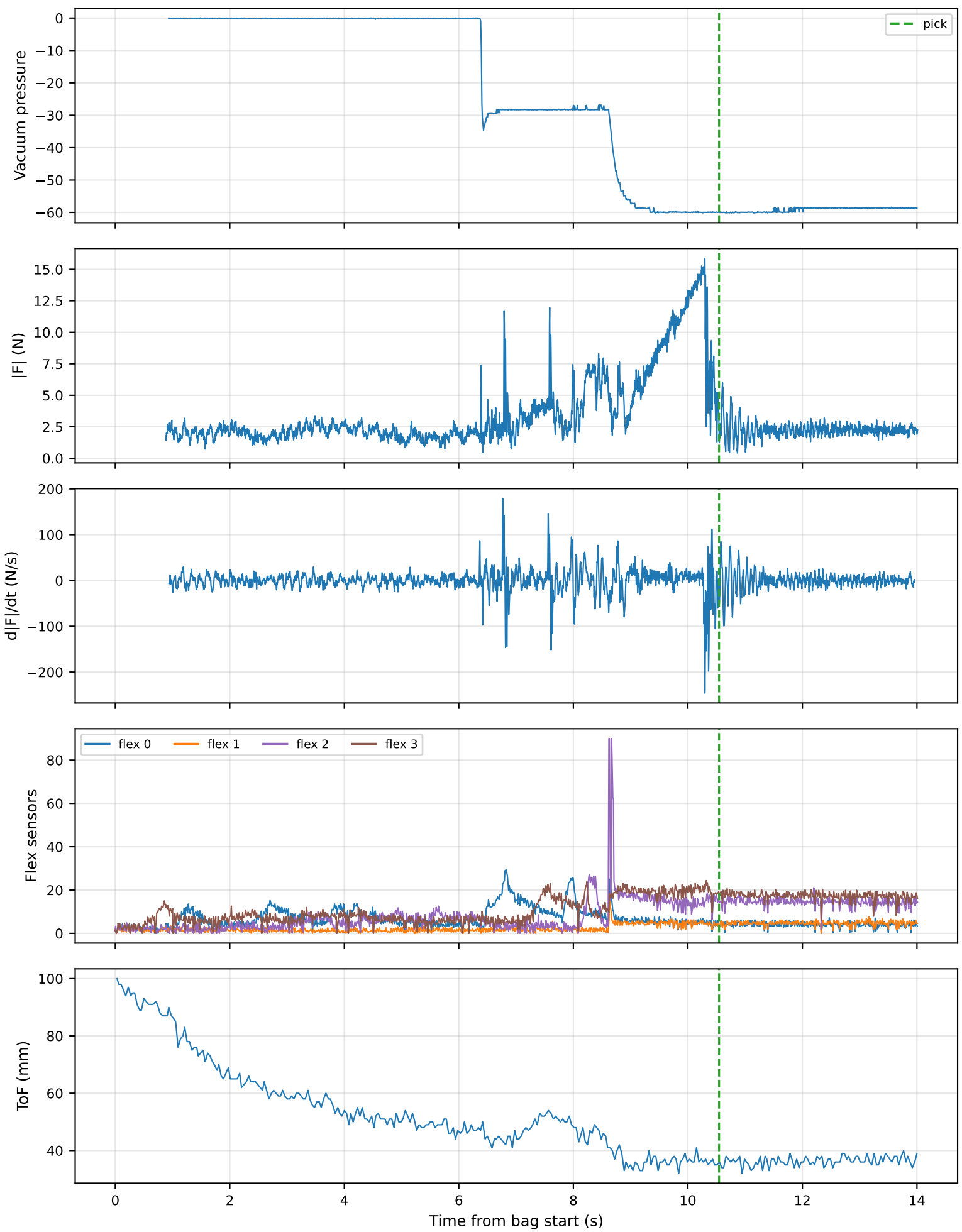


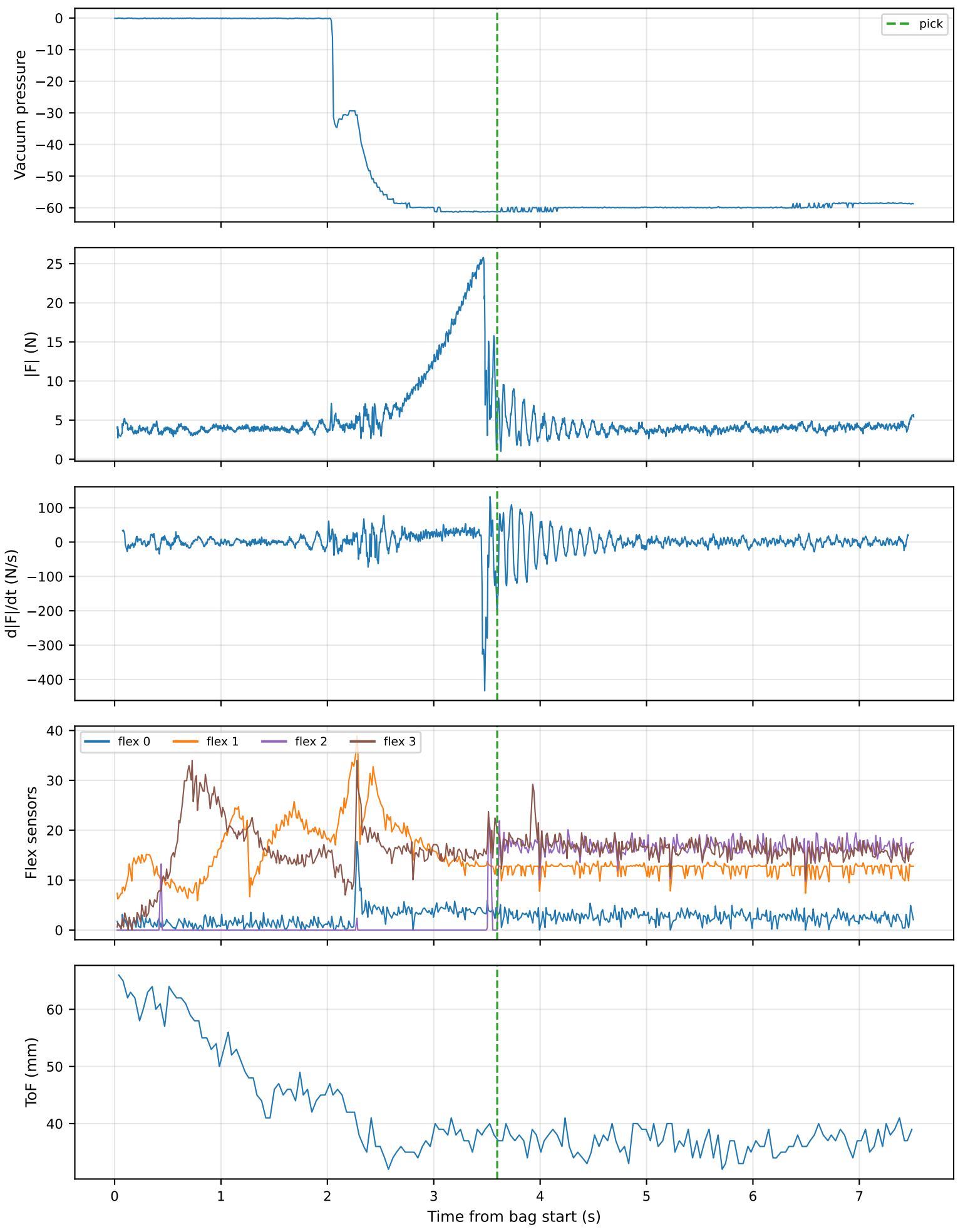

final_approach_and_pick_20261007_164117.db3
PICK at 5.35 s | 2026-10-07 16:41:18 | batch 157, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

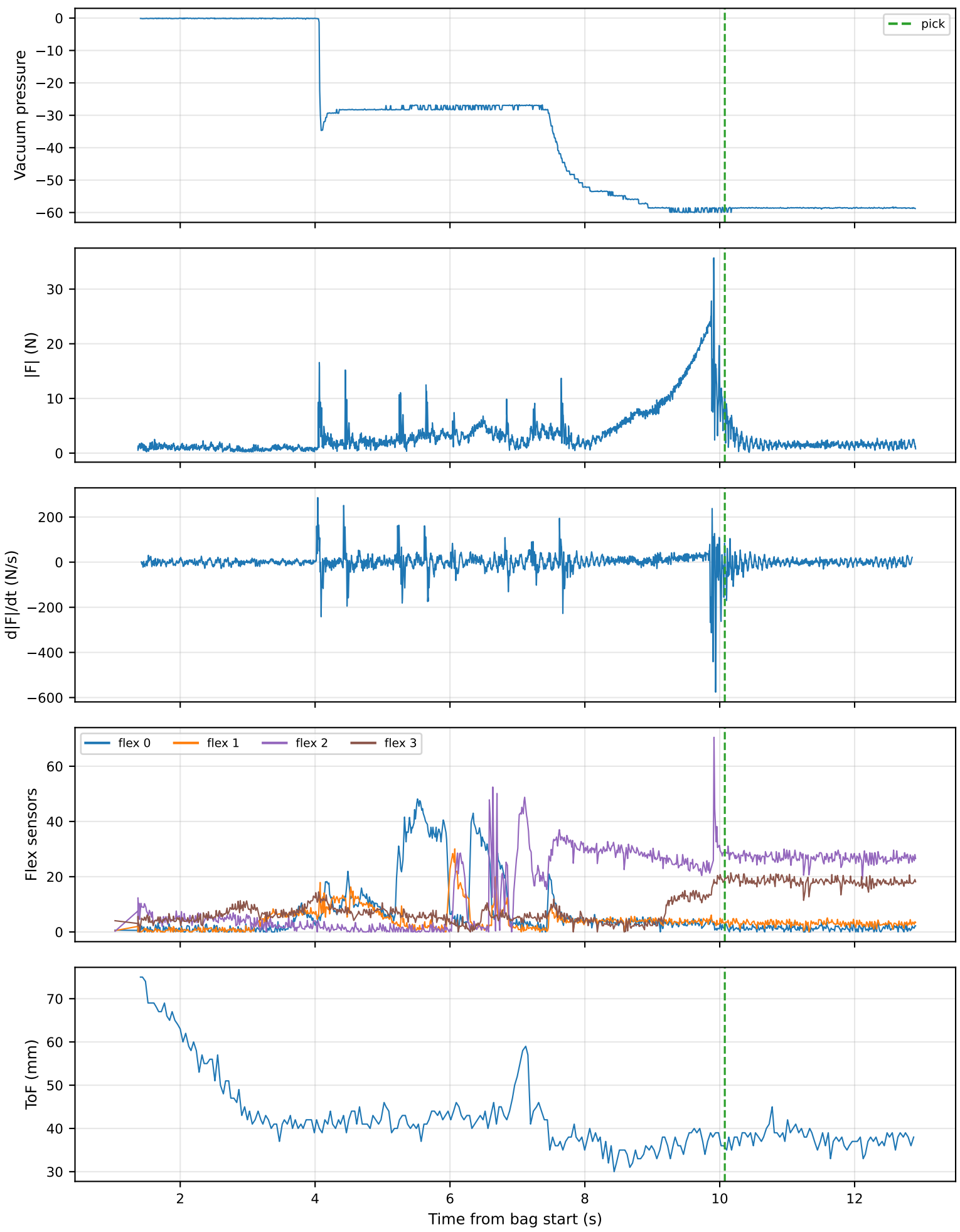

final_approach_and_pick_20261007_164230.db3
PICK at 8.57 s | 2026-10-07 16:42:32 | batch 158, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

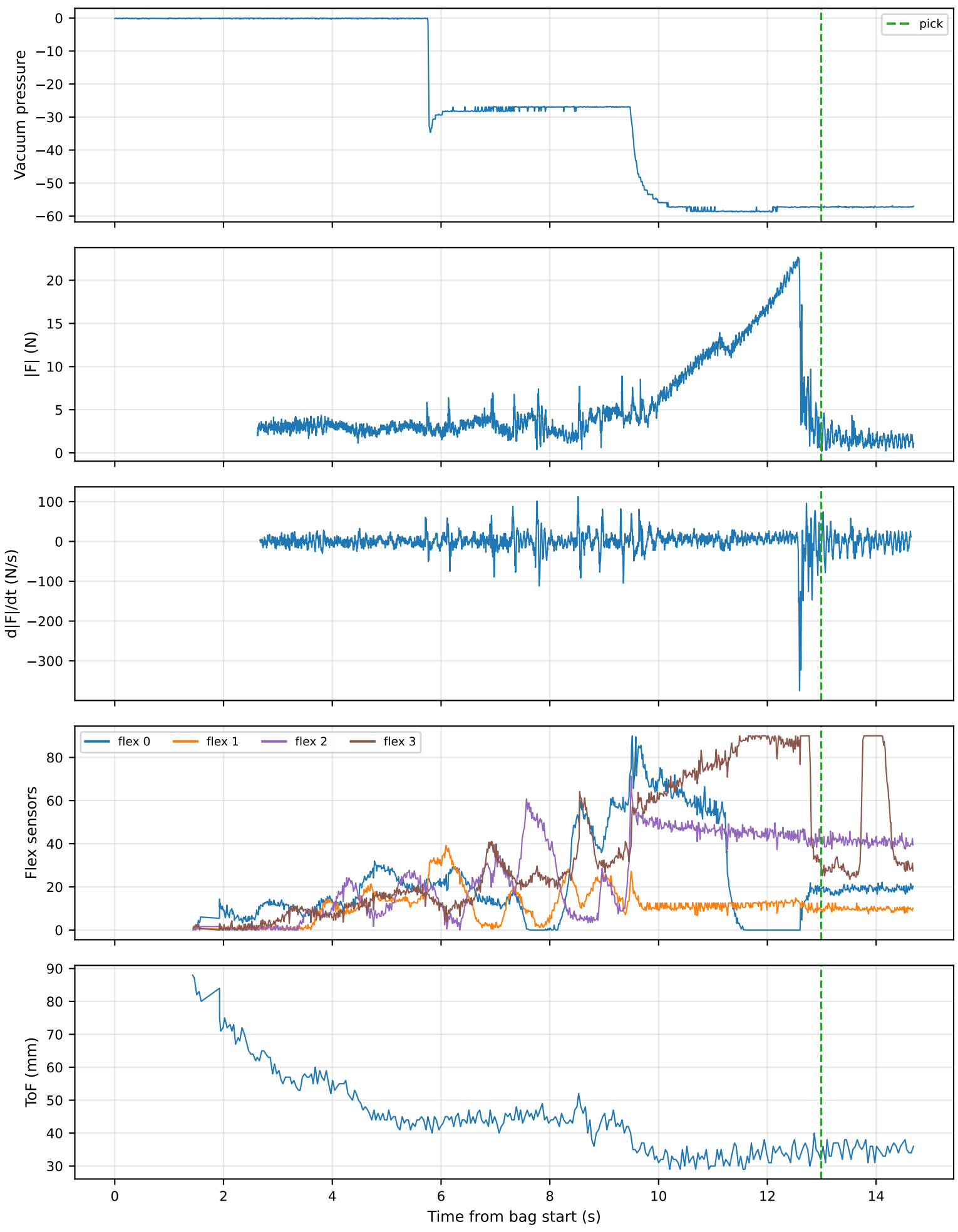

final_approach_and_pick_20261007_172046.db3
PICK at 16.54 s | 2026-10-07 17:20:47 | batch 159, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

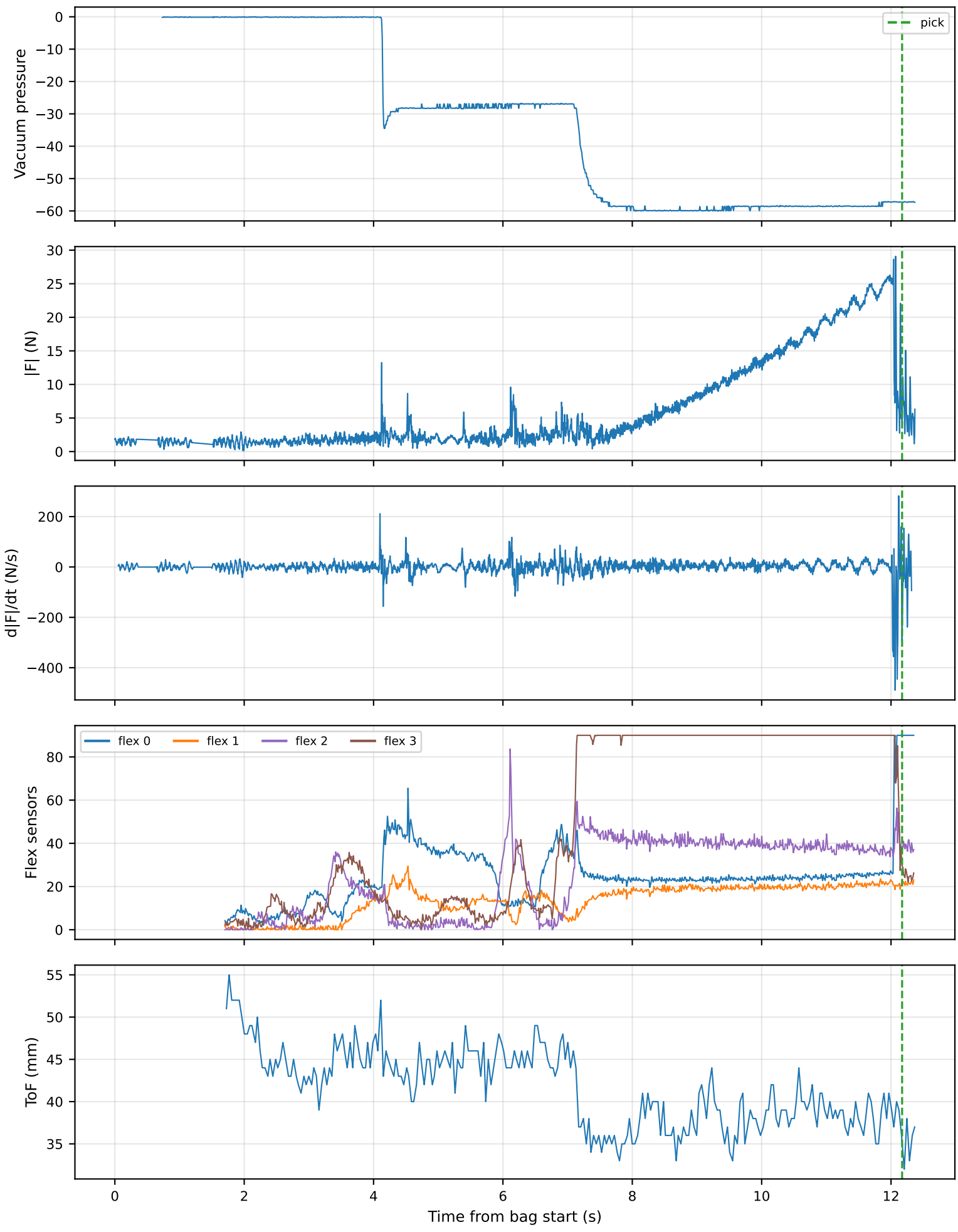

final_approach_and_pick_20261007_172433.db3
PICK at 6.28 s | 2026-10-07 17:24:34 | batch 160, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

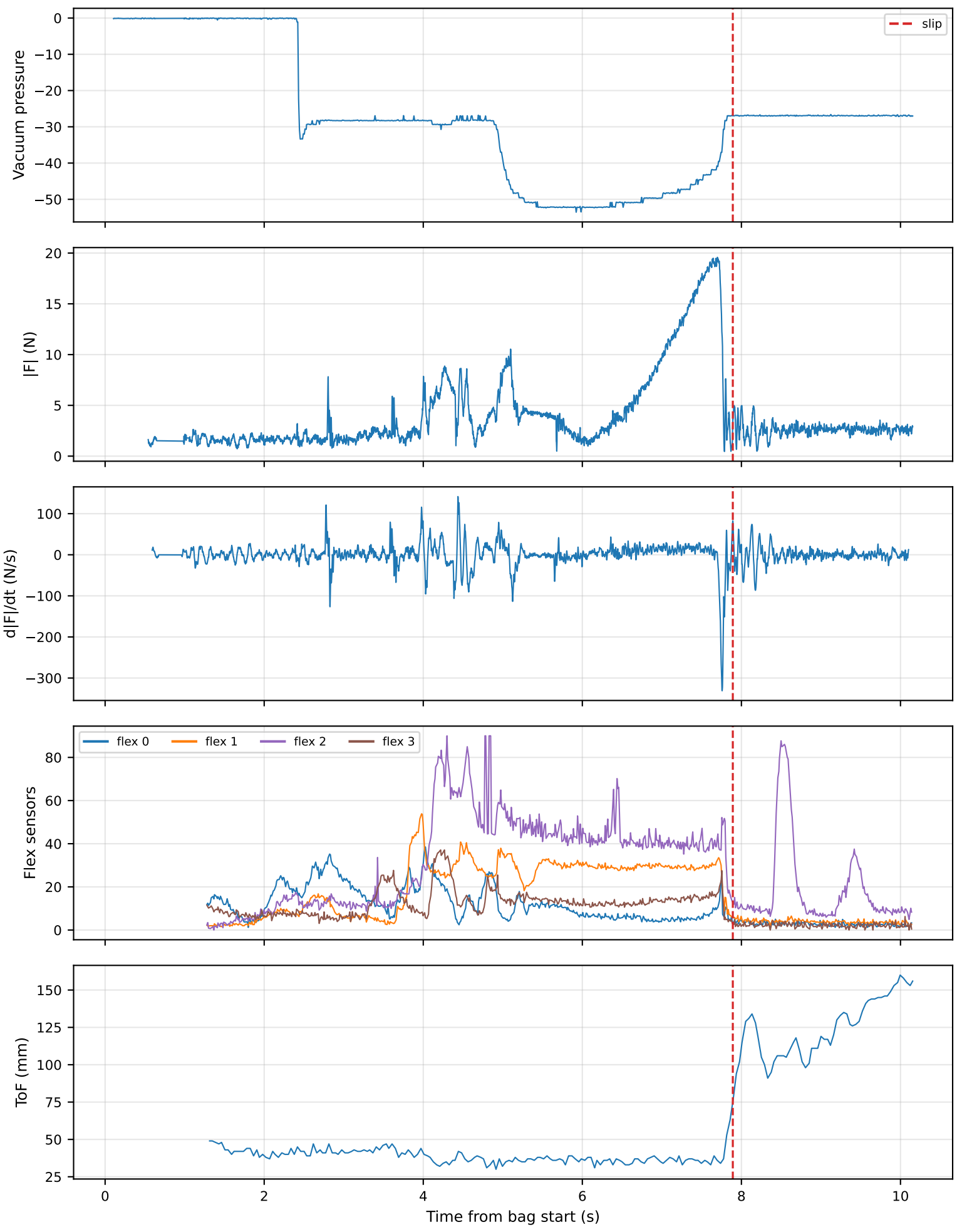

final_approach_and_pick_20261007_172541.db3
PICK at 10.54 s | 2026-10-07 17:25:42 | batch 161, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
flex 0
flex 1
flex 2
flex 3
Flex sensors
ToF (mm)
Time from bag start (s)

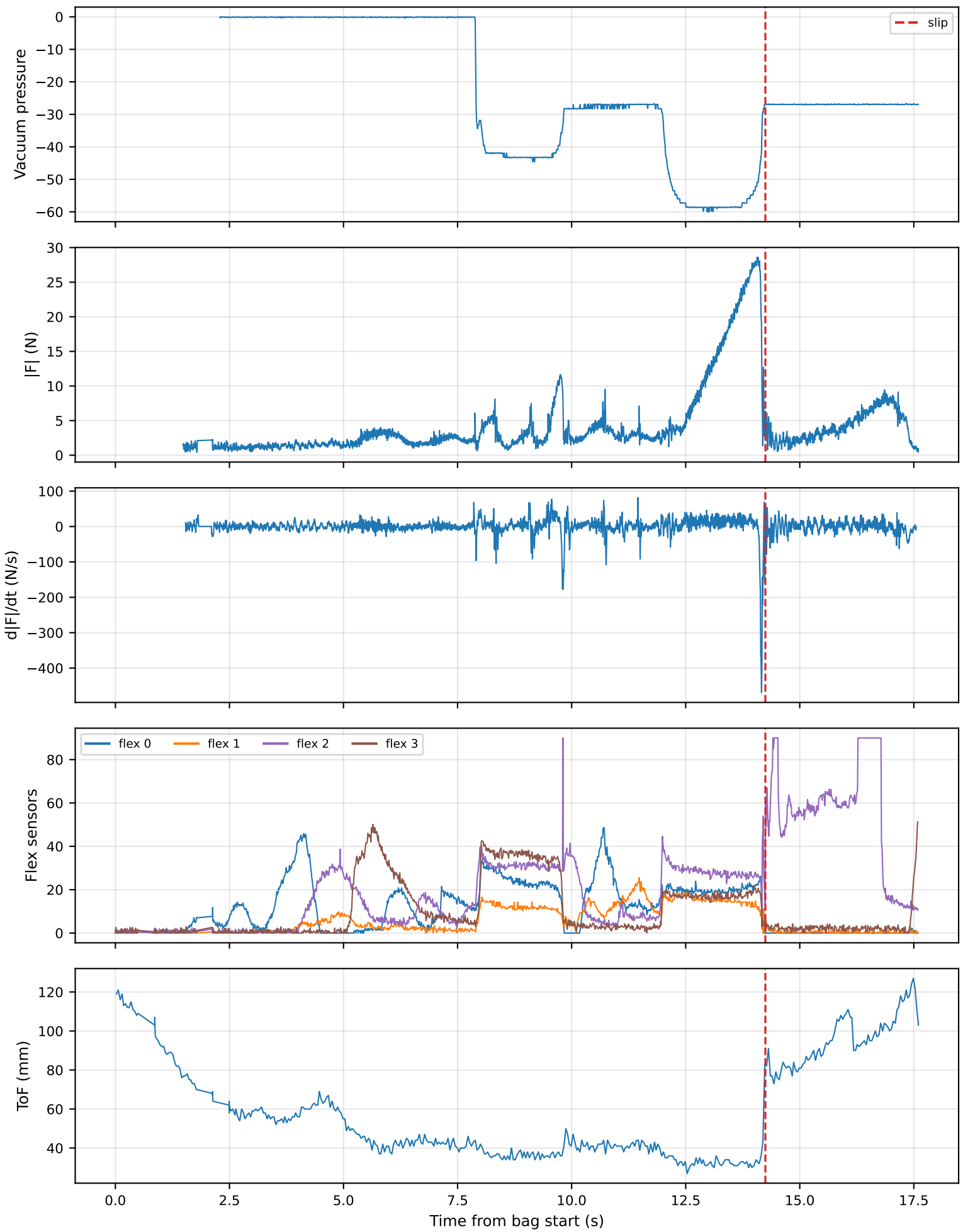

final_approach_and_pick_20261007_172743.db3
PICK at 3.60 s | 2026-10-07 17:27:45 | batch 163, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

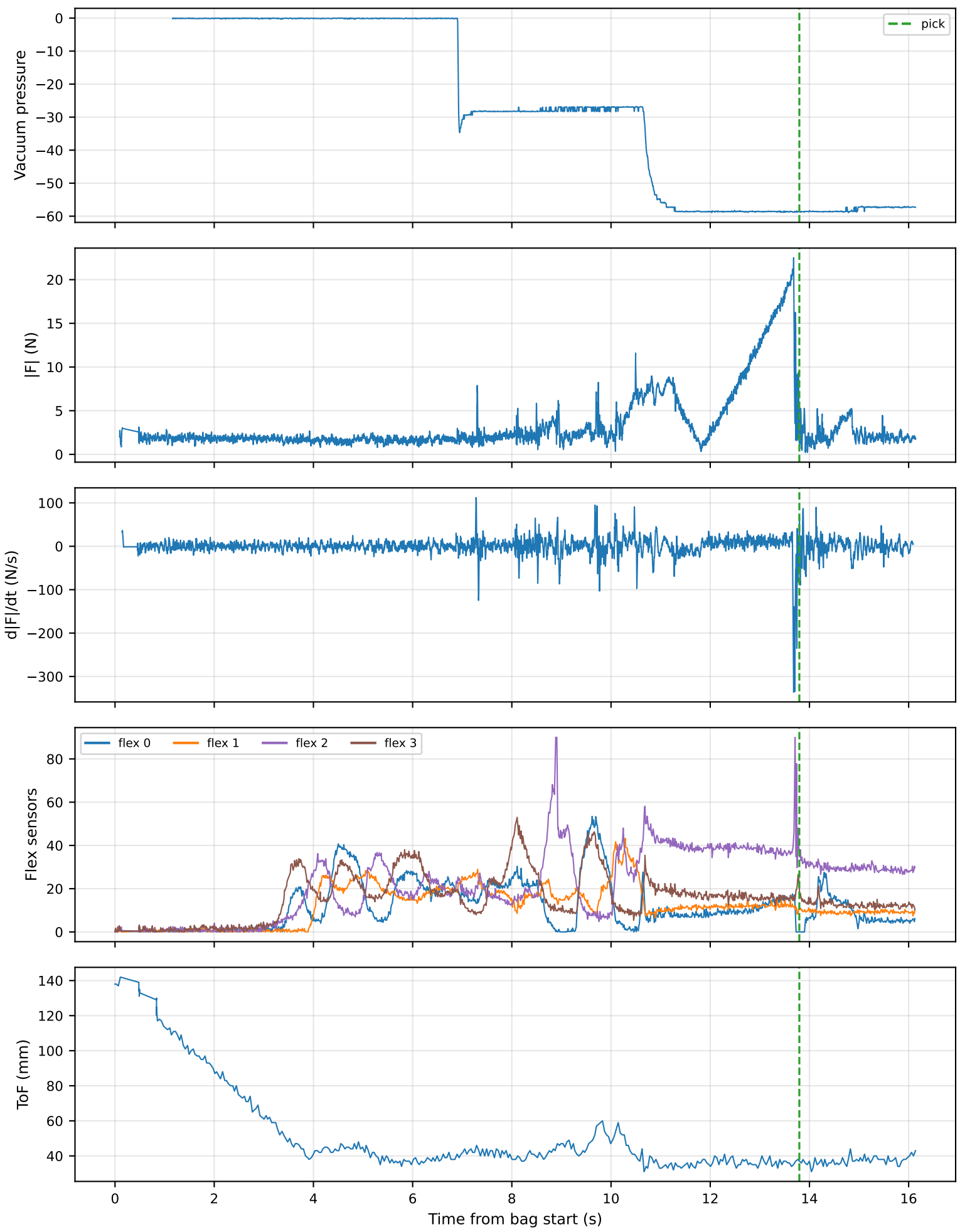

final_approach_and_pick_20261007_172839.db3
PICK at 10.08 s | 2026-10-07 17:28:40 | batch 164, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_172932.db3
PICK at 12.99 s | 2026-10-07 17:29:33 | batch 165, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_173025.db3
PICK at 12.17 s | 2026-10-07 17:30:27 | batch 166, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

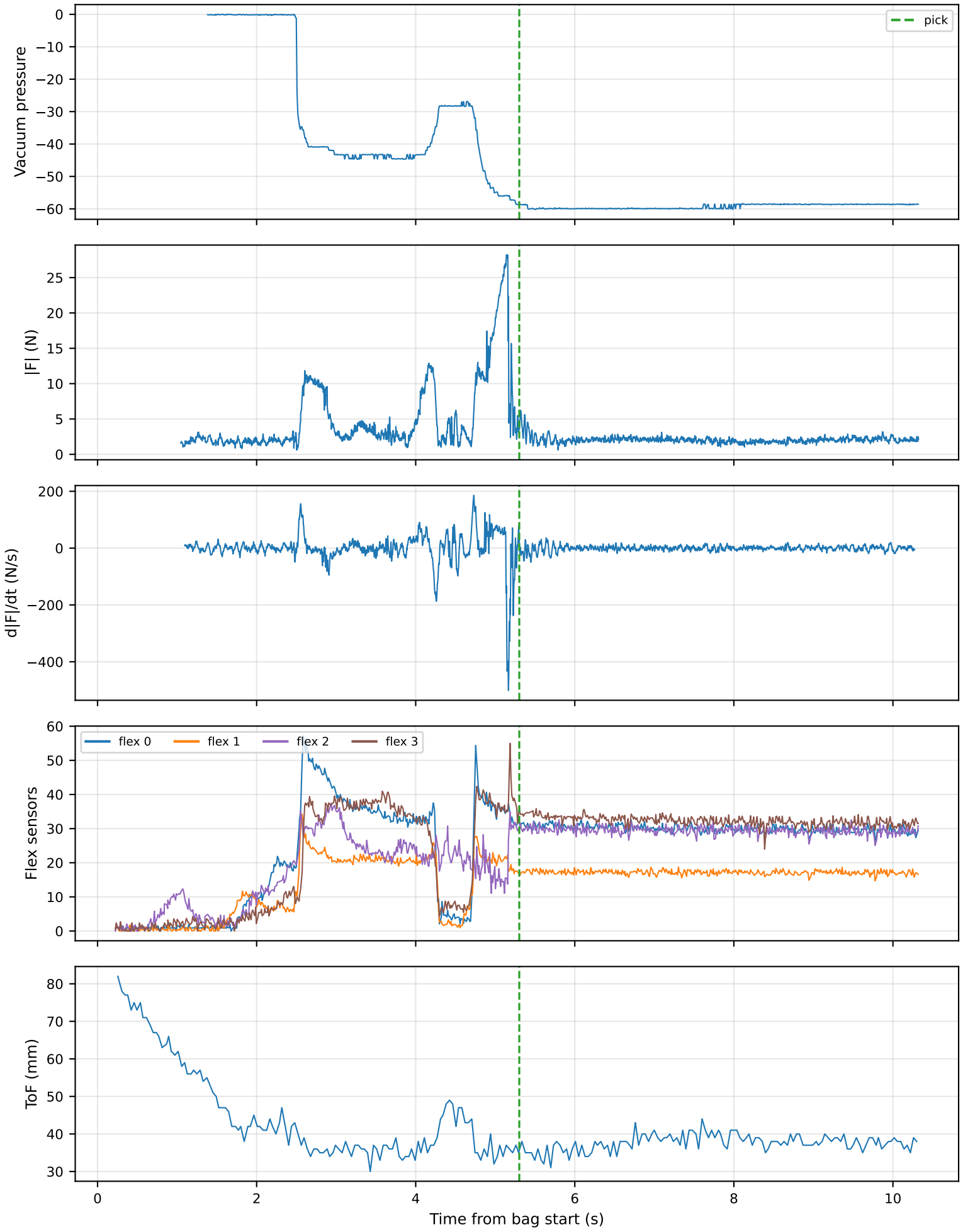

final_approach_and_pick_20261007_174828.db3
SLIP at 7.89 s | 2026-10-07 17:48:30 | batch 167, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

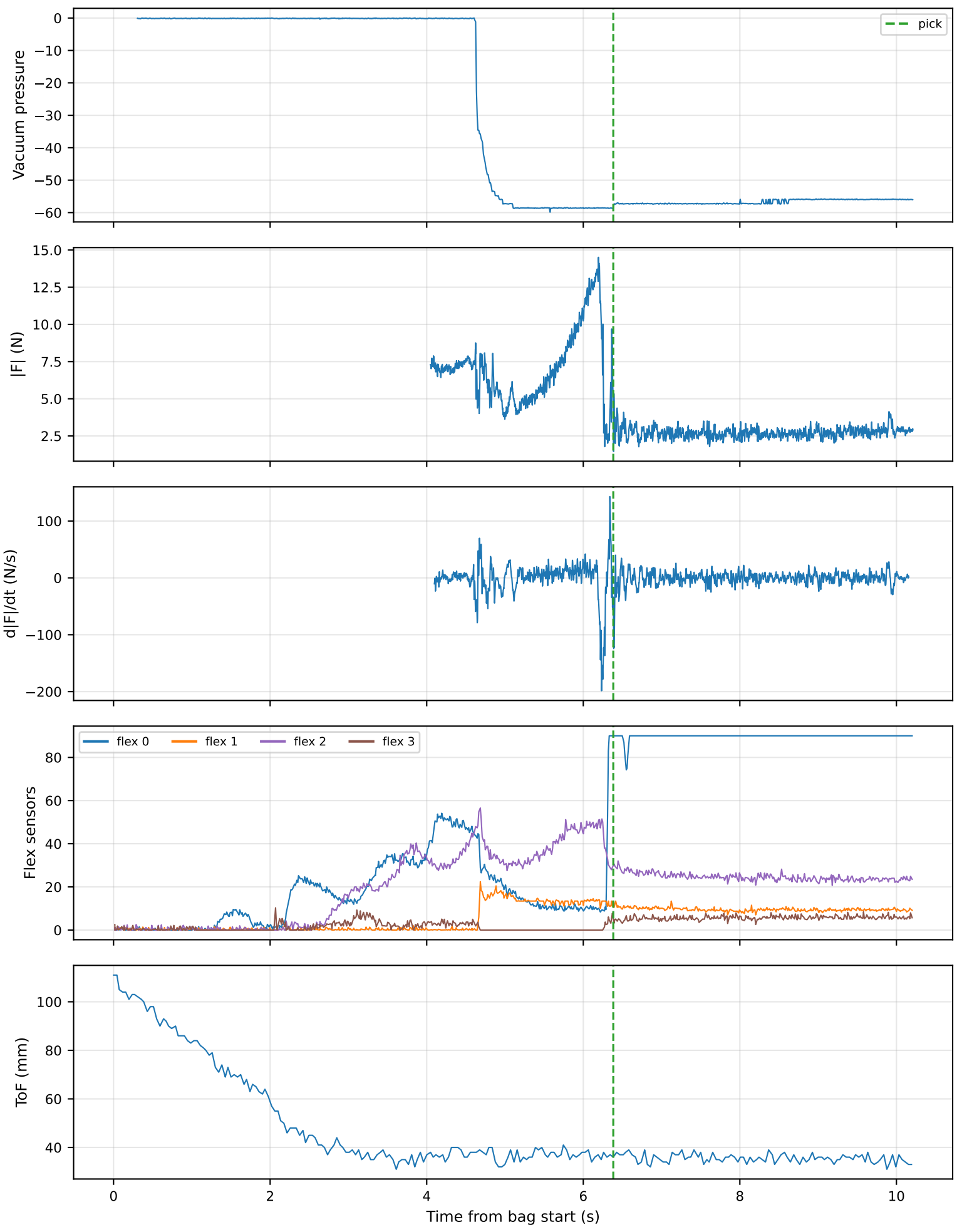

final_approach_and_pick_20261007_175220.db3
SLIP at 14.25 s | 2026-10-07 17:52:20 | batch 169, apple 0 | stiffness_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

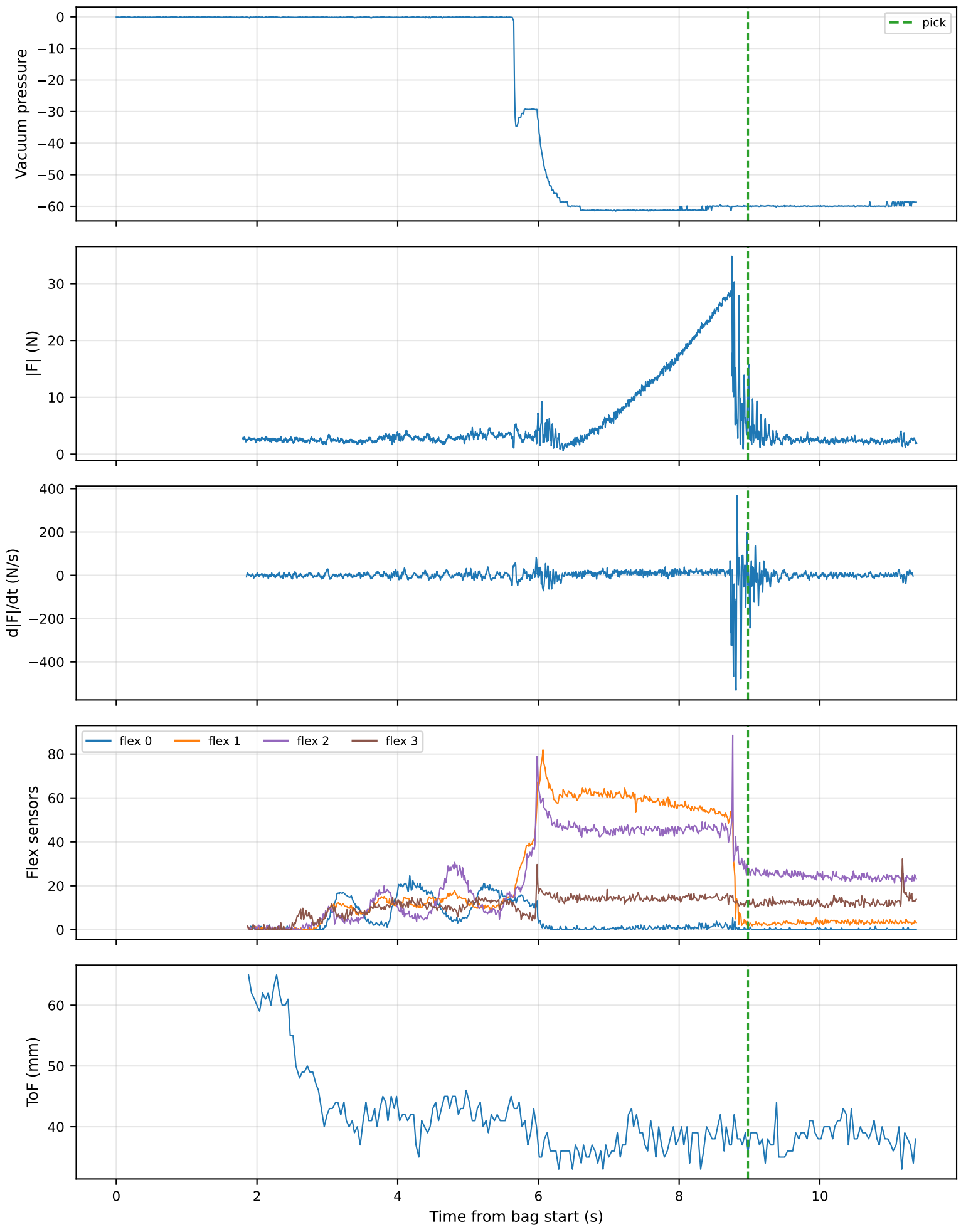

final_approach_and_pick_20261007_175632.db3
PICK at 13.80 s | 2026-10-07 17:56:33 | batch 171, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

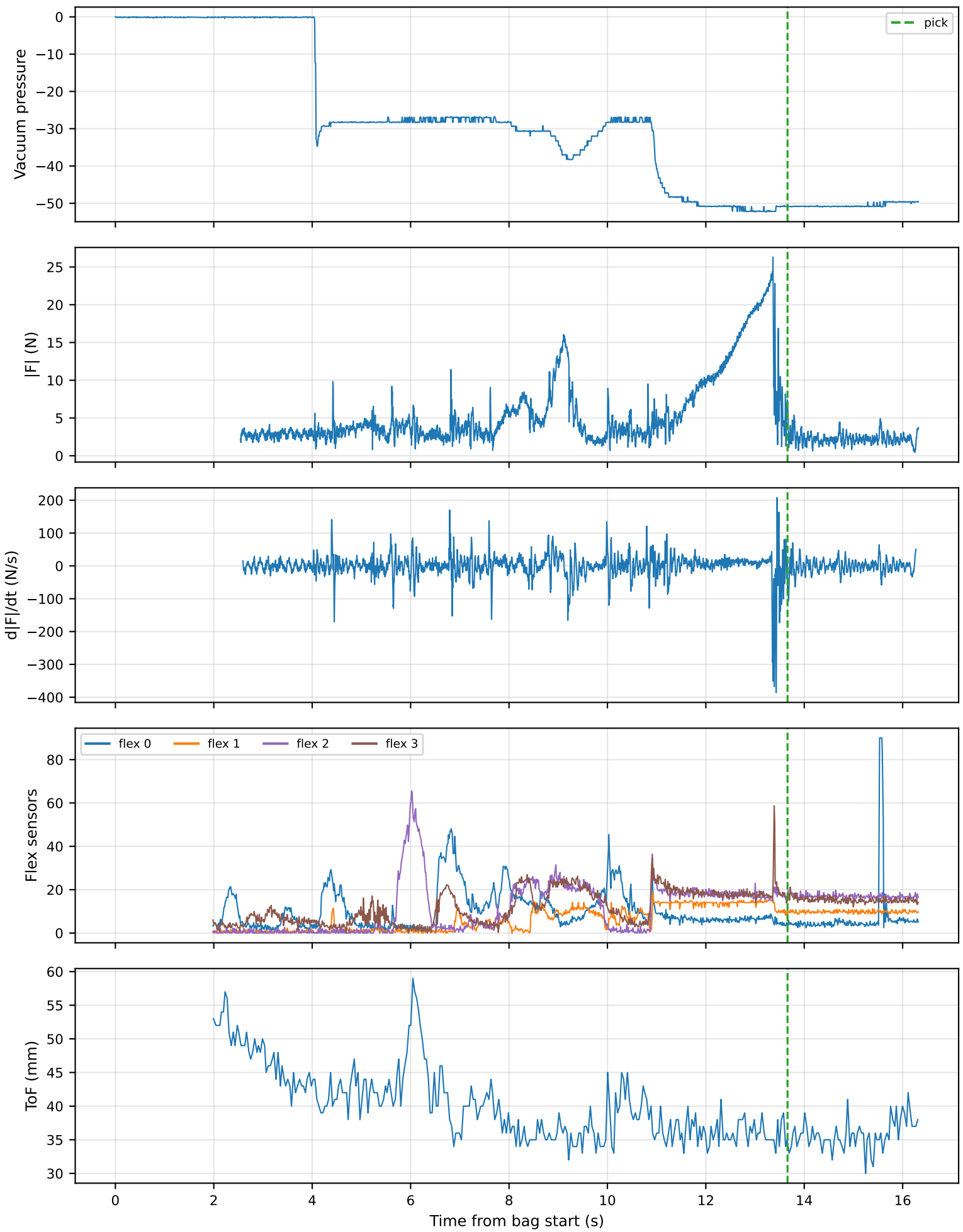
final_approach_and_pick_20261007_175759.db3

PICK at 5.30 s | 2026-10-07 17:58:00 | batch 172, apple 0 | stiffness_controller



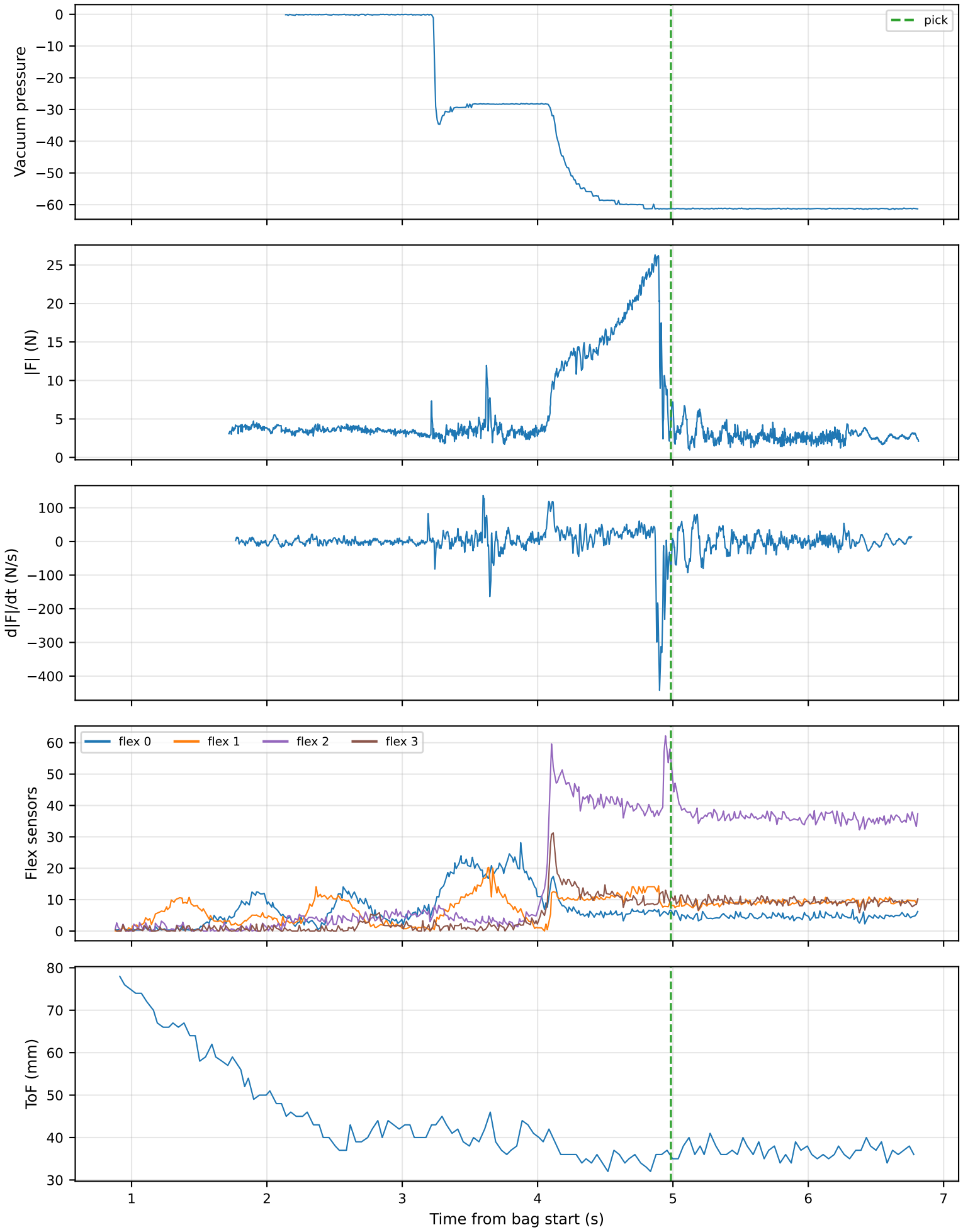

final_approach_and_pick_20261007_180009.db3
PICK at 6.38 s | 2026-10-07 18:00:10 | batch 173, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

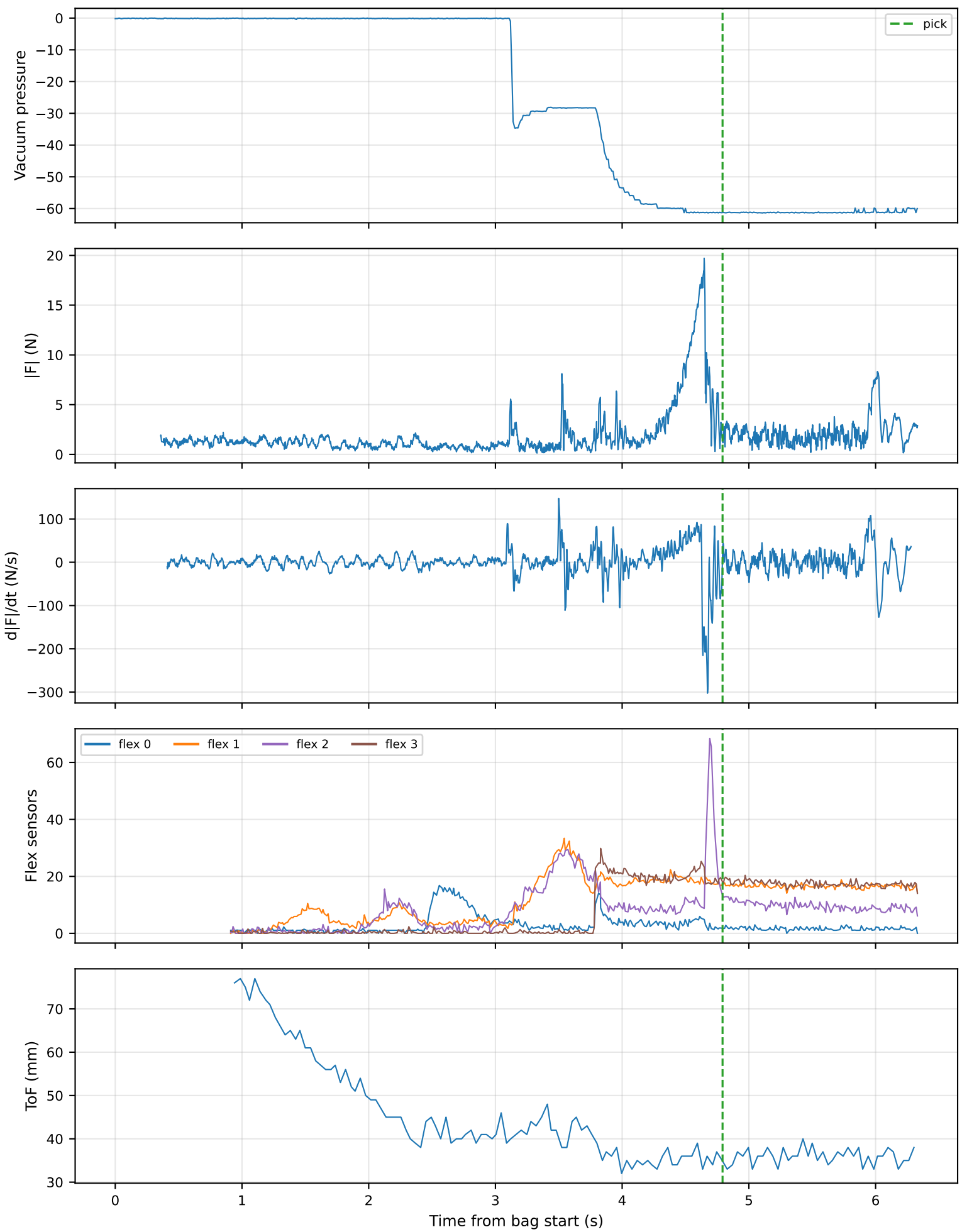

final_approach_and_pick_20261007_180434.db3
PICK at 8.98 s | 2026-10-07 18:04:35 | batch 176, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

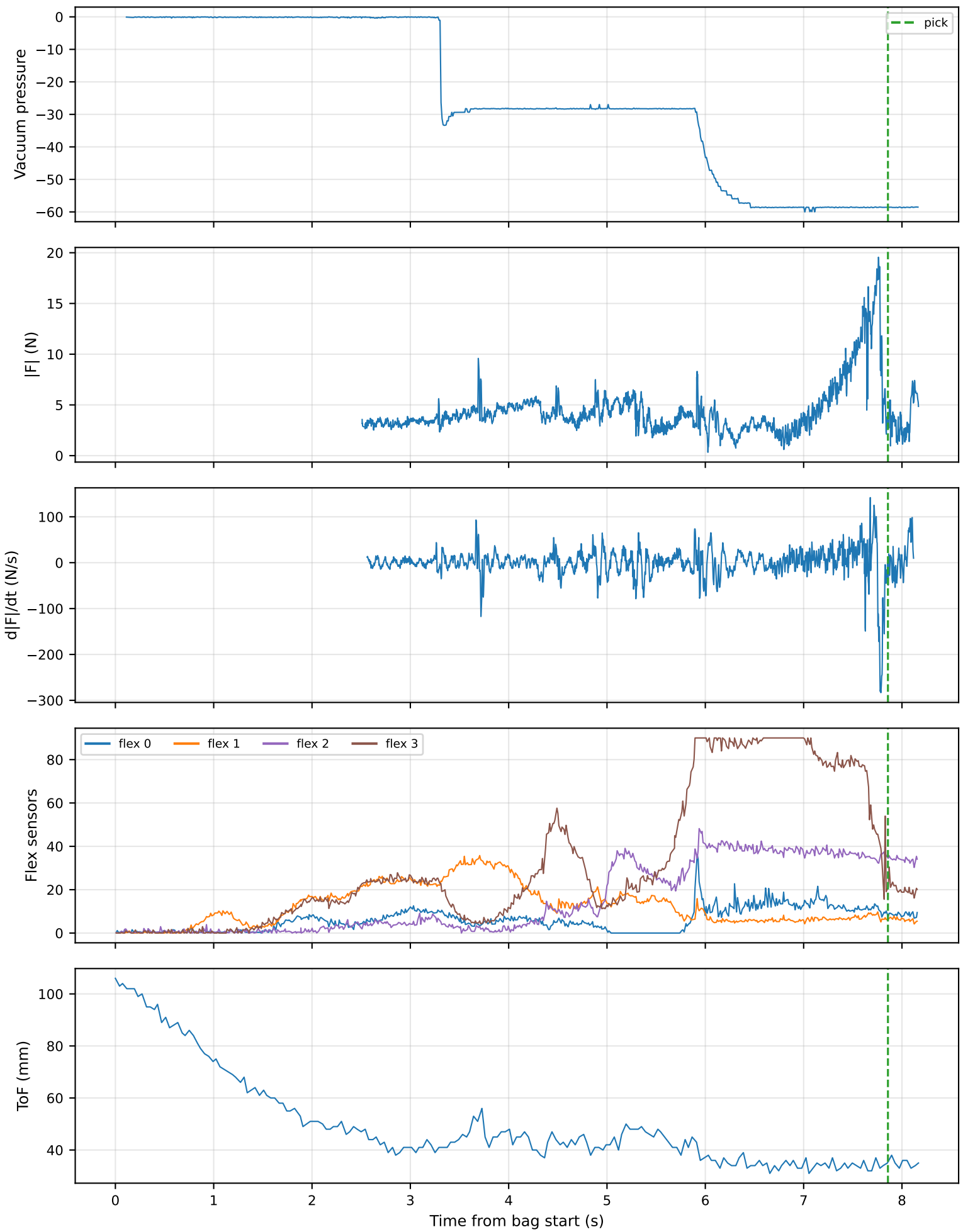

final_approach_and_pick_20261007_180545.db3
PICK at 13.66 s | 2026-10-07 18:05:46 | batch 177, apple 0 | stiffness_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261007_180704.db3

PICK at 4.99 s | 2026-10-07 18:07:05 | batch 178, apple 0 | pull_twist_controller

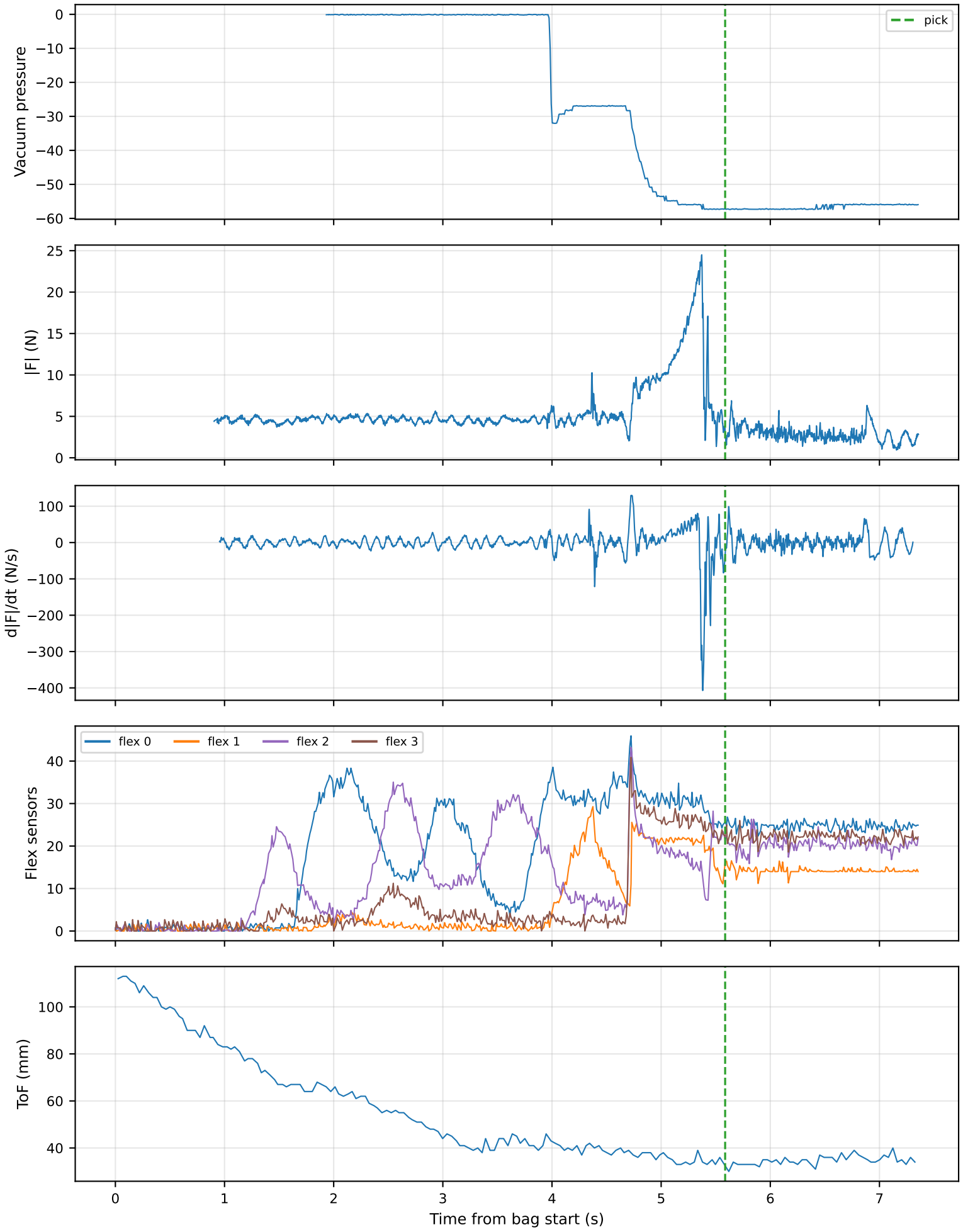


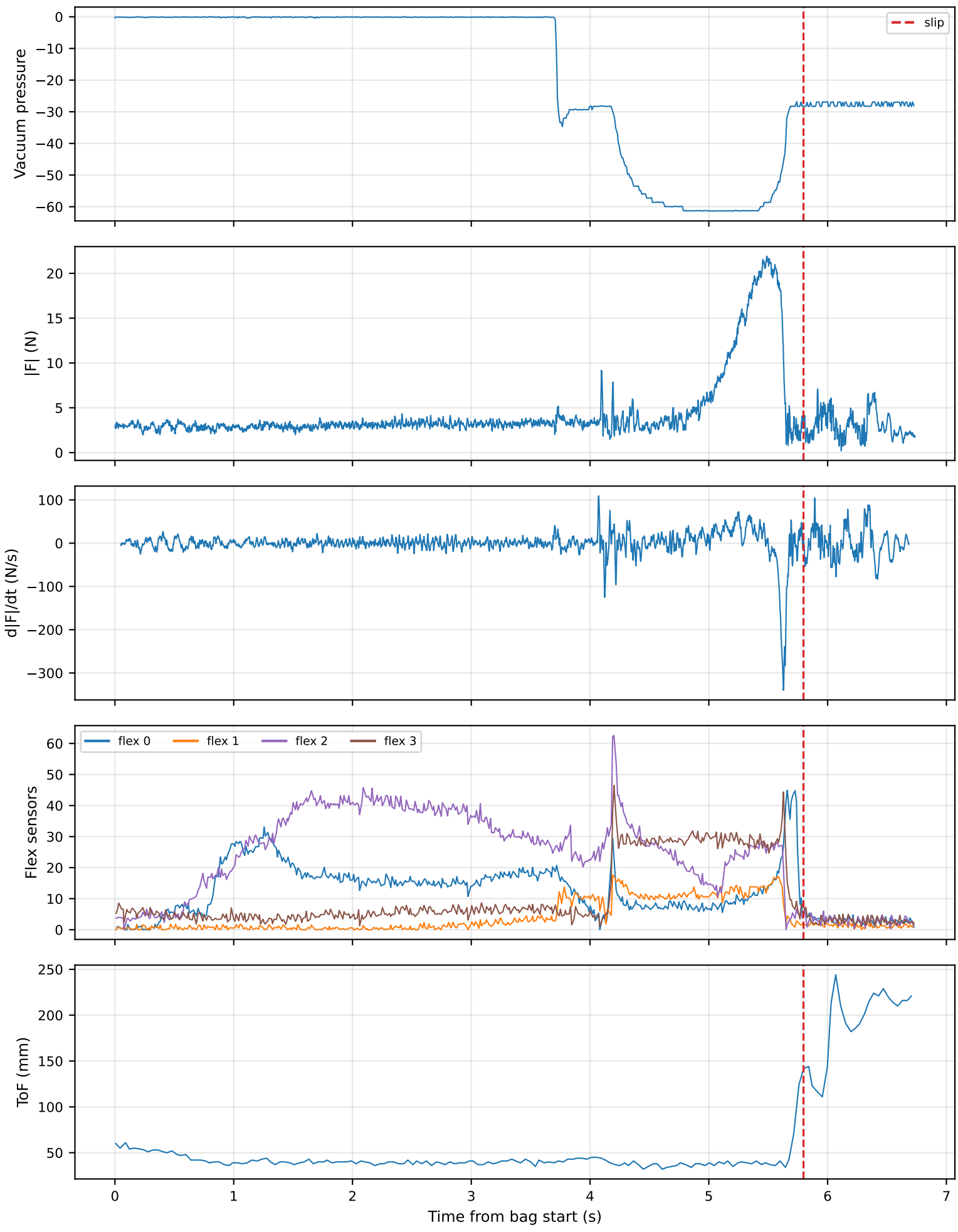

final_approach_and_pick_20261007_180916.db3
PICK at 4.79 s | 2026-10-07 18:09:17 | batch 180, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

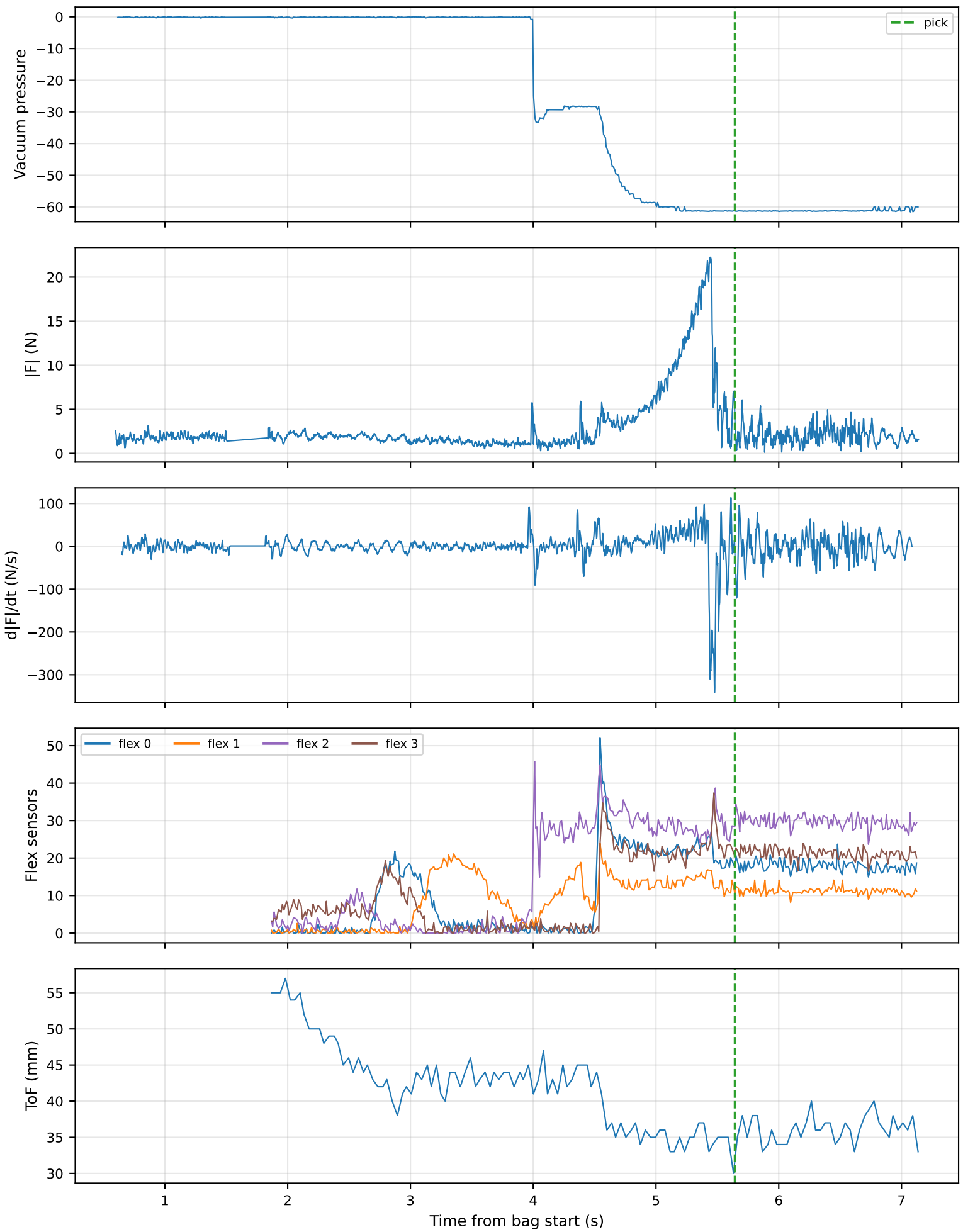

final_approach_and_pick_20261007_181012.db3
PICK at 7.86 s | 2026-10-07 18:10:13 | batch 181, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261007_181157.db3

PICK at 5.59 s | 2026-10-07 18:11:57 | batch 182, apple 0 | pull_twist_controller

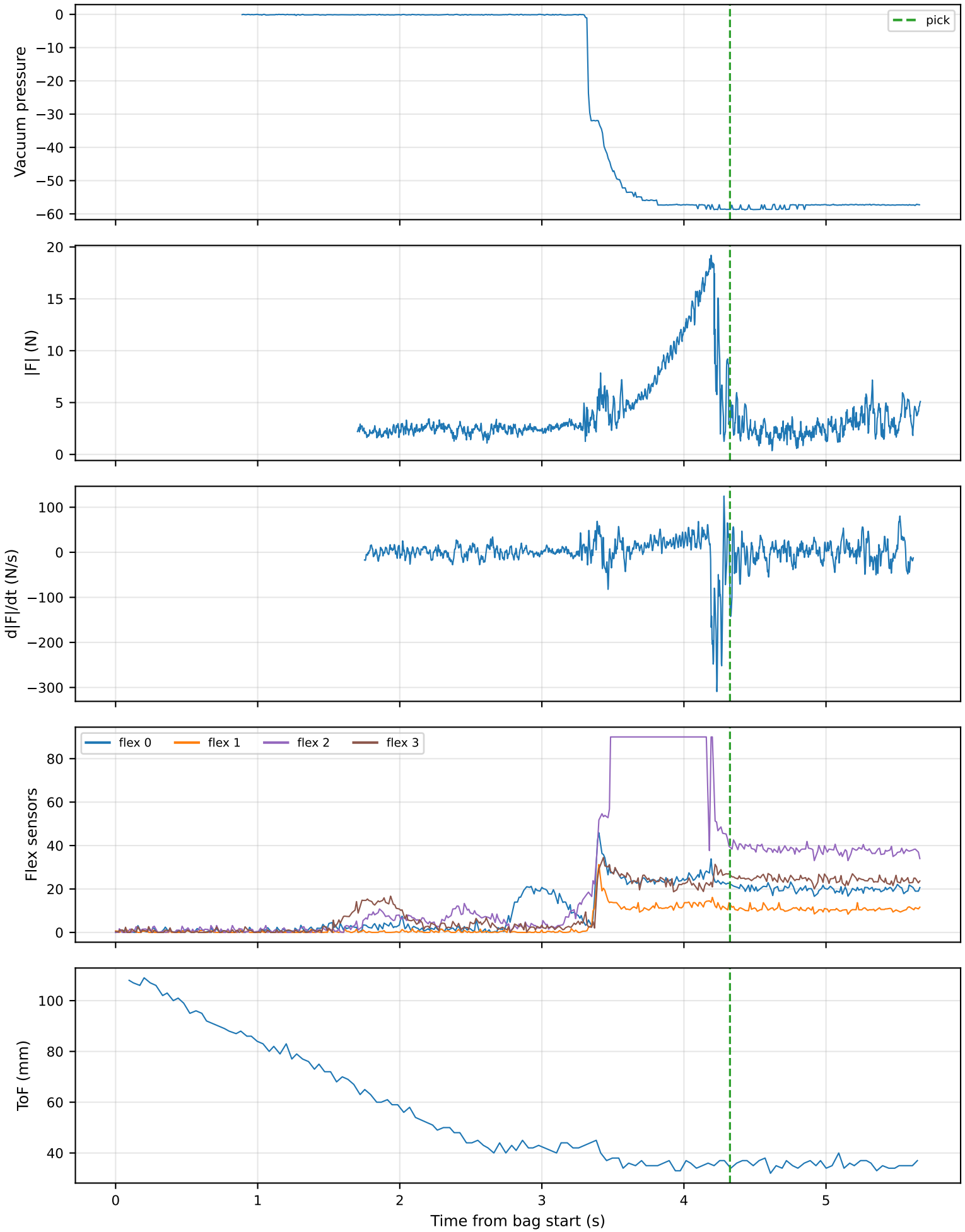


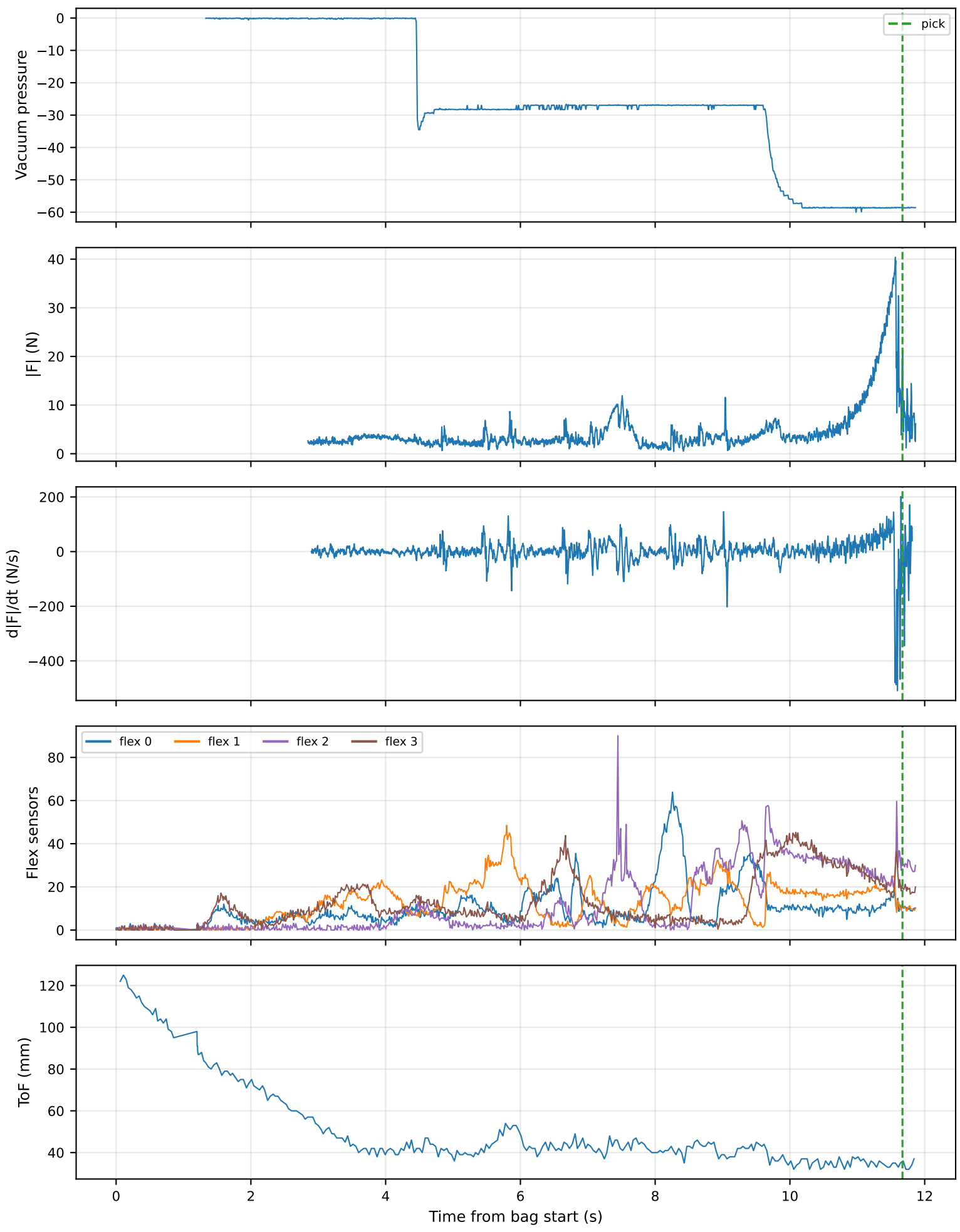

final_approach_and_pick_20261007_181750.db3
SLIP at 5.80 s | 2026-10-07 18:17:53 | batch 184, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

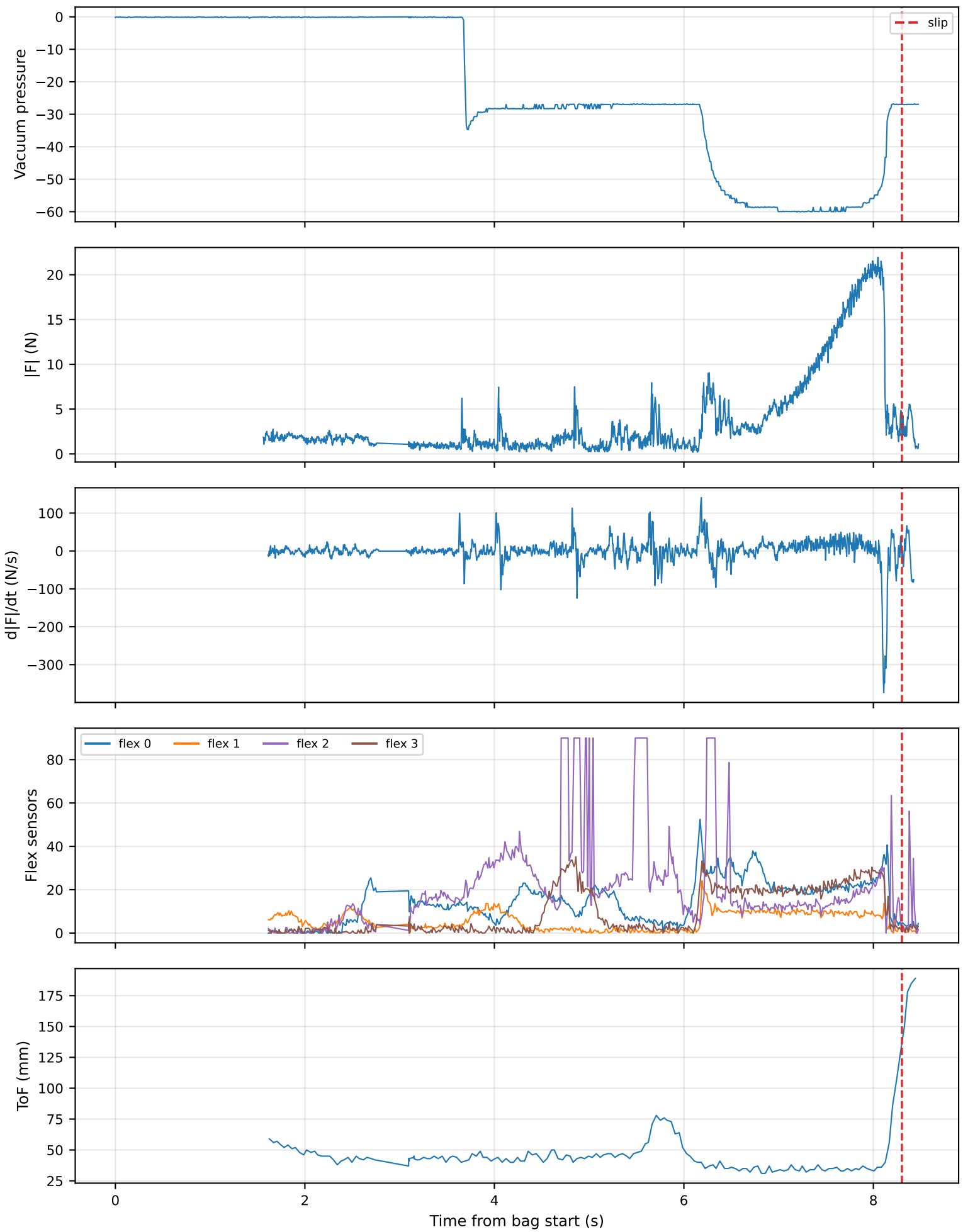

final_approach_and_pick_20261007_182519.db3
PICK at 5.64 s | 2026-10-07 18:25:20 | batch 186, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261007_182614.db3

PICK at 4.32 s | 2026-10-07 18:26:15 | batch 187, apple 0 | pull_twist_controller

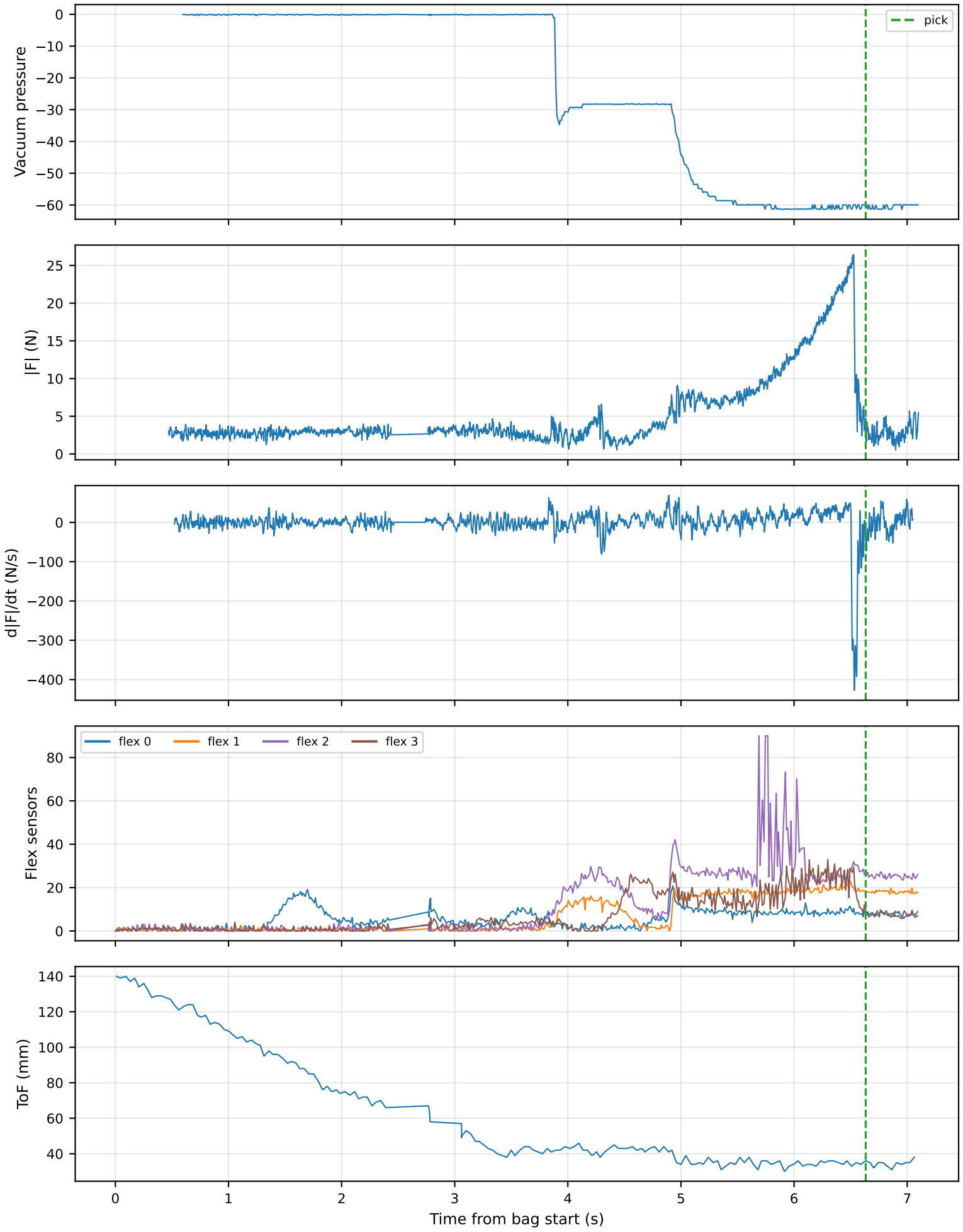


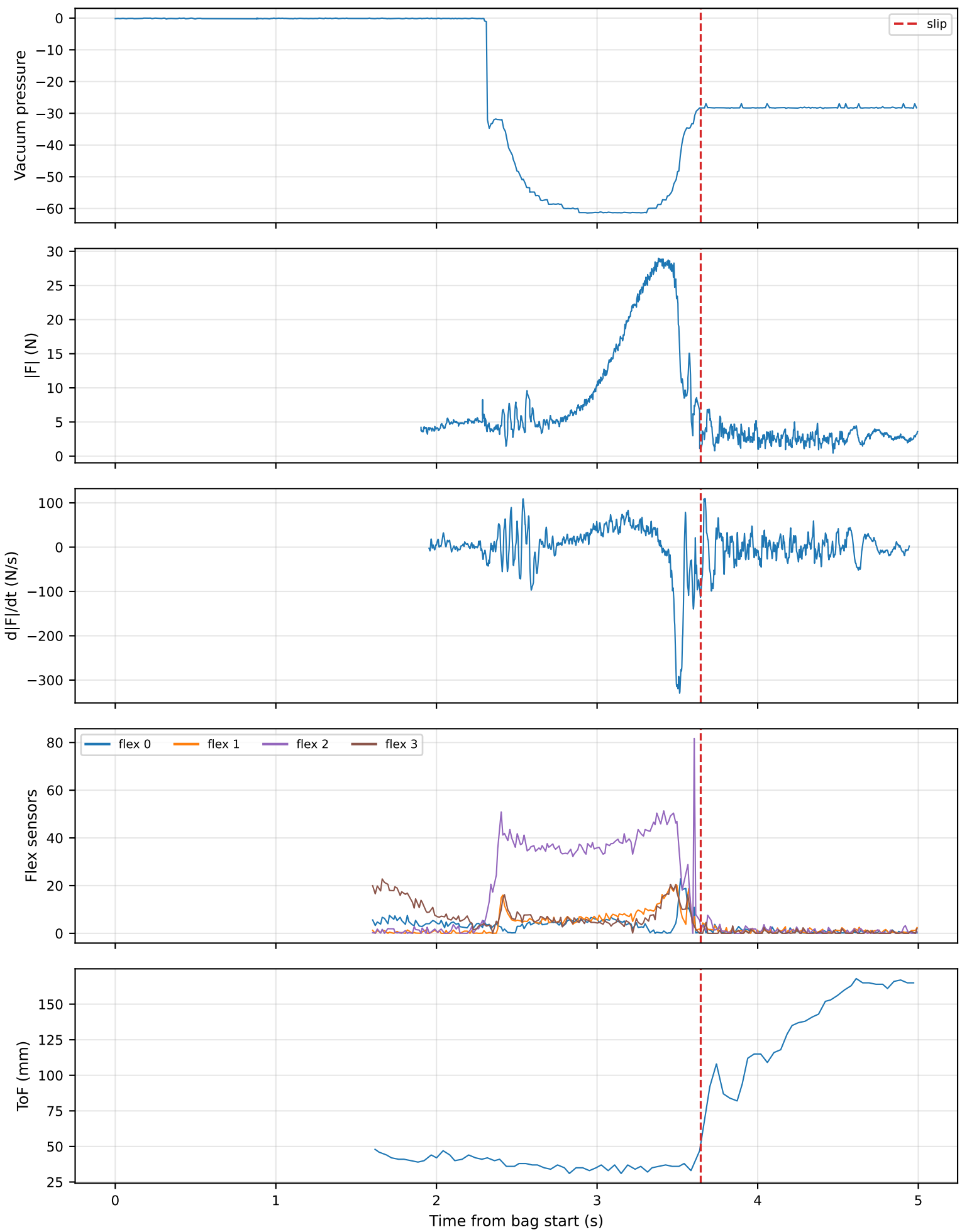

final_approach_and_pick_20261007_182705.db3
PICK at 11.67 s | 2026-10-07 18:27:06 | batch 188, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_182750.db3
SLIP at 8.30 s | 2026-10-07 18:27:50 | batch 189, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261007_182834.db3

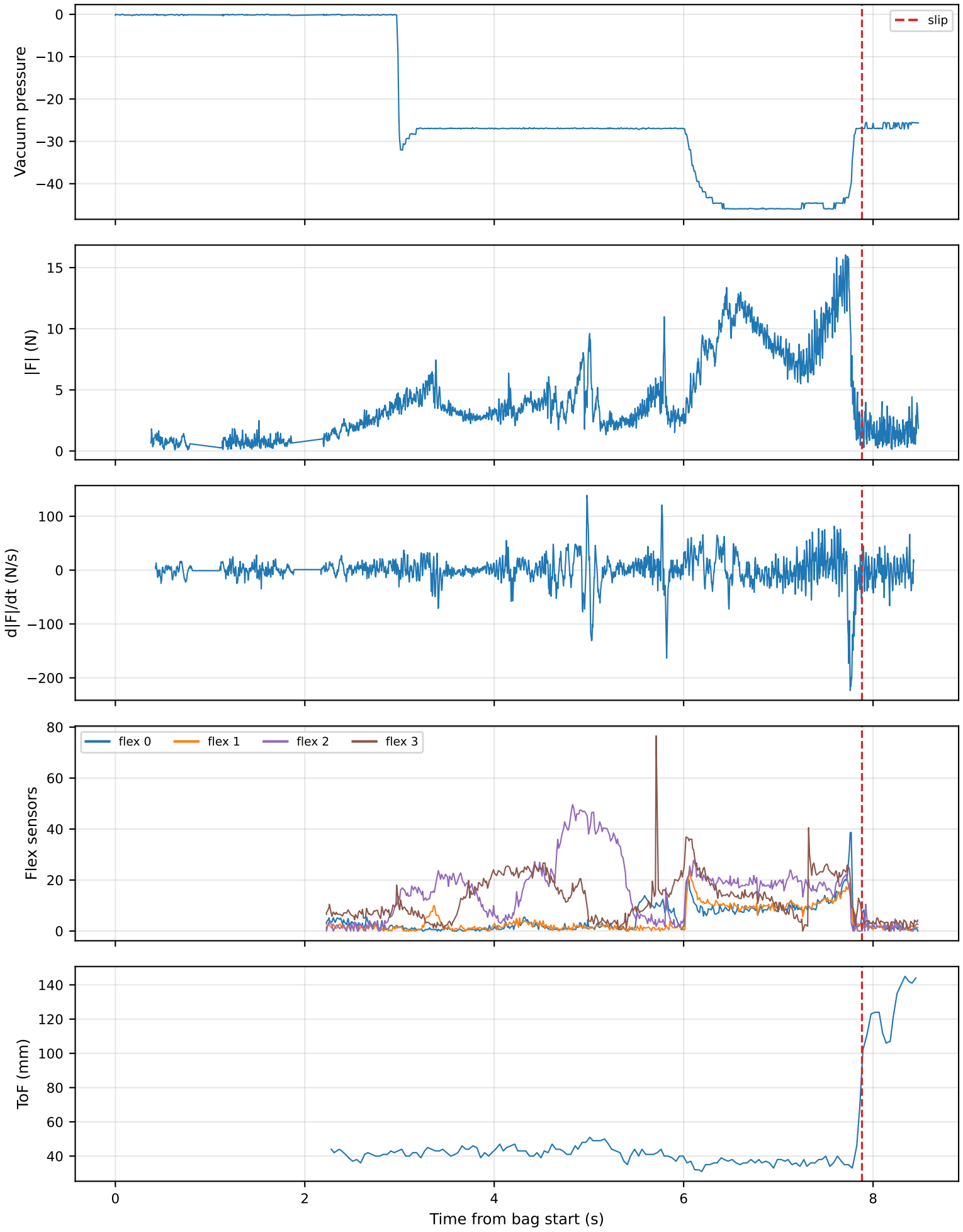
PICK at 6.63 s | 2026-10-07 18:28:35 | batch 190, apple 0 | pull_twist_controller

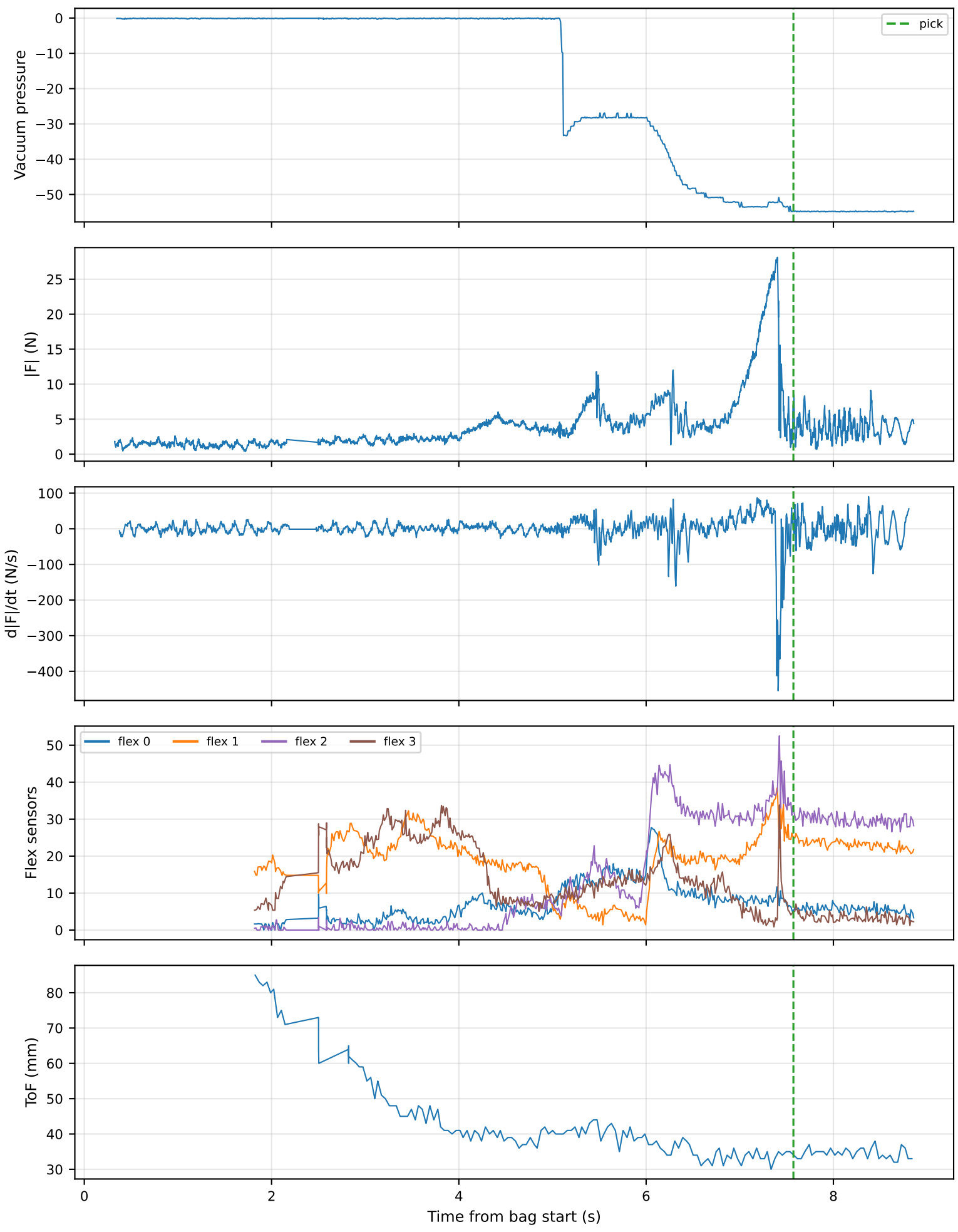


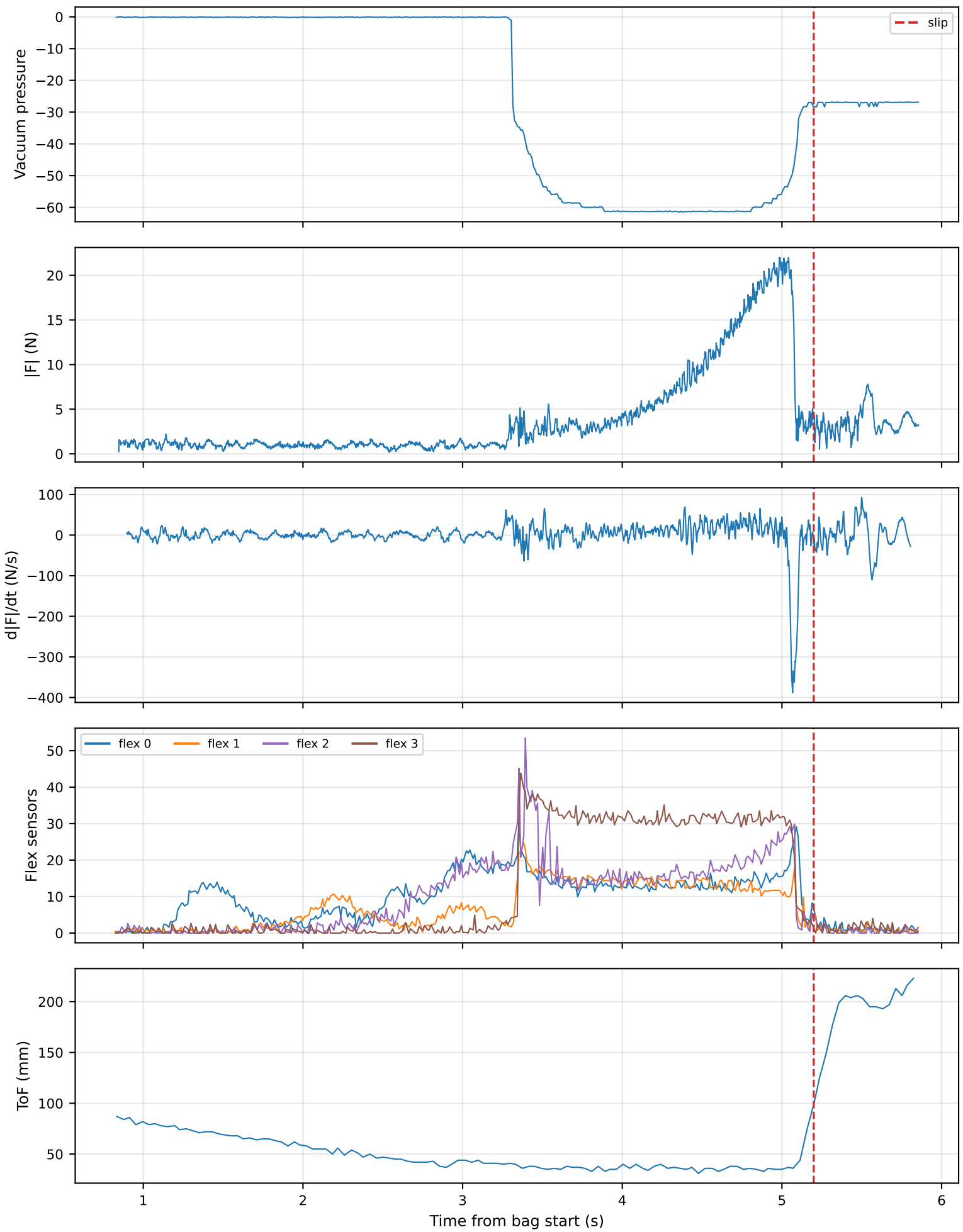

final_approach_and_pick_20261007_183043.db3
SLIP at 3.65 s | 2026-10-07 18:30:44 | batch 191, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

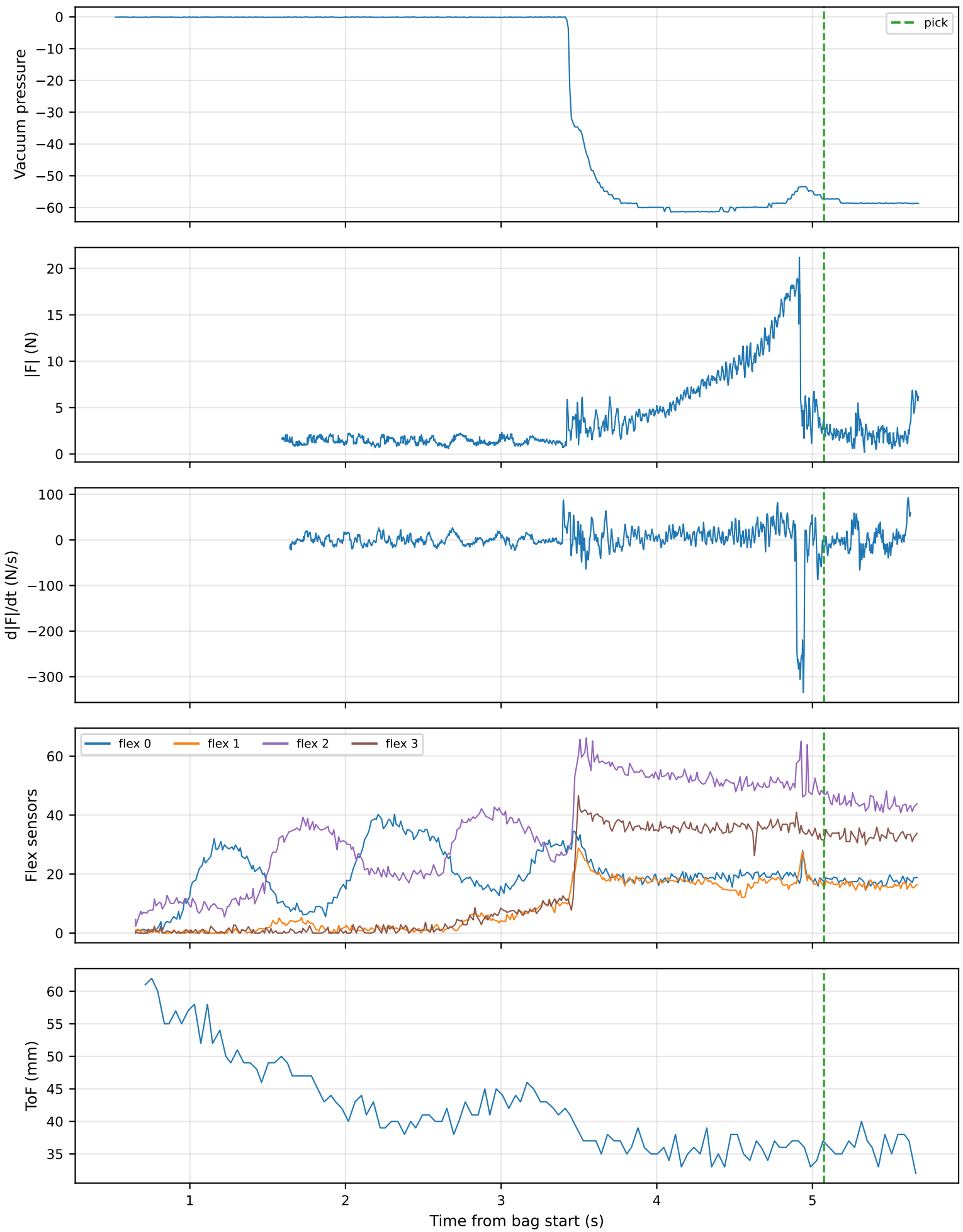
final_approach_and_pick_20261007_183134.db3

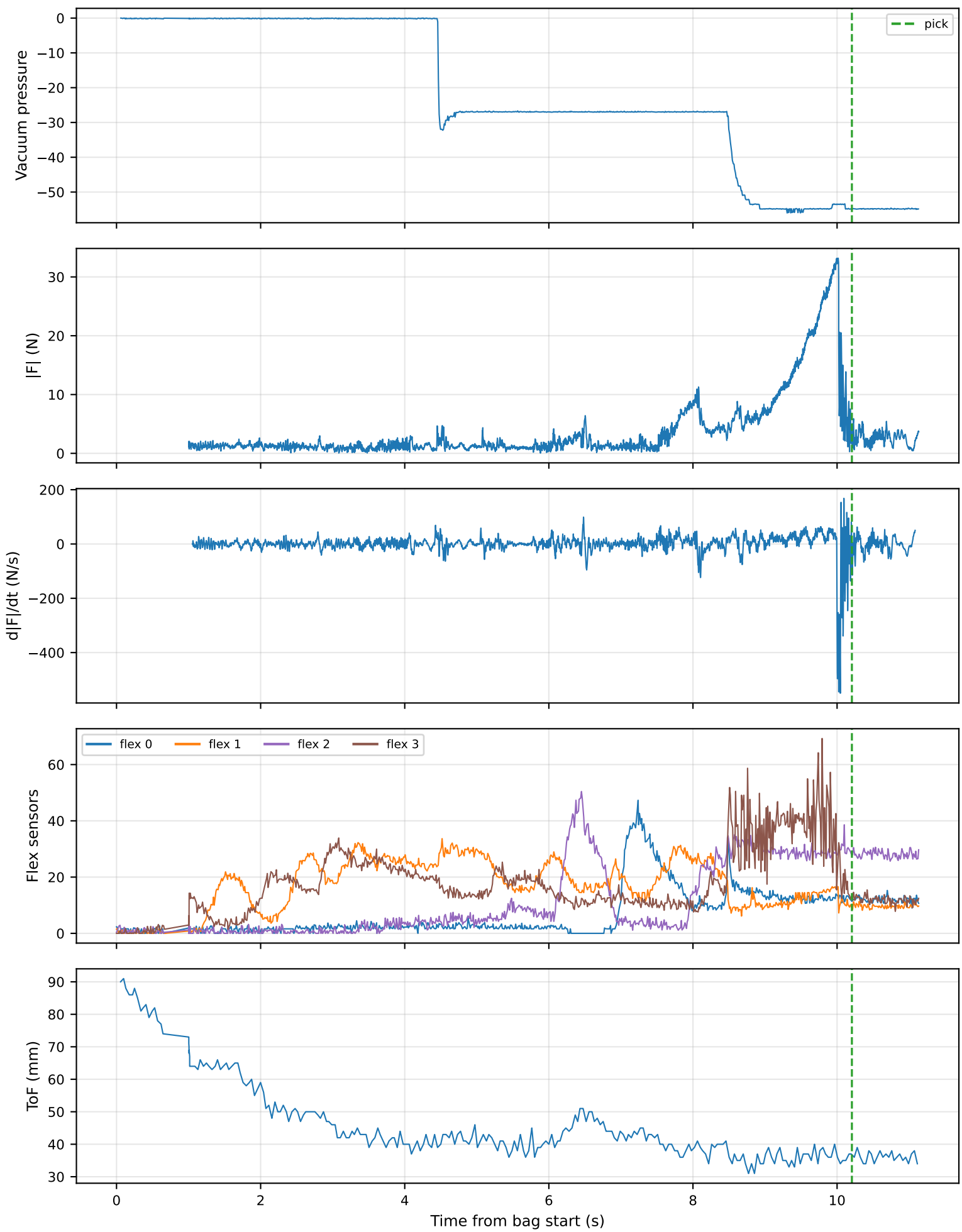
SLIP at 7.88 s | 2026-10-07 18:31:35 | batch 192, apple 0 | pull_twist_controller

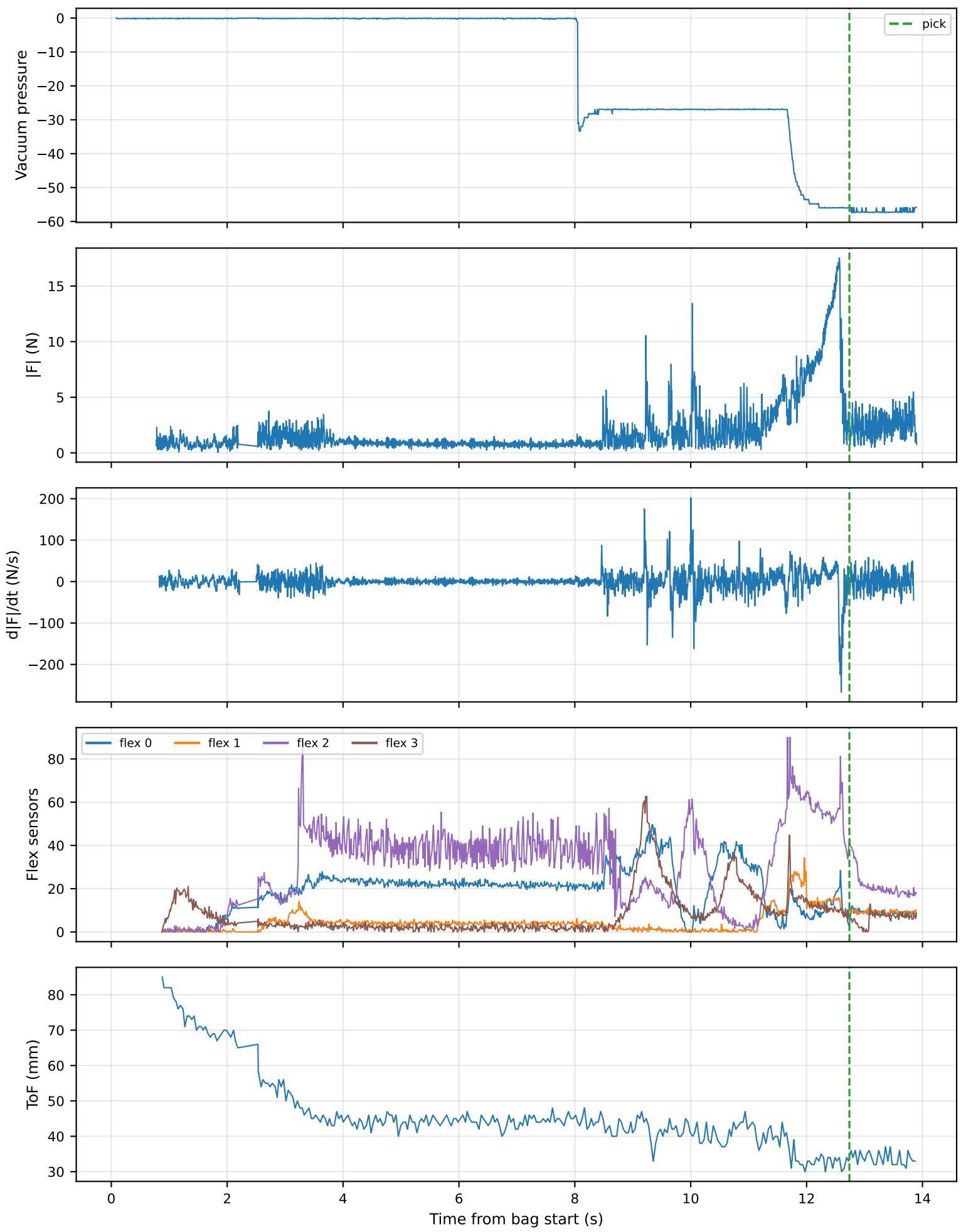


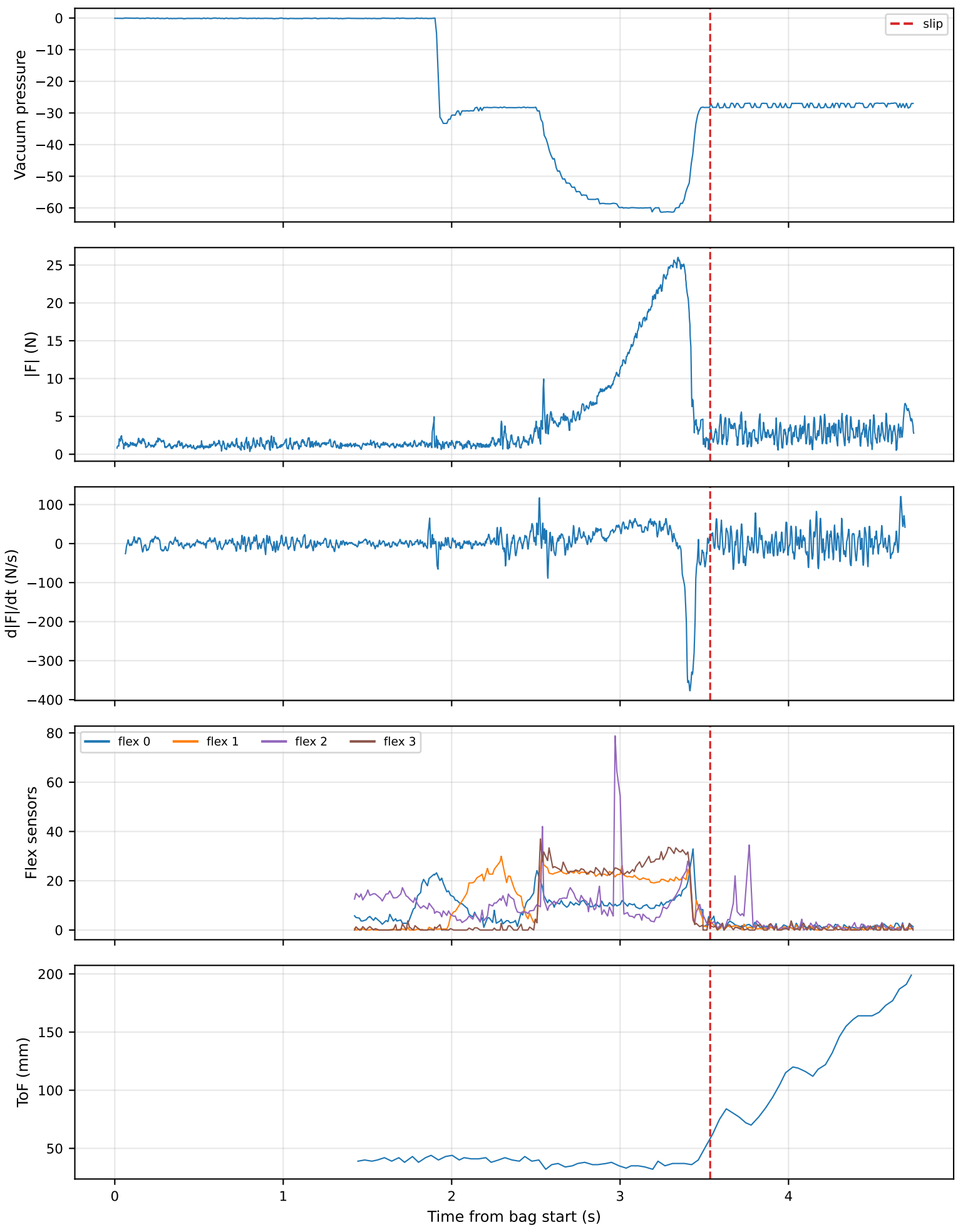

final_approach_and_pick_20261007_183222.db3
PICK at 7.57 s | 2026-10-07 18:32:23 | batch 193, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_183321.db3
SLIP at 5.20 s | 2026-10-07 18:33:22 | batch 194, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)


final_approach_and_pick_20261007_183438.db3
PICK at 5.07 s | 2026-10-07 18:34:40 | batch 195, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

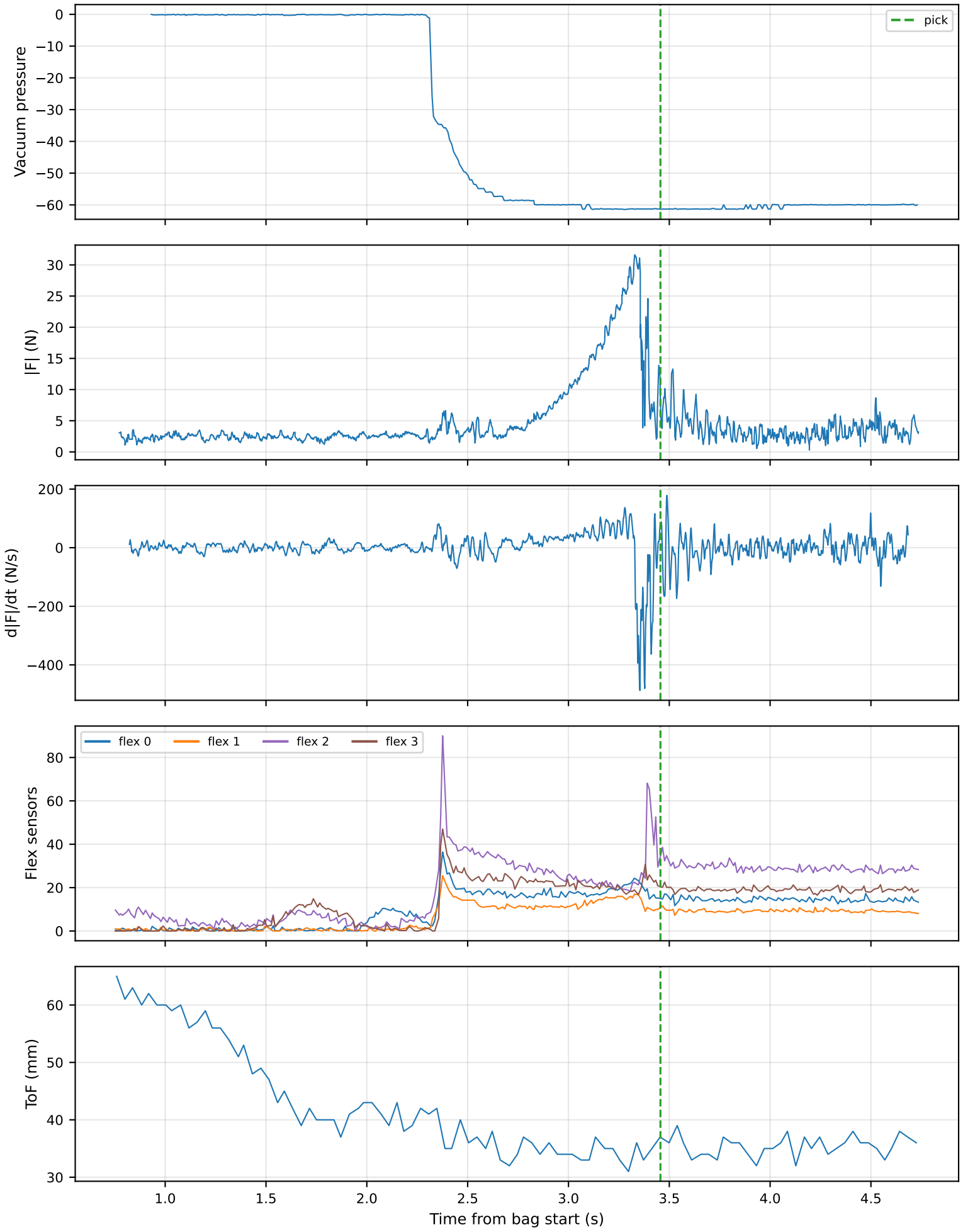

final_approach_and_pick_20261007_183605.db3
PICK at 10.21 s | 2026-10-07 18:36:06 | batch 196, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

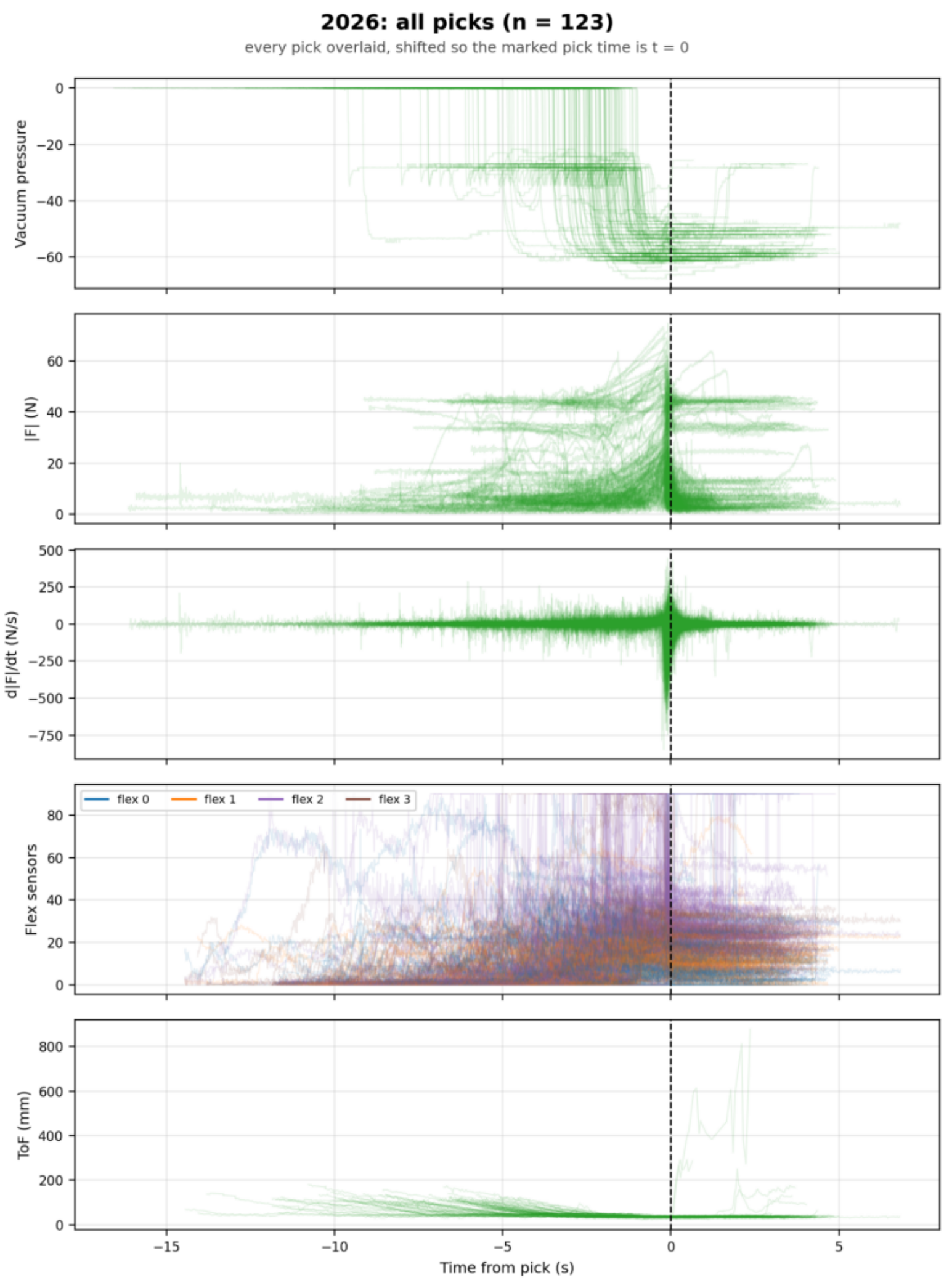

final_approach_and_pick_20261007_183700.db3
PICK at 12.74 s | 2026-10-07 18:37:01 | batch 197, apple 0 | pull_twist_controller
Vacuum pressure
pick
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

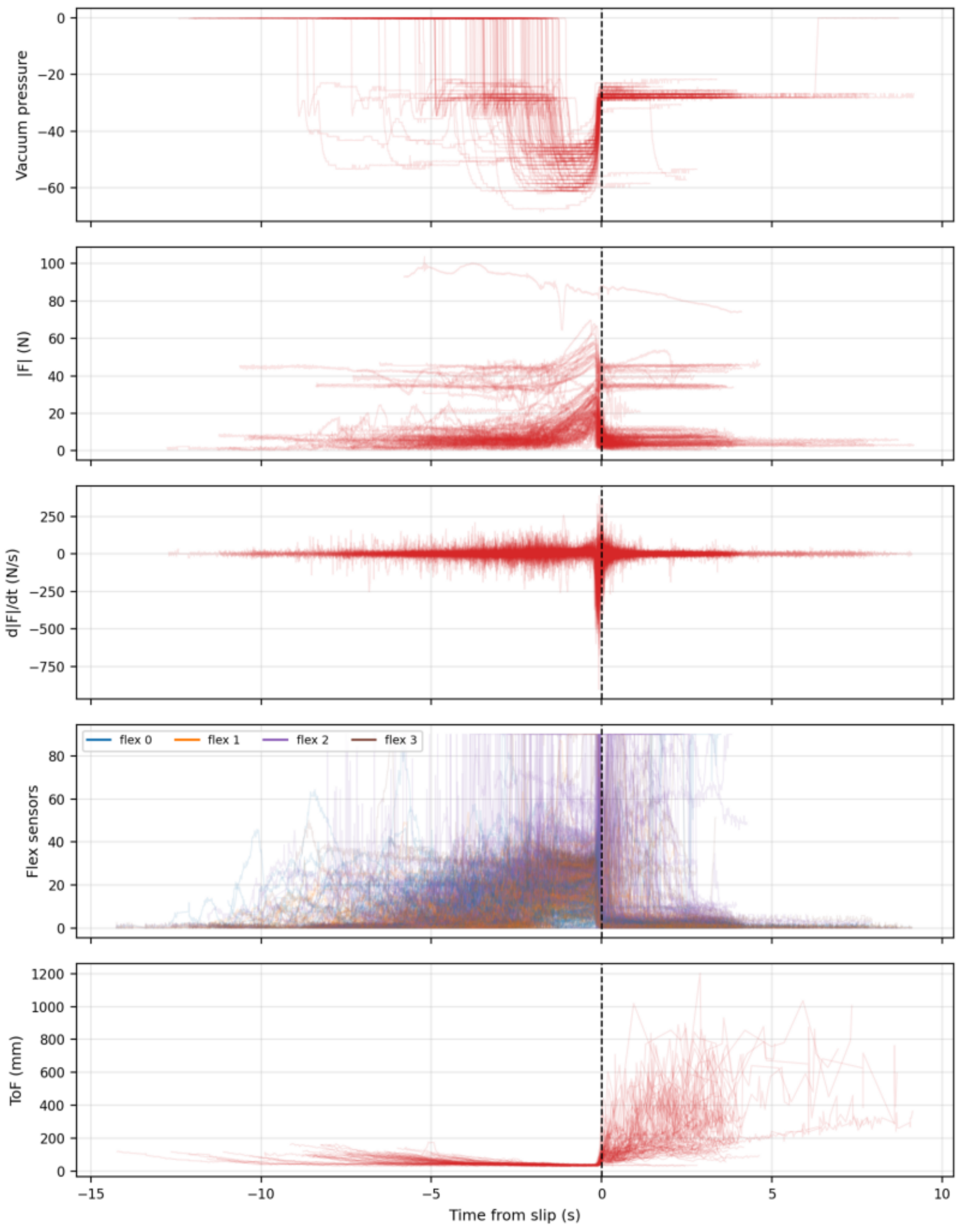

final_approach_and_pick_20261007_184032.db3
SLIP at 3.53 s | 2026-10-07 18:40:34 | batch 198, apple 0 | pull_twist_controller
Vacuum pressure
slip
|F| (N)
d|F|/dt (N/s)
Flex sensors
flex 0
flex 1
flex 2
flex 3
ToF (mm)
Time from bag start (s)

final_approach_and_pick_20261007_184120.db3

PICK at 3.46 s | 2026-10-07 18:41:21 | batch 199, apple 0 | pull_twist_controller




2026: all picks (n = 123)
every pick overlaid, shifted so the marked pick time is t = 0
Vacuum pressure
0
−20
−40
−60
|F| (N)
60
40
20
0
d|F|/dt (N/s)
500
250
0
−250
−500
−750
flex 0
flex 1
flex 2
flex 3
Flex sensors
80
60
40
20
0
ToF (mm)
800
600
400
200
0
−15
−10
−5
0
5
Time from pick (s)


2026: all slips (n = 104)
every slip overlaid, shifted so the marked slip time is t = 0
Vacuum pressure
0
−20
−40
−60
|F| (N)
100
80
60
40
20
0
d|F|/dt (N/s)
250
0
−250
−500
−750
flex 0
flex 1
flex 2
flex 3
Flex sensors
80
60
40
20
0
ToF (mm)
1200
1000
800
600
400
200
0
−15
−10
−5
0
5
10
Time from slip (s)